II श्री गणेश प्रसन्न II

CAMPBELL-WALSH-WEIN · SUPPLEMENT

# Extras from Handwritten Notes

*Arranged by book, chapter and subject*

Points from your handwritten notes that are **not** in the 572-page condensed books or the Paper 1–4 answers, grouped by book, chapter and subject. Where the notes disagreed with the books or papers, the book/paper version was kept and the point left out. "(check)" marks a point where the handwriting or figure was doubtful.

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# BP Basic Principles

**Other topics in this book**

## Tissue engineering: strategies and stem cells

▸ **Strategies**

- Three ways to create a tissue substitute: cell-based therapy alone, unseeded biomaterials, and biomaterials seeded with cells.
- In cell-based therapy, stem cells are preferred over native (differentiated) cells.

▸ **Stem cells**

- **Source:** fetal or neonatal amniotic fluid and placenta.
- **Defining property:** they easily form a clonal population.
- **Totipotent:** produce all the different cells of the organism, for example the zygote.
- **Pluripotent:** can differentiate into all three germ layers.
- **Multipotent:** produce only a closely related family of cells.

## Biomaterials

▸ **Ideal biomaterial**

- Biocompatible, with proper mechanical and physical properties.
- Also promotes cellular interaction and tissue development, so that new functional tissue develops.

▸ **Naturally derived materials (collagen, alginate, or collagen–alginate combination)**

- **Problem:** high degradation and uncontrollable mechanical properties.
- **Bioprinting:** alginate and collagen are used as bio-inks. They are printed into the desired scaffold shape and serve as the backbone for the new tissue or organ.

▸ **Collagen**

- The most abundant and ubiquitous material in the body.
- Easily purified from both animal and human tissue.
- Minimal inflammatory and antigenic responses.
- Can be processed into sponges, fibres, films and many other structures.
- Intermolecular cross-linking reduces its degradation rate.
- **Main limitation as a scaffold:** high degradation rate.

▸ **Alginate**

- A polysaccharide isolated from seaweed.
- Copolymer of D-mannuronate and L-guluronate.
- Gentle gelling in the presence of divalent cations such as calcium ($Ca^{2+}$).
- **Uses:** injectable cell-delivery vehicle and cell-immobilisation matrix.
- **Limitation:** mechanical properties are not controllable.

▸ **Acellular tissue matrices**

- Collagen-rich.
- Degrade after implantation and are replaced by extracellular matrix (ECM) proteins.

- No evidence of immunogenic rejection.

▸ **Synthetic polymers: PGA, PLA and PLGA**

- Polyesters of naturally occurring alpha-hydroxy acids.
- Thermoplastics, so easily formed into 3D scaffolds.
- Degradation products are non-toxic and are eliminated as carbon dioxide (CO2) and water (H2O).

## Tissue engineering in urology

▸ **Ureter**

- Tissue-engineered ureter is still experimental and not yet used in humans.

▸ **VUR and urinary incontinence**

- Endoscopic injection of stem cells has been used.

▸ **Testis and female infertility**

- **Testicular dysfunction (androgen deficiency):** microencapsulated Leydig cells with long-term testosterone release.
- **Female infertility:** oocytes derived from stem cells have been implanted (experimental).

▸ **Kidney**

- One of the most difficult tissues to replicate in the laboratory.
- **Selective cell therapy** (isolating one cell type with a specific function) is a promising approach, for example cultured erythropoietin-producing cells to treat anaemia in ESRD (end-stage renal disease).
- Proximal tubule, distal tubule and glomerular cells have been isolated separately in vitro, expanded and seeded onto polyglycolic acid scaffolds. This shows renal cells can be harvested and cultured.
- Embryonic renal cells harvested and expanded in vitro produced functional renal units with unidirectional urine flow.

## Iodinated contrast media

▸ **Ionic versus non-ionic agents**

| Feature | Ionic | Non-ionic |
|---|---|---|
| In blood | Dissociate into an anion and a cation | Do not dissociate (also described as "insoluble") (check) |
| Cost | Less expensive (their one advantage) | More expensive (their one disadvantage) |
| Viscosity | More | Less |

▸ **Classification and brand names**

- **Ionic high-osmolality:** iothalamate (Conray) and ioxitalamate. Diatrizoate is sold as Hypaque.
- **Non-ionic low-osmolality:** iobitridol, iopromide, ioversol, iohexol and iopamidol. Iohexol is sold as Omnipaque.
- **Non-ionic iso-osmolal:** iotrolan.

## Adverse reactions to intravascular iodinated contrast

### ▸ Classification

- Four groups: idiosyncratic anaphylactoid reactions, non-idiosyncratic reactions, delayed contrast reactions and contrast-induced nephropathy.

### ▸ Moderate reaction

- **Incidence:** 0.5–2% of patients.
- **Signs:** tachycardia or bradycardia, hypertension or hypotension, dyspnoea and pulmonary oedema, bronchospasm, laryngeal oedema and skin reactions.
- **Treatment:** close observation and hydrocortisone 100 mg to 500 mg IV or IM, plus a beta-agonist for bronchospasm.

### ▸ Severe reaction

- **Drug of choice:** epinephrine 0.01 mg/kg in 1:10,000 dilution IV. With no IV access, give the same dose as 1:1000 dilution IM.
- **Hypotension:** fluids, and dopamine may be needed, with careful monitoring in an ICU.

### ▸ Delayed reaction

- Occurs from 3 hours to 7 days after contrast and usually resolves spontaneously.

### ▸ Prevention

- Suggested drugs: H2 antagonists (with H1 antihistamines), ephedrine and corticosteroids.

## Contrast-induced nephropathy (CIN)

### ▸ Diagnosis

- There is no standard criterion for diagnosing CIN.

### ▸ Pathophysiology

- Contrast in the renal vasculature causes adenosine release from the macula densa, endothelin release and prostaglandin dysregulation. Together these produce sustained intrarenal vasoconstriction lasting hours.
- The vasoconstriction prolongs contrast transit, increasing tubular exposure to contrast.
- Oxidative stress and inflammation contribute to the cellular injury and death that lead to acute kidney injury.

### ▸ Risk factors

- **Patient-related:** hypertension; advancing age (threshold 75 years); reduced cardiac output and hypotension; class IV heart failure; hypoalbuminaemia; diuretic use.
- **Non-patient-related:** increased contrast viscosity.

### ▸ Prevention and precautions

- **Before the procedure (steps 1–5):** correct modifiable risk factors, including anaemia and hypoalbuminaemia. Discuss patients with GFR below 30 mL/min with a nephrologist.
- **During the procedure:** non-ionic contrast at the lowest dose.
- **After the procedure:** check serum creatinine and continue hydration.
- **Other agents used:** ascorbic acid and statins.

## Haemostatic agents

▸ **Classification and brands**

| Type | Brands | Action |
|---|---|---|
| Fibrin-based glues or sealants | Tisseel (Tisseel VH), Evicel, Crosseal, Vivostat | Sealant and adhesive in a dry field |
| Gelatin-based | Floseal and Surgiflo (injector-type); Surgiflo is flowable | Topical haemostasis only, no sealant effect |
| Cellulose-based | Surgicel, Oxycel, Tabotamp | Topical haemostasis only, no sealant effect |
| Microfibrillar collagen | Avitene | Topical haemostasis only, no sealant effect |
| Glutaraldehyde-based | BioGlue | Tissue sealant, not useful for active bleeding |
| PEG-polymer tissue sealants | Coseal, Advaseal | Tissue sealant, not useful for active bleeding |

▸ **Fibrin sealants**

- **Active ingredients:** fibrinogen, calcium chloride, aprotinin (except Evicel) and thrombin.
- Tisseel takes 20 minutes to set up.

▸ **Gelatin-based agents**

- **Mechanism (gelatin–thrombin):** blood percolates through the gelatin matrix and the granules swell. The patient's fibrinogen and the agent's thrombin then form the clot.
- Reabsorbed within 6–8 weeks.

▸ **Collagen-based agents**

- Come as a pre-loaded system.

▸ **BioGlue**

- Glutaraldehyde cross-links bovine serum albumin and tissue proteins.
- Needs a dry field to work.

## 3D laparoscopy

▸ **Terms**

- **Depth perception:** the visual ability to judge the relative distance of objects.
- **Stereopsis:** the brain combining two views into a single 3D image through retinal disparity.
- **Stereoscopy:** the technique of creating or enhancing the illusion of depth in an image.

▸ **Advantages**

- Besides depth perception, allows correct measurement of the dimensions of anatomical spaces.
- Retains tactile feedback.
- Low maintenance cost.
- Conventional laparoscopic instruments can be used.

## MRI

▸ **Principle**

- Protons (hydrogen nuclei) are positively charged and have magnetic spin. MRI uses this property to produce images.
- In the neutral state, protons spin haphazardly.
- The magnet's field aligns the protons in a straight line in the direction of the field.
- A radiofrequency pulse applied perpendicular to the field pushes the aligned protons to a higher energy level.
- When the pulse is removed (relaxation), the protons lose energy. The machine processes this released energy into the image.

▸ **Advantages**

- Much anatomical detail even without contrast.
- Multiple planes without moving the patient.
- Contrast agents carry a considerably smaller risk of allergic reaction than CT contrast.

▸ **Disadvantage**

- The radiofrequency transmitter can cause burns if mishandled.

▸ **Types**

- Conventional (T1- and T2-weighted), diffusion-weighted, dynamic contrast-enhanced, and MR spectroscopy or chemical shift MRI.

## Nanotechnology in urology

▸ **Definition and aims**

- **Nanomedicine:** applying nanotechnology to medicine, that is, monitoring, repair, construction and control of human biological systems at the molecular level using engineered nanodevices and nanostructures.
- **Imaging limitations it may overcome:** diagnosing malignancy early, detecting micrometastasis, and detecting recurrence at a curable stage.

▸ **Lymphotropic nanoparticle enhanced MRI (LNMRI)**

- Magnetic nanoparticles, being small, extravasate into the interstitial space, travel to lymph nodes and are taken up by macrophages there. They change the tissue's magnetic properties, which MRI detects, for example in prostate cancer nodal metastases.
- **Agents:** intravenous iron oxide or superparamagnetic iron oxide with gold nanoparticles.

▸ **Diagnostic assays**

- **Gold nanoparticle PSA assay:** detects PSA recurrence after radical prostatectomy; 300 times more sensitive than conventional assays.
- **Nanoparticle DNA assay:** detects polycystic kidney disease single nucleotide polymorphisms (PKD-SNPs) using only 0.02-0.05 ml of blood.
- **UTI:** nanosensors are used, and nanoparticle-based specific E. coli antibodies have been developed for improved detection.

▸ **Cancer therapy**

- Liposomal doxorubicin is an example of nanoparticle chemotherapy delivery exploiting the enhanced permeability and retention effect.

- **Prostate cancer drug delivery:** gelatin nanoparticles carrying paclitaxel, or transferrin-conjugated paclitaxel.
- **Prostate cancer thermal therapy:** nanocarriers have applied thermal energy (cryotherapy and HIFU) with encouraging animal results.

▸ **Bladder, OAB/IC and ED**

- **Bladder reconstruction:** 3D porous polylactide-co-glycolide and polyether urethane scaffolds have grown bladder muscle, avoiding the complications of intestinal segments.
- **OAB/IC:** weekly intravesical empty liposomes (80 mg in 40 cc water) were superior to oral pentosan polysulfate.
- **ED:** nanoparticles encapsulating tadalafil applied as a gel on the glans; liposomal PGE1 formulations for intraurethral use.

▸ **Human nephron filter (HNF)**

- A portable artificial kidney worn by the patient: two membranes in a cartridge.
- **G-membrane:** mimics the glomerulus and produces an ultrafiltrate with low molecular weight solutes.
- **T-membrane:** performs tubular reabsorption.
- **Advantages:** GFR of 30 ml/min, patient mobility and better quality of life.

▸ **Biodegradable ureteric stent**

- Poly-epsilon-caprolactone / polylactide-co-glycolide stent with nanofibres and micropores degraded gradually over 10 weeks in a porcine model.

▸ **Emerging tools**

- Nanotweezers for microsurgical procedures, nano-urobots for cystoscopy and ureteroscopy, and the nano laser (a laser with nanoscale dimensions).

▸ **Limitations**

- Ethical, socioeconomic, political and environmental concerns must be fully addressed before clinical use.
- National and international regulations are needed to prevent misuse, such as "genetic colonialism" and terrorist activities.

## Nuclear medicine: renal radionuclide agents

▸ **Classification**

- **Tubular-secreted:** iodine-131 or iodine-123 orthoiodohippurate (OIH); Tc-99m ethylenedicysteine (EC).
- **Cortical-binding:** Tc-99m glucoheptonate (GHA), like DMSA.

▸ **DMSA**

- **Excretion:** some glomerular filtration; about 60% fixed in the tubules (tubular fixation).
- Also indicated to evaluate renal space-occupying lesions.
- In acute pyelonephritis may show diffusely decreased uptake.
- **Drawbacks:** higher overall radiation dose and time-consuming.

▸ **MAG3**

- Limited availability.
- High hepatobiliary uptake.

▸ **Tc-99m ethylenedicysteine (EC)**

- Less radiation and less time-consuming than DMSA.
- Reliable differential renal function; can substitute for MAG3.
- Lower sensitivity than DMSA for cortical lesions.

▸ **DTPA and diuretic renogram**

- DTPA is the most commonly used renal agent.
- **Rationale of diuretic renogram:** to increase the sensitivity of the dynamic renal study.
- Maximal diuretic response after furosemide occurs within 15–20 minutes.
- **Advantages of renography:** short and non-nephrotoxic.
- **Pitfalls:** poor cortical imaging (DTPA); low-grade obstruction can give a false negative.

▸ **O'Reilly curves**

- **Type IIIb (equivocal, slow response to diuretic):** repeat with the F−15 protocol.
- **Type IV ("delayed compensation", Homsy's sign):** double-peak response to the diuretic. The F−15 protocol usually shows an obstructed system.

## Pneumoperitoneum: metabolic and physiological changes

▸ **Choice of gas**

- CO2 is also preferred because it is readily available.

▸ **Heart**

- Fall in cardiac output and cardiac index is compensated by a rise in heart rate, which increases myocardial oxygen demand. Young patients tolerate this; the elderly can have a myocardial infarction.

▸ **Brain**

- Raised intra-abdominal pressure compresses the IVC, reducing venous return from the brain and CSF absorption, so intracranial pressure rises.
- Absorbed CO2 causes cerebral vasodilatation and more arterial inflow, also raising intracranial pressure.
- This can be harmful in brain tumour and head injury.

▸ **Lungs**

- Diaphragmatic splinting causes poor ventilation and hypoxaemia, particularly in obese and COPD patients.

▸ **CO2 absorption**

- CO2 combines with water to form carbonic acid, which dissociates into H+ and HCO3- (CO2 + H2O, in equilibrium with H2CO3, in equilibrium with H+ + HCO3-). More H+ in blood produces acidosis.
- **Uncompensated acidosis** (when respiration is compromised, as in obesity or COPD): decreased myocardial contractility and vasodilatation.
- **Direct sympathetic effect:** peripheral vasoconstriction and raised SVR increase myocardial workload and oxygen demand, which can cause myocardial infarction in the elderly with poor cardiac reserve.

## Robotics

▸ **Surgeon console (da Vinci)**

- Resting the head on the viewport interrupts the head sensors and activates the masters, so instruments move only while the surgeon looks into the viewer.
- Two masters sit directly under the magnified 3D display.
- Foot switches control the clutch, camera, focus and cautery.

▸ **Vision cart and patient-side cart**

- **Vision cart:** also carries the flat-screen monitor, insufflator and xenon light source.
- **Patient-side cart (robotic tower):** set-up joints hold and position the camera arm and instrument arms.

▸ **Current use in adult urology**

- **Renal:** includes renal artery aneurysm repair.
- **Ureteric:** includes ileal ureter replacement and megaureter repair.

▸ **Paediatric urology**

- Benefits remain controversial.
- Rapid gastric emptying increases small-bowel distension, compromising access and visualisation.
- Lax abdominal wall gives a higher risk of bowel or vascular injury during access.
- Limited surgeon experience and few training centres.
- **Procedures performed:** include urachal cyst excision and bladder neck reconstruction.

## Virtual simulators in urology

▸ **Definition**

- A simulator is a device that lets the operator reproduce or represent test conditions or phenomena likely to occur in actual performance.
- Simulation is a "dress rehearsal" for the finale.

▸ **Why simulators are needed**

- Medicine is safety-critical, with real lives at stake, so widespread simulator use is an important adjunct to training.
- Limited resident work hours, financial constraints, fear of litigation, and an increasing number of complex cases.

▸ **Barriers**

- Small market for surgical simulators.
- Few validated for teaching.
- Can never substitute for real patient-based training.

▸ **Available simulators**

| Procedure | Simulator |
|---|---|
| Cystoscopy and ureteroscopy | UroMentor |
| Ureteroscopy | Uro-Scopic Trainer |
| TURP and TURBT | SurgicalSIM TURP, UroTrainer VR |

| | |
|---|---|
| PCNL and renal access | PERC Mentor, SimPORTAL trainer |
| Laser prostate surgery | UroSim, HoLEP simulator |
| Laparoscopic urology | Procedicus MIST |
| Robot-assisted surgery | da Vinci Skills Simulator |

## Surgical navigation, virtual reality and augmented reality

### ▸ Surgical navigation

- **Definition:** guiding surgery with computer-based technology; also called computer-aided or image-guided surgery.
- The surgeon uses special instruments tracked by the system. This matters most when the instrument tip cannot be seen, as in minimally invasive surgery.
- **Examples:**
  - iPad-assisted puncture of the pelvicalyceal system, which reduces radiation exposure.
  - Electromagnetic tracking of the puncture.
  - Computer-based tracking of ultrasound-guided kidney biopsy or puncture.
  - Marker-based endoscopic tracking during laparoscopic radical prostatectomy or partial nephrectomy, to reduce positive margins.
  - Stereotactic radiosurgery (CyberKnife), a navigation-based treatment.
- **Advantages:** more surgical precision, better ergonomics, lower risk of surgical errors.
- **Disadvantage:** huge, bulky equipment.

### ▸ Virtual reality (VR)

- On the computer, experienced mainly through two of the five senses: vision and sound.
- **Pre-operative planning:** data must be accurate and the surgical model must be built quickly.
- **VR-assisted robotic surgery:** VR gives visual access to areas not originally visible, while haptic feedback gives the impression that the surgeon's hands are inside the patient.
- Users in theory cannot tell what is real from what is virtual.

### ▸ Augmented reality (AR)

- Mainly tested in the urological and cardiovascular domains.
- **ARAUS:** augmented reality assisted urological surgery.
- Google Glass with an application showing penile prosthesis placement from start to finish has been used in a teaching programme.
- Does not need a headset.
- Users can distinguish what is real from what is virtual.

## Types of research studies

### ▸ Classification

- Epidemiological studies fall into four groups: descriptive, analytical, experimental and integrative.
- **Analytical:** ecological, case-control, cohort and cross-sectional studies.
- **Experimental:** RCTs, non-randomised controlled trials, and uncontrolled trials.
- **Integrative:** systematic reviews and meta-analyses.

▸ **Case-control study**

- An observational study of persons with the disease of interest (or another variable) and a suitable control group without it.
- Being observational, no intervention is attempted and no attempt is made to alter the course of the disease.
- Retrospective.
- **Advantages:** good for rare diseases, less time consuming, multiple risk factors studied simultaneously.
- **Disadvantages:** data quality is a problem because it is retrospective; a suitable control group can be hard to find.

▸ **Cohort study**

- An analytical study done to get more evidence to refute or support an association between a suspected cause and a disease.
- **Other names:** longitudinal, incidence and forward-looking study.

▸ **Case-control versus cohort**

| Feature | Case-control | Cohort |
|---|---|---|
| Direction | Effect to cause | Cause to effect |
| Question tested | Does the suspected cause occur more often in those with the disease than without? | Does the disease occur more often in exposed than unexposed individuals? |
| Timing | Retrospective | Retrospective or prospective |
| Subjects | Few only | Large number |
| Speed | Quick results | Long follow-up |
| Cost | Relatively inexpensive | Expensive |
| Rare diseases | Suitable | |
| Yields | Generally only an estimate of relative risk (odds ratio) | Incidence rate, relative risk (RR) and attributable risk (AR) |
| Role | Usually the first approach to testing a hypothesis | Reserved for testing a precisely formulated hypothesis |

▸ **Level of evidence and grade of recommendation (EAU 2018 / Campbell)**

- **Level 3:** also includes well-designed non-experimental comparative studies: case-control, correlation studies, case reports and case series.
- **Grade B:** directly based on level II evidence, or extrapolated from level 1 evidence.
- **Grade C:** directly based on level III evidence, or extrapolated from level 1-2 evidence.

## Randomised controlled trial (RCT)

▸ **Basic design**

- **Order of steps:** protocol, selection of a suitable population (reference or target population), selection of a suitable sample (study population), necessary exclusions, randomisation into experimental and control groups, manipulation and follow-up, and assessment.
- **Protocol:** fixes the aims and objectives, inclusion and exclusion criteria, and the intervention.

- **Reference (target) population:** the population to which the findings will apply if the trial succeeds.
- **Study (experimental) population:** drawn from the reference population, chosen at random and representative of it.
- **Exclusions before randomisation:** those not eligible and those not willing.
- **Manipulation:** deliberate application, withdrawal or reduction of the suspected causal factor, such as a drug or vaccine.
- **Follow-up:** examining study and control groups at different time intervals. Some losses, for example from death or migration, are inevitable.
- **Assessment:** in terms of positive and negative results.

▸ **Advantages, disadvantages and types**

- **Advantage:** multiple outcome variables can be measured in a single study.
- **Disadvantages:** time and cost, danger of conflict of interest, ethical restrictions, and non-compliance of controls, which threatens validity.
- **By outcome of interest:** explanatory or pragmatic trials.

▸ **Randomisation**

- Best done using random number tables.
- Simple randomisation is the easiest method.
- **Two processes:** choosing a randomisation procedure, and allocation concealment.
- **Advantages:** makes blinding possible; lets probability theory express how likely it is that any outcome difference between groups is due to chance alone.

▸ **Allocation concealment**

- **Definition:** strict precautions so that patients' group assignments are not revealed before they are finally allocated. The person randomising does not know the next allocation.
- **Methods:** sequentially numbered opaque sealed envelopes (SNOSE), numbered or coded containers, a secured computer method, a pharmacy-controlled method, and a centralised service, where the researcher calls the trial office for the allocation sequence.
- **Best method:** centralised service (central or web-based).
- Trials using allocation concealment yield up to 40% larger estimates of effect and greater heterogeneity in results (check).
- **Example:** a provider who knows which subjects will go to the intervention group may steer a subject they think will benefit more. This selection bias is what allocation concealment helps eliminate.

▸ **Blinding (masking)**

- **Definition:** concealing group allocation from one or more people involved in a study.
- **Why needed:** randomisation minimises differences between groups but does not prevent differential treatment of groups later in the trial or differential assessment of outcomes. Either can bias the treatment effect estimate; blinding minimises both.
- **Double-blind, participant:** does not know their group, so beliefs about treatment are less likely to influence the outcome.
- **Double-blind, researcher:** does not know who gets the real treatment, so is less likely to accidentally reveal subtle clues that might influence the outcome.

## Bias

▸ **Definition**

- A systematic error: disproportionate weight for or against one thing, person or group compared with another, usually in a way considered unfair.

▸ **Types**

- **Selection bias:** some individuals are more likely than others to be selected for a study. Also called "Berksonian bias" (check).
- **Funding bias:** outcomes favour the study's financial sponsor.
- **Analytical bias:** from the way results are evaluated.
- **Exclusion bias:** systematic exclusion of certain individuals.
- **Recall bias:** participants cannot recall past events accurately.
- **Observer bias:** the researcher's subconscious cognitive bias in judgement.

▸ **Confounding**

- **Positive:** observed association biased away from the null.
- **Negative:** observed association biased towards the null.

## Kaplan–Meier plots

▸ **Definition and use**

- A non-parametric method for estimating the survival function from lifetime data.
- Often used to measure the fraction of patients still alive at a certain time after treatment.
- Named after Edward L. Kaplan and Paul Meier.

▸ **Construction**

- A series of declining horizontal steps. With a large enough sample, it approaches the true survival function for that population.
- **Data needed per patient (at least two):** status at last observation, and time to the event (or to censoring).
- **x-axis:** length of each horizontal line is the survival duration for that interval.
- **y-axis:** estimated probability of survival.
- **Worked example:** percent survival against years; patients with a "Gene B signature" have lower survival than those with a "Gene A signature".

▸ **Advantages**

- **Main advantage:** includes censored data, so information about patients lost at any time, for any reason, can still be used.
- Improves professionals' understanding of disease processes.
- Useful in clinical practice for counselling patients with different diseases.

## Data types and statistical tests

▸ **Data types**

| Type | Group | Features and examples | Tests |
|---|---|---|---|
| Nominal | Qualitative (categorical) | Cannot be arranged in ascending or descending order, for example sex or eye colour | Non-parametric |

| | | | |
|---|---|---|---|
| Ordinal | Qualitative (categorical) | Can be counted or ordered ascending or descending, for example cancer stages (I, II, III, IV) or house numbers | Non-parametric |
| Interval | Quantitative (numerical) | Measured on a scale with each portion equidistant from the next, for example pain rated 1 to 10 | Parametric |
| Ratio | Quantitative (numerical) | Can be compared as multiples of one another, and multiplied or divided | Parametric |

▸ **Parametric and non-parametric tests**

- **Non-parametric test:** used for non-metric independent variables. Examples include chi-square, Mann-Whitney U, Fisher's exact and Wilcoxon tests.
- **Parametric test:** makes assumptions about the population parameter. Examples: z-test for large samples; for small samples, paired t-test, unpaired t-test, one-way ANOVA and two-way ANOVA.

▸ **Student t-test**

- **Definition:** any hypothesis test whose test statistic follows Student's t-distribution under the null hypothesis.
- **Requirements:** random sample, unknown population standard deviation, normally distributed population.
- **One-sample:** compares a population mean with a theoretical value.
- **Unpaired two-sample:** compares means of two independent samples, for example blood pressure of patients versus normal persons.
- **Paired:** compares means of two related groups, for example blood pressure before and after a drug in the same patients (n1 = n2 = n).
- **Formulas** (x is the sample mean, mu the population mean, n the sample size):
  - Significance of a sample mean (whether its difference from the population mean is significant or due to chance): t = (x - mu) / (SD x square root of 1/n).
  - Unpaired: t = (x1 - x2) / (SD x square root of (1/n1 + 1/n2)), where x1 and x2 are sample means and n1 and n2 sample sizes.
  - Paired: t = (x1 - x2) / (SD x square root of (1/n + 1/n)).

▸ **P value**

- **Definition:** the probability of finding the difference between two results by chance, that is, the probability of the null hypothesis being true.
- **Null hypothesis:** assumes no significant difference between specified populations.
- **Example:** tossing a coin, the probability of head or tail is 50%, so P = 0.5.
- **Diagram:** on a probability density curve, P is the shaded tail area beyond the observed data point, in the region of very unlikely observations, with the confidence interval marked on the set of possible results.
- The smaller the P value, the smaller the probability that the event occurred by chance.
- As the confidence interval widens, a previously significant value may become insignificant.

▸ **Sample size**

- **Definition:** the number of participants or observations in a study, usually written n.
- **Formula:** n = Z squared x p x (1 - p) / c squared, where Z = 1.96 for a 95% CI, c is the confidence interval and p the prevalence in previous studies.
- **Influences:** precision of the estimates and the power of the study to draw conclusions.

- **Power:** the probability of finding a statistically significant result.
- Adequate sample size minimises sampling error; larger samples tend to have a smaller margin of error.
- **Law of diminishing returns:** beyond a certain point, increasing sample size no longer reduces sampling error.

# 2 Oncology, DSD, Gender & Ageing

## Ch 24 · Basic Principles of Immunology and Immunotherapy in Urologic Oncology

### Angiogenesis

▸ **Definition**

- Physiological formation of new blood vessels from pre-existing vessels.
- **Versus vasculogenesis:** vasculogenesis is the differentiation of precursor cells (angioblasts) into endothelial cells and the de novo formation of a primitive vascular network.

▸ **Steps of tumour angiogenesis (4)**

- Injury to the tissue basement membrane, causing destruction and hypoxia.
- Activation of endothelial cells.
- Proliferation and stabilisation of endothelial cells.
- Continued action of angiogenic factors, forming new blood vessels.
- Angiogenic factors from tumour cells act on nearby vessels, so the tumour grows with its own blood supply and can then metastasise.

▸ **Activators**

- **Endogenous growth factors:** VEGF, PDGF and TGF, plus FGF, angiogenin, hepatocyte growth factor, epidermal growth factor, placental growth factor, TNF-alpha and granulocyte colony-stimulating factor.
- **Small molecules:** adenosine, nicotinamide, prostaglandins E1 and E2, butyryl glycerol.

▸ **Inhibitors**

- Endogenous or exogenous (drugs).
- **Matrix-derived endogenous:** arresten, canstatin, endorepellin, endostatin, fibronectin fragments, fibulin, thrombospondin 1 and 2, TIMP 1, 2 and 3.
- **Non-matrix-derived endogenous:** interferons, interleukins, platelet factor 4, angiostatin, cleaved antithrombin III.

### Apoptosis

▸ **Definition and caspases**

- Programmed cell death (cellular suicide): a process of sequential biochemical events that results in programmed cell death.
- **Caspases:** a family of protease enzymes essential in both programmed cell death and inflammation. Once activated, executioner caspases attack several intracellular target proteins.

▸ **Intrinsic pathway**

- A pro-apoptotic signal shifts the BCL-2 family balance towards the pro-apoptotic proteins: BAX, BAK, BID, BAD, BIK and BOK.
- Outer mitochondrial membrane permeability rises, releasing cytochrome c into the cytoplasm.
- Caspase-9 activates executioner caspase-3, which cleaves cellular proteins and activates nucleases, causing apoptosis.

▸ **Extrinsic pathway**

- The death receptor signal passes into the cell through its death domain; the cytosolic adaptor proteins TRADD and FADD and pro-caspase-8 attach to it to form active caspase-8.

▸ **Apoptosis versus necrosis**

| Feature | Apoptosis | Necrosis |
|---|---|---|
| Nature | Programmed, coordinated cell death | Cell degradation by hydrolytic enzymes with tissue degradation |
| Cause | Physiological and pathological processes | Hypoxia and toxins |
| Inflammation | None | Always present |
| Extent | Single cells | Many adjacent cells |
| Lysosomes and organelles | Intact | Lysosomes break down and release hydrolytic enzymes |
| Mechanism | Triggered by extrinsic and intrinsic pathways, executed by caspases | ATP-dependent membrane damage and free-radical injury |

▸ **Apoptosis and cancer**

- An intact apoptosis cascade is essential for response to immunotherapy.
- Advanced cancer cells can protect themselves from apoptosis.

## Immunotherapy

▸ **Definition and types**

- Treatment that stimulates or restores the immune system's ability to fight disease.
- **Act directly against cancer:** immune checkpoint inhibitors, adoptive cell transfer (CAR-T and CIK cells), monoclonal antibodies, cancer vaccines.
- **Enhance the body's own response:** cytokines (interferons and interleukins), BCG.

▸ **Immune checkpoint inhibitors**

- In prostate cancer, the CTLA-4 inhibitor ipilimumab benefited men with bone disease but not men with visceral metastases.

▸ **CAR-T cell therapy**

- **Production:** draw blood to collect the patient's T cells, insert the CAR gene (for example, against CD19) in the laboratory, grow millions of CAR-T cells in culture, then infuse them to bind and kill cancer cells.
- Called a "living drug" because the cells keep working inside the patient.
- **Source:** allogeneic (healthy donor's T cells) or autologous (patient's own blood).
- **Mechanism:** extensive stimulated proliferation, increased cytotoxicity, and increased secretion of cytokines, interleukins and growth factors.
- **Toxicities:**
  - Cytokine release syndrome: fever, fatigue, myalgia, tachycardia and capillary leak; can progress to cardiac, hepatic and renal failure.
  - Neurological toxicity: delirium, aphasia and seizures due to cerebral oedema.
  - Anaphylaxis.
- **Pregnancy:** no long-term safety data.

▸ **CIK (cytokine-induced killer) cell therapy**

- **Preparation:** monocytes are isolated from a blood sample and incubated with interferon-gamma, anti-CD3 antibody and recombinant human IL-1 and IL-2. The CIK cells are amplified, quality-tested and infused, where they selectively kill infected or cancer cells.
- Can be autologous or allogeneic.

▸ **Monoclonal antibodies**

| Type | Human content | Suffix |
|---|---|---|
| Murine | 0% | -omab |
| Chimeric | 65% | -ximab |
| Humanised | More than 90% | -zumab |
| Fully human | 100% | -umab |

- **Immunogenicity:** falls from high (murine) to low (fully human).
- **Common side effects:** allergic reactions, flu-like symptoms, nausea.
- **Rare serious reactions:** infusion reactions, leucopenia, skin problems, bleeding.

▸ **Cancer vaccines**

- Anti-idiotype vaccines are one type.
- **Oncophage** (HSPPC-96, heat shock protein–peptide complex-96 vaccine): approved in Russia in 2008.
- **GVAX-Prostate:** based on two prostate cancer cell lines, one androgen-sensitive and one castration-resistant; given intradermally.

## Ch 25 · Molecular Genetics and Cancer Biology

### Cancer biology basics

▸ **Attributes needed to become cancer (at least 8)**

- Genetic instability and mutation; autonomous growth; angiogenesis; resistance to apoptosis; insensitivity to internal and external antiproliferative signals; local invasion with or without metastasis; unlimited cell-division potential; immune-system evasion.

▸ **Tumour suppressor genes**

- Homozygous gene deletion is one way a tumour suppressor gene loses function.

▸ **TP53 ("guardian of the genome")**

- **Activated by:** DNA damage and oncogene activation, plus replication stress, hypoxia, viral infection and translational stress.
- **Transcriptional targets:** FAS/APO-1 mediates apoptosis; BAI1 (brain angiogenesis inhibitor 1) acts on angiogenesis.

### Circulating tumour cells (CTCs)

▸ **Biology**

- Shed from tumours as an early step in blood-borne metastasis; scarce in most tumours (less than 10/mL).
- **Entry into vessels:** EMT-mediated (epithelial–mesenchymal transition) active entry, or non-EMT-mediated passive entry.

- **Metastatic cascade:** intravasation, transit to distant organs, extravasation and invasion, then adaptation and proliferation into a metastasis.
- **Colonisation:** CTCs escape immune surveillance, extravasate and invade the foreign tissue parenchyma to form a metastasis.

▸ **Detection and use**

- Could be used for early tumour detection (CTC-positive versus CTC-negative).
- **Physical methods:** size (filtration), density (gradient centrifugation), electrical properties (dielectrophoresis).
- **Biological methods:** CellSearch system, CTC-chip.

## Gene therapy in urology

▸ **Basics**

- Genes are specific base sequences encoding instructions for proteins; these proteins carry out most life functions and make up most cell structures.
- **Roles in urology:** repair and cytoreduction. Cytoreduction works by increasing cytotoxic gene products or increasing the immune response against cancer cells.

▸ **Vectors**

- **Viral vector delivery:** virus adheres to the cell surface, is internalised in endosomes, released into the cytoplasm, transported along microtubules to the nucleus, releases its DNA, then protein is produced.
- **Liposome:** artificial lipid sphere (lipid bilayer) with an aqueous core holding the DNA and an outer protective layer against immune detection; can carry DNA across the target cell membrane.

▸ **Improving specificity**

- **Transductional targeting:** directs the vector to the target cell surface for selective adhesion and gene delivery.
- **Transcriptional targeting:** controls transcription so the therapeutic gene is produced selectively in the nucleus of target cells.
- **Targeted transgenes:** aimed at tumour-specific dysregulated metabolic pathways.
- Ideally the three approaches are combined.

▸ **Organ-specific use**

- **Kidney:**
  - Adenovirus vector for transductional targeting of the G250 tumour antigen (in vitro).
  - Ultrasound-directed non-viral gene therapy using intravenous microbubbles, with ultrasound increasing cell membrane permeability (in vitro).
- **Bladder cancer:** aims to reduce recurrence or progression; systemic delivery uses non-viral vectors to target p53.
- **Prostate cancer:** non-targeted suicide-gene (enzyme–prodrug) approach uses thymidine kinase alone or with cytosine deaminase. Specificity can be enhanced by hybrid promoters from the TARP promoter and/or PSMA and PSA enhancer elements.
- **Testis cancer:** not needed, because current treatments are excellent.
- **Erectile dysfunction:** mesenchymal stem cells transfected with VEGF have been implanted. Nitric oxide synthase gene therapy to the penile corpora stops the gradual loss of smooth

muscle cells to fibrosis.

## MicroRNA (miRNA)

### ▸ Structure and nomenclature

- Found only in eukaryotic cells; important in gene regulation (gene silencing).
- Single stranded; binds the 3' untranslated region (3' UTR) of its target mRNA.
- **Nomenclature:** "mir" is the precursor; "miR" is the mature form.
- **Gene location:** in introns, or within exons of non-coding genes.

### ▸ Biogenesis and action

- **Biogenesis:** RNA polymerase II transcribes the miRNA gene into pri-miRNA, processed to pre-miRNA in the nucleus, then to mature miRNA in the cytoplasm.
- Mature miRNA joins the RNA-induced silencing complex (RISC) to form the RNA-induced silencing unit, which targets the 3' UTR of mRNA and degrades it.
- Silences genes by mRNA degradation and inhibiting translation.
- Can act as a tumour suppressor gene or as an oncogene.
- Can modulate an entire gene programme, affecting hundreds of transcripts and shutting down entire deregulated pathways.

### ▸ Diagnostic and prognostic roles

- May have a role in disease diagnosis and management.
- May help define cancer subtypes (for example, basal versus luminal breast cancer) and the developmental origin of tumours.
- **Prostate cancer:** miR-143 and miR-187 are potential urine-based markers; miR-187 is an independent predictor of prostate cancer on biopsy.
- **Prognosis:** for example in bronchogenic carcinoma; miR-141 has a role in tumour prognosis.
- **Other diseases:** miR-122 in sepsis; roles also in cardiovascular disease, inflammation, diabetes and perioperative organ protection.

### ▸ RNA-based therapy

- **RNA interference therapeutics:** siRNA and shRNA (target a single transcript), and miRNA.
- **Single-transcript example:** PSMA-based RNA therapy in prostate cancer.
- **miRNA replacement therapy:** drug mimics of the endogenous miRNA, for example miR-16.
- **miRNA inhibitor therapy:** drugs that target the miRNA, for example miR-122.

# Ch 59 · Differences of Sexual Development

## Sex chromosome DSD

### ▸ Klinefelter syndrome

- **Karyotype variants:** 48,XXXY, 48,XXYY, 49,XXXYY, 49,XXXXY.
- **Tumour predisposition:** breast cancer, extragonadal germ cell tumour, Leydig cell tumour, Sertoli cell tumour.
- **Surveillance:** post-pubertal patients need malignancy surveillance with scrotal ultrasound every 4 months.

▸ **Turner syndrome**

- **Incidence:** about 1 in 2500 live births.
- **Cause:** non-disjunction during gametogenesis, or an error in mitosis.
- **Mosaicism:** 45,X/46,XX or 45,X/46,XY.
- **Features:**
  - Facial: ptosis, hypertelorism, retrognathia, ear malformations.
  - Body: webbed neck, broad chest with widely spaced nipples, lymphoedema and peripheral oedema at birth, pigmentation, hypoplastic nails, short 4th metacarpal, cubitus valgus.
  - Cardiac: mitral valve prolapse (a further association).
  - Renal: horseshoe kidney, plus duplication, renal agenesis, malrotation and multiple renal arteries.
  - Neurophysiological and cognitive deficits.
- **Consider it in:** any infant with lymphoedema; any young woman with short stature and primary amenorrhoea.
- **Hormones:** elevated gonadotropins with low oestrogen (and testosterone).
- **Prenatal diagnosis:** from ultrasound findings (renal, cardiac) or abnormal fetal karyotype; most affected pregnancies end in spontaneous abortion.
- **Assisted reproduction:** high rate of miscarriage, stillbirth and malformed infants.

## Ovotesticular DSD (true hermaphroditism)

▸ **Features**

- **Karyotype:** 46,XX in 60%, mosaic in 33%, 46,XY in 7%.
- **Gonads:** one or two ovotestes is more common than a separate ovary and testis. The ovary is more often on the left; the testis or ovotestis more often on the right.
- **Ducts:** an ovotestis has a fallopian tube in two-thirds and a vas in one-third.
- **Genitalia:** typically clitoromegaly or hypospadias with chordee.
- **Tumour risk:** about 3% in 46,XY patients; rare with 46,XX.

▸ **Management by sex of rearing**

- **Raised female:** potential for fertility, unlike other forms of gonadal dysgenesis. Remove all testicular and Wolffian tissue; gonadal tumour surveillance is still advisable.
- **Raised male:** historically the most common. Remove all ovarian and Müllerian tissue. Consider gonadectomy at puberty with androgen replacement, because malignancy risk is high and male fertility unlikely; if gonads are kept, long-term scrotal ultrasound surveillance is needed.

## Bilateral vanishing testis

- Must be distinguished from pure gonadal dysgenesis, in which there is no evidence of testicular function in utero.
- **Causes:** bilateral torsion, genetic mutation, teratogenesis.

## Congenital adrenal hyperplasia (CAH)

### ▸ General

- Cholesterol side-chain cleavage enzyme deficiency is one of the five enzyme defects causing CAH.

### ▸ 21-hydroxylase deficiency

- **Gene:** chromosome 6, within the HLA complex.
- **Non-salt-wasting boys:** isosexual precocity from about 2–3 years (enlarged penis, scrotum and prostate, deepening voice, pubic hair, acne, muscle development). Long term: short stature and infertility.
- **Salt-losing form:** presents in the first few weeks with progressive weight loss, dehydration and vomiting.

### ▸ 11-beta-hydroxylase deficiency

- **Gene:** chromosome 8, not HLA-linked.
- **Features:** hypertension due to raised deoxycorticosterone; severe forms show marked virilisation.
- **Diagnosis:** raised plasma 11-deoxycortisol and deoxycorticosterone (DOC).

### ▸ 3-beta-HSD deficiency

- Autosomal recessive; diagnosed by raised serum 17-OH pregnenolone (with DHEA).

### ▸ Prenatal diagnosis and treatment

- **Diagnosis:** formerly amniotic fluid 17-OH progesterone at 16–17 weeks; now HLA genotyping or DNA analysis of chorionic villus cells at 9–11 weeks.
- **Treatment:** dexamethasone to the mother, started no later than 9 weeks after the LMP (last menstrual period). It crosses the placenta, suppresses fetal ACTH and prevents virilisation.
- Only a female fetus needs it; detecting the Y chromosome by PCR of free fetal DNA as early as 9 weeks avoids treating male fetuses.

### ▸ Postnatal treatment

- **Fludrocortisone (salt-losing CAH):** 0.1–0.2 mg/day.
- **Prophylactic adrenalectomy:** an option in salt losers and extremely virilised females when adrenal suppression is harder to maintain than adrenal crisis is to prevent; gives better quality of life and fertility on replacement therapy.

## Müllerian anomalies

### ▸ Persistent Müllerian duct syndrome (PMDS)

- Uterus and upper vagina drain into the prostatic utricle.
- MIS (AMH) gene is on chromosome 19.

### ▸ MRKH syndrome

- About 10% are atypical, with a greater chance of urinary tract anomaly.

## Evaluation of the newborn with ambiguous genitalia

### ▸ Examination and imaging

- **Rectal examination:** a uterus may be palpable as an anterior midline cord-like structure.

- **Ultrasound:** a cyst within a gonad suggests an ovotestis.

▸ **Investigations**

- **Further test:** PCR characterisation of the androgen receptor.
- Impalpable gonads, Müllerian structures, normal 17-OH progesterone and 46,XX karyotype: consider non-adrenal masculinisation such as aromatase deficiency.

**Other topics in this book**

## Next-generation sequencing (NGS)

- Also called second-generation or massively parallel sequencing.
- A DNA and RNA sequencing technique distinct from first-generation (Sanger) sequencing.
- **DNA sequencing:** determining the precise order of nucleotides in a DNA molecule.
- **Advantages:** low cost, speed, high yield.
- **Clinical uses:** sequencing entire cancer genomes, detecting pathogens, discovering new tumour markers, providing accurate prognostic and diagnostic data.

# 3 Infections & Inflammation

## Ch 26 · Infections of the Urinary Tract

### General considerations

- **Factors making a UTI complicated:** renal concentrating problems (a functional abnormality); a history of childhood UTI.
- **Community-acquired organisms** include Bacteroides.
- **Hospital-acquired organisms** include Providencia.

### Pathogenesis: UPEC virulence and bacterial adherence

▸ **Virulence factors**

- Intrinsic to the bacterium (chromosomally encoded) or obtained from mobile genetic elements such as plasmids.
- **Sat:** UPEC-specific autotransporter, toxic to urinary tract cells.
- **Iron acquisition:** the IroN system as well as aerobactin.

▸ **Adherence**

- Depends on the adhesive capacity of the bacterium, the receptive capacity of the host epithelium and the fluid bathing both surfaces.
- Adhesins are fimbrial (pili, made of pilin: type 1, P, S and F1C) or afimbrial (the Dr adhesin family).

| Adhesin | Features |
|---|---|
| Type 1 pili | Repeating FimA subunits; haemagglutinate guinea pig red cells, inhibited by mannose |
| P pili | Haemagglutinate human red cells, not inhibited by mannose |
| S pili | Bind sialic acid residues; linked to both bladder and kidney infection |
| F1C pili | Bind glycosphingolipids in renal epithelial cells |
| Dr adhesins | Bind many sites throughout the urinary tract |

▸ **Phase variation**

- Has a cost: type 1 pili help adherence to and colonisation of the bladder mucosa, but also enhance phagocytosis and killing.

### Natural defence against UTI

▸ **Normal flora**

- Introitus, periurethral area and urethra: lactobacilli, corynebacteria, coagulase-negative staphylococci and streptococci.
- Changed by cervical IgA, vaginal pH, oestrogen, antimicrobials and spermicides.

▸ **Uromodulin (Tamm–Horsfall protein)**

- Saturates the mannose-binding sites of type 1 pili, blocking bacterial binding to the urothelium.

▸ **Innate immunity**

- **Toll-like receptors:** TLR-4 and TLR-11 on superficial bladder cells recognise bacterial lipopolysaccharide and peptidoglycan, triggering innate and inflammatory responses that kill bacteria.
- **Cells:** PMN leucocytes (neutrophils), macrophages, eosinophils, NK cells, mast cells and dendritic cells.
- **Nitric oxide:** PMN leucocytes increase transcription of inducible nitric oxide synthase (iNOS); high NO and its breakdown products are toxic to bacteria.

▸ **Exfoliation**

- Type 1 piliated bacteria induce exfoliation of superficial cells by programmed cell death. This can allow later invasion of deeper tissue, so it may be a bacterial offence manoeuvre rather than a host defence.
- Strong exfoliation leaves no intracellular bacterial communities (IBCs). Mild exfoliation allows IBCs and biofilms, leading to recurrent UTI.

## Conditions altering host defence

- **Renal papillary necrosis:** causes include amyloidosis and contrast media injection.
- **Diabetes:** increases asymptomatic and symptomatic UTI in women (data in men doubtful); infections are more severe and due to atypical organisms.
- **HIV:** UTIs 5-fold more common than in the normal population; complicated, recurrent and need longer treatment.

## Antimicrobial resistance

- **Inherited chromosomal:** for example, Proteus and Pseudomonas are always resistant to nitrofurantoin.
- **Transferable plasmid-mediated (R-factor):** possible with all agents except nitrofurantoin (not reported). Fluoroquinolone resistance is rarely transmitted by plasmids.
- **Nitrofurantoin:** resistance is uncommon and its mechanism not fully known.

## Clinical features of UTI

- Lower tract symptoms appear several days before upper tract symptoms.
- Clinical symptoms correlate poorly with the site of infection.

## Diagnosis: urine collection

▸ **Men**

- Circumcised: no preparation needed. Uncircumcised: retract the foreskin and clean the glans with soap and water.
- Catheterisation for a sample is not indicated unless the patient cannot void.

▸ **Women**

- Spread (separate) the labia, clean the vulva and periurethral area with moist gauze without antiseptics (they give false negatives), then take a clean-catch midstream sample.
- **Mid-catheter specimen:** if voided samples are contaminated. Because of the risk of iatrogenic infection, give a single antibiotic dose (such as oral TMP-SMX) in high-risk cases.

▸ **Suprapubic aspiration**

- Highly accurate but slightly invasive; used in neonates and SCI patients.
- **Technique:** full bladder, shave the skin, clean with an alcohol sponge, withdraw 20 mL with an LP needle (5 mL for culture, 15 mL for urinalysis).

▸ **Handling**

- Infant bag specimen: examine within 1 hour.
- If examination is delayed, store at 5 °C.
- **False negatives:** early infection (few bacteria and white cells); urine diluted by high fluid intake and diuresis.

## Urinalysis (CUE)

▸ **Physical examination**

- **Colour:** normal pale yellow from urochrome; brown-black suggests alkaptonuria.
- **Turbidity:** often phosphaturia, which clears with acetic acid. Pyuria gives turbid, pungent urine and needs microscopy. Other causes: lipiduria, hyperoxaluria, hyperuricosuria.
- **Specific gravity:** normal 1.001–1.035. Fixed at 1.010 means renal insufficiency (isosthenuria).
- **Osmolality:** 50–1200 mOsm/L; not measured on dipstick but a better indicator of renal function.
- **pH:** normal 4.5–8 (4.5–5.5 acidic, 6.5–8 alkaline). Usually reflects blood pH, except in renal tubular acidosis.

▸ **Dipstick: general**

- Tests blood, protein, glucose, ketones, urobilinogen, bilirubin and white cells on a fresh uncentrifuged specimen.
- Pyridium (phenazopyridine) makes it unreliable.

▸ **Dipstick: blood**

- **Principle:** peroxidase activity of haemoglobin oxidises orthotoluidine, which changes colour.
- **Reading:** discrete dots mean intact red cells (below about 250 RBC/mL); dots coalesce above 250/mL; a uniform field change means haemoglobinuria or myoglobinuria.
- **Separating the two:** centrifuge blood. Pink plasma means haemoglobinuria (free haemoglobin binds haptoglobin); clear plasma means myoglobinuria.
- **Normal red cell excretion:** about 1000 RBC/mL.
- **False positives:** menstrual blood, first-morning sample, dehydration, vigorous exercise.

▸ **Dipstick: glucose**

- Appears once blood glucose exceeds 180 mg/dL.
- **Principle:** double oxidation. Glucose oxidase forms gluconic acid and hydrogen peroxide, then peroxidase gives a colour change. No cross-reaction with other sugars.

▸ **Dipstick: ketones**

- Ketone bodies: acetoacetic acid, acetone and β-hydroxybutyric acid; marked as not detected on the dipstick (standard sources say only β-hydroxybutyric acid is missed) (check).
- **Principle:** sodium nitroprusside turns purple with acetoacetic acid.

### ▸ Dipstick: bilirubin and urobilinogen

- Bilirubin is absent from normal urine; a small amount of urobilinogen is normal.
- Both detected by a diazonium salt colorimetric reaction.

### ▸ Dipstick: leucocyte esterase

- **ChemStrip-LN:** neutrophil esterase converts an indoxyl carbonic acid ester to indoxyl within about 5 minutes; indoxyl oxidises the diazonium salt to change colour.
- False positive: contamination.

### ▸ Microscopy: technique

- Centrifuge 10–15 mL for 5 minutes at 3000 rpm; place the sediment (0.01–0.02 mL) on a slide under a coverslip.
- Examine at low power (100x) and high power (400x). Scan the whole coverslip area at low power, especially the edges, where casts collect.
- **Low power:** red cells, white cells, casts, parasites, cystine crystals, macrophages.
- **High power:** circular versus dysmorphic red cells, other crystals, bacteria, yeast.
- One HPF is about 1/20,000 to 1/50,000 mL of urine.
- Cytology: by Pap smear or automated flow cytometry.

### ▸ Microscopy: cells

- **Red cells:** morphology on high power or phase contrast. Can be confused with yeast (buds) or fat droplets (refractile).
- **White cells:** normal 1–2/HPF in males, up to 5/HPF in females.
- **Glitter cells:** white cells whose cytoplasmic granules show glittering movement when specific gravity is below 1.019.

### ▸ Microscopy: casts

- A coagulum of Tamm–Horsfall protein in the renal tubule that traps any luminal content.
- **Hyaline (mucoprotein):** no pathological significance.
- **Broad granular or waxy:** chronic kidney disease.
- **Fatty:** nephrotic syndrome.

### ▸ Microscopy: bacteria

- 1 bacterium/HPF means more than 30,000/mL; 5/HPF means more than 100,000 ($10^5$)/mL, the criterion used in women.
- Seen in more than 90% of infections with counts of $10^5$ CFU/mL.
- False negatives at $10^2$–$10^4$ CFU/mL, because only a limited volume is seen. A negative result never excludes 3000/mL or fewer.
- Sensitivity 40–70%, specificity 85–95% for UTI.
- Any bacteria in a male sample need urine culture.

### ▸ Microscopy: fungi and parasites

- **Candida albicans:** confused with red cells and calcium oxalate crystals, but buds. Indicates contamination, vaginal candidiasis or diabetes.
- **Parasites:** Trichomonas vaginalis, Schistosoma haematobium.

### ▸ Other findings

- Microscopic haematuria in 40–60% of cystitis.

- Sterile pyuria causes also include glomerulonephritis and interstitial cystitis.

## Urine culture and localisation

- **Direct surface plating:** split agar plate. Blood agar grows Gram-positive and Gram-negative; EMB or deoxycholate agar grows Gram-negative only.
- **Dip-slides:** dipped in the sample; MacConkey or EMB agar on one side (Gram-negative), soy agar on the other (all bacteria).
- **Timing:** usually overnight incubation; refrigerate and culture within 24 hours of collection.
- 20–40% of women with symptomatic UTI have $10^2$–$10^4$ CFU/mL.
- **Localisation by ureteral catheterisation:** irrigate the bladder thoroughly; pass two ureteric catheters into a small residual of irrigant and sample through each; then pass them into the renal pelvis and take four serial cultures from each kidney.

## Imaging in UTI

- Indications in acute pyelonephritis include a history of calculi, especially struvite, and infection with urea-splitting organisms such as Proteus.

## Uncomplicated cystitis

### ▸ Risk factors

- Dehydration (reduced urine flow).
- Bacterial ascent is helped by faecal incontinence, residual urine with bladder-wall ischaemia, catheterisation and urinary incontinence.

### ▸ Features and diagnosis

- A superficial infection of the bladder mucosa by definition, so fever, chills and other signs of spread are absent.
- Foul-smelling urine is a presenting feature.
- **Culture:** not needed in everyone; indicated for doubtful diagnosis or recent antibiotics.
- **Differential:** miscellaneous non-inflammatory causes of urethral discomfort.

### ▸ Algorithm

- No pyuria or bacteriuria: assess for vaginitis or urethritis.
- Complicating factors: culture before treatment.
- Symptoms persisting on therapy: culture on therapy. Negative needs nothing further. Positive means unresolved infection; do susceptibility testing (new antimicrobials if resistant; urological evaluation if susceptible and compliant).
- Symptoms with positive urinalysis after therapy stops: recurrence.

### ▸ Regimens

- **Women (fluoroquinolone):** ciprofloxacin 250 mg twice daily or levofloxacin 250 mg once daily for 3 days.
- **Healthy men under 50:** TMP-SMX double-strength, ciprofloxacin 500 mg twice daily or levofloxacin 500 mg once daily, each for 7 days.

### ▸ Nitrofurantoin

- Best choice after recent antibiotics, because resistance is minimal.
- Does not eliminate E. coli from the vagina.

▸ **Cephalosporins**

- Resistance among cystitis pathogens below 5%.
- Eradicate vaginal E. coli with minimal change to normal vaginal flora.

## Complicated cystitis

- Ranges from mild cystitis to life-threatening kidney infection and urosepsis.
- **Mild to moderate, outpatient, no nausea or vomiting** (E. coli, Proteus, Klebsiella, Pseudomonas): ciprofloxacin 500 mg twice daily or TMP-SMX one double-strength tablet twice daily for 10–14 days.
- **Severe, possible urosepsis or hospitalised** (Serratia, enterococci, staphylococci): parenteral ampicillin 1 g 6-hourly plus gentamicin 1 mg/kg 8-hourly.
  - Other parenteral options: ciprofloxacin 400 mg twice daily, levofloxacin 250 mg once daily, aztreonam 1 g two or three times daily, imipenem 250–500 mg three or four times daily, ticarcillin–clavulanate 3.1 g 6-hourly.
  - Complete with an oral cephalosporin or TMP-SMX for 10–14 days.

## Unresolved UTI

- **Definition:** initial therapy fails to eliminate symptoms and/or growth, with the same pathogen and resistance profile.
- **Causes:** in azotaemia and papillary necrosis, urinary antibiotic levels may fall below the MIC. In Munchausen syndrome, patients secretly inoculate the bladder and often miss doses.
- **Empirical therapy:** a fluoroquinolone for 7 days.
- **Follow-up cultures:** during therapy and 7 days after.

## Recurrent UTI

▸ **Bacterial persistence**

- Further correctable causes: ureteric duplication, infected periurethral glands, unilateral medullary sponge kidney, infected urachal cyst, infected communicating calyceal cyst, papillary necrosis, perivesical abscess with bladder fistula.
- The infected ureteric stump after nephrectomy is typically normal-appearing and non-refluxing.

▸ **Reinfection**

- Occurs at varying intervals, mostly in women and girls, from bowel flora colonisation.
- Diaphragms and nonoxynol-9 spermicides increase E. coli adherence to vaginal cells and reduce lactobacilli.
- Abnormalities reducing formation, transmission and elimination of urine increase susceptibility.

▸ **Management pathway**

- Negative culture: evaluate for vaginitis, urethritis and interstitial cystitis.
- Reinfection without risk factors and not coital: low-dose prophylaxis or self-start therapy. With risk factors: urological evaluation.
- Other strategies (no conclusive evidence): hygiene, timing and frequency of voiding, avoiding hot tubs.

▸ **Low-dose continuous prophylaxis**

- **Rationale:** bactericidal urine levels and elimination of pathogens from introital and bowel reservoirs, using agents that should not cause resistance.
- **TMP-SMX:** 40–200 mg nightly; eradicates Gram-negative aerobic flora from bowel and vagina.
- **Nitrofurantoin:** repeatedly clears urine, but introital colonisation by Enterobacteriaceae continues.
- **Cephalexin:** 250 mg once nightly.
- **Other cephalosporins:** expensive; only when the other three cannot be used (resistance or intolerance).
- **Schedule:** nightly or alternate nights; usually stopped after 6 months.
- **Effect:** recurrences fall by 95%.
- **After stopping:** 30% have spontaneous remissions of up to 6 months, but many then reinfect; treat with low-dose prophylaxis again or another strategy.
- **Monitoring:** every 1–3 months. Breakthrough infection: full dose of the same or a culture-directed drug, then restart prophylaxis.

▸ **Self-start intermittent therapy**

- Patient has a dip-slide, cultures urine when symptoms occur, then starts a 3-day full-dose empirical course.
- Fluoroquinolones ideal; nitrofurantoin or TMP-SMX alternatives. Next culture 7–10 days after therapy.

▸ **Post-coital prophylaxis**

- A single dose of nitrofurantoin, cephalexin, TMP-SMX or a fluoroquinolone.

## Acute pyelonephritis

▸ **Definition and features**

- Fever, chills and flank pain with bacteriuria and pyuria are reasonably specific for acute kidney infection; do not use the term if flank pain is absent.
- **Spectrum:** Gram-negative sepsis to a cystitis-like illness with mild flank pain.
- **Classical:** abrupt chills, fever of 100.3 °F or more, unilateral or bilateral flank pain and/or costovertebral angle tenderness.
- Rarely progresses to chronic pyelonephritis without overt symptoms. Chronic pyelonephritis develops in a complicated urinary system.

▸ **Organisms**

- **Gram-negative besides E. coli:** Proteus, Klebsiella, Pseudomonas, Enterobacter, Citrobacter, Serratia, especially with recurrent UTI, hospitalisation, catheters or instrumentation.
- **Gram-positive:** Enterococcus faecalis, Staphylococcus aureus, Staphylococcus epidermidis.

▸ **Investigations**

- About 20% have colony counts below $10^5$ CFU/mL.
- Blood cultures positive in 25%.
- Raised ESR and CRP; urine may show granular as well as leucocyte casts.

- **CT or ultrasound:** renal enlargement, hypoechoic or attenuated parenchyma, compressed collecting system, perirenal fat stranding.
- **Differential:** acute appendicitis, diverticulitis, pancreatitis, herpes zoster.

▸ **Management**

- Uncomplicated but ill (sepsis, raised white count, high fever, vomiting): admit and correct dehydration and electrolytes.
- No improvement on parenteral therapy: review cultures and sensitivities, give IV treatment for 14–21 days, drain any obstruction or abscess.
- **Follow-up cultures:** day 5–7 of therapy and 10–14 days after stopping.
- **Relapse:** 10–30% after a 14-day course; treat with a second 14-day course, rarely 6 weeks.

## Chronic pyelonephritis

- Occurs with an underlying structural or functional abnormality; uncomplicated chronic bacterial pyelonephritis is very rare.
- **Symptoms:** none unless renal insufficiency develops, then those of chronic renal failure. May have a history of recurrent acute pyelonephritis.
- **Gross:** diffusely contracted, scarred, pitted kidney with Y-shaped scars; thin parenchyma with loss of corticomedullary differentiation.
- **Histology:** patchy interstitial infiltrate of lymphocytes, plasma cells and polymorphs, with periglomerular fibrosis. Leucocyte and hyaline casts in tubules resemble thyroid colloid (renal thyroidization).
- **Pyelography:** asymmetric, irregular outlines with blunting and dilatation of one or more calyces.
- **CT:** parenchymal thinning or atrophy, calyceal clubbing or asymmetry.
- **Function:** nuclear renogram.
- **Management:** treat infection with prolonged therapy, prevent future infection, preserve renal function. Relieve obstruction and remove stones, which may enhance damage. Arrange urological and nephrological evaluation.

## Emphysematous pyelonephritis

- **Risk factors:** obstruction from papillary necrosis (as well as stones); debilitated or immunocompromised state.
- **Organisms:** Clostridium is another gas-former.
- **Features:** a sick patient with severe acute pyelonephritis: fever, flank pain and vomiting. Pneumaturia if the pelvicalyceal system is involved.
- **X-ray KUB:** mottled gas over the perirenal area; crescent of gas over the upper pole.
- **Ultrasound:** focal echoes suggest gas.
- **CT class 4 (bilateral or solitary kidney):** drain with bilateral PCN or DJ stents; follow up if responding, nephrectomy if not.
- **Additional poor prognostic factors:** diabetic ketoacidosis; raised total leucocyte count.

## Renal and perinephric abscess

### ▸ Renal abscess

- **Risk factors:** ascending Gram-negative: neurogenic bladder, bladder outlet obstruction. Haematogenous Gram-positive (staphylococcal): IV drug abuse, HIV, multiple skin carbuncles.
- **Presentation:** chills, weight loss and malaise, besides fever and flank pain.
- **Ultrasound:** low-echodensity space-occupying lesion with increased transmission.
- **CT:** acute: renal enlargement with focal rounded low-attenuation areas. Chronic: echo-free or slightly echogenic mass with a higher-attenuation wall that rim-enhances.
- **Radionuclide:** gallium or indium.
- **Small abscess:** unstable patient: drain (percutaneous or surgical) rather than antibiotics alone. Stable: antibiotics with careful observation, drainage if CT shows no resolution.

### ▸ Perinephric abscess

- **Risk:** immunocompromised patients.
- Leucocytosis, raised creatinine and pyuria in 75%.
- **Ultrasound:** anechoic mass displacing the kidney, or an echogenic collection.
- After drainage, treat the underlying problem, such as an enteric communication or pyonephrosis.

## Pyonephrosis

- Total or near-total loss of renal function.
- **Routes:** ascending and haematogenous.
- **Risk factors:** immunosuppression or AIDS, diabetes, obstruction, stones.
- **Presentation:** usually fever, chills, flank pain and tenderness; occasionally only mild symptoms; may progress to sepsis.
- **Investigations:** raised total leucocyte count; culture urine and blood.
- **Imaging:** diagnosis rests on internal echoes in the dependent part of the dilated pelvicalyceal system. CT or ultrasound may show non-specific thickened pelvis, perirenal fat stranding and hydronephrosis.
- Once stable, identify and treat the underlying source.

## Xanthogranulomatous pyelonephritis (XGP)

- **Organisms:** Proteus (commonest), E. coli, anaerobes.
- **Types:** segmental involves one or more calyceal areas. Diffuse: kidney enlarged with shape preserved; may show papillary necrosis and cortical thinning.
- **Histology:** foamy lipid-laden histiocytes with small dark nuclei and clear cytoplasm; not specific for XGP.
- **History:** usually UTI, previous calculi or urological intervention.
- **Urinalysis:** pus and protein.
- Azotaemia or renal failure is rare, as disease is usually unilateral.
- **CT triad (50–80%):** unilateral enlarged kidney, large renal pelvic calculus, little or no function.
- **Differentials:** pyonephrosis usually has a dilated pelvis; renal lymphoma is usually bilateral without stones.

- **Treatment:** incorrect diagnosis is the main obstacle. Open nephrectomy is best. If RCC versus XGP was in doubt, review the histopathology after nephrectomy.

## Malacoplakia

- Can also affect bones and mesenteric lymph nodes.
- **Michaelis–Gutmann bodies:** basophilic inclusions, intracytoplasmic or extracytoplasmic.
- **IVU:** filling defects in renal pelvis, ureter and bladder.
- **CT:** masses less dense than surrounding parenchyma; shows extension beyond the kidney.

## Urosepsis

- SIRS and sepsis-like states also have non-infectious causes such as trauma and burns.
- **Receptors:** Gram-negative lipopolysaccharide binds CD14 with TLR4 on macrophages; Gram-positive teichoic and lipoteichoic acid bind CD14 with TLR2.
- **Mediators:** pro-inflammatory cytokines include IL-2 and interferon-gamma; platelet-activating factor is also released.
- **Laboratory:** leucocytosis or leucopenia, hyperbilirubinaemia, hyperglycaemia, raised CRP. Respiratory alkalosis early; metabolic acidosis in advanced shock.
- **Organisms:** Gram-negative also include Providencia, Serratia and Enterobacter; Gram-positive are staphylococci and streptococci.
- **Vasoactive agents:** dopamine is listed with noradrenaline.
- **Antibiotic duration:** until afebrile for 3–4 days and clinically stable.
- **Source control:** may need nephrectomy for pyonephrosis or emphysematous pyelonephritis.

## Asymptomatic and catheter-associated bacteriuria

### ▸ Asymptomatic bacteriuria

- In institutionalised patients, commonly Pseudomonas and Proteus.
- Treatment may be considered for asymptomatic women whose catheter-associated bacteriuria persists 48 hours after catheter removal.

### ▸ Catheter-associated bacteriuria and CAUTI

- **Short-term catheters:** 10–30% of those with bacteriuria have typical UTI symptoms.
- **Long-term catheters:** symptomatic infection at about 1 episode per 100 catheter-days.
- **Prevention:** silver oxide (as well as silver alloy) on the catheter and hydrogen peroxide in the bag reportedly lower incidence.
- **Symptomatic CAUTI:** stop antibiotics within 48 hours of resolution.

### ▸ Catheter removal with likely bacteriuria

- When bacteriuria is likely or dipstick positive, send a culture 24 hours before removal.
- Start empirical TMP-SMX or a fluoroquinolone just before decatheterisation and continue for 2 days.
- Post-therapy culture 7–10 days later to document eradication.

## UTI in spinal cord injury (SCI)

### ▸ Prevalence and catheters

- About 33% are bacteriuric at any time; eventually nearly all are.

- **Suprapubic catheter:** bacteriuria may start later than with an indwelling urethral catheter, but the eventual rate is the same.
- **Condom drainage:** bacteriuria in 60–100%.

▸ **Risk factors**

- Impaired voiding, bladder overdistension, raised intravesical pressure, obstruction, vesicoureteric reflux, instrumentation, stones.
- Reduced fluid intake, poor hygiene, perineal colonisation, decubitus ulcers, local trauma, reduced immunity of chronic illness.

▸ **Features and organisms**

- **Other symptoms:** flank, back or abdominal discomfort, malaise or lethargy, malodorous urine.
- **Common pathogens:** enterococci, Proteus mirabilis and Pseudomonas, then E. coli, Klebsiella, staphylococci, Candida and Serratia.
- **Most persistent:** E. coli and Providencia stuartii.

▸ **Treatment and follow-up**

- **Afebrile:** oral fluoroquinolone.
- **Fever or chills:** a penicillin plus an aminoglycoside, or a third-generation cephalosporin.
- **No improvement in 24–48 hours:** re-culture, adjust to the previous culture report, and image to exclude obstruction, stones or abscess.
- **Post-therapy cultures:** usually not needed, as asymptomatic recolonisation is common and insignificant; but if a urea-splitting organism was found, culture to confirm eradication.

## Bacteriuria in the elderly

▸ **Prevalence and pathogenesis**

- After 65 years, about 20% of women and 10% of men; rises with age.
- Multifactorial and poorly understood. Probable factors: age-related decline in cell-mediated immunity, neurogenic bladder dysfunction, perineal soiling from incontinence, more catheter use, and in women oestrogen depletion with vaginal changes.
- **Organisms:** E. coli remains commonest, but Proteus, Klebsiella, Enterobacter, Serratia, Pseudomonas and enterococci become significantly more frequent.

▸ **Diagnosis and management**

- Concomitant diseases can mask or mimic UTI; keep a high index of suspicion.
- Urinary tract abnormalities are common; evaluate thoroughly (ultrasound, CT, urodynamics, cystoscopy).
- **Goal:** eliminate symptoms, not sterilise the urine.
- **Prompt therapy needed** with structural abnormalities, systemic disease such as diabetes, or a urea-splitting organism, as sepsis and its sequelae are common.
- Control systemic diseases such as diabetes and structural problems such as obstruction.
- **Fluoroquinolones:** work well, but use with caution with QT prolongation or uncorrected hypokalaemia or hypomagnesaemia.

## Bacteriuria in pregnancy

### ▸ Renal changes in pregnancy

- Renal length increases by about 1 cm (more vascular and interstitial volume); histology unchanged.
- Increased renal plasma flow, increased urinary protein excretion, RAAS upregulation.
- Hyperuricosuria as well as hypercalciuria.
- **BUN:** average falls from 13.0 to 9.0. **Creatinine:** from 0.7 to 0.5. So values normal outside pregnancy may mean renal insufficiency in pregnancy.

### ▸ Ureter, bladder and urethra changes

- Ureteric peristalsis decreases.
- Right-sided hydroureteronephrosis is commoner, because the right ureter crosses the iliac and ovarian vessels at an angle before entering the pelvis.
- **Bladder:** displaced superiorly and anteriorly; hyperaemic, possibly congested. Capacity increases (progesterone relaxes smooth muscle). Hypertrophy may be from oestrogen.
- **Urethra:** oestrogen may cause squamous changes.

### ▸ Clinical course and screening

- Pyelonephritis may also cause nausea and vomiting.
- 10–20% with pyelonephritis have another episode before or just after delivery.
- Stillbirth was a complication mainly in the pre-antibiotic era; the link with maternal anaemia is inconsistent.
- If the first-trimester screening culture is sterile, repeat cultures are unnecessary.

### ▸ Prophylaxis

- **Low-dose:** for recurrent UTI and VUR before conception.
- **Post-coital:** cephalexin 250 mg or nitrofurantoin 50 mg, once the treatment course is complete.
- **Positive follow-up culture:** unresolved infection needs a properly chosen alternative drug; persistence or rapid reinfection needs prophylaxis or suppression.

### ▸ Oral antibiotic doses

| Drug | Dose |
|---|---|
| Ampicillin (extensively used) | 500 mg four times daily |
| Amoxicillin | 250 mg three times daily |
| Penicillin V | 500 mg four times daily |
| Cephalexin (extensively used) | 500 mg four times daily |
| Cefaclor | 500 mg four times daily |
| Nitrofurantoin | 100 mg four times daily |
| Chloramphenicol | Avoid: grey baby syndrome |

## Fungal infections

### ▸ Organ involvement and risk factors

| Fungus | Main sites |
|---|---|

| | |
|---|---|
| Candidiasis | Bladder and kidney; prostate less often |
| Mucormycosis, aspergillosis | Kidney; occasionally prostate or bladder |
| Blastomycosis | Prostate, then kidney; occasionally bladder |
| Cryptococcosis | Kidney, then prostate |
| Histoplasmosis | Kidney; less often prostate |

- **Further risk factors:** old age, recent surgery, dialysis, central venous catheters.

▸ **Candidiasis**

- Candida albicans causes 50–70%.
- Biopsy gives the definitive diagnosis of fungal infection.
- **Prostatitis or epididymo-orchitis:** surgical drainage plus fluconazole 400 mg daily until imaging shows resolution.
- **Fungal ball with partial obstruction:** systemic antifungal for 6 weeks. Streptokinase or urokinase in the irrigation stops the ball blocking the drainage tube.

## Ch 27 · Inflammatory and Pain Conditions of the Lower Genitourinary Tract: Prostatitis, Orchitis, Epididymitis and Scrotal Pain

### Epidemiology

- Commonest urological diagnosis in men under 50; third commonest over 50, after BPH and prostate cancer.
- Prostatitis-like symptoms: prevalence 2.2–16%. Chronic prostatitis/CPPS: 7%.

### Aetiology

▸ **Organisms**

- **Gram-negative:** Klebsiella is among the commonest, with E. coli and Pseudomonas.
- **Gram-positive (5–10%):** S. aureus, haemolytic streptococci, coagulase-negative staphylococci.
- **Others:** Candida, viruses. Role of anaerobes unknown.

▸ **Risk factors for bacterial prostatitis**

- Altered prostatic host defence, allowing colonisation or infection.
- Phimosis, acute epididymitis, condom and indwelling catheters, dysfunction of prostatic secretion.
- **Intraprostatic ductal reflux:** causes inflammation and prostatic calculi.

▸ **CP/CPPS factors**

- **High-pressure dysfunctional voiding:** acts by autonomic overstimulation of perineal and pelvic nerves (chronic neuropathic pain) and by intraprostatic ductal reflux.
- Chemical irritation.

### Pathogenesis of CP/CPPS

- **Initiation:** infection, immunogen, toxin, trauma or stress.
- **Response:** inflammation and neurological injury.
- **Facilitation:** neuroendocrine mechanisms.

- **Propagation:** immunologic and neurologic.
- **Outcome:** neuropathic pain.

## Classification and clinical features

### ▸ Drach (traditional) classification

- Acute bacterial, chronic bacterial, non-bacterial prostatitis and prostatodynia.
- NIH IIIA corresponds to non-bacterial prostatitis; IIIB to prostatodynia.

### ▸ Category I: acute bacterial prostatitis

- Can also cause suprapubic pain, external genital pain and hypotension.
- Examination: fever, tachycardia, hypotension; hot prostate; DRE may be painful.
- About 5% progress to chronic bacterial prostatitis.

### ▸ Categories II and III

- CP/CPPS pain can also be in the groin and back.
- Examination usually unremarkable; prostate normal or enlarged and boggy, with or without nodules.

### ▸ Category IV

- Asymptomatic inflammatory prostatitis, found in asymptomatic men secondary to BPH, prostate cancer and similar conditions.

## Evaluation

### ▸ Tiered evaluation of CP/CPPS

- **Mandatory:** history, examination including DRE and pelvic floor, urinalysis, midstream culture.
- **Recommended:** 2-glass test, cystoscopy, flow rate and residual urine, cytology, NIH-CPSI, sexual history.
- **Optional:** TRUS, CT or MRI when other tests suggest an abnormality.
- Urethral symptoms (discharge, dysuria, penile pain): STI assessment. Major psychopathology: psychological evaluation.

### ▸ NIH-CPSI

- 9 questions; used for research, clinical care and follow-up.
- **Domains:** pain 4 (location, severity, frequency, relation to ejaculation); urinary 2 (voiding and storage); impact 2 (effect on work, thinking about symptoms); quality of life 1.
- **Pain score (0–21):** mild 0–7, moderate 8–13, severe 14–21.

### ▸ Localisation tests

- **Meares–Stamey 4-glass test:** gold standard but time-consuming and expensive. Follows when the 2-glass test localises bacteria.
  - VB1: first 10 mL voided; the urethral specimen.
  - EPS: collected directly into a sterile container after massage.
  - VB3: first 10 mL after massage; includes EPS trapped in the prostatic urethra.
- **2-glass test:** simple and cost-effective.
- **Category IIIA:** EPS has more than 5–10 WBC/HPF.

▸ **Imaging and markers**

- **TRUS:** detects a medial (midline) prostatic cyst; diagnoses and drains obstructed seminal vesicles.
- **Markers:** MCP-1 (monocyte chemoattractant protein-1) and MIP-1α (macrophage inflammatory protein-1 alpha) are raised in IIIA and IIIB.

## Phenotype-directed management

- **Pelvic floor muscle tenderness:** heat therapy; resolves inflammation quickly.
- **Depression and anxiety:** anxiolytics and antidepressants with behavioural therapy.
- **Neuropathic or extrapelvic:** pain clinic referral.
- **Prostate tenderness or inflammation:** prostatic massage; Cernilton (pollen extract). Phytotherapy has 5-ARI, alpha-blocker and anti-inflammatory effects.
- **Bladder:** pentosan polysulfate, quercetin, intravesical therapy.

## Minimally invasive therapy and surgery

- **Minimally invasive for CP/CPPS:** electro-acupuncture, TUMT (microwave thermotherapy), intraprostatic botulinum toxin A; almost all need further study.
- **Retention in acute bacterial prostatitis:** suprapubic catheter or a short-term (12 hours) small-calibre urethral catheter; a large or long-term catheter blocks prostatic ducts and can cause an abscess.
- **Prostatic abscess beyond the capsule:** transperineal incision and drainage through the levator ani.
- **Seminal vesicle abscess not responding to conservative therapy:** laparoscopic seminal vesiculectomy.

# Ch 28 · Interstitial Cystitis/Bladder Pain Syndrome and Related Disorders

## Definition and epidemiology

- **ICS definition (painful bladder syndrome):** suprapubic pain related to bladder filling, with other symptoms such as increased day or night frequency, without proven infection or other obvious pathology.
- Affects about 3% of females.
- No harmful effect on pregnancy.
- Men are diagnosed older and have more Hunner lesions.

## Associated disorders and aetiology

- **Associated disorders:** vulvodynia (including focal vulvitis), allergies, depression.
- **Initiating insults:** bladder trauma, overdistension and pelvic floor dysfunction damage the epithelium.
- **Antiproliferative factor:** secreted by bladder epithelial cells; stops repair, so urine constituents such as potassium leak into the interstitium.
- **Cycle:** leaked constituents activate C fibres, releasing substance P, which interacts with mast-cell histamine release and immune and allergic responses, causing progressive injury and further leakage.

## Pathology

- Biopsy rules out diagnoses including eosinophilic and radiation cystitis.
- **Patterns:** completely normal; non-ulcerative inflammation; ulcerative inflammation with common epithelial sloughing.
- **Lamina propria:** oedema, haemorrhage, congested venules, increased mast cells, inflammatory infiltrates.
- Detrusor fibrosis in 10%.

## Diagnosis

- **Mandatory:** history, examination, urinalysis.
- **Recommended:** urine culture and cytology, symptom and quality-of-life scores, frequency–volume chart, PSA, PVR, cystoscopy with hydrodistension.
- **Optional:** ultrasound, urodynamics, X-ray, potassium test, biopsy.
- **Examination standing:** kyphosis, scars, hernia. **Supine:** hip abduction and adduction, hyperaesthesia.
- **Men:** scrotum and pelvic floor muscles as well as DRE. Pain mapping of the lower limbs in both sexes.

## Conservative treatment

- **Empirical antibiotic:** a single course of doxycycline if none given before.
- **General:** exercise, warm tub baths, stress reduction.
- **Pelvic therapy for muscle relaxation:** biofeedback, soft tissue massage.
- **Diet (unsupported by literature):** avoid artificial sweeteners, hot pepper, urine-acidifying drinks, caffeine and alcohol.
- Urine alkalinisation may be worth trying.

## Oral therapy

- Does not change the natural course of BPS.

### ▸ Amitriptyline

- **Action:** central and peripheral anticholinergic, serotonin–noradrenaline reuptake inhibition, antihistaminic.
- **Side effects:** sedation, constipation, dry mouth, blurred vision, dizziness, weight gain, cardiovascular effects.
- **Contraindications also include:** unstable angina, premature ventricular contractions.

### ▸ Pentosan polysulfate

- Marketed as Cystopen-100; corrects the epithelial permeability defect.
- **Side effects:** nausea, diarrhoea, rash, reversible alopecia.

### ▸ Other oral agents

- **Immunomodulators:** suplatast tosilate, azathioprine, chloroquine derivatives, mycophenolate mofetil, adalimumab, ciclosporin.
- **Miscellaneous (no strong evidence):** L-arginine (raises urinary NOS activity), quercetin (anti-inflammatory), methotrexate, montelukast (leukotriene inhibitor), nifedipine, misoprostol (cytoprotective), PDE inhibitors, dextroamphetamine.

- **Analgesia:** NSAIDs and acetaminophen; analgesia is integral to managing chronic pain syndromes.

## Intravesical and intradetrusor therapy

▸ **DMSO (RIMSO-50)**

- Organosulfur compound from lignin (wood pulp industry); freely miscible with water, lipids and organic solvents.
- Analgesic, anti-inflammatory and muscle relaxant; desensitises nociceptive (C-fibre) pathways.
- **Cocktail:** 50 mL RIMSO-50, 10 mg triamcinolone (Kenalog), 44 mEq sodium bicarbonate and 20,000–40,000 U heparin; weekly for 6 weeks, then monthly for 6 months if the response is good.

▸ **Botulinum toxin A**

- Analgesic as well as paralytic: reduces peripheral sensitisation and affects the sensory feedback loop.

▸ **Other intravesical agents**

- Doxorubicin and capsaicin.

## Neuromodulation and surgery

- **Neuromodulation:** includes TENS; uncontrolled trials report 60–80% success for chronic pelvic pain.
- **Therapeutic hydrodistension:** damages mucosal nerve endings. Complications: haematuria, hypogastric pain, low backache, possible bladder rupture.
- **Hunner ulcers:** transurethral resection.
- **Supratrigonal cystectomy with bowel augmentation (intractable IC):** may need CIC afterwards.

## Treatment algorithm

- **Complicated PBS:** BPS with incontinence, haematuria, sterile pyuria, gynaecological symptoms or GI signs and symptoms. Needs urine cytology, urodynamics, imaging, cystoscopy and referrals. If normal, enter the first-line pathway; if abnormal, treat the other abnormalities.
- **Older six-line algorithm** (pain management throughout):
  - First: stress reduction, pain management, education, behavioural modification.
  - Second: oral and intravesical drugs.
  - Third: hydrodistension under anaesthesia and treatment of Hunner lesions.
  - Fourth: neuromodulation.
  - Fifth: ciclosporin or botulinum toxin.
  - Sixth: surgery for intractable IC.

# Ch 29 · Sexually Transmitted Diseases

## Epididymitis and orchitis: classification

- Acute bacterial: secondary to UTI or to STDs.

- Non-bacterial infectious: viral, fungal or parasitic.
- Non-infectious orchitis: idiopathic, traumatic or autoimmune.
- Chronic orchialgia or epididymalgia.
- Chronic orchitis or epididymitis.

## Renal mucormycosis

▸ **Source and spread**

- The Mucorales fungi are found in decaying vegetables and organic matter.
- Routes: inhalation, ingestion and traumatic inoculation.
- Further predisposing conditions: chronic renal failure and viral hepatitis.
- Isolated renal disease is rare; disseminated disease is the more common form.
- Disseminated disease: mortality 50–80%; kidney involved in 22% of cases.

▸ **Presentation and diagnosis**

- Can present with hypotension.
- **Laboratory:** raised total leucocyte count; work-up should include blood glucose and an ELISA for HIV.
- **Ultrasound:** enlarged kidney with a heterogeneous renal mass.
- **Contrast CT:** large hypodense mass, loss of fat planes with adjacent structures, lack of contrast excretion.
- **Renal biopsy:** broad aseptate hyphae shown with PAS stain.
- **Main differential:** renal cell carcinoma.

▸ **Treatment**

- Reversal of immunosuppression can include immune stimulation with GM-CSF.
- **Amphotericin B:** cumulative dose about 3 g; the liposomal form is less toxic than conventional.
- Other antifungals: posaconazole, itraconazole.

## HIV and Urologist: Basics and Diagnosis

▸ **Virology and epidemiology**

- HIV infects dendritic cells as well as T cells.
- HIV-1 is much more common than HIV-2.
- **Age groups:** 25–35 years commonest, then 15–25 years, then 35–45 years.

▸ **Diagnosis**

- **Screening:** conventional or rapid enzyme immunoassay (EIA), results in 30 minutes.
- **Confirmation of a positive screen:** supplemental antibody test (Western blot or indirect immunofluorescence) or an HIV-1 RNA assay.
- **Window period:** the first 3 months, when the antibody screen may still be negative.

## Interaction with Other STDs

- STDs raise HIV transmission and acquisition:
  - bleeding from genital ulcers during intercourse raises infectiousness;
  - HIV is present in genital ulcer exudates;

    - loss of mucosal integrity raises susceptibility.
- **HSV:** makes keratinocytes vulnerable to HIV and increases HIV replication.
- **Non-ulcerative STDs:** increase HIV shedding from infected people.

## Genitourinary infections in HIV

### ▸ Renal infections

- **Tuberculosis:** arises from reactivation of a latent focus, causing renal and other extrapulmonary disease.
- Rifampicin is an enzyme inducer and lowers levels of NNRTIs and protease inhibitors.
- Other renal infections: CMV, Aspergillus, Toxoplasma.

### ▸ Prostatitis

- E. coli is still the commonest cause.
- Other organisms in HIV, especially at CD4 count below 200 cells/μL: Staphylococcus aureus, Klebsiella, Pseudomonas, Serratia, Salmonella, mycobacteria, CMV and fungi (Candida, Aspergillus, Cryptococcus, Histoplasma).
- **Cultures:** cover usual organisms, atypical bacteria, fungi and Mycobacterium tuberculosis.
- **After 4–6 weeks of antibiotics:** low-dose antimicrobial suppression may be needed to prevent recurrence; lifelong suppression if immunocompromised despite ART.

### ▸ UTI

- About 40% of HIV patients with bacteriuria are asymptomatic.
- **Unusual pathogens:** Enterococcus, Proteus, parvovirus B19, Toxoplasma gondii, Aspergillus, CMV, adenovirus, Cryptococcus, Candida, mycobacteria.
- Treatment may be complicated by antibiotics already given as prophylaxis for other infections.

### ▸ Testis, epididymis and seminal vesicle

- Testis and seminal vesicles appear to be HIV reservoirs; HIV in semen is the main vector of transmission and can persist despite ART.
- HIV can cause reduced spermatogenesis and maturation arrest.
- **Testicular opportunistic infections:** CMV, Toxoplasma gondii, Mycobacterium avium-intracellulare complex.

## Renal Function (HIVAN)

- **Course:** rapid decline in renal function to irreversible renal failure.
- **Pathology:** global or focal segmental glomerulosclerosis, "fetal variant" glomerulopathy and collapsing glomerulopathy.
- **Cause:** HIV infection of renal epithelial cells (podocytes, glomerular parietal epithelial cells, tubular cells); viral genes vpr and nef.
- Lowering viral load reduces incidence; ART reduces mortality by 57%.

## Voiding, haematuria, erectile dysfunction and stones in HIV

### ▸ Voiding dysfunction

- Direct toxic effects of HIV or ART can cause LUTS; urodynamics show overactive bladder and detrusor underactivity.

### ▸ Haematuria

- Reported higher incidence in HIV; investigated as in other patients.

### ▸ Erectile dysfunction

- **Causes:** may be a manifestation of HIV-induced early ageing; psychological stress and hypogonadism contribute.
- **PDE5 inhibitors:** cleared by CYP3A, which NNRTIs inhibit, so start at the lowest possible dose.

### ▸ Stones

- Lopinavir and nelfinavir are also associated with stones, to a lesser extent than indinavir.

## Neoplasms

### ▸ Kaposi sarcoma

- Neoplasm of vascular origin.
- **Four types:** classical, African endemic, iatrogenic, epidemic (AIDS).
- Second commonest tumour in HIV; typically at CD4 counts below 150 cells/µL.
- **Pathogenesis:** HHV-8 drives proliferation of endothelial and spindle cells and angiogenesis.
- **Lesions:** pink to purple or brown pigmented, often with oedema and lymph node and visceral involvement.
- **Treatment** (depends on type):
  - ART;
  - first-line chemotherapy: pegylated liposomal doxorubicin or daunorubicin;
  - second-line: paclitaxel or docetaxel.

### ▸ Other urological cancers

- **RCC:** risk increased.
- **Testicular cancer:** risk increased, particularly seminoma and non-Hodgkin lymphoma.
- **Prostate and bladder cancer:** risk the same as the general population or even lower.
- **BCG for bladder cancer:** theoretical risk of disseminated infection.

## Post-Exposure Prophylaxis (PEP)

### ▸ Risk assessment

- **Factors affecting transmission risk:** other STDs, plasma and anogenital viral load of the source, circumcision status.
- If the source's HIV status is difficult to establish, start PEP without delay.

### ▸ Exposures

- **Fluids that may warrant PEP:** blood, blood-stained saliva, breast milk, genital secretions, CSF, amniotic, peritoneal, synovial, pericardial and pleural fluid.
- **Mucous-membrane exposures:** sexual exposure, splash to the eye, oral cavity.
- **PEP not needed:** exposed person already HIV positive; source established HIV negative; fluids of no significant risk (tears, urine, sweat, non-blood-stained saliva).

### ▸ Regimen

- Two drugs are effective, but three drugs are preferred.
- **Children:** zidovudine, lamivudine and lopinavir for 28 days.

## Classification of Anti-HIV Drugs

| Class | Drugs |
|---|---|
| Nucleoside reverse transcriptase inhibitors (NRTIs) | Zidovudine, lamivudine, stavudine, didanosine, zalcitabine, abacavir |
| Nucleotide reverse transcriptase inhibitor | Tenofovir |
| Non-nucleoside reverse transcriptase inhibitors (NNRTIs) | Nevirapine, delavirdine, efavirenz |
| Protease inhibitors | Nelfinavir, amprenavir, saquinavir, indinavir, ritonavir, lopinavir, atazanavir, fosamprenavir |
| Entry (fusion) inhibitor | Enfuvirtide |

# Ch 31 · Tuberculosis and Parasitic Infections of the Genitourinary Tract

## Genitourinary Tuberculosis (GUTB): Epidemiology

- About one-third of the world's population has latent TB.
- In developed countries, GUTB occurs in 2-10% of patients with pulmonary TB.
- Two-thirds of patients are men, usually adults.
- **Follow-up after treatment:** about 2 years for pulmonary TB, 10 years for GUTB.

## Pathogenesis

### ▸ Infection

- Inhalation of MTBC-infected aerosols into the alveoli (rarely direct inoculation), then phagocytosis by alveolar macrophages.
- If macrophages kill all inhaled bacilli, there is no infection and no immune response.

### ▸ Granuloma

- **Formation:** macrophages secrete IL-6, IL-12, IL-1 beta and TNF-alpha as chemoattractants.
- **Contents:** macrophages, foamy macrophages, multinucleated giant cells, epithelioid cells, lymphocytes.
- **Maintenance:** activated T cells secrete IL-2, TNF-alpha and IFN-gamma, which maintain it and induce killing of bacilli inside macrophages.

### ▸ Progression and reactivation

- About 5% progress to active TB within 1 year of infection.
- Malignancy is a further trigger for reactivation of latent TB.

## Development of GUTB

- **Four routes:** haematogenous, ascending (retrograde), contiguous, direct inoculation.
- Becomes evident after a prolonged latency that can be as long as 46 years.

## Pathology by organ

### ▸ Kidney

- Chronic pyelonephritis is one pathological form.

- **Caseo-cavernous autonephrectomy:** viable tissue replaced by granulomas and cavities filled with inflammatory exudate, with or without calcification.
- **Fibrotic type:** severe scarring and calcification produce a shrunken kidney.

▸ **Bladder**

- Usually begins at the ureteric orifices and spreads along lymphatics, passing through mucosal oedema, ulceration and healing by fibrosis.

▸ **Urethra**

- Very rare, because the urethra is relatively resistant to TB.

▸ **Penis**

- Can present as inflamed papules or a keratotic plaque (lupus vulgaris).
- Orificial tuberculosis carries a poor prognosis.

▸ **Prostate**

- Arises by haematogenous spread or urinary contamination.
- May present as hard nodules that must be differentiated from malignancy.

▸ **Seminal vesicle**

- Can cause low-volume ejaculate, azoospermia and oligospermia.

▸ **Epididymis**

- May present as a hard nodular epididymis containing large granulomas.

## Diagnosis: Criteria and Laboratory Tests

▸ **Diagnostic criteria**

- **Major:** granulomatous lesion, positive AFB test, positive PCR.
- **Minor:** radiological signs of TB, raised ESR, weight loss, pulmonary changes.
- Diagnosis needs 1 major or 2 minor criteria.

▸ **Culture media**

| Medium | Features |
|---|---|
| Lowenstein-Jensen (LJ) | Solid, egg-based |
| Middlebrook 7H10 | Solid, agar-based; results about 1 week earlier than LJ |
| BACTEC, MGIT | Liquid culture systems |

▸ **Nucleic acid amplification test (NAAT)**

- Detects TB even with a low bacterial load.
- **Sensitivity:** 85-95%.
- Urine can give false negatives, because it contains natural inhibitors of DNA or RNA amplification.

▸ **GeneXpert MTB/RIF**

- Detects TB even in sputum-negative cases; needs only one sample.
- **Specificity:** 99%. **Sensitivity:** 92%.
- **Limitations:** uninterrupted power supply, cost, culture and sensitivity still needed, ambient temperature below 30 C, annual calibration of Xpert modules.

▸ **Histopathology**

- The bacteria themselves are often not seen.

▸ **Tuberculin skin test (TST) and interferon test**

- **Induration 10 mm or more positive in:** recent arrivals (within 5 years) from high-prevalence countries, laboratory personnel, diabetes mellitus, chronic renal failure.
- **Induration 5 mm or more positive in:** organ transplant recipients.
- The QuantiFERON test is specifically the QuantiFERON-TB Gold In-Tube test (QFT-GIT).

## Imaging and Endoscopy

▸ **Plain film**

- Papillary necrosis: triangular, ring-like calcification.
- TB-infected renal or ureteric calculi.
- Calcification of the bladder wall, seminal vesicles and prostate.
- Erosion of vertebral bodies.

▸ **Intravenous urography (IVU)**

- Gold standard for imaging early renal TB.
- **Earliest finding:** loss of sharpness or irregularity of the calyceal edges.
- **Filling defects:** tuberculomas rupturing into a calyx, or papillary necrosis.
- **Commonest findings:** calyceal obliteration, infundibular narrowing, hydrocalycosis, segmental or total hydronephrosis, hydroureter.
- **Kerr's kink:** hiked-up renal pelvis with sharp angulation at the pelviureteric junction.
- **Tuberculous ureter:** rigid, calcified, straightened and tubular (pipe-stem), without peristalsis.
- A non-functioning tuberculous kidney does not take up contrast.

▸ **CT urography**

- Has largely replaced IVU, but early changes may be better seen on IVU.
- Detects lesions of 3-4 mm.

▸ **Other imaging**

- **Retrograde or antegrade pyelography:** when CT or IVU cannot be done, for example renal insufficiency or contrast allergy.
- **MRI:** not commonly used.

▸ **Cystoscopy and ureteroscopy**

- Limited diagnostic role; hyperaemia, mucosal erosion and oedema are non-specific.

## Medical Treatment: Principles and Regimen

- **Intensive phase:** targets rapidly multiplying bacilli.
- **Continuation phase:** eradicates slow, sporadic multipliers and persistent bacilli.
- **Beyond 6 months (usually 9 months)** for:
  - extensive pockets of infection;
  - concurrent smear-positive cavitary pulmonary TB;
  - CNS involvement;
  - delayed conversion of cultures from positive to negative;

    - inability to take pyrazinamide.
- **Second-line drugs:** reserved for failure, side effects or resistance with first-line agents.
- **Monitoring:** repeat liver function tests monthly.

## Drugs in GUTB and Dosage

▸ **Drug lists**

- **First-line** also includes rifabutin and rifapentine.
- **Second-line:** streptomycin, capreomycin, kanamycin, amikacin, levofloxacin, moxifloxacin, cycloserine, ethionamide, linezolid, bedaquiline, aminosalicylic acid.

▸ **Adult doses**

| Drug | Intermittent dose (three times weekly) |
|---|---|
| Isoniazid | 15 mg/kg (maximum 900 mg) |
| Rifampicin | 10 mg/kg (maximum 600 mg) |
| Pyrazinamide | 2500 mg |
| Ethambutol | 2000 mg |
| Streptomycin | 15 mg/kg IM or IV (maximum 1 g) |

▸ **Isoniazid**

- **Action:** prodrug activated by mycobacterial catalase-peroxidase; generates oxygen-derived free radicals and inhibits mycolic acid synthesis in the cell wall, causing DNA damage and bacterial death (bactericidal).
- Acts on extracellular and intracellular organisms; essential component of all regimens.
- **Metabolism:** hepatic acetylation.
- **Neuritis:** paraesthesia, numbness, convulsions.
- **Hepatitis:** dose-related, reversible on stopping.
- **Other side effects:** arthralgia, acne, pruritus, nausea, vomiting, epigastric pain, headache, insomnia, agitation.

▸ **Rifampicin**

- **Action:** bactericidal against intracellular and extracellular organisms; blocks DNA-dependent RNA polymerase, inhibiting gene transcription and mRNA and protein synthesis.
- **Hepatitis:** dose-related, more frequent with pre-existing hepatitis, reversible on stopping.
- **Cutaneous syndrome:** rash, flushing, pruritus.
- **Flu-like syndrome:** chills, fever, bone pain, myalgia.
- **Abdominal syndrome:** nausea, vomiting, pain, diarrhoea.
- **Also:** orange-coloured body fluids, thrombocytopenia, leukopenia.

▸ **Pyrazinamide**

- **Action:** active form pyrazinoic acid impairs mycolic acid biosynthesis by inhibiting enzymes such as fatty acid synthase I.
- **Metabolism:** liver.
- **Other side effects:** nausea, vomiting, anorexia, pruritus, rash, dermatitis, rhabdomyolysis.

▸ **Ethambutol**

- **Action:** inhibits arabinosyl transferase, blocking conversion of arabinose to arabinogalactan, the principal polysaccharide of the mycobacterial cell wall; tuberculostatic.
- **Excretion:** renal, so caution when GFR is reduced.
- Side effects are dose- and time-dependent.
- **Retrobulbar neuritis:** reversible; blurred vision, reduced visual acuity, scotomas, red-green colour loss.
- **Other side effects:** peripheral neuritis, GI symptoms (nausea, vomiting, pain), neutropenia, thrombocytopenia, headache, dizziness, confusion, hyperuricaemia or gout, hypersensitivity.

▸ **Streptomycin**

- **Action:** binds 16S rRNA of the 30S ribosomal subunit, causing codon misreading and inhibiting protein synthesis (bactericidal).
- **Side effects:** vestibular and auditory ototoxicity, nephrotoxicity, injection-site abscess, thrombocytopenia, aplastic anaemia, haemolytic anaemia, agranulocytosis.

## Surgical Treatment

- **Timing:** waiting 4-6 weeks after starting medical therapy also lets lesions stabilise.
- **Upper ureteric stricture:** endoscopic (with or without balloon dilatation, plus a double-J stent) or ureteroureterostomy.
- **Bladder neck contracture:** dilatation or transurethral incision.
- **Urethral stricture:** endoscopic, usually needing repeated procedures.
- **Urethral fistula:** medical therapy and suprapubic catheter, then delayed reconstruction.

## GUTB in special groups

▸ **Renal transplant recipients: features**

- Rare (0.35-15%), but risk of TB is 50-100 times greater.
- Shorter interval from infection to presentation, usually within 6 months, though it can occur after more than 7 years.
- Typical symptoms may be absent because of immunosuppression; usually present with fever; urinary symptoms in under 20%.
- **Chest radiograph:** commonly bilateral; fibrosis and cavitation uncommon because of immune deficiency.
- Culture and sensitivity are required for diagnosis; imaging shows little or no change because immunosuppression prevents its development.

▸ **Renal transplant recipients: treatment**

- **Rifampicin:** increases metabolism of sirolimus, ciclosporin and prednisolone; graft loss in 20-30%. Usually not given with ciclosporin; if used, double the prednisolone dose.
- **Suggested regimen:** 3 months pyrazinamide, ofloxacin, isoniazid and ethambutol; then 6 months ofloxacin, isoniazid and ethambutol; then 9 months isoniazid and ethambutol (3 ZOHE plus 6 OHE plus 9 HE).
- **Isoniazid prophylaxis:** 10 mg/kg, duration without consensus; consider after 18 months of immunosuppression in recipients with prior TB or whose donor had prior TB.

- **Side effects:** more common than in the general population; more neuropathy (calcineurin inhibitors plus isoniazid) and more isoniazid hepatotoxicity.
- Ciclosporin aggravates the hyperuricaemia caused by pyrazinamide and ethambutol.

▸ **MDR-TB and XDR-TB**

- **MDR-TB cure rate:** 50-60%.
- **XDR-TB:** extremely difficult to cure; cure rate 30-50%.

▸ **Pregnancy**

- Women of child-bearing age with GUTB should avoid pregnancy.
- GUTB found in pregnancy should be treated promptly; the fetal risk from TB outweighs the risk of drug adverse effects.

▸ **HIV**

- About 25% of HIV deaths are due to TB.
- More lymph node enlargement and more bilateral renal disease.
- Start treatment promptly; mortality is high.

## Hydatid Disease

▸ **Hosts and life cycle (Echinococcus granulosus)**

- Humans are an accidental, dead-end host; sheep and goats are the true intermediate hosts.
- **Life cycle:**
  - adult worm lives in the small intestine of dogs and passes embryonated eggs in faeces;
  - sheep and goats ingest the eggs while grazing;
  - the larva (oncosphere) hatches in the small intestine, penetrates the wall and enters the bloodstream;
  - dogs are infected by eating sheep or goat organ meat containing protoscolices from the cyst.
- **Human infection:** contact with infected dogs or eating raw, uncooked vegetables; the oncosphere penetrates the intestine (duodenum) and enters the circulation.
- The liver is the first filter and the lung the second, but larvae can reach any organ and form a hydatid cyst.

▸ **Clinical features**

- Cysts grow slowly; may take 5-20 years to become symptomatic.
- **Renal hydatid:** palpable mass, flank pain, haematuria, malaise, fever.
- Symptoms depend on location, for example adrenal, adnexal or pelvic cysts.

▸ **Investigations**

- **CT:** more sensitive than ultrasound; helps surgical planning.
- **Others:** IVU, plain X-ray KUB, MRI.
- **Eosinophilia:** 20-50% of cases.
- **Serology:** ELISA is commonly used; others are complement fixation, immunoelectrophoresis and indirect haemagglutination. Used in follow-up: titres should fall after definitive medical or surgical therapy.

▸ **Ultrasound classification**

| Gharbi type | Features | WHO equivalent |
|---|---|---|
| Type 1 | Pure fluid collection (unilocular cyst) | |
| Type 2 | Fluid collection with split wall and floating endocyst (water-lily sign) | CE3 |
| Type 3 | Multivesicular, multiseptate honeycomb pattern | CE2 |
| Type 4 (inactive) | Heterogeneous hypo- or hyperechoic cyst without daughter vesicles | CE4 and CE5 |
| Type 5 (inactive) | Partial or complete wall calcification | CE4 and CE5 |

- **WHO CL stage:** unilocular cyst without internal echoes or septa.

▸ **Medical therapy**

- **Indications:** adjunct to surgical excision, or primary therapy in poor surgical candidates or disseminated echinococcosis.
- **Preoperative:** albendazole with or without praziquantel (50 mg/kg) for 4 weeks; kills scolices, decreases cyst antigenicity, renders the cyst inactive and reduces risk of intraoperative spillage.
- **Postoperative:** praziquantel 40 mg/kg/day for 1-2 weeks to eradicate infective particles spilled at surgery.
- **Albendazole side effects:** hepatotoxicity, leucopenia, allergy, alopecia.

▸ **Surgery**

- Mainstay specifically for active, expanding, symptomatic and infected cysts.
- **Approach:** transperitoneal or retroperitoneal, open or laparoscopic.
- **Scolicidal agents:** hypertonic saline, 10% povidone-iodine, 95% ethanol, 10% cetrimide, 0.5% silver nitrate. When injected before aspiration, left for 10 minutes.
- **Palanivelu hydatid system:** laparoscopic controlled suction-irrigation system, described for hepatobiliary hydatid; removes cyst contents and endocyst without excising the whole cyst, with minimal risk of intraperitoneal spillage.
- **Deroofing:** enucleation of the contents, with removal of the endocyst and ectocyst.

## Schistosomiasis

▸ **Epidemiology and life cycle**

- Associated with poor sanitation and poverty.
- **Hosts:** snail is intermediate host; man is definitive host.
- Cercariae lose their tail after penetrating skin and become schistosomulae.
- Schistosomulae migrate to the portal blood in the liver and become adults over 1-4 weeks.
- S. haematobium eggs are shed in urine and some also in faeces.
- Eggs hatch 1-3 hours after reaching fresh water, releasing miracidia.
- In the snail: miracidia become primary sporocysts, then daughter sporocysts, then cercariae.
- Cercariae released into water live 1-4 days.

▸ **Acute schistosomiasis and Katayama fever**

- May present with a maculopapular rash; the acute phase may be silent, especially in endemic residents, or unrecognised because features are non-specific.

- **Katayama fever features:** fever, dry cough, fatigue, headache, diarrhoea, neck pain, urticaria.
- **Onset:** 2-8 weeks after primary infection.
- **Frequency:** commonest with S. japonicum, then S. mansoni; much less common with S. haematobium.
- **Treatment:** corticosteroids.

▸ **Chronic schistosomiasis**

- **Urinary:** haematuria, dysuria, frequency, proteinuria.
- **Late chronic active stage:** tissue egg burdens peak; bladder may contract to less than 50 mL from fibrosis, with chronic suprapubic and pelvic pain and irritative LUTS. Over years becomes a chronic inactive phase.
- **Female genital:** eggs in ovaries, tubes, uterus, cervix, vagina and vulva; dyspareunia, pelvic or abdominal pain, vaginal bleeding, infertility.
- **Male genital:** high egg numbers in seminal vesicles and ejaculatory ducts; testicular mass, scrotal pain, infertility.

▸ **Diagnosis**

- **Eggs:** usually seen in urine or stool 8-12 weeks after infection.
- **Tissue biopsy:** when eggs are not found but suspicion is high and serology is positive.
- **PCR:** on serum, urine or stool.
- **Circulating antigens:** circulating anodic antigen (CAA) and circulating cathodic antigen (CCA); both specific for active infection because only adult worms release them.
- **Antibodies:** Falcon Assay Screening Test-ELISA (FAST-ELISA) with Western blot, 90% sensitivity. Detectable 4-6 weeks after infection, so not useful in endemic areas.
- **Voiding cystourethrogram (VCUG):** vesicoureteric reflux in 25%.
- **Ureteric calcification:** schistosomal is intramural with a dilated ureter; tuberculous is in a non-dilated ureter.

▸ **Drugs**

- **Praziquantel side effects:** abdominal pain, nausea, headache, dizziness; mild and self-limited. Pills are large and bitter.
- **Artemisinin and analogues (artemether, artesunate):** specifically target schistosomules but are poor treatment for established infection.
- Praziquantel and artemisinin are not usually combined, because artemisinin use induces malaria resistance.

▸ **Surgery**

- **Immediate:** severe bladder haemorrhage.
- **Delayed:** complications not responding to medical therapy.
- **End-stage bladder:** options also include hydrodistension and vesical denervation; cancer surgery for malignancy.

▸ **Prevention and prognosis**

- **Prevention:** improved sanitation and hygiene; heating water (above 125 degrees for 5 minutes) and chlorination kill cercariae; mass drug treatment is the WHO strategy.
- **Prognosis:** most have mild infection with good prognosis; severe disease carries 50% mortality in 2-5 years.

## Filariasis

### ▸ Organisms and epidemiology

- Vector-borne disease; humans are the definitive host.
- **Eight filarial nematodes with humans as definitive host:**
  - lymphatic: Wuchereria bancrofti, Brugia malayi, Brugia timori;
  - subcutaneous: Loa loa, Onchocerca volvulus, Mansonella streptocerca;
  - serous cavity: Mansonella perstans, Mansonella ozzardi.
- Commoner in Africa and Asia; one third of infected patients develop clinical disease.

### ▸ Life cycle of W. bancrofti

- Adult worms in human lymphatics produce sheathed microfilariae.
- A mosquito ingests them with a blood meal.
- Microfilariae shed their sheaths, penetrate the mosquito midgut and migrate to the thoracic muscles.
- They develop from L1 to L3 larvae, migrate to the head and proboscis, and enter human skin at the next blood meal.
- Adult worms live in afferent lymphatics, especially of the lower limbs.

### ▸ Clinical features

- **Subclinical:** dilated lymphatics, scrotal lymphangiectasia, microscopic haematuria or proteinuria.
- **Acute adenolymphangitis (ADL):** often the first manifestation; fever, lymphadenitis, lymphangitis and oedema for days to a week; may cause funiculitis, epididymitis and scrotal pain and tenderness.
- **Dermatolymphangioadenitis (DLA):** fever, chills, myalgia, headache, with skin plaques, vesicles or ulcers.
- **Lymphoedema:** commoner in lower limbs, usually asymmetrical; from progressive scarring and obliteration of lymphatics.
- **Genitourinary:** commoner in males and with W. bancrofti.
- **Funiculoepididymitis:** usually self-limiting but can recur or lead to chronic lymphoedema; no sterility or orchitis.
- **Hydroceles:** usually large.
- **Scrotal and penile elephantiasis:** present late, in those with poor medical care.
- **Tropical pulmonary eosinophilia:** cough, fever, wheeze, adenopathy, eosinophilia and raised IgE; allergic response; microfilaraemia usually absent.

### ▸ Diagnosis

- **Microfilariae:** best detected on Giemsa-stained blood smear; microfilaraemia in only 30-40% of all infections.
- **Antigen detection:** ELISA, PCR, immunochromatographic card test (ICT).
- **Antibody detection:** IgG4 antibodies.
- **Lymphoscintigraphy:** shows lymphatic anatomy but is not specific.

### ▸ Diethylcarbamazine (DEC) and doxycycline

- DEC is marketed as Hetrazan-100.
- **Tropical pulmonary eosinophilia:** DEC for 2-3 weeks.

- **DEC side effects:** fever, chills, arthralgia, headache, nausea, vomiting, skin nodules.
- **Doxycycline:** augments DEC and other anthelminthics; benefits lymphoedema and hydrocele even without active infection.

▸ **Management of chronic disease**

- **Chronic lymphoedema:** good hygiene, prevention of secondary infection, elastic stockings, elevation, physiotherapy; antiparasitic therapy only for active infection.
- **Lymphoedema surgery** (lymphovenous or nodovenous anastomosis): challenging and usually unnecessary.
- **Genital elephantiasis:** rarely amenable to surgery.
- **Recurrent, painful funiculoepididymitis:** needs more radical surgery.

**Other topics in this book**

## Dimethyl Sulfoxide (DMSO)

- **Properties:** membrane penetrant, bacteriostatic, diuretic, cholinesterase inhibitor, collagen solvent, vasodilator.
- Good results only in interstitial cystitis; not useful in neurogenic or idiopathic detrusor overactivity.

## Fournier Gangrene

▸ **Names, organisms and risk factors**

- **Other names:** idiopathic gangrene of the scrotum, streptococcal scrotal gangrene, perineal phlegmon, spontaneous fulminant gangrene of the scrotum.
- **Mixed organisms:** facultative (E. coli, Klebsiella, Enterococci) with anaerobes (Bacteroides, Clostridium, microaerophilic streptococci, Fusobacterium).
- **Further risk factors:** local trauma, paraphimosis, perirectal or perianal infections, circumcision or herniorrhaphy.

▸ **Clinical features and diagnosis**

- Starts as cellulitis next to the portal of entry (swelling, erythema, tenderness), then dark purple areas and frank gangrene.
- **Systemic:** tachycardia, tachypnoea, mental status alterations.
- **Bloods:** commonly anaemia, hyponatraemia, hypocalcaemia; usually raised serum creatinine.
- Scrotal ultrasound can be useful.
- Cellulitis and necrotising fasciitis may be hard to tell apart early; systemic toxicity out of proportion to local findings should raise suspicion.

▸ **Treatment**

- **Antibiotics:** ampicillin-sulbactam is an alternative broad-spectrum option to piperacillin-tazobactam.
- **Hyperbaric oxygen** (with antibiotics and debridement): shortens hospital stay, improves wound healing, reduces gangrenous spread.
- **Reconstructive flap cover:** once the patient is stable and wound healing is complete.

## Haemorrhagic Cystitis

▸ **Causes**

- Often occurs in elderly patients with medical comorbidities.
- **Infective:** bacterial, viral, fungal, parasitic.
- **Trauma:** external, or after TURBT.
- **Malignancy:** bladder cancer; invasion or metastasis from other tumours.
- **Vascular malformations:** arteriovenous malformation, telangiectasia.
- **Chemicals:** aniline dyes, cyclophosphamide, ifosfamide, busulfan.
- **Radiation:** typically after radiotherapy for carcinoma of the prostate or cervix.
- **Drugs:** bleomycin, danazol, allopurinol; penicillin through an immune reaction.
- **Systemic illness:** rheumatoid arthritis, Crohn's disease, amyloidosis.
- **Coagulation disorders.**

▸ **Management algorithm**

- **Intravesical agent by renal function:** alum if no renal failure; aminocaproic acid (Amicar), silver nitrate or prostaglandin if renal failure.
- **Stable patient, continued bleeding:** with history of radiotherapy or cyclophosphamide, hyperbaric oxygen with or without formalin; without such history, formalin.
- **Bleeding after angioembolisation:** temporary bilateral nephrostomies, then conduit diversion with or without cystectomy.
- **Other options:** hydrostatic pressure (bladder distension), pentosan polysulfate.
- Cystopen is also listed as an option (check).

## Antibiotics

▸ **Quinolones**

- **Other side effects:** bad taste, dizziness, insomnia, anxiety, skin rash or hypersensitivity.

▸ **Cotrimoxazole**

- **Action:** sequential blockade of folate metabolism. Sulfamethoxazole inhibits folate synthase (para-aminobenzoic acid to dihydrofolic acid); trimethoprim inhibits dihydrofolate reductase (dihydrofolic acid to tetrahydrofolic acid).
- Sulfonamides are bacteriostatic, but bactericidal concentrations may be reached in urine.
- **Spectrum:** Gram-positive and Gram-negative, not anaerobes.
- **Side effects:** nausea, vomiting, stomatitis; rarely megaloblastic anaemia and blood dyscrasias. In AIDS patients treated for Pneumocystis: fever, rash, bone-marrow hypoplasia.
- **Tablets (TMP:SMX):** 80:400 mg single strength; 160:800 mg DS.

▸ **Beta-lactams**

- Have a beta-lactam ring: penicillins, cephalosporins, monobactams, carbapenems.
- **Beta-lactamase inhibitors:** clavulanic acid, sulbactam, tazobactam.

▸ **Penicillins**

- **Action:** bind penicillin-binding proteins and inhibit transpeptidase, blocking cross-linking in cell wall synthesis.
- **Adverse effects:** local irritation; hypersensitivity in 1-10% including anaphylaxis; rarely superinfection.

| Group | Drugs |
|---|---|
| Natural prototype | Penicillin G |
| Acid-resistant alternative to penicillin G | Phenoxymethylpenicillin (penicillin V) |
| Penicillinase-resistant | Methicillin, cloxacillin, dicloxacillin |
| Extended-spectrum: aminopenicillins | Ampicillin, amoxycillin, bacampicillin |
| Extended-spectrum: carboxypenicillins | Carbenicillin, ticarcillin |
| Extended-spectrum: ureidopenicillins | Piperacillin, mezlocillin |

### ▸ Piperacillin–tazobactam

- **Piperacillin spectrum:** E. coli, Klebsiella, some Bacteroides, Gram-positive organisms; frequently used for serious Gram-negative infections.
- Concurrent gentamicin or tobramycin is advised with piperacillin.
- Tazobactam is paired because its pharmacokinetics match piperacillin's.
- **Dose:** 4.5 g (4 g/500 mg) IV every 6–8 hours; adjust when creatinine clearance is below 40 mL/min.
- **Side effects:** nausea, constipation or diarrhoea, allergic reactions including anaphylaxis, thrombocytopenia.

### ▸ Cephalosporins

- Semisynthetic, from the fungus Cephalosporium.
- **Action:** inhibit cell wall synthesis like penicillins, but bind different proteins.
- **Second generation** (including oral cefuroxime axetil): weaker than first generation against Gram-positive organisms.
- **Third generation:** resistant to beta-lactamases, less active against Gram-positive organisms; includes ceftizoxime, cefoperazone (anti-pseudomonal) and cefdinir.
- **Fourth generation:** cefepime and cefpirome; spectrum similar to the third generation.
- **Side effects:** diarrhoea, disulfiram-like reaction, thrombophlebitis, hypoprothrombinaemia, pain on injection, nephrotoxicity, fungal superinfection.

### ▸ Imipenem

- Rapidly hydrolysed by dehydropeptidase-I on the brush border of renal tubular cells.
- Usually given with cilastatin, a reversible dehydropeptidase-1 inhibitor that prevents its renal metabolism.
- **Spectrum:** Gram-positive cocci, Enterobacteriaceae, Pseudomonas, Listeria, and anaerobes such as B. fragilis and C. difficile.
- **Dose:** imipenem–cilastatin 0.5 g IV every 6 hours; add gentamicin for Pseudomonas.
- **Side effects:** nausea, vomiting, diarrhoea, skin rashes, hypersensitivity; seizures at higher doses in predisposed people.

### ▸ Meropenem and other carbapenems

- Meropenem is not hydrolysed by renal peptidase, so needs no cilastatin.
- Reserve drug for serious nosocomial infections; less likely than imipenem to cause seizures.
- **Dose:** 0.5–2 g IV every 8 hours.
- **Other carbapenems:** faropenem (oral), doripenem.

▸ **Aminoglycosides**

- **Action, two steps:** transport through the bacterial cell wall and cytoplasmic membrane, then ribosome binding (16S rRNA of the 30S subunit), causing codon misreading and inhibiting protein synthesis.
- **Side effects:** vestibular or cochlear ototoxicity; nephrotoxicity from tubular damage; neuromuscular blockade.
- **Renal impairment:** 70% of adult dose at creatinine clearance 70, 50% at 50, 30% at 30; below 30, reduced doses on alternate days.

▸ **Nitrofurantoin**

- Rapidly metabolised in liver and other tissues, but excreted unchanged in urine.
- **Acute reaction:** chills, fever, leucopenia.

# 4 Paediatric Urology 1

## Ch 41 · Management Strategies for Vesicoureteral Reflux

### Demographics and inheritance

- **Prevalence in UTI, by age:** under 1 year 75%; 4 years 25%; 12 years 15%; adults 5.2%.
- **Sex:** in infants about 75% are male; in older children with persisting reflux 85% are female.
- **Fetal hydronephrosis:** common, often resolves, low specificity for postnatal VUR; about 35% have reflux after birth.
- **Inheritance:** may be autosomal dominant.

### Functional anatomy of the antireflux mechanism

- At the extravesical hiatus the ureter has three muscle layers.
- Some fibres of the inner ureteral muscle (which forms the superficial trigone) pass medially to form Mercier's bar.
- No true space separates the ureter from the bladder.
- **Opening:** active contraction of the longitudinal muscle in the tunnel draws the extravesical and intravesical points of the intramural ureter together, shortening and widening the tunnel.
- **Closure:** compression of the intramural ureter, plus return to full tunnel length as the ureteral muscle relaxes.

### Pathophysiology of reflux

- **Contributing abnormalities:** ureteral smooth muscle, extracellular matrix composition, neural function.
- **Secondary VUR:** a prolapsing ureterocele in females is an anatomical cause.

### Renal scarring and its consequences

- **Ultrasound:** a larger scar is a hyperechoic, shrunken area.
- **Risk factors:** age under 4 years after UTI; bacterial virulence; host susceptibility and response.
- **Hypertension:** from post-infection scarring nephropathy, congenital dysmorphism associated with reflux, or both.
- **Growth:** reflux correction is a poor predictor of catch-up renal growth; the somatic growth curve is one of the best predictors of renal function in children.

### Associated anomalies

▸ **Pelviureteric junction obstruction (PUJO)**

- **VUR in PUJO:** 9–18%; mostly incidental, low grade and resolves spontaneously; ureter minimally dilated but pelvis significantly dilated.
- **PUJO in VUR:** 0.75–3.6%; usually with high-grade reflux.
- **VCUG signs of coexisting PUJO:** little or no filling of the pelvis; contrast diluted in the pelvis with reduced radiodensity.
- **Hollowell categories:**

    - Primary PUJO with incidental low-grade reflux.
    - Secondary PUJO with high-grade reflux (pyeloplasty recommended in both of these).
    - Significant upper tract dilatation only.
- **Why pyeloplasty comes first:**
    - Reflux correction alone risks amplifying upper tract dilatation during postoperative oedema.
    - Reflux correction risks introducing infection into an obstructed pelvis.
    - Sterile reflux can be observed, whereas obstruction can jeopardise renal function.
    - The secondary insult to the UPJ from reflux is an evolving process not corrected by repairing the reflux.
- Pyeloplasty and endoscopic injection can be done at the same time.

▸ **Ureteral duplication**

- VUR is the most common anomaly with complete duplication.
- Mostly low grade, with no extra risk of infection or scarring.
- High-grade duplex reflux in females is more prone to infection and scarring and may need aggressive management.

▸ **Bladder diverticulum**

- A diverticulum projecting into the bladder neck can incite secondary reflux.
- Manage the reflux by its own indications, irrespective of the diverticulum.

▸ **Renal agenesis**

- Contralateral renal pathology in about 46%.
- Contralateral VUR resolves spontaneously less often than with MCDK (multicystic dysplastic kidney).

▸ **Megacystis–megaureter syndrome**

- Non-obstructive, unlike PUV.
- High chance of infection.
- Vesicostomy can be considered before ureteric reimplantation.

▸ **Syndromes**

- **VACTERL:** vertebral, anal, cardiac, tracheo-oesophageal, renal and limb anomalies.
- **CHARGE:** coloboma, heart disease, atresia of the choanae, retarded development, genital hypoplasia, ear anomalies.

## Pregnancy and reflux

- Bladder tone falls in pregnancy, predisposing to bacteriuria; prior urinary infection raises its incidence.
- **With renal scarring:** 3.3-fold higher incidence of hypertension, 7.6-fold higher risk of pre-eclampsia, more obstetric intervention.
- **Reflux with normal kidneys:** increased risk of hypertension in the third trimester.

## Diagnosis and evaluation

▸ **Confirming UTI and who to investigate**

- Store and deliver the urine sample at 4 °C.

- Colonisation (bacteriuria without pyuria) with reflux still threatens the upper tracts.
- **Traditional indications for imaging:** all children with febrile UTI; any boy with UTI, regardless of age or fever; all children under 5 years with UTI.
- In older children without fever, detecting reflux is of questionable significance.

▸ **Cystography (VCUG and ultrasound)**

- Direct cystographic ultrasound can use colour Doppler or contrast-enhanced ultrasound.
- Indirect cystography has a high false-positive rate.
- **Factors affecting direct cystography:** bladder contraction during voiding; volume instilled; infection and inflammation at the VUJ.
- During VCUG, bladder capacity is recorded when contrast inflow stops.

▸ **Radionuclide cystography (RNC)**

- **Grading against VCUG:** RNC 1 equals VCUG 1; RNC 2 (low grade) equals VCUG 2–3; RNC 3 (high grade) equals VCUG 4–5.
- **Radiation:** about 1% of the VCUG dose.
- **Uses:** screening, monitoring natural history, surgical follow-up.
- **Agent:** direct RNC uses Tc-99m pertechnetate with gamma camera detection.
- Cyclic cystography (2nd and 3rd fill–void cycles) is commonly used.

▸ **Cystoscopy and PIC (positioning the instillation of contrast)**

- Not routine. Just before surgery it shows orifice position, duplication, proximity to a diverticulum and urethral patency.
- **PIC technique:** empty the bladder; place the beak of a 9.5–14 F rigid cystoscope at the orifice; instil contrast from 1 m height with simultaneous fluoroscopic spot imaging.
- PIC is not yet standardised; high pressures may create iatrogenic reflux.

▸ **Upper tract imaging**

- **Renal ultrasound:** shows dimensions, corticomedullary differentiation and cortical abnormalities; good for large defects, may miss small ones.
- **Doppler:** may show perfusion abnormalities, with raised resistive index in higher-grade reflux and pyelonephritis.
- **DMSA:** gives a proportional representation of GFR (check).
- **Top-down approach:** normal DMSA needs no further work-up unless UTIs recur, then VCUG. Detects high-grade VUR poorly: sensitivity 79%, specificity 53%.

## Management

▸ **Principles and natural history**

- **Walker's tenets:** spontaneous resolution is very common; high-grade reflux is less likely to resolve; sterile reflux is benign; extended prophylactic antibiotics are benign; open correction is very successful.
- **Spontaneous resolution:** grade I 100%; grades IV/V 9–25%; when it occurs, it is within 5 years.

▸ **Antibiotic prophylaxis (watchful waiting)**

- Given at night, so the drug stays concentrated in bladder urine during the longest physiological retention, when infection is most likely.

- Many withdraw it at age 5. After documented radiographic resolution it is usually stopped a few days after the VCUG.
- **Breakthrough with sensitive organism:** non-compliance or dose too low.
- **Breakthrough with resistant organism:** dose too high, or residual bladder volume too high (check).

▸ **International Reflux Study in Children**

- A cooperative study with a North American arm and a European arm.

▸ **By sex and age**

- **Girls:** wait until 5 years on low-dose prophylaxis unless an indication arises; global dysmorphism of one or both kidneys indicates earlier correction.
- **Girls over 5:** correct surgically even if asymptomatic, to reduce fetal and maternal morbidity in pregnancy.
- **Boys over 5:** bilateral high-grade reflux and recurrent pyelonephritis are indications for surgery (surgery for the asymptomatic older boy with bilateral high-grade VUR is controversial).

▸ **Sibling reflux**

- Older siblings have a prevalence of about 7%.
- **Siblings over 5:** ultrasound. Normal kidneys need no further screening. Renal abnormalities prompt review of voiding habits and urological history, with screening only if these are positive.

## Surgical management

- **Principles:** attention to the ureter's entry point, the tunnel direction and ureteromucosal anastomosis, and the muscular backing of the ureter.
- Handle the bladder gently to reduce postoperative haematuria and spasm.
- **Extravesical:** Lich–Gregoir and modified Lich–Gregoir.
- **Laparoscopic:** Gil-Vernet is listed among them.

▸ **Intravesical reimplant: steps**

- Open the bladder up to 2 cm proximal to the bladder neck.
- Place a 3–5 Fr feeding tube in the ureter, sutured to the mucosa at the inferior edge of the orifice.
- Needle-point cautery makes a circumscribing incision about 1–2 mm from the orifice; mobilise the ureter with tenotomy scissors and gentle traction on the tube.
- **Politano–Leadbetter:** new hiatus about 2.5 cm superior and somewhat medial to the original.

# Ch 49 · Anomalies of the Kidney and Collecting System

## Calyceal diverticulum

- **Definition:** non-secretory, smooth-walled, spherical cavity in the corticomedullary region, usually upper or lower pole. Also called pyelogenic cyst, especially when arising from the pelvis.

- **Congenital origin:** 5 mm embryo stage, when third- and fourth-generation ureteric branches that normally degenerate persist as isolated branches.
- **Acquired causes:** obstruction from stones or infection in a calyx; spasm or dysfunction of supposed sphincters around a minor calyx.
- **IVP:** spherical contrast collection beside a papilla; the channel is hard to show; upright films show a contrast–fluid level.
- **Indications for surgery:** progressive enlargement with pain or infection; urosepsis.

| Dretler type | Mouth and neck | Suits |
|---|---|---|
| I | Open mouth, short neck | SWL |
| II | Closed mouth, short neck | URS |
| III | Closed mouth, long neck | PCNL |
| IV | Obliterated neck | PCNL |

## Bilateral renal agenesis (BRA)

### ▸ Epidemiology and genetics

- Described extensively by Potter; only about 500 cases reported.
- **Risk factor:** increasing maternal age.
- **Inheritance:** possibly autosomal recessive; autosomal dominant with variable penetrance also suggested.
- **Families of an index child:** 4.5% of siblings and parents had unilateral agenesis and 3.5% had BRA.
- **Syndromes:** oesophageal atresia (VATER), Fraser (cryptophthalmos), Klinefelter, Kallmann.

### ▸ Pathogenesis

- Absent nephrogenic ridge or failure of the ureteric bud to develop from the nephric duct, from altered molecular development or mutation of control genes on both sides.
- Renal vessels completely absent in about 25%.
- **Bladder absent or hypoplastic:** no stimulation by fetal urine, and the ureteric bud and Wolffian duct migrating to the cloaca are needed for bladder development.
- **Lungs:** in the first step of lung development the kidneys make proline, needed for collagen in the bronchiolar tree.

### ▸ Clinical features

- Low birth weight or intrauterine growth restriction.
- **Potter facies:** prematurely senile look; prominent fold from each eye curving in a semicircle over the inner canthus onto the cheek; blunted nose; prominent depression between lower lip and chin; low-set, forward-drawn ears pressed to the head with broad, large lobes; periauricular pits and tags may occur.
- **Skin and limbs:** dry, loose skin; large claw-like hands; bowed, clubbed legs with excess hip and knee flexion; occasional fused lower limbs (sirenomelia); meningocele.
- **Genitalia:** penis and vas usually normal; absent scrotum and clitoral hypertrophy described.
- **Müllerian anomalies (high incidence):** absent or hypoplastic ovaries; rudimentary or bicornuate uterus; absent vagina or short blind pouch.

- **Amnion nodosum:** small white keratinised nodules on the amnion; placental hallmark of severe oligohydramnios, not pathognomonic of BRA.

▸ **Diagnosis and screening**

- DMSA if ultrasound is inconclusive; no uptake above background in both renal fossae and any ectopic site confirms BRA.
- Prenatal and postnatal ultrasound when either parent or a first-degree relative has a solitary congenital kidney.

## Unilateral renal agenesis

- More often left-sided; autosomal dominant familial tendency.
- **Syndromes:** Poland, Fraser; about 30% of VACTERL patients have it.
- **Contralateral kidney:** ectopia or malrotation.
- **Adrenal:** ipsilateral gland may be flattened; agenesis in under 10%.
- **Men:** ampulla and ejaculatory duct may be absent; Wolffian structures sometimes ectopic or rudimentary.
- **Women:** 43% of those with genital abnormalities have it. Atypical (type II) MRKH is commoner than typical and is associated with unilateral agenesis, ectopia of one or both kidneys and horseshoe kidney.
- **Other anomalies:** cardiovascular 15%, gastrointestinal 9%, haematological 6%, neurological 3%.
- **Involuted kidneys:** about 5% of apparent cases are dysplastic kidneys or MCDKs that involuted before birth. On plain film the splenic or hepatic flexure in its normal place (not in the empty fossa, as in true agenesis) suggests involution. MCDK shows curvilinear calcification on X-ray or CT.
- Complex anomalies need imaging of all organs, including the spine.
- Postnatal colour Doppler shows absent ipsilateral renal vessels.
- **Solitary kidney prognosis:** 50% probability of dialysis by age 30.

## Supernumerary kidney

- Usually smaller; only about 100 cases; sexes equal; left more often.
- **Embryology:** a second ureteric bud or extra branch of the first forms, then the nephrogenic anlage divides into two metanephric tails.
- A distinct mass, completely separate or attached only by loose areolar tissue.
- **Position:** caudal to the main kidney in 60%; cranial if it has a separate, distinct ureter.
- **Ureter:** joins the main ureter to a single orifice in 50% ("bud off a bud"); fully independent in 50%.
- **Blood supply:** anomalous, depends on position.
- **Presentation:** rarely symptomatic; pain, fever, hypertension or mass; usually found on imaging for other reasons.
- **Pre-operative work-up:** MRU or retrograde pyelography for anatomy, radionuclide studies for function, cystoscopy for orifices.

## Crossed renal ectopia

### ▸ Presentation by age

- Children: as part of other malformations. Young adults: during evaluation at menarche. Elderly: incidental.

### ▸ Embryology

- Cause unknown.
- **Cook and Stephens:** malalignment and abnormal rotation of the caudal fetus, with the distal curled vertebral column displaced to one side, so the Wolffian duct structures or cloaca lie to one side. One ureter then crosses the midline into the opposite blastema, or the kidney and ureter are carried across during normal ascent.
- **Kelalis:** teratogenic factors; genetic factors may also play a role.
- Both kidneys start ascent together, but the ectopic one lags because of the time taken to cross.
- Ascent continues until the uncrossed kidney reaches its normal position or a retroperitoneal structure stops it.
- Final shape of a fused kidney depends on time and extent of fusion and degree of rotation.

### ▸ Non-fused forms

- **Without fusion:** male to female 2:1. Uncrossed kidney usually normal in position; ectopic kidney inferior, diagonal or horizontal, with anterior pelvis. Each has its own capsule and Gerota's fascia.
- **Solitary crossed ectopia:** very rare (about 34 reported); male to female 2:1. Left kidney to right with absent right kidney is twice as common as the reverse. Usually incomplete rotation and ascent.
- **Bilateral crossed ectopia:** only about 5 patients. Kidneys and pelves may look normal, but ureters cross the midline at the lower lumbar vertebrae.

### ▸ Crossed fused forms

| Form | Features |
|---|---|
| S-shaped (sigmoid) | Crossed kidney inferior, fused to lower pole of uncrossed kidney. Late fusion, after rotation. Convex lateral borders face opposite ways. |
| Lump (cake) | Usually ascends only to the sacral promontory. Both pelves anterior, draining separate areas of parenchyma. |
| L-shaped | Crossed kidney in the midline or contralateral paramedian space. |
| Disc | More extensive medial fusion gives a disc or shield shape. Both pelves anterior. |
| Superior ectopia | Both pelves anterior, indicating very early fusion. |

### ▸ Blood supply

- Ectopic kidney: one or more branches of the aorta or common iliac artery.
- Uncrossed kidney: often anomalous, with multiple renal arteries from various aortic levels.

### ▸ Associated anomalies

- Trigone normal, except in solitary crossed ectopia.
- Ureterocele, ectopic orifice, imperforate anus, vertebral or orthopaedic and cardiovascular anomalies.

- **Solitary crossed ectopia:** highest rate; 40–50% have skeletal and genital abnormalities, most often cryptorchidism, absent vas, vaginal atresia, unilateral uterine abnormality and imperforate anus.

▸ **Symptoms, diagnosis and prognosis**

- Mostly asymptomatic and incidental; about one-third have an asymptomatic abdominal mass.
- **Investigations:** ultrasound, DMSA, CT urography, retrograde pyelography; MRU or MRA for pre-operative planning.
- Most have normal longevity; prognosis also depends on associated anomalies.

## Ch 50 · Renal Dysgenesis and Cystic Disease of the Kidney

### Cystic kidney diseases

▸ **Autosomal recessive polycystic kidney disease (ARPKD)**

- **Gene:** PKHD1 product fibrocystin, expressed in kidney, liver and pancreas.
- Enlarged kidneys keep a reniform shape; increased echogenicity is due to diffuse microcysts.
- **Diagnosis:** rests on overall clinical data: family history with recessive inheritance, positive liver biopsy, and absence of extrarenal malformations seen in other cystic diseases.
- CT is more sensitive than ultrasound and can be used if in doubt.
- Once strongly suspected: genetic evaluation and counselling, with a family history of at least 3 generations.
- **Ultrasound differentials:** bilateral hydronephrosis (enlarged kidneys, hypoechoic calyces); multicystic kidney (hypoechoic cysts, non-reniform mass). Wilms tumour and bilateral mesoblastic nephroma are also differentials.
- **Treatment:** respiratory care can extend life. Survivors may need treatment for congestive cardiac failure, hypertension, renal and hepatic failure, and portal hypertension.

▸ **Medullary sponge kidney (MSK)**

- Usually non-hereditary; some forms autosomal dominant.
- **Embryology:** disruption at the ureteric bud–metanephric blastema (RET/GDNF) interface.
- **Associations:** Beckwith–Wiedemann, Caroli disease, Ehlers–Danlos.
- **Clinical forms:** typical (recurrent calcium stones, renal tubular acidosis, metabolic bone disease); indolent (few or no stones, no tubular defects); rare very severe form dominated by excruciating pain.
- **Histology:** dilated intrapapillary collecting ducts and medullary cysts 1–8 mm, giving a sponge-like cut surface. Concretions are mostly calcium phosphate.
- Bilateral in 70%.
- **Imaging:** single-slice helical CT, MRI and ultrasound cannot show the specific findings; classical IVU and 3D volume-rendered urographic-phase multidetector CT are sensitive.
- Some patients with infections need long-term prophylaxis.

▸ **Multicystic dysplastic kidney (MCDK)**

- No reniform shape, little stroma, no connection between glomeruli and calyces. Size from small nubbin to large palpable mass.
- **Types:** infundibulopelvic (pyeloinfundibular atresia), with atresia of pelvis and ureter; less common hydronephrotic type, with only the upper ureter atretic.

- More common in males.
- **Theories:** extreme obstructive hydronephrosis from ureteric or pelvic atresia; abnormal ureteric bud–metanephric mesenchyme interaction (GDNF, WNT, PAX2 mutations); failure of union of bud and mesenchyme.
- **Angiography:** absent or small renal artery; rarely indicated.
- **Cystoscopy:** hemitrigone with absent orifice, or, if present, retrograde pyelography shows an atretic ureter; rarely indicated.
- Can be ignored until its bulk becomes inconvenient.

## Genes in kidney and ureter development

- More than 30 genes are known in mammalian kidney development, but only a few (Agtr2, Bmp4, FoxC1, Pax2, Eya1) are known to cause kidney and ureteric anomalies.

| Gene | Condition and features |
|---|---|
| PAX2 | Renal–coloboma syndrome, autosomal dominant: optic nerve coloboma, VUR, renal anomalies. |
| EYA1 | Branchio-oto-renal syndrome, autosomal dominant: duplex system, hypoplasia, dysplasia, renal agenesis. |
| FOXC1 | Duplex kidney with dilated upper pole ureters, which may open into the seminal vesicle or vas. |
| BMP4, BMP5 | Hydronephrosis and hydroureter; they control smooth muscle formation in the proximal ureter and pelvis. |

▸ **Renin–angiotensin system (AGTR2)**

- Active in fetal life through AGTR2 (angiotensin II receptor type 2); maintains fetal glomerular filtration, ensures urine production and supports normal kidney and ureter development.
- AT1 and AT2 receptors are both in the developing mesonephros and metanephros. AT2 acts in embryonic renal development; AT1 modulates later vascular development and classical angiotensin II effects (vasoconstriction, sodium reabsorption).
- **ACE inhibitors in pregnancy:** oligohydramnios, fetal hypotension, anuria.

# Ch 52 · Ectopic Ureter, Ureterocele and Ureteral Anomalies

## Embryology and aetiology

- The specific mechanisms of ectopic ureter and ureterocele remain unidentified.
- **Trigone:** classical view, the common nephric duct (mesoderm) forms it; revised model, the common nephric duct regresses by apoptosis and the trigone forms from endoderm, like the bladder.
- **Ureterovesical connection:** depends on ureteral maturation and apoptosis of the common nephric duct. A loop of remaining common nephric duct and distal ureter contacts the urogenital sinus, setting the future orifice position. Merger with the sinus and apoptotic removal of the remaining duct creates the patent connection, at first at the level of the verumontanum; later bladder and urethral growth separates the mesonephric duct from the distal ureter.
- **Ret gene:** controls common nephric duct remodelling, which sets the final ureter position. No remodelling: ureter stays joined to the sex ducts. Partial remodelling: joins the urogenital sinus near the nephric duct, at the verumontanum.

- The ureter is a solid cord that canalises from its centre towards both ends.
- **Chwalla membrane:** formed by cells at the distal ureter; undergoes apoptosis shortly before birth. Persistence or abnormality leads to ureterocele.

▸ **Single-system ectopic ureter: causes**

- Misalignment of the loop with the distal urogenital sinus, altering orifice insertion.
- Delayed or defective insertion of the nephric duct into the cloaca.
- Defective ureteral maturation (under-rotation).
- Partial regression of the common nephric duct: ureter joins the bladder at an abnormal posterior site, near or within the urethra.
- Abnormal apoptosis preserving the ureter–Wolffian duct link: drainage into vas or ejaculatory duct in males, Gartner duct cyst in females.

▸ **Ectopic vas**

- A vas joined to the ureter (anomalous common excretory duct) should be considered in males with recurrent epididymitis and ipsilateral hydronephrosis.

## Ureterocele classification

- **Extravesical (ectopic):** orifice may be in the bladder, at the bladder neck or in the urethra.
- **Stephens:** cecoureterocele is the most important subgroup; orifice within the bladder, cavity extends beyond the bladder neck into the urethra.
- **Ureterocele disproportion:** non-obstructive ureterocele with duplication; upper pole non-dilated and so dysplastic that most imaging misses it.
- **Churchill:** contralateral system at risk from reflux or bladder outlet obstruction (BOO).

## Evaluation

- **Three parts:** anatomical (examination, ultrasound, MRI), functional and endoscopic.
- **Examination:** palpable dilated upper pole possible.
- **Ultrasound:** parenchymal thickness and corticomedullary differentiation (CMD) of the affected moiety.
- **IVP:** only qualitative function; less useful baseline.
- **Endoscopy:** not always essential.

## Treatment of duplex-system ureterocele

▸ **Conservative**

- Stable child, no hydronephrosis, normal function, low-grade reflux: antibiotic prophylaxis and follow-up.

▸ **Upper pole partial nephrectomy**

- A second (Gibson) incision may be needed to remove the whole upper pole ureter when it refluxes.
- **Follow-up:** Doppler sonography is best; shows normal anatomy, no urinoma and normal lower pole blood flow.
- **Outcome:** pre-existing lower pole reflux resolves in 20%.
- **Complications:** IVC (inferior vena cava) laceration; duodenal perforation; peritoneal tears; conversion to (or loss of the kidney needing) total nephrectomy.

▸ **Pyeloureterostomy or ureteroureterostomy**

- When the upper pole draining into a ureterocele is preserved this way, ureterocele resection may be avoided if it is well decompressed.

▸ **Ureterocele excision and common sheath reimplant**

- Transvesical; excision starts with a transverse incision over the ureterocele.
- **Total reconstruction:** upper pole partial nephrectomy, ureterocele excision and common sheath reimplant of the lower pole ureter. Definitive but extensive.

▸ **Marsupialisation**

- Removes the thin intravesical wall and sutures the edges.
- Satisfactory results, but once open surgery is under way, definitive correction is better.

## Ch 53 · Surgery of the Ureter in Children

### Ureteral reimplantation

▸ **Politano–Leadbetter**

- A second submucosal tunnel can bring the opening closer to the bladder neck, especially when the original orifice is too lateral.
- **Patency check:** the sure sign is a urine jet from the orifice; a 5 Fr feeding tube can also be passed up the ureter.

▸ **Glenn–Anderson (infrahiatal)**

- Success about 98%, similar to other reimplants.
- Main problem is inadequate tunnel length.

▸ **Cohen cross-trigonal**

- Overcomes the short tunnel of Glenn–Anderson.
- **Bilateral repair:** the more lateral ureter's tunnel goes superior to the contralateral orifice; the second tunnel goes towards the inferior edge of the laterally displaced ureter.
- **Later access to the superolateral orifice:** wire or ureteric catheter placed via suprapubic cystostomy (SPC) and directed cystoscopically; curved-tip vascular access catheter; flexible ureteroscopy (for example for nephrolithiasis).

▸ **Extravesical reimplantation**

- Simple to learn.
- Distal ureter found where it crosses medial to the origin of the obliterated umbilical artery from the internal iliac.
- **Transient voiding inefficiency:** after bilateral repair, especially in boys under 3 years; usually normal voiding within 1–2 weeks.

▸ **Laparoscopic and endoscopic**

- **Gil-Vernet:** least successful laparoscopic procedure.
- **Laparoscopic extravesical:** the most common. Steep learning curve: hard to make the tunnel without injuring urothelium, long operative times, difficult detrusor flap elevation.
- **Endoscopic transvesical cross-trigonal:** similar to Cohen, with $CO_2$ insufflation of the bladder.

## Follow-up and complications of reimplantation

- **At 3 months:** ultrasound and VCUG. No reflux: stop prophylaxis, no further VCUG.
- **Persistent at 3 months:** repeat VCUG at 1 year; persistent reflux and/or recurrent UTI then means correcting the failed surgery.

### ▸ Persistent and new contralateral reflux

- Persistent reflux is commoner in high-grade cases; high grade is one cause of long-term persistence.
- Cause of new contralateral reflux after unilateral repair is not clearly known.

### ▸ Late obstruction

- **By site:** suprahiatal (twist, ischaemia); hiatal (angulation); tunnel (improper creation); orifice.
- At the orifice, the apex of the spatulation is most vulnerable.
- Tunnel obstruction can be treated by endoscopic unroofing of the distal few millimetres.

## Endoscopic injection treatment

### ▸ Technique and follow-up

- Inject more until the orifice is crescent- or slit-shaped.
- Keep the needle in for 1 minute to avoid dislodging the implant; not needed with Deflux.
- Persistent reflux: repeat injection at 6 months, then surgery if still unresolved.

### ▸ Injectable materials

- **Ideal:** non-toxic, easy to inject, minimal inflammation, encapsulated by normal fibrous tissue, viscous enough to avoid leakage, keeps its volume.
- **Classification:** autologous or non-autologous; particulate or degradable. Particulate agents raise concern about migration.
- **Non-autologous agents used for VUR:** calcium hydroxyapatite (Coaptite), cross-linked bovine collagen.
- **Autologous:** fat, collagen, muscle, chondrocytes. Need harvesting and expansion; main drawback is volume loss.

### ▸ Deflux

- A natural product.
- Carrier gel is reabsorbed; dextranomer microspheres are encapsulated by fibroblasts and collagen.
- Overall success 65–90%.
- No ratcheted syringe needed, because a small-gauge needle is used.

# Ch 56 · Hypospadias

## Epidemiology and associations

- First-degree relatives have 13 times greater risk.
- About 90% isolated (clinical diagnosis, no imaging needed); about 10% with syndromes or other abnormalities.

- **Syndromes:** WAGR, G syndrome, Wolf–Hirschhorn, 13q deletion, hand–foot–uterus, Smith–Lemli–Opitz.
- **With undescended testis in DSD:** commonest causes are mixed gonadal dysgenesis or ovotesticular DSD.

## Pre-operative androgen stimulation

- Increases penile length and glans circumference.
- Results controversial; some studies report more, not fewer, complications.

## Ventral curvature (chordee)

- **Distal:** only about 10% have chordee after degloving, usually under 30°.
- **Proximal:** about 50% have none or under 30°; 50% over 30°.
- **Over 30°, ventral corporotomies:** three transverse cuts (4 to 8 o'clock) without grafting, at the point of greatest curvature and 4 mm above and below it.
- **Multiple plications:** results may not last into adulthood.
- **Single 3-to-9 o'clock corporotomy:** needs a graft, which leaves only a flap urethroplasty for the urethra (a graft will not take over a graft).

## Distal hypospadias repairs

### ▸ MAGPI

- **Indication:** glanular or coronal meatus, flat or convex glans, healthy thick skin proximal to the meatus, no chordee.
- **Steps:**
  - Subcoronal incision and circumferential degloving.
  - Meatal advancement: vertical incision in the glanular groove closed transversely, advancing the dorsal meatus.
  - Glanuloplasty: ventral meatal lip pulled distally and glans closed beneath it.
  - Skin closed as for circumcision.

### ▸ Mathieu (flip-flap)

- **Indication:** distal hypospadias too proximal for MAGPI.
- **Steps:**
  - Measure plate from meatus to glans tip; mark the same length on proximal shaft skin.
  - Incise glans and flap; deglove the shaft.
  - Dissect flap subcutaneous tissue so it flips over the plate, with a feeding tube between them into the bladder.
  - Suture flap edges to plate edges.
  - Cover suture lines with a dorsal dartos flap, split in the midline and brought round each side.
  - Reapproximate glans wings; close skin.

### ▸ TIP (Snodgrass)

- Degloving by a ventral incision 2 mm below the meatus.
- Ventral dartos flap split, when possible, into two longitudinal segments crossed over the neourethra for two-layer cover.

- **Variations:** single- or two-layer neourethra; epithelial or subepithelial suturing.

## Proximal hypospadias repairs

### ▸ Onlay preputial flap

- **Steps:**
  - Stay sutures at the corners of the dorsal prepuce.
  - Harvest a 10 mm strip of inner prepuce, keeping its dartos supply.
  - Extend pedicle dissection to the penopubic junction to avoid tension when moving the flap ventrally.
  - 6 Fr stent into the bladder; sew flap to each side of the plate with subepithelial sutures.
  - Glansplasty and skin closure.
- **Complications:** 27–45%.

### ▸ Single-stage tubularised preputial flap

- After degloving, raise a 12–15 mm inner preputial flap, keeping its dartos supply.
- **Option 1:** tubularise over a 6 Fr stent with two layers of 7-0 Vicryl subepithelial sutures, move ventrally, sew to the native meatus proximally and glans wings distally.
- **Option 2:** bring the flap ventrally first, sew one edge to the corpora to make a pseudoplate, then tubularise.
- **Complications:** 14–33%.

### ▸ Two-stage graft repair

- **Graft choice:** prepuce if circumcision is wanted; oral mucosa if preputioplasty is desired.
- **Stage 1:**
  - Transect the plate distally at the corona and excise it off the corpora, proximally to the meatus and beyond towards the membranous part.
  - Stretch the native urethra distally and anchor it to the corpora, reducing graft length needed.
  - Complete proximal urethrostomy: meatus sutured to corpora at 10, 12 and 2 o'clock and to penile skin or scrotum at 4, 6 and 8 o'clock.
  - Dorsal shaft skin sutured to the preputial collar; inner preputial graft placed ventrally, stitched to the glans at the corona and to skin each side.
  - Graft's proximal end split in the midline around the urethrostomy, sutured at 10, 12 and 2 o'clock medially and to shaft skin or scrotum laterally.
  - After quilting, catheterise and apply a tie-over bandage for 7 days.
- **Stage 2:** close penile and scrotal skin in the midline with subepithelial sutures.

### ▸ Byars flaps

- Few outcomes reported; Snodgrass had complications in all 9 cases and does not recommend it.

## Reoperations

- **Dorsal inlay graft:** plate removed but a strip of grossly healthy skin remains; also for neourethral strictures.
- **Two-stage oral mucosa graft:** strictures nearly obliterating the lumen; hair in the neourethra.

## Complications

▸ **Risk factors**

- Glans width under 14 mm calls for extended dissection of the glans wings.

▸ **Glans dehiscence**

- Complete separation of the glans wings, with or without a skin bridge; causes a splaying stream.
- **Prevention:** polyglactin sutures (chromic abandoned).

▸ **Meatal stenosis**

- **Prevention:** keep the plate incision within the plate, not into the glans; do not suture the glans wings to the neourethra.
- **Treatment:** dorsal incision of the neomeatus, which enlarges it without recreating the defect.

▸ **Neourethral stricture**

- Under 1 cm: direct vision internal urethrotomy, but not when the neourethra is a tubularised flap or graft.

▸ **Diverticulum**

- Expose through a ventral median raphe incision, open, de-epithelialise excess tissue, close the neourethra in two layers over a catheter, vest-over-pants.

▸ **BXO (balanitis xerotica obliterans)**

- Medical treatment: topical steroid or tacrolimus.

# Ch 57 · Etiology, Diagnosis and Management of the Undescended Testis

## Testicular descent

▸ **Phases**

| Phase | Timing | Events |
|---|---|---|
| I | About 5 weeks | Future gubernaculum contacts the mesonephros at the internal inguinal ring. |
| II | 8–10 weeks | Genitofemoral nerve accompanies the new gubernaculum and processus vaginalis; gubernaculum grows, processus deepens, cremaster extends into gubernaculum. |
| III | 10–14 weeks | Testis grows; Müllerian ducts and mesonephros regress; gubernaculum starts swelling. |
| IV | 14–20 weeks | Continued gubernacular swelling, further cremaster development, processus vaginalis migration widening the inguinal canal. |

▸ **Regulation**

- **Inguinal phase:** androgens, genitofemoral nerve/CGRP, and neuromuscular mechanisms.
- **Factors in descent:** inguinal bursa formation, differential body wall growth, processus vaginalis.

## Incidence and aetiology

- More common on the right.

- **Non-syndromic risk factors:** breech presentation; small for gestational age; maternal obesity; maternal cola consumption; maternal paracetamol (acetaminophen) or other analgesics in pregnancy.
- **Familial:** may be autosomal dominant with reduced penetrance. Recurrence risk ratio highest in twins (10), then brothers (3.5), then father (2.3).
- **Syndromic:** conditions with reduced androgen production and/or action, such as DSD; about 500 syndromes and 200 genes. Includes Down syndrome.

## Diagnosis

- **Superficial inguinal pouch:** anterior to rectus abdominis and medial to the pubic tubercle (check).
- **Ectopic sites:** perirenal is listed among sites off the normal line of descent (check).
- Note scrotal asymmetry.
- An extra-abdominal testis may be impalpable because of small size, obesity or an uncooperative child.
- **MRI:** when an ectopic testis is not localised by laparoscopy.

## Risks and aims of surgery

- **Catch-up growth after orchidopexy:** may occur before 9 months; variable from 9 months to 3 years; none after 3 years.
- **Torsion risk:** anomalous tunica vaginalis attachment and partial or complete testis–epididymis non-fusion.
- **Aims:** preserve fertility, reduce malignancy risk, lower hernia and torsion risk, cosmesis, earlier detection of malignancy by placing the testis where it can be examined.
- **Men over 50:** malignancy risk not defined.
- **Pubertal or post-pubertal boys:** orchidectomy prevents malignancy and torsion, as fertility potential is already lost.

## Palpable testis

### ▸ Inguinal orchiopexy

- If length is short, incise the transversalis fascia along the medial border of the internal ring, with medialisation of the cord beneath the inferior epigastric vessels (Prentiss manoeuvre).

### ▸ Testis not reaching the scrotum

- Fix to the scrotum with a non-absorbable suture and apply gentle pull at regular intervals.
- In two-stage surgery, a PTFE membrane wrapped round the testis helps find it at stage two.

### ▸ Transscrotal orchiopexy incisions

- Bianchi (superior scrotal), transverse low scrotal, midline scrotal.

## Non-palpable testis: laparoscopy

- **Diagnostic yield:** laparoscopy about 95%; MRI sensitivity about 65%; ultrasound about 45%.
- **Ports, unilateral:** 5 mm infraumbilical plus 2 or 3 mm trocars in right and left lower quadrants.
- **Ports, bilateral:** infraumbilical plus one each side at umbilical level in the midclavicular line.
- **Position:** supine, 10 degrees head-down.

- **Anaesthesia:** avoid nitrous oxide, which distends bowel.
- **Difficulty in children:** small abdominal space, hard port placement, miniature instruments needed.
- **Vanishing testis:** vas and vessels end blindly near the psoas. With a closed internal ring, 97% have a vanished testis.
- **Intra-abdominal testis:** mobilisation is adequate when the testis reaches towards the contralateral internal ring. The new hiatus is made medial to the inferior epigastric vessels and lateral to the medial umbilical ligament.
- **Microsurgical (autotransplant) orchiopexy:** experts only; can be used for a solitary intra-abdominal testis.

## Post-pubertal undescended testis

▸ **Unilateral**

- **Palpable:** counsel, then inguinal exploration and orchidectomy, especially if hypoplastic. If declined, orchiopexy after explaining malignancy risk and lack of fertility contribution.
- **Non-palpable, vas and vessels entering the canal at laparoscopy:** orchidectomy, or orchiopexy if declined, after explaining malignancy and fertility issues.

▸ **Bilateral**

- **Considerations:** no fertility potential; torsion and malignancy risk; endocrine dysfunction if both are removed.
- **Both palpable:** bilateral inguinal exploration. Both healthy: bilateral orchiopexy. One healthy: orchidectomy of the hypoplastic side, orchiopexy of the healthy side.
- **Both hypoplastic:** karyotype and testicular biopsy, then bilateral orchidectomy with hormone replacement, or keep the better-functioning testis.
- **Both non-palpable:** once DSD is excluded, bilateral orchidectomy with hormone replacement.
- **One palpable, one non-palpable:** remove the abdominal testis (more loss of function and malignancy); orchiopexy of the inguinal testis to keep some endocrine function, explaining malignancy and fertility issues.

# Ch 63 · Paediatric Urologic Oncology: Renal and Adrenal

## Neuroblastoma: origin and epidemiology

- **Origin:** neural crest cells, at sites in the neck, thorax, retroperitoneum, pelvis or adrenal.
- **Share:** 8–10% of all childhood cancers.
- **Incidence:** 10 per 1 million live births a year, so more common than Wilms tumour.

## Neuroblastoma: genetics

- Mostly sporadic; less than 1% familial. Familial cases may be bilateral and present early.
- **Risk to siblings and offspring:** less than 6%.
- **Hereditary predisposition genes:** PHOX2B and ALK, annotated as "chromosome 11" (check).

## Neuroblastoma: pathology

- **Spectrum of maturation and differentiation:** neuroblastoma, ganglioneuroblastoma, ganglioneuroma.

### ▸ Classification

- Shimada classification and the International Neuroblastoma Pathology Classification (INPC, the modified Shimada).
- Shimada is age-linked. Stroma-poor tumours (less than 50% stroma) have a poor prognosis.

### ▸ Ganglioneuroma

- Diagnosed in older children; uncommon in the adrenal.
- Resect only when symptomatic. Survival is not influenced by extent of resection, so avoid aggressive resection.

## Neuroblastoma: clinical presentation and spread

### ▸ Metastatic disease

- Bone or joint pain.
- Periorbital ecchymosis and proptosis (raccoon eyes).
- Cutaneous metastases (blueberry muffin lesions).

### ▸ Direct extension

- Spinal cord: neurological signs.
- Thoracic tumours: cough and dyspnoea.
- Pelvic tumours at the organ of Zuckerkandl: urinary retention or constipation.

### ▸ Paraneoplastic features

- **Opsoclonus–myoclonus:** "dancing eyes, dancing feet" (acute myoclonic encephalopathy).
- **Catecholamines:** paroxysmal hypertension and palpitations.
- **Vasoactive intestinal peptide (VIP):** watery diarrhoea plus hypokalaemia.

## Neuroblastoma: investigations

- **Bloods:** CBC, LFT and RFT plus serum calcium and BUN.
- **Bone marrow:** anaemia results from marrow involvement; two aspirates and two biopsies are recommended.
- **Plain X-ray:** stippled calcification in an abdominal or posterior mediastinal mass.
- **Spinal canal extension:** also helps to distinguish it from Wilms tumour.
- **MIBG:** with 131I or 123I; 123I is more sensitive.

## Neuroblastoma: staging (INSS)

| Stage | Features |
|---|---|
| 1 | Localised, complete gross resection. |
| 2A | Localised, incomplete gross resection. |
| 2B | Positive ipsilateral nodes, with or without complete gross resection. |
| 3 | Unresectable unilateral tumour infiltrating across the midline, with or without regional nodes; or localised unilateral tumour with contralateral nodes; or midline tumour with bilateral extension by infiltration or nodes. |

| | |
|---|---|
| 4 | Dissemination to distant nodes, bone, bone marrow, liver, skin or other organs. |
| 4S | Primary localised (stage 1, 2A or 2B). |

## Neuroblastoma: treatment

▸ **Low risk**

- **Includes:** stage 1, stage 4S, and stage 2 without MYCN amplification or unfavourable histology.
- **Stage 4S:** chemotherapy for extensive liver disease ("more than 2 months" of age) (check) or MYCN amplification; otherwise observe.
- **Stage 2:** radiotherapy for residual tumour-bed disease or relapse.

▸ **Intermediate and high risk**

- **Includes:** stage 3, stage 4, and stage 2 with MYCN amplification and unfavourable histology.
- **Induction:** etoposide, cisplatin, cyclophosphamide, doxorubicin.
- **Still inoperable after induction:** radiotherapy.
- **Consolidation:** busulfan plus melphalan.
- **Minimal residual disease:** fenretinide, a biological therapy that induces apoptosis.

▸ **Spinal cord compression**

- Avoid radiotherapy, because it impairs spinal growth.

## Neuroblastoma: prognostic factors

| Factor | Favourable | Unfavourable |
|---|---|---|
| Serum ferritin | Below 143 ng/mL | Above 143 ng/mL |
| Urine VMA/HVA ratio | Below 1 | Above 1 |
| Neuron-specific enolase (NSE) | Below 100 | Above 100 |

- **16q/1p loss of heterozygosity (LOH):** raises relapse risk independent of histology and stage.
- **Other unfavourable biological factors:** high telomerase activity, increased VEGF expression, aneuploidy.

## Wilms tumour: epidemiology and clinical presentation

- **Incidence:** 8 per million children.
- **Typical presentation:** about 90% are an incidental, asymptomatic abdominal mass at 2–5 years.
- **Classic triad:** fever, lump and haematuria.
- **Constitutional symptoms:** fever, anorexia, weight loss.
- **Acute abdominal pain:** from tumour rupture.
- **Mass:** firm, non-tender, does not cross the midline.
- **Venous involvement:** congestive cardiac failure, with hypertension, varicocele and hepatomegaly.

## Wilms tumour: genetics and syndromes

- **Familial:** earlier onset, more often bilateral. Familial genes are FWT1 and FWT2.

- About 10% are associated with syndromes or congenital anomalies.
- **Somatic and germline mutations:** WT1 (11p13), WT2 (11p15), TP53 (chromosome 17).
- **Somatic only:** WTX and CTNNB1.
- **Beckwith–Wiedemann syndrome:** overgrowth disorder with macroglossia, abdominal wall defects and "microcephaly" (check).
- **Contralateral screening after nephrectomy for unilateral tumour:** ultrasound every 3–4 months until 5 years of age, especially under 12 months with nephrogenic rests in the specimen (greatest risk).

## Nephrogenic rests

### ▸ Definition and background

- **Nephrogenesis:** ureteric bud meets metanephric blastema at 5 weeks; fetal lobulations by 28 weeks; complete at 36 weeks.
- **Definition:** metanephric blastema persisting beyond 36 weeks of gestation, when nephrogenesis is almost complete.
- Found in 1% of infant autopsies.
- Usually clinically occult; found incidentally with a Wilms tumour.

### ▸ Diffuse nephroblastomatosis

- Multiple rests that enlarge the kidney but preserve its shape.
- Prone to Wilms tumour; bilateral lesions are common.

### ▸ Perilobar versus intralobar rests

| | Perilobar | Intralobar |
|---|---|---|
| Share | 90% | 10% |
| Number and margin | Multiple, well circumscribed, usually smooth | Single or multiple, irregular margins that may interdigitate with adjacent normal interstitium |
| Main tissue | Blastema | Stroma and epithelium |
| Malignant transformation | Less chance | More chance |
| Associations | Beckwith–Wiedemann, Perlman syndrome, trisomy 18 | WAGR, Denys–Drash, sporadic aniridia |

### ▸ Pathology

- **Gross:** tan nodules or white plaques.
- **Histology:** sclerosing, hyperplastic, dormant or neoplastic (transforming into Wilms tumour).

### ▸ Imaging

- **Ultrasound:** nodular appearance, loss of corticomedullary differentiation, diffusely echogenic kidney.
- **CT:** enhance less than parenchyma; surface may look lobulated.
- **MRI (best):** hypointense on T1, hyperintense on T2.
- **Angiography:** rests do not opacify; contrast may puddle in intervening normal parenchyma.

### ▸ Risk and treatment

- The more rests, the more likely a Wilms tumour.

- **Highest risk:** diffuse perilobar rests, about 35% (check).
- **Treatment:** controversial. Some give chemotherapy; others hold that close radiological surveillance of an enlarging mass is enough.

## Wilms tumour: pathology

- **Favourable histology:** 90%; no anaplasia and no sarcomatous features.
- **Unfavourable (anaplastic) histology:** 10%.
- **Epithelial component of the triphasic tumour:** tubules, rosettes and glomeruloid structures.
- **After pre-operative chemotherapy:** proportions change; stromal and epithelial tumours are seen more often, and blastemal components decrease.

## Wilms tumour: clinical staging (pre-chemotherapy, NWTS/COG)

- **Distribution:** stage I 43%, II 20%, III 21%, IV 11%, V 5%.

## Wilms tumour: surgery and approaches

### ▸ NWTSG/COG

- No contralateral exploration if pre-operative CT or MRI shows a normal contralateral kidney.
- **Pre-operative chemotherapy indications:** planned renal-sparing surgery, bilateral tumours, IVC extension above the hepatic veins, and tumour inoperable at exploration.
- **After pre-operative chemotherapy:** repeat imaging at 6 weeks. Adequate shrinkage leads to surgery, then chemotherapy with or without radiotherapy. Progressive disease needs more intense chemotherapy and has a poor prognosis.
- **Laparoscopic nephrectomy:** usually combined with pre-operative chemotherapy.

### ▸ SIOP

- Pre-operative chemotherapy lowers morbidity, particularly late effects of radiotherapy.

### ▸ UKCCSG (United Kingdom Children's Cancer Study Group)

- Pre-operative chemotherapy like SIOP, but biopsy before starting. This avoids chemotherapy for benign tumours and other histologies; about 12% of presumed Wilms tumours are benign.
- **Disadvantage of pre-treatment biopsy:** tumour track seeding.

### ▸ Bilateral Wilms tumour and renal-sparing surgery

- Operate first on the kidney with the lower tumour burden.
- **Renal-sparing options:** partial nephrectomy or wedge excision, aiming for negative margins.
- **Partial nephrectomy, past:** reserved for bilateral tumour, solitary kidney and impaired renal function.
- **Partial nephrectomy, now:** after neoadjuvant chemotherapy, oncological outcomes equal radical nephrectomy, so it is extended to unilateral tumours.

### ▸ Upfront surgery (NWTS) versus pre-operative chemotherapy (SIOP)

| | Upfront surgery (NWTS) | Pre-operative chemotherapy (SIOP) |
|---|---|---|
| Advantages | Avoids chemotherapy for benign disease and chemotherapy-resistant histologies | Micrometastases treated early |

| | | |
|---|---|---|
| Disadvantages | Surgical complications may delay chemotherapy | May treat benign tumours and other histologies; modifies histology; loses staging information |

## Wilms tumour: post-operative chemotherapy (NWTSG/COG)

- **Abbreviations:** VA is vincristine and actinomycin D; VDA is vincristine, doxorubicin and actinomycin D.

| Histology and stage | Treatment |
|---|---|
| Favourable, I/II | VA for 18 weeks |
| Favourable, III/IV | VDA for 24 weeks plus 10.8 Gy to the tumour site |
| Focal anaplasia, I | VA for 18 weeks, with or without doxorubicin |
| Focal anaplasia, II–IV | VDA for 24 weeks plus 10.8 Gy to the tumour bed |
| Diffuse anaplasia, I | VA for 18 weeks, with or without doxorubicin |
| Diffuse anaplasia, II–IV | VDA plus etoposide and cyclophosphamide for 24 weeks plus radiotherapy |

### ▸ Radiotherapy

- **Stage IV with lung metastases:** extra 12 Gy to the chest ("one dozen to chest").
- **Whole abdomen:** for tumour rupture or spill and peritoneal metastasis.
- **Timing:** start within 14 days of nephrectomy.

## Wilms tumour: SIOP 93-01 (favourable histology)

| Stage | Pre-operative | Post-operative | Radiotherapy |
|---|---|---|---|
| I | VA, 4 weeks | VA, 4 weeks | None |
| II | VA, 4 weeks | VAD, 27 weeks | 15 Gy if nodes positive |
| III | VA, 4 weeks | VDA, 27 weeks | 15 Gy |
| IV | VDA, 6 weeks | "VCA", 27 weeks (check); if no complete response after 9 weeks, ICED (ifosfamide, carboplatin, etoposide, doxorubicin) for 34 weeks | Only if the metastasis has not gone by 9 weeks |

## Wilms tumour: recent advances

- **List:** screening protocols, pre-treatment biopsy, risk stratification, management (laparoscopic partial nephrectomy, new chemotherapy protocols) and tumour markers.

### ▸ Risk stratification

- **NWTSG/COG by histology:** low is mesoblastic nephroma; intermediate is favourable histology; high is anaplastic, clear cell sarcoma and rhabdoid tumour.
- **SIOP:** clear cell sarcoma and rhabdoid tumour are high risk with diffuse anaplasia.

### ▸ Tumour markers

- Serum amyloid A.
- Apolipoprotein C-III.
- MicroRNA markers allow a targeted biopsy before chemotherapy.

- LOH at 11p15 and WT1 mutation (11p13) predict relapse in very-low-risk Wilms tumour.

**Other topics in this book**

## Paediatric versus adult adrenocortical carcinoma (ACC)

- **Female-to-male ratio:** 1:1 under 12 years; 6:1 at 12–20 years.
- **Commonest association:** Li–Fraumeni syndrome.
- **Course:** paediatric ACC presents at low stage with improved survival of 54.2%; adult ACC presents at advanced stage.

### ▸ Paediatric staging

- **I:** less than 200 g and less than 5 cm, completely resected.
- **II:** more than 200 g or more than 5 cm, completely resected.
- **III:** nodal, surrounding-organ or vessel involvement, or incomplete resection.
- **IV:** haematogenous metastasis.

## Micturating cystourethrogram (MCUG)

### ▸ Indications and complications

- **Further indications:** enuresis, prune belly syndrome, urethral diverticulum.
- **Complications also include:** haematuria, rare bladder perforation, rare contrast reaction from absorption.

### ▸ Antibiotic prophylaxis

- If not already on antibiotics: 3 days, on day minus 1, day 0 and day plus 1 (trimethoprim or cephalexin).
- **Cephalexin syrup:** 125 mg/5 mL and 250 mg/5 mL (Sporidex); 25–50 mg/kg/day in 3–4 divided doses.

### ▸ Procedure

- Ask the child to empty the bladder first if possible.
- **Catheter:** 5–8 Fr feeding tube, sterile, with 2% xylocaine (lignocaine) jelly. Measure and collect any residual urine.
- **Contrast:** 10 mL diluted 1:5.
- **Filling:** until contrast stops dripping or the child wants to void. This volume is the bladder capacity; compare with the formula-based expected capacity for age.
- **Sedation or anaesthesia:** avoid if possible; if needed, it must not interfere with voiding.
- **Infants:** may need tapping over the bladder (with a wooden object or scale) to start voiding.

### ▸ Images

- **Early filling:** shows a ureterocele.
- **Full bladder:** bladder wall appearance and capacity.
- **Voiding:** females supine (2–4 images); males turned to right or left anterior oblique (2–4 images).

### ▸ Cyclic MCUG

- At least 2 voiding cycles. Also helps identify a possible ectopic ureter or intermittent VUR; the child can void around the catheter for the first void.

▸ **How to read it**

- **Scout:** bones (especially symphysis, spine and sacrum), other radio-opaque densities, bowel gas pattern.
- **Cystogram phase:** bladder shape and contour, filling defects, trabeculation, diverticula, capacity, bladder neck position.
- **Voiding phase:** unilateral or bilateral reflux, bladder neck, posterior urethra, reflux into ejaculatory ducts.

# 5 Paediatric Urology 2

## Ch 32 · Embryology of the Human Genitourinary Tract

### Kidney: origin from intermediate mesoderm

- As the notochord and neural tube develop, the mesoderm on each side of the midline divides into paraxial (somite), intermediate and lateral plate mesoderm.
- With transverse folding of the embryo, the intermediate mesoderm migrates towards the intra-abdominal coelom (future peritoneum).
- It forms bilateral longitudinal masses, the nephrogenic cords, which develop progressively in a craniocaudal direction.
- Each nephrogenic cord bulges from the posterior wall of the coelomic cavity to produce the urogenital ridge.

### Pronephros and mesonephros

▸ **Pronephros**

- Develops in the region of the future neck and thorax as five to seven paired segments.

▸ **Nephric (mesonephric, Wolffian) duct**

- **Day 24:** appears as a pair of solid longitudinal tissue condensations parallel to the nephrogenic cords, in the dorsolateral embryo, before the mesonephric tubules develop.
- **Day 28:** its blind distal end fuses with the cloaca. Canalisation then progresses cranially (in reverse direction), giving definitive ducts with excretory capability.

▸ **Mesonephros**

- Serves as the excretory organ of the embryo while the metanephros begins its development.
- Develops at about the 9th to 10th somite levels.
- Vesicles begin to form in the 4th week; 40–42 pairs of tubules form in all.
- Only about 30 pairs are seen at any one time, because the cranial tubules start to degenerate at about the 5th week.
- Each tubule becomes an excretory unit: the medial end forms a renal corpuscle and the lateral end joins the nephric duct.
- By the end of the 5th week it has almost completely disappeared, apart from its derivatives.
- **Female remnants:** small non-functional mesosalpingeal structures, the epoophoron and the paroophoron.

### Metanephros and nephron development

▸ **Reciprocal induction**

- The tip of each dividing ureteric bud branch is the ampulla; it interacts with the metanephric mesenchyme to induce future nephrons.
- Each new ampulla acquires a cap-like condensation of metanephric mesenchyme, giving the developing kidney a lobulated appearance.

▸ **Cadherins**

- Proper mesenchymal condensation and tubulogenesis need expression of R-cadherin, E-cadherin and cadherin 6, and suppression of cadherin 11.

▸ **Larsson stages of nephron development**

| Stage | Features |
|---|---|
| I | Mesenchymal condensation forms a pretubular aggregate (a detached oval mass), which becomes the renal vesicle, still unconnected to the ureteric bud. |
| II | Two clefts develop. |
| III | Glomerular (Bowman) capsule oval; proximal and distal tubules identifiable; endothelial and mesangial cells recruited to form the glomerulus. |
| IV | Glomerulus round, resembling a mature renal corpuscle; proximal tubule appears mature; distal segments still primitive. |

- In humans, all nephrons at birth are at varying steps of stage IV.
- More differentiated nephrons lie near the juxtamedullary region; less differentiated ones lie at the periphery.
- Renal maturation continues after birth, although nephrogenesis is complete before birth.

## Collecting system

- Dichotomous branching of the ureteric bud determines the eventual pelvicalyceal pattern. It is complete by 20–22 weeks; cortical and medullary domains are established by 22–24 weeks.
- **At birth:** 70% cortex and 30% medulla.
- **Papilla:** the apex of each medullary cone, formed by convergence of the collecting ducts.
- **Calyces and pelvis:** form from the most central segments, derived from the first five generations of ureteric bud branching.

## Renal ascent

- Attributed to differential growth of the lumbar and sacral regions; the exact mechanism is not known.
- Transient aortic sprouts arise at progressively higher levels and degenerate. The final pair forms in the upper lumbar region and becomes the definitive renal arteries.

## Urogenital sinus and trigone

▸ **Cloaca**

- In the 4th week, embryonic folding carries the cloacal membrane to the ventral aspect; the terminal endoderm-lined yolk sac dilates to become the cloaca.
- **Classic Rathke and Tourneux theory:** two lateral ridges fuse in the midline while the descending urorectal septum fuses with the cloacal membrane.
- **Challenge to the classic view:** there are no lateral ridges and no descending septum.

▸ **Trigone**

- **Common excretory duct stage:** about day 33.
- **Classical view:** right and left common excretory ducts fuse in the midline as a triangular area, forming a primitive trigone structurally different from bladder and urethra.

- **Day 37:** ureteric orifices open into the bladder and migrate cranially and laterally. Meanwhile the nephric duct orifices move caudally and flank the paramesonephric ducts at the urogenital sinus, at the future verumontanum (or vaginal canal).

## Ureter

- Begins as a simple cuboidal epithelial tube surrounded by loose mesenchymal cells.
- **Lumen:** complete by day 28; transiently obstructed between days 37 and 40, then bidirectional recanalisation beginning in the mid ureter.
- **Chwalla membrane:** present at about days 37–39; a two-cell-thick layer over the ureteric orifice that may cause physiological obstruction.
- **Epithelium:** transitional by 14 weeks.
- **Muscle:** muscularisation and elastic fibre development begin at 12 weeks. Bladder smooth muscle develops before ureteric muscle.
- **Direction of muscle growth:**
  - Bladder: first in the subserosal region of the dome, extending towards the base and urethra.
  - Ureter: starts in the subepithelial region at the vesicoureteric junction and ascends towards the intrarenal collecting system.

## Bladder and continence mechanism

▸ **Bladder**

- **7–12 weeks:** surrounding connective tissue condenses and smooth muscle fibres begin to appear.
- **10 weeks:** a cylindrical tube lined by a single layer of cuboidal cells in loose connective tissue; the apex tapers into the urachus, continuous with the allantois.
- **21 weeks:** epithelium has four to five cell layers, like fully differentiated urothelium.
- **Compliance:** very low in early gestation and gradually increases (exact cause not known). It changes with fetal urine production (mechanical distension), increasing wall muscle thickness and a falling ratio of thick to thin collagen fibres.

▸ **Continence mechanism**

- There are no detailed studies on the development of continence.
- After division of the cloaca and rupture of the urogenital membrane, a mesenchymal condensation forms around the caudal urogenital sinus.
- **15 weeks:** striated muscle fibres are seen, and smooth muscle becomes thicker at the bladder neck.
- **Urethral sphincter** (central smooth muscle, peripheral striated fibres): develops first in the anterior urethral wall, later extending to the posterior wall. Fibres project to the lateral wall of the prostate in males and of the vagina in females.

# Ch 43 · Exstrophy–Epispadias Complex

## Incidence and inheritance

- **Recurrence within a family:** about 1 in 100.
- **Offspring of an affected parent:** 1 in 70, about 500 times the general population risk.

- **Genetics:** a breakpoint mutation at the 5' region of the CASPR3 gene on chromosome 9 was the first finding to suggest a genetic basis.
- **Suspected risk factor:** advanced maternal age.

## Classic bladder exstrophy: musculoskeletal and abdominal wall defects

### ▸ Skeletal defects (3D CT)

- Coronal rotation of the sacroiliac joint gives an angle about 10 degrees larger than normal.
- Convergence of the iliac wings is one of the rotational pelvic abnormalities.

### ▸ Pelvic floor

- **Puborectalis sling:** supports about two times more body cavity area than normal and is more flattened; length and thickness are normal.
- **Levator ani group:** less dome-shaped and more irregular; length and thickness usually the same as in children without exstrophy.
- **Effect of closure:** reshapes the box-like pelvic floor into a more inwardly rotated hammock, redistributes a significant proportion of the levator ani into the anterior compartment, and gives a smooth, uniform contour.

### ▸ Fascial defect

- Premature rupture of the abnormal cloacal membrane leaves a triangular fascial defect occupied by the exstrophied bladder and posterior urethra.
- **Inferior limit:** the intersymphyseal band, representing the divergent urogenital membrane.
- The anterior rectus sheath has a fan-like extension behind the urethra and bladder neck that inserts into the intersymphyseal band.
- Bladder musculature extends laterally and interdigitates with rectus fascia fibres. These must be incised lateral to the urethral plate down to the inferior pubic ramus and levator hiatus.

### ▸ Umbilical hernia and omphalocele

- A small umbilical hernia may be present; it is repaired at initial closure.
- A small omphalocele occurs rarely and is closed at bladder closure.

### ▸ Inguinal hernia

- **Incidence:** indirect hernias in about 80% of boys and 10% of girls; bilateral in 81.8%.
- **Causes:** persistent processus vaginalis, large internal and external rings, and lack of obliquity of the inguinal canal.

## Classic bladder exstrophy: anorectal and genital defects

### ▸ Anorectal defects

- 72-fold more likely in exstrophy, with an incidence of 1.8%.
- The anus lies directly behind the urogenital diaphragm; the perineum is short and broad.
- **Commonest anomalies:** imperforate anus (most common), rectal stenosis, congenital rectal prolapse.
- **Rectal prolapse:** due to divergent levator ani and puborectalis; its appearance is an indication for definitive repair of the bladder.

### ▸ Male genital defects

- The most severe and troublesome aspect of reconstruction.

- **Corpora:** posterior corporal length is normal; the angle between the corpora cavernosa is unchanged.
- **Prostate:** normal; PSA below the upper limit of normal. The urethra usually lies anterior to it.
- **Vas, ejaculatory ducts, seminal vesicles:** normal.
- **Testes:** mostly retractile.

▸ **Female genital defects**

- Reconstruction is less complex than in males.
- **Vagina:** hardly more than 6 cm deep, normal calibre.
- **Labia, mons pubis, clitoris:** divergent.
- **Ovaries and tubes:** normal.
- **Levator hiatus:** normal anteroposterior length but two times wider than normal.

## Classic bladder exstrophy: urinary tract

▸ **Bladder: structure**

- **Mucosa:** usually normal at birth but may carry hamartomatous polyps.
- **Muscarinic receptors:** normal density and binding affinity.
- **Collagen:** collagen-to-smooth-muscle ratio increased, with a threefold rise in type III collagen.
- **Nerves:** myelinated fibres per field significantly reduced. This may mean the newborn bladder is at an earlier stage of development, with potential for normal development after successful initial closure.
- **Caveolae** (intracellular structures for cell-to-cell signalling): usually normal after a successful closure; significantly lacking in those who eventually need augmentation cystoplasty.
- **Detrusor:** lower intracellular calcium.

▸ **Bladder: polyps and metaplasia**

- Polyps are of two types, fibrotic and oedematous; both show squamous metaplasia in about 50%.
- Cystitis cystica and von Brunn nests can occur, in addition to cystitis glandularis. Surveillance is by cytology and cystoscopy.

▸ **Upper tract**

- **Anomalies:** duplicated collecting system (most common), hypoplastic or absent kidney, pelvic kidney, PUJ (pelviureteric junction) obstruction, multicystic dysplastic kidney.
- The pouch of Douglas is unusually deep and enlarged, so the distal ureter approaches the bladder from inferior and lateral to the orifice, with little or no obliquity.

## Exstrophy variants

- **Superior vesical fissure:** the resulting superior vesical fistula resembles a vesicostomy.
- **Anteroposterior duplication (duplicate exstrophy):** the ureters attach to the closed pelvic bladder, so the superficial everted mucosa is dry.
- **Side-by-side duplication with exstrophy:** left and right halves with a muscle-containing midline septum; each has its own ureter and intact sphincter.

## Primary closure

### ▸ Selection for newborn closure

- Ectopic bowel within the exstrophied bladder is a relative contraindication.

### ▸ Osteotomy

- **Types:** vertical iliac; combined anterior innominate and vertical iliac; bilateral superior pubic ramus; diagonal iliac wing.
- **Vertical iliac:** made just lateral to the sacroiliac joint, leaving the posterior iliac cortex intact.
- **Combined anterior innominate and vertical iliac:** allows external fixator and intrafragmentary pins to be placed under direct vision; superior cosmetic appearance.
- **Timing:** can be done at any stage if the diastasis prevents urological goals. Usually not needed in newborns, because sacroiliac ligament laxity allows tension-free closure in the first 48–72 hours.
- **Benefits:** removes the need for fascial flaps at initial closure; brings the large pelvic floor muscles near the midline to support the bladder neck and improve continence.
- **Pubic suture:** No. 2 nylon horizontal mattress between the fibrous cartilage of the pubic rami; No. 2 nylon has the strongest load-to-failure ratio.
- **Aftercare:** tunnelled epidural catheters for 2–3 weeks control pain and movement, preventing the effects of inadequate immobilisation.

## Surgical approaches in the newborn

### ▸ Warsaw approach

- **Primary closure:** bladder, posterior urethra, pubis and abdominal wall, with or without osteotomy; spica cast for 3 weeks, then elastic bandage for 3 weeks.
- **Second stage:** bladder neck and epispadias repair together, once bladder capacity is suitable and the child is interested in continence.
- **Complications:** fistula or urethral stricture in 10%.

### ▸ Kelly repair

- **Newborn stage:** bladder dome closure with hernia repair.
- **Radical soft-tissue mobilisation:** incision taken onto the pubis to raise a flake of bone and carried down through the levator muscles, which are wrapped around the reconstructed proximal urethra for continence.
- **Penis:** dorsum dissected from the corporal bodies as in penile disassembly; meatus brought to the penoscrotal junction; the resulting penoscrotal hypospadias is repaired at 3 years.

### ▸ Mitchell complete repair

- 30–70% need a later hypospadias repair.

### ▸ Mainz repair

- Mainz ureterosigmoidostomy is used for failed cases and small bladder templates.
- The penis is reconstructed in the same procedure; one further operation may be needed for cosmetic correction.

▸ **Erlangen repair**

- **Classical repair:** bladder closure with bilateral reimplantation, bladder neck repair, bilateral groin exploration, epispadias repair and pubic closure without osteotomy; epidural catheter for 5 days.
- Osteotomy only for cloacal exstrophy or a failed repair.

## Follow-up after primary closure

- **Renal ultrasound:** if the first scan shows good drainage, repeat 3 months after discharge, then every 6 months to 1 year for the next 2–3 years, to detect upper-tract changes from reflux, infection or obstruction.
- **Urine culture:** before discharge and at later intervals.
- Abdominal distension is one cause of dehiscence.

## Failed closure and failed reconstruction

▸ **Bladder prolapse or dehiscence**

- Intramuscular testosterone is given 5 weeks and 2 weeks before repair.
- Major soft-tissue loss at reclosure may need buccal mucosal or full-thickness skin grafts.

▸ **Neourethral stricture**

- Usually associated with paraexstrophy skin flaps.
- **Warning signs:** prolonged dry intervals and a detectable post-void residual.
- Significant outlet obstruction with upper-tract deterioration may need upper-tract diversion if dilatation, incision or open urethroplasty is not enough.
- Bladder neck repair should wait until a posterior urethral stricture has clearly been repaired.

▸ **Vesicocutaneous fistula**

- Fistulous tract in the midline lower abdomen.
- Pubic diastasis wider than the preclosure measurement.
- Cystoscopy shows the bladder lying anteriorly just under the abdominal skin.

▸ **Failed bladder neck repair**

- **Causes of incontinence:** inadequate outlet resistance, small capacity, decreased compliance, or a combination.
- **Bulking agents:** when the dry interval is nearly acceptable (more than 2 hours), for example repeated collagen or a dextranomer-based agent.
- **Artificial urinary sphincter:** used in some patients with good bladder capacity.

▸ **Failed genital reconstruction**

- After loss of glans, corpora, penile skin or urethral plate from penile disassembly: tissue expansion, skin grafting and buccal mucosal grafting before neophalloplasty.
- Penile scars: excision followed by flaps, grafts or tissue expanders.

## Urinary diversion

▸ **Standard ureterosigmoidostomy**

- The first form of diversion used in exstrophy.
- **Complications:** rectal incontinence and ureteral obstruction.

- **Lifelong surveillance:** yearly renal ultrasound, plus colonoscopy in adult life.

▸ **Mainz sigma pouch**

- Non-refluxing ureteric reimplantation, abdominal wall reconstruction, bladder closed as a seminal receptacle above the prostate, penis reconstructed later (check).

▸ **Non-refluxing colonic conduit**

- Protects the kidneys from reflux and allows undiversion at an older age when clinically indicated.

## Sexuality and fertility

▸ **Male sexuality**

- **Concerns:** penile length, appearance and chordee.
- **Lengthening:** degloving, with resection and release of all residual scar tissue.
- **Skin cover:** tissue expanders for 4–6 weeks, or full-thickness grafts.
- In cloacal exstrophy the corpora are very small, so flap phalloplasty may be the only option.
- Men with exstrophy can have a sexual life as satisfactory as their peers.

▸ **Female sexuality**

- **Concerns:** appearance of the external genitalia, adequacy of the vaginal opening, uterine prolapse.
- **Timing:** initial external genital correction at birth; "touch-up" surgery may be needed at puberty.
- **Options:** tissue expanders, an inverted U flap of perineal skin to widen the vaginal opening, slings for prolapse.

▸ **Fertility**

- **Male infertility:** can follow iatrogenic injury to the verumontanum during functional closure or bladder neck reconstruction. Gamete intrafallopian transfer (GIFT) is an assisted reproduction option.
- **Women:** most can complete a pregnancy. Vaginal delivery and caesarean section have both been done, but deliveries are high risk for mother and baby and should be in a tertiary centre.

## Epispadias

▸ **Male epispadias: features**

- **Male-to-female ratio:** 3:1 to 5:1.
- **Frequency:** penopubic, then penile, then glanular.
- **Associated anomalies:** usually confined to the external genitalia, pubic diastasis and continence mechanism.
- **Renal anomaly:** the only one seen in some cases is agenesis of the left kidney.
- **Reflux** in complete epispadias: due to an inherently deficient ureterovesical junction.
- Inguinal hernias are significantly less common than in exstrophy.

▸ **Male epispadias: repair**

- **Urethral reconstruction:** options include a transverse island flap.

- Complete penile disassembly can be combined with reapproximation of the perineal muscle complex, to try to avoid bladder neck reconstruction.
- **Bladder neck:** endoscopic submucosal injection of plastic microspheres is an option.

▸ **Female epispadias: features**

- **Davis types:**
  - Intermediate (subsymphyseal): urethra split dorsally along most of its length.
  - Most severe (retrosymphyseal): cleft involves the entire urethra and the sphincter mechanism.
- **Labia minora:** usually poorly developed, ending anteriorly at the corresponding half of the clitoris.
- **Symphysis:** may be a narrow fibrous band.
- **Vagina and internal genitalia:** usually normal.

▸ **Female epispadias: repair**

- Inverting urethral closure starts near the bladder neck and progresses distally.
- Medial halves of the labia minora and clitoris are denuded before being brought together, adding urethral resistance.
- With simultaneous bladder neck reconstruction, the Foley catheter is not left in the urethra.
- **Older continence operations** (transvaginal plication of urethra and bladder neck, muscle transplantation, urethral twisting, urethral cauterisation, bladder flap, Marshall–Marchetti vesicourethral suspension): raise urethral resistance but do not correct the incontinence or the anatomy; results are disappointing.
- **Small bladder:** augmentation with PTFE injection at the bladder neck, or simultaneous bladder neck reconstruction and augmentation.

## Cloacal exstrophy

- Vaginal reconstruction is a planned step of modern functional reconstruction.

# Ch 44 · Prune-Belly Syndrome

## Synonyms, genetics and epidemiology

- **Synonyms:** abdominal musculature syndrome, triad syndrome, Eagle–Barrett syndrome.
- **Chromosomes:** abnormality is the exception. Reported associations: Turner syndrome, trisomy 13 and 18, monosomy 16, Beckwith–Wiedemann syndrome.
- **Incidence:** similar to bladder exstrophy; higher in children of younger mothers.

## Genitourinary anomalies

▸ **Kidneys, ureters and bladder**

- Pelviureteric junction obstruction can occur, primary or secondary.
- **Poor ureteric peristalsis:** ineffective coaptation from a reduced number of thick and thin myofibrils.
- **Splayed trigone:** ureteric orifices lie laterally and superiorly, possibly the reason for reflux.
- During voiding the bladder neck opens widely into the dilated prostatic urethra.

### ▸ Prostate and accessory sex organs

- **Prostatic histology:** more connective tissue, fewer epithelial and smooth muscle cells.
- **Distal posterior urethral obstruction:** atresia, valve, stenosis, membrane, diverticulum.
- **Vas and seminal vesicles:** often atretic.
- **Epididymis:** poorly attached to the testis.
- **Retrograde ejaculation:** due to an incompetent bladder neck.

### ▸ Anterior urethra and megalourethra

- The anterior urethra is usually normal.

| Megalourethra | On voiding | Cause |
|---|---|---|
| Fusiform | Whole penis dilates | Mesenchymal deficiency of the urethral folds |
| Scaphoid | Ventral dilatation | Mesenchymal deficiency of the urethral supportive tissue |

### ▸ Testis and infertility

- Intra-abdominal testes lie next to the dilated ureters.
- Undescent is probably caused by the distended bladder and raised intra-abdominal pressure.
- Histology is usually similar to other undescended testes.
- **Infertility:** a combination of testicular histological abnormalities, ductal structural defects and prostatic abnormalities.

## Abdominal wall and extra-genitourinary anomalies

### ▸ Abdominal wall

- Medial and inferior musculature is most commonly deficient; complete absence is rare.
- **Appearance:** neonate has wrinkled, redundant skin with flank bulges; in the growing child the wrinkles lessen and the abdomen becomes pot-bellied.
- Costal margin flaring from poor lower chest wall support contributes to respiratory illness.
- Wound healing is good, with no tendency to wound infection or incisional hernia.

### ▸ Cardiac and pulmonary

- **Cardiac:** patent ductus arteriosus, atrial septal defect, ventricular septal defect, tetralogy of Fallot.
- **Pulmonary:** pneumomediastinum and pneumothorax, with or without pulmonary hypoplasia. Restrictive lung disease can develop secondary to musculoskeletal disorders.

### ▸ Gastrointestinal

- Incomplete midgut rotation gives malrotation, volvulus, atresia and stenosis.
- **Associated:** omphalocele, gastroschisis, anorectal abnormalities.
- Splenic torsion can occur from abnormal mesenteric fixation.

## Presentation and diagnosis

### ▸ Prenatal

- Fetal hydronephrosis can be correctly diagnosed in the second trimester.
- Routine termination is not recommended, partly because the cause of prenatal hydronephrosis cannot be diagnosed precisely.

▸ **Woodard categories**

- Woodard described the spectrum (categories I–III) in 1985.
- **Category I:** Potter features plus dysplasia, oligohydramnios, pulmonary hypoplasia and urethral atresia.

▸ **Late presentation**

- Incomplete forms, and patients without the abdominal wall anomaly, may present in adulthood with urinary infection, hypertension or renal failure.

▸ **Evaluation**

- Serum BUN (blood urea nitrogen) and electrolytes detect systemic acidosis or electrolyte imbalance.

## Surgical management

▸ **Supravesical diversion**

- Used only in rare cases of pelviureteric or vesicoureteric junction obstruction.

▸ **Internal urethrotomy**

- **Indications:** high post-void residual, increasing hydroureteronephrosis, or VUR (vesicoureteric reflux) with recurrent upper tract infections.
- Done at the distal end of the prostatic urethra; does not cause incontinence in this group.

▸ **Reduction cystoplasty**

- Remodels the bladder into a more spherical shape to improve contractility and emptying.
- Ranges from simple excision of the urachal diverticulum to excision of redundant mucosa with an overlap between flaps.

▸ **Urethral surgery**

- **Atresia or hypoplasia:** simple dilatation, urethroplasty or hydrodistension.
- **Megalourethra repair:** subcoronal circumferential incision with penile degloving, excision or infolding of redundant urethra, reconstruction over a catheter.

▸ **Orchidopexy**

- Early surgery is justified partly by the potential for testicular carcinoma.
- When a transabdominal approach is not possible: microvascular autotransplantation, or single-stage or staged Fowler–Stephens orchidopexy.

▸ **Abdominoplasty**

- **Indication:** moderate to severe abdominal wall laxity.
- **Randolph technique:** based on electromyographic mapping showing infraumbilical regions most affected and supraumbilical and lateral regions least affected. Incision from the 12th rib to the pubic symphysis and on to the opposite 12th rib; full-thickness removal of skin, lower abdominal musculature and peritoneum; healthy fascia approximated to the anterior iliac spines, pubic tubercle and inferior fascia.
- **Ehrlich procedure:** umbilicus preserved on the inferior epigastric artery; skin and subcutaneous tissue elevated off muscle and fascia; each side advanced vest-over-pants to the opposite flank.

## Meningomyelocele: embryology and risk factors

▸ **Embryology**

- Spinal cord and vertebral column formation begins at day 18; canal closure proceeds caudally.
- The mechanism of neural tube closure, and what causes dysraphism, are not known.

▸ **Risk factors**

- **Maternal:** young and advanced age, obesity, diabetes, caffeine, smoking, low education and socioeconomic status, occupational exposures, a preconceptional stress event, low weight gain.
- **Folate:** deficiency, including folic acid antagonists such as valproate.
- **Obstetric:** higher birth order and previous maternal miscarriage.

## Spinal level and bladder effects

- **Level:** cervical to high thoracic 2%; low thoracic 5%.
- About 10% have no urological deficit.
- The bulbocavernosus reflex (S2–S4) helps to distinguish upper from lower motor neuron lesions.

| Feature | Upper motor neuron lesion | Lower motor neuron lesion |
|---|---|---|
| Detrusor | Overactive | Acontractile |
| Sacral reflexes | Exaggerated | Absent |
| Bladder wall | Thick-walled | Small, smooth-walled |
| Bladder neck | Closed | Open |
| External sphincter | No voluntary control; detrusor–sphincter dyssynergia | Partial or complete denervation |

## Neonatal assessment

- **Lower urinary tract dynamics:** synergic 26%; dyssynergic with or without poor compliance 37%; complete denervation 36%.
- Dyssynergic and complete denervation are the at-risk categories.

## Management

▸ **Surveillance**

- After puberty, ultrasound every 2 years.
- Repeat VCUG (voiding cystourethrogram) if infections recur.

▸ **Reflux**

- Cannot empty the bladder: CIC (clean intermittent catheterisation), whatever the grade.
- Grade 1–3 and able to void: antibiotic prophylaxis.
- Grade 4–5: CIC.
- With poor detrusor compliance: anticholinergics.
- **Antireflux surgery indications:** reflux persisting into puberty; reflux in a child having surgery to increase outlet resistance.

▸ **Vesicostomy**

- Indicated when the child is a poor candidate for augmentation.

▸ **Artificial somatic–autonomic reflex pathway**

- A novel approach, with limited data, aiming to restore bladder function in children.
- Made by limited laminectomy and microanastomosis of a lumbar ventral root to the S3 ventral root.

## Ch 47 · Functional Disorders of the Bladder and Bowel in Children

### Nocturnal enuresis: definition and epidemiology

- **American Psychiatric Association definition:** repeated urination into bed or clothes at least twice a week for at least 3 consecutive months, in a child of at least 5 years, not due to a drug side effect or medical condition.
- **Boys to girls:** 2:1.
- **Prevalence:** 15% at 5 years; 1% at 15 years.

### Nocturnal enuresis: causes

▸ **Genetics**

- Autosomal dominant inheritance reported; chromosomes 12, 13 and 22 implicated.
- Concordance is higher in monozygotic than dizygotic twins.
- **Risk in offspring:** 43% if one parent had prolonged enuresis, 77% if both, 15% if neither.

▸ **Bladder and maturation**

- Thought to stem from a maturational delay in bladder control.
- Sleep urodynamics show more bladder contractions. With pelvic floor activity there is no enuresis; without it the child wets.

▸ **Nocturnal polyuria**

- May also be due to increased evening fluid intake, increased evening dietary solute load, abnormal renal sodium handling, abnormal circadian rhythm of GFR, and abnormal sodium and calcium excretion.

▸ **Sleep**

- Parents often describe these children as deep sleepers.
- Episodes may occur primarily during non-REM sleep.

▸ **Secondary enuresis**

- **Stressful triggers:** parental divorce, sexual abuse.
- **Psychological causes:** conduct disorder, ADHD.

### Nocturnal enuresis: evaluation

- **Examination:** includes the lower limbs and the sacral reflex arc.
- **Rarely needed:** investigations such as MRI and cystoscopy.

## Nocturnal enuresis: treatment

### ▸ General principles

- Start treatment according to the concern and motivation of the child, not the parents.
- **Behavioural:** void regularly during the day and just before bed, 6–7 times daily in total.
- Avoid high-sugar and caffeine-based drinks.

### ▸ Enuresis alarm

- **Proposed mechanisms:** suppression of bladder emptying during sleep, increased nocturnal bladder volume, and waking the child to void by signalling when they urinate.
- **Device:** moisture detector wired on a bed pad or wireless in the undergarments; auditory alarm with or without a vibratory belt.
- **Results:** about two-thirds become dry during treatment; about 50% stay dry after it ends.
- **Use:** every night, without holidays, for 6 weeks. No improvement in 6 weeks: shift to desmopressin or combination therapy.
- **Relapse:** more than 2 wet nights in 2 weeks; treated by reusing the alarm with overlearning.

### ▸ Desmopressin

- **Half-life (oral):** 2–3 hours.
- **Response:** significant decrease in night wetting in 40%; 60–70% relapse after stopping.
- **Daily use:** 1-week drug holiday every 3–6 months, then reassess.
- **Negative response:** increase the dose, use combination therapy, or move to second- or third-line treatment.

### ▸ Anticholinergics (oxybutynin, tolterodine)

- Not effective as monotherapy; act by reducing nocturnal detrusor overactivity.

### ▸ Tricyclic antidepressants (imipramine)

- **Place:** after failure of combined alarm, desmopressin or anticholinergic therapy; given with desmopressin or alone.
- **Assessment:** response checked after 1 month; if no improvement after 3 months, taper and stop.
- **Course:** drug holiday every 3–6 months; dose tapered gradually over 2 weeks.

### ▸ Investigational therapies

- Indomethacin, diclofenac, diazepam, hypnosis, psychotherapy, acupuncture.

# 6 Female Urology

## Ch 101 · Surgical, Radiographic and Endoscopic Anatomy of the Female Pelvis

### Female Urethra

▸ **Layers**

- **Lumen:** urothelium proximally, non-keratinised stratified squamous epithelium distally.
- **Muscularis propria:** inner longitudinal smooth muscle, outer circular smooth muscle, then skeletal muscle (the sphincter).
- **Outermost layer:** serosa, or periurethral fascia.

▸ **Blood supply**

- **Arterial:** internal pudendal and vaginal arteries.

### Continence in Females

- **Compressor urethrae and urethrovaginal sphincter:** longitudinally arranged striated muscle fibres arising from the vagina and the perineal membrane.
- **Pubourethral ligament:** anteriorly fixes the mid-urethra and prevents transmission of raised intra-abdominal force to the rest of the urethra.

## Ch 108 · Urinary Incontinence and Pelvic Prolapse: Epidemiology and Pathophysiology

### Risk Factors

- **Obesity:** risk factor for primary pelvic organ prolapse (POP) and for early recurrence after POP surgery.

### Pathophysiology of POP (DeLancey levels)

- **Level I vertical fibres:** the critical factor distinguishing vaginal eversion from a post-hysterectomy cystocele, rectocele or enterocele, in which the vaginal apex stays well suspended.

## Ch 105 · Evaluation and Management of Women With Urinary Incontinence and Pelvic Organ Prolapse

### Risk Factors: Female Incontinence

- **Prevalence in adults:** stress urinary incontinence (SUI) 10–25%, mixed (MUI) 5–10%, urgency (UUI) 3–10%.
- **Diet:** carbonated beverages, like caffeine, are a risk factor for UUI but not for SUI.
- **Medical conditions:** depression is listed with diabetes as a risk factor.

### SUI Pathophysiology in Women

- **Hammock hypothesis:** loss of posterior supports with preserved anterior supports creates a shearing effect, so the urethra can no longer be compressed.

## Evaluation of Incontinence and Prolapse

▸ **Questionnaires**

- Complement the history and avoid physician bias in assessing symptoms.

▸ **Dye testing**

- Confirms the leaked fluid is urine, not vaginal discharge or peritoneal fluid.

▸ **Voiding cystourethrogram (VCUG) in incontinence**

- **Indications:** recurrent UTI, fluoroscopic demonstration of leakage, urethral diverticulum, vesicoureteric reflux (VUR) and bladder contour.

▸ **Prolapse assessment**

- Careful POP-Q examination, reassessed at surgery if necessary, remains the cornerstone.
- Investigations are rarely needed; options are VCUG, MRI and defecography.

# Ch 114 · Conservative Management of Urinary Incontinence: Behavioural and Pelvic Floor Therapy, Urethral and Pelvic Devices

## Behavioural Treatments: Overview and Assessment

- **Three most established programmes:** behavioural training with urge and stress strategies, bladder training with urge suppression, and pelvic floor muscle training (PFMT).
- A non-relaxing striated sphincter is among the conditions that benefit.
- Based on the principles of learning, so patient education comes first.
- **Assessment:** urine examination (CUE) to rule out infection.

## Lifestyle Modifications

▸ **Fluids and caffeine**

- Restrict fluids at times when toilet access will be limited.
- Caffeine reduction improves both stress and urge incontinence; the ICI (International Consultation on Incontinence) recommends it.

▸ **Other dietary advice**

- Reduce alcohol, spicy foods and citrus products; evidence is weak.

▸ **Constipation**

- Chronic constipation (fewer than 3 stools per week) is a risk factor through chronic straining.

## Pelvic Floor Muscle Training

▸ **Assessment**

- **Methods:** digital (vaginal or rectal) examination, visual observation, EMG, manometry and ultrasound; digital is the most common.
- **Laycock PERFECT:** digital scoring scheme, used alongside the Brink score.

▸ **Regimen**

- Daily practice increases muscle strength and improves motor skills.
- Empty the bladder before each session.

▸ **Outcomes**

- **Cure:** 16–27%; **improvement:** 48–80%.
- Little research on PFMT for long-term incontinence after prostatectomy.

▸ **Vaginal cones**

- **Method:** insert the heaviest cone she can retain standing, then walking about; on success, move to the next heaviest.
- May be better than no active treatment; further studies needed.

## Urge Suppression and Bladder Training

- **Urge suppression:** squeeze the pelvic floor repeatedly, wait about a minute, and use distraction if urgency continues.
- **Other urgency-suppression techniques:** deep breathing with relaxation, quick pelvic floor contractions and distraction.
- **Reinforcement:** consistent encouragement and positive feedback.

## Biofeedback

- Teaches control over physical responses through precise, instantaneous feedback; used outside urology for headache, muscle spasm and hypertension.
- **Perineometer:** first pelvic floor biofeedback device, developed by Kegel.
- **Computerised:** visual feedback on a monitor (from vaginal or anal probes or skin electrodes) or auditory feedback through speakers.

## Voiding and Pelvic Floor Dysfunction

- Up-training (teaching the muscle to contract) is easier than down-training (teaching it to relax).
- **Down-training:** biofeedback teaches patients to tell muscle tension apart from an active muscle contraction.
- **High-tone pelvic floor dysfunction:** create a relaxing environment and plan adequate time for voiding.

## Mechanical Vaginal and Urethral Devices

▸ **Intravaginal devices and pessaries**

- **Mechanism:** restore the urethra above the pelvic floor, where intra-abdominal pressure helps closure.
- **Uses:** stress, urgency and mixed incontinence.
- **Pessary material:** silicone or plastic; ring and dish pessaries each come with or without support.

▸ **Intraurethral devices**

- **Contraindications:** urge incontinence (device not retained), active UTI, urethral stricture; not useful in a fibrotic urethra.

▸ **FemSoft insert**

- **Design:** single-use silicone tube in a mineral-oil-filled silicone sleeve.
- **Use:** remove for voiding; change every 6 hours.

- **Sizes:** 16, 18 and 20 Fr; lengths 3.5 and 4.5 cm.

## Ch 116 · Retropubic Suspension Surgery for Incontinence in Females

### Indications and Contraindications

- In SUI with pan-pelvic floor weakness, colposuspension does not always correct the vaginal prolapse; a pelvic floor repair may also be needed.
- **Potential contraindication:** inadequate vaginal length or mobility from prior radiotherapy or prior surgery.
- Bladder overactivity coexists with SUI preoperatively in 30%.

### Procedures

▸ **Marshall–Marchetti–Krantz (MMK)**

- Risk of urethral sphincter dysfunction; no longer recommended for SUI.

▸ **Vagino-obturator shelf**

- Paravaginal variant combining paravaginal repair and Burch principles; sutures go laterally into the internal obturator fascia instead of hitching the vagina to the iliopectineal line.
- **Iliopectineal line:** compound of the arcuate line of the ilium and the pectineal line of the pubis.

▸ **Burch colposuspension**

- **Hybrid modification:** vaginal wall approximated to Cooper's ligament and vaginal tissue also to the internal obturator fascia.
- Prophylactic obliteration of the pouch of Douglas can be considered to prevent posterior vaginal wall weakness and enterocele.
- **Aftercare:** a drain may be placed; voiding trial on postoperative day 5.

▸ **Laparoscopic colposuspension**

- Most popular laparoscopic incontinence procedure.
- **Problems:** technical difficulty, longer operating time, higher cost.

### Complications of Colposuspension

- **Voiding difficulty:** detrusor denervation, associated detrusor dysfunction, over-correction of the urethral axis, over-tight sutures.
- **Systemic:** DVT, PE, atelectasis, wound infection.

## Ch 118 · Slings: Autologous, Biologic, Synthetic and Mid- Urethral

### Injection Therapy for UI (Urethral Bulking Agents)

▸ **Agents**

- **Heterologous biological:** bovine and porcine collagen; collagen discontinued in about 2010 (check).
- **Uroving (Urovive) microballoons:** balloon-type injectable, no longer used.
- **Investigational cell-based:** autologous muscle-derived stem cells, adipose tissue-derived stem cells, minced autologous striated muscle.

▸ **Indications and contraindications**

- **Indications:** mild SUI on exercise; women with SUI who cannot stop anticoagulants.
- **Contraindications:** severe urethral mucosal scarring (tissue cannot expand); hypersensitivity to the agent.

▸ **Mechanism**

- Augments or restores mucosal coaptation, giving a "hermetic seal" (airtight seal) effect.
- Improves ISD by preventing bladder neck opening during stress.
- Rapid response.

▸ **Technique**

- Transurethral route is more common than periurethral.
- Implanted circumferentially into the submucosa/lamina propria at 3 and 9 o'clock (or 4 and 8 o'clock).

▸ **Complications**

- **Migration:** commonest with Teflon.
- Urethral mucosal prolapse.

▸ **Stamey grading of incontinence outcome**

| Grade | Description |
|---|---|
| 0 | Continent |
| 1 | Urine loss with sudden increase in abdominal pressure, not when supine |
| 2 | Urine loss with physical stress such as walking or change of position |
| 3 | Total incontinence unrelated to physical activity |

## Adjustable Continence Therapy (ACT / ProACT)

- Developed as a non-migrating alternative to bulking agents.
- **Device:** two inflatable silicone balloons on silicone tubing with a titanium and silicone port.
- **Placement in women:** periurethral space at the bladder neck, with an introducer through two 1 cm incisions in the labial sulci at the level of the introitus.
- Balloons inflated after placement; ports buried subcutaneously.
- **Re-adjustment (reinjection):** may be needed at 6-weekly intervals.
- Better suited after surgical failure of SUI treatment; long-term data needed.
- **Complications:** infection, voiding problems (retention, urgency), perforation, migration.

## Graft and Sling Materials

▸ **Ideal graft**

- Biocompatible, chemically inert, non-carcinogenic, mechanically strong, sterile, easily available, affordable, minimal allergic reaction, resists shrinkage.

▸ **Autologous fascia**

- **Disadvantages:** longer operative time and hospital stay.
- **Harvest-site problems:** seroma, postoperative pain, hernia.

▸ **Allografts**

- **Examples:** cadaveric fascia lata, cadaveric dermis (acellular human dermis), dura mater.

- Pliable, easy to use, available in many sizes.
- **Risk:** potential transmission of HIV, hepatitis and Creutzfeldt–Jakob disease.

▸ **Xenografts**

- **Examples:** porcine small intestinal submucosa (SIS), porcine dermis, bovine dermis, bovine pericardium.
- **Sterilisation of allografts and xenografts:** freeze-drying, solvent dehydration or gamma irradiation.

▸ **Synthetic materials**

- **Advantages:** uniform, more consistent, more durable, sterile, biocompatible, least degradation or disruption of the graft.
- **Absorbable synthetics:** promote fibroblast proliferation, fewer infections, not rejected, less harmful to viscera.
- **Pore size** determines flexibility, infection risk and fibroblast ingrowth.
- Knitted and woven meshes are multifilament with small pores and interstices.

▸ **Amid classification**

| Type | Pores | Brands |
|---|---|---|
| I | Macroporous, more than 75 µm | Prolene, Natural Mesh, Marlex |
| II | Microporous, less than 10 µm | Gore-Tex, Surgical Membrane, Dual Mesh (all expanded PTFE, ePTFE) |
| III | Macroporous with microporous or multifilamentous components | Teflon (PTFE), Surgipro (polypropylene), Mycromesh (ePTFE), Mersilene (Dacron) |
| IV | Submicronic | Silastic (silicone), Celgard (polypropylene–Silastic sheet) |

## Urethral Slings: Pre-operative Assessment

- Urgency on preoperative history predicts a poor outcome.
- **Additional indications for urodynamics:** negative stress test, unexplained abnormal urinalysis, evidence of dysfunctional voiding.

## Pubovaginal Slings (PVS)

▸ **Indications and concept**

- Can aid urethral reconstruction after iatrogenic hypospadias, trauma or mesh perforation.
- Present concept is a modified McGuire technique with a shorter free graft whose tension can be adjusted.

▸ **Procedure**

- **Anaesthesia:** spinal or general.
- 18 Fr Foley catheter to empty the bladder.
- **Incision:** inverted U starting 2 cm from the meatus, extending to the bladder neck.
- Endopelvic fascia perforated with Metzenbaum scissors aimed at the ipsilateral shoulder, just medial to and immediately under the ischiopubic ramus.
- Fascial graft sutured to periurethral tissue before vaginal closure.
- Oestrogen-soaked vaginal gauze pack placed; removed on postoperative day 1 with the Foley.

▸ **Outcomes**

- **Autologous:** 15-year continence 60–97%.
- **Allograft:** limited data; failure 6–36% for frozen or freeze-dried grafts.
- **Synthetic:** comparable to modified Burch colposuspension, but with mesh or sling related problems.

▸ **Complications**

- **Vaginal incision:** urethral injury, bladder injury, infection.
- **Abdominal incision:** seroma, infection, hernia.
- **Synthetic sling:** perforation, exposure, infection.
- **General:** DVT, PE.

## Mid-Urethral Slings (MUS)

▸ **Mechanics**

- Impedes movement of the posterior urethral wall above the sling and directs it antero-inferiorly or anteriorly.
- Inward movement of the posterior wall against the sling narrows the lumen.

▸ **Retropubic MUS**

- **Kit:** two specially curved 5 mm trocars attached to a 40 cm, 1.1 cm wide polypropylene mesh.
- **Bottom-up:** 18 Fr Foley passed and bladder emptied at the start.
- **Top-down:** the tips of the surgeon's opposite fingers in the vagina guide the trocar.

▸ **Transobturator tape (TOT), outside-in**

- **Skin entry:** at the level of the clitoris, laterally at the adductor longus insertion.

▸ **Single-incision sling**

- Polypropylene mesh with two harder pointed ends anchoring in obturator internus on each side.
- Placed in a hammock or U-position.
- Decreased efficacy on longer follow-up.

▸ **Mixed urinary incontinence**

- Reasonable efficacy, but lower cure rates than in pure SUI.

▸ **Complications**

- **Vaginal mesh exposure:** management not standardised; one option is reapproximating vaginal mucosa over the mesh after irrigation with antibiotic saline.
- **Urethral or bladder perforation:** conservative management is not an option.
- **Voiding dysfunction:** can occur without obstruction; no clear diagnostic guidelines; cystoscopy looks for a hypersuspended bladder neck, bladder pathology and urethral mesh perforation.

# Ch 117 · Vaginal and Abdominal Reconstructive Surgery for Pelvic Organ Prolapse

## Types of POP Surgeries

- **Restorative surgery:** suits discrete defects in the fibromuscular layer of the endopelvic fascia, each corrected individually.

- **Transvaginal mesh (TVM) kits:** trocar-based or non-trocar-based.

## Anterior Compartment

- **Anterior colporrhaphy:** also called Kelly's (Kelley's) procedure.
- **Mesh-augmented repair materials:** polypropylene mesh (better results), cadaveric fascia, porcine dermis.
- **Abdominal paravaginal repair:** low midline or transverse incision.

## Apical Compartment

### ▸ Uterosacral ligament (high uterosacral vault suspension)

- Runs from S1–S3 to the cervical os and proximal posterior vagina, spreading fan-like.
- **Parts:** sacral, intermediate, cervical.

### ▸ Iliococcygeus suspension

- Keeps the vagina in a more normal axis than sacrospinous fixation.

### ▸ Abdominal sacrocolpopexy

- **Indication to consider:** active lifestyle.
- **Open approach:** lower midline or Pfannenstiel incision; peritoneum entered and bowel packed away.
- **Halban culdoplasty:** obliterates the cul-de-sac with sutures from the posterior peritoneum onto the outer rectal surface up to the vaginal cuff.

### ▸ Total colpocleisis

- Circumferential purse-string sutures around the prolapse, incorporating pubocervical and rectovaginal fascia.

### ▸ Hysteropexy

- **Vaginal uterus-sparing options:** mesh hysteropexy, sacrospinous hysteropexy.

# Ch 119 · Genitourinary Complications of Mesh and Their Repair

## Complications Related to Mesh in Pelvic Reconstructive Surgery

### ▸ Causes

- Classified as modifiable and non-modifiable factors.
- Poorly selected sutures cause wound dehiscence and mesh exposure.
- Bacterial contamination is non-modifiable and may not be adequately prevented by sterile technique.

### ▸ Types

- **Anatomical grouping:** neurologic, musculoskeletal, organ injury.
- **Nerve injury from improper placement:** ilioinguinal nerve and genital branch of the genitofemoral nerve.
- Osteomyelitis is recognised.

### ▸ Evaluation

- Vaginoscopy and colonoscopy as required.
- VCUG shows the location of an anatomic obstruction.

▸ **Treatment algorithm**

- After history and examination, the patient goes into one of four groups:
  - **Normal postoperative findings:** reassurance and routine follow-up.
  - **Bleeding, de novo voiding or defecatory dysfunction, or obstruction:** urine culture, post-void residual (PVR) and evaluation for perforation.
  - **Mesh exposure, extrusion or vaginal atrophy:** vaginal oestrogen, revision or excision.
  - **Pain alone:** symptom control and follow-up.
- Persistent or additional symptoms, or failed treatment, return to continued evaluation and re-enter the algorithm.

## Ch 121 · Urinary Tract Fistulae

### Vesicovaginal Fistula (VVF)

▸ **Causes**

- **Congenital:** listed separately.
- **Post-surgical:** colporrhaphy, vaginal biopsy, bladder biopsy, endoscopic or laser bladder therapy, vascular or pelvic rectal surgery.
- 3 times more common after total abdominal hysterectomy than after vaginal hysterectomy.
- **Obstetric:** obstructed labour, forceps laceration, uterine rupture, bladder injury at caesarean section (LSCS).
- **External trauma:** penetrating injury, pelvic fracture, sexual trauma.

▸ **Clinical features**

- **Pain:** from skin irritation by leaking urine, or in radiation fistulae.

▸ **Dye tests**

- **Double dye test:** oral pyridium (phenazopyridine) 200 mg three times daily; an IV dye can replace the oral dye.
- **Three-swab test:** swabs removed after 10 minutes; staining of only the lowest swab means a urethrovaginal fistula or dye backflow into the introitus.

▸ **Investigations**

- Urine routine examination, culture and, when indicated, cytology.
- **Doubtful leak:** vaginal fluid creatinine and potassium are both higher than serum.
- **Cystoscopy:** distinguishes immature from mature fistula.
- **VCUG:** voiding images are necessary.
- **Algorithm:** if not located on examination and dye testing, do VCUG or cystography; if still not found, look for a ureterovaginal fistula; if absent, consider other causes of wetness such as incontinence or vaginitis.

▸ **Conservative and minimally invasive treatment**

- Small fistulae (under 3–5 mm) suitable for a trial are typically oblique.
- Anticholinergics during the 2–3 week catheter trial.
- **Minimally invasive options:** Nd:YAG laser, fibrin sealant, bovine collagen.

▸ **Timing and approach**

- Best chance is early initial repair if the injury is recognised within 24–48 hours.

- Delay optimises tissue pliability.
- **Transvaginal:** as soon as 2–3 weeks after injury; shorter operating time.
- **Abdominal:** often after a 3–6 month delay.

▸ **Vaginal repairs**

- **Flap-splitting:** perivesical fascia closed with Lembert-type sutures perpendicular to the first suture line.
- **Vaginal cuff excision:** the funnel-shaped defect after tract excision is closed in 3 or 4 layers.

▸ **Abdominal repairs**

- **O'Conor (suprapubic):** vagina closed with a running absorbable suture.
- **Transvesical:** vertical anterior wall cystotomy without bivalving; tract excised transvesically; vaginal edges mobilised from the bladder; vagina then bladder closed separately; suprapubic catheter (SPC) placed before closing the anterior bladder wall.

▸ **Martius flap**

- Named after Heinrich Martius.
- The lateral (obturator) blood supply is sacrificed during mobilisation.

▸ **Omentum and other interposition**

- **Omentum advantages:** inherent lymphatic properties, heals even in infection, easy epithelialisation over its surface.
- **Other options:** labial myocutaneous flaps, seromuscular intestinal flaps, bladder mucosa free graft.

## Ureterovaginal Fistula

- **Gynaecological causes:** radical hysterectomy, LSCS, cystocele repair.
- **Other pelvic surgery:** vascular surgery, colon surgery, bladder neck suspension.
- **Further causes:** pelvic trauma, pelvic inflammatory disease, actinomycosis.

## Vesicouterine Fistula

- **Usual cause:** simultaneous injury to bladder and uterus.
- **Other causes:** vaginal operative delivery, vaginal birth after caesarean (VBAC), placenta percreta, IUD (foreign body), brachytherapy, uterine artery embolisation, obstructed labour (uterine rupture with bladder injury or ischaemia).
- **Conservative:** small immature fistulae may resolve with prolonged catheter drainage, with or without fulguration; hormonal induction of menopause causes involution of the puerperal uterus and may close the fistula.
- **Repair:** intraoperative ureteric stents help identify the ureters.

## Urethrovaginal Fistula

- **Cause:** Behçet syndrome (vasculitis and necrosis).
- **Leakage:** minimal if small, continuous if large.
- **Harder than VVF repair:** extensive soft tissue defects, lack of local viable tissue for multilayer closure, need for rotational vaginal flaps or bladder flap tubes.
- **Timing:** controversial; delayed repair at 2–6 months, or immediate once vaginal tissues are healthy.

- **Distal fistula without incontinence:** observation or extended meatotomy.
- **Vaginal flap repair:** inverted-U incision with the base of the U at the proximal fistula margin.
- **Adjuncts:** fibrin glue, free labial skin grafts.

## Vesicoenteric Fistula

▸ **Causes and presentation**

- **Other causes:** trauma, foreign body.
- **Presentation:** fever, chills, haematuria, non-specific GI symptoms, orchitis.

▸ **CT**

- **Triad:** bladder wall thickening next to a thickened colon loop, air in the bladder, colonic diverticula.
- Do after oral contrast and before IV contrast so enteric contrast in the bladder can be seen.

▸ **Other tests**

- **Cystography and barium enema:** low yield; adjuncts to assess malignancy or other colonic disease.
- **Bourne test:** after a non-diagnostic barium enema, the first voided urine is centrifuged at once and X-rayed; radiodense particles mean a positive test.
- **Oral activated charcoal:** black particles in the urine indicate a fistula.

▸ **Treatment**

- **Non-operative:** IV total parenteral nutrition (TPN), bowel rest, antibiotics, treatment of Crohn disease.
- **Preoperative preparation:** bowel preparation, nutritional supplementation, appropriate antibiotics.
- **Three-stage procedure for complex fistulae:**
  - **Stage I:** proximal defunctioning and distal drainage of both GI and urinary fistula, with eradication of sepsis.
  - **Stage II:** recovery (TPN, nursing, radiological planning).
  - **Stage III:** joint urological and GI reconstruction.

## Ureteroenteric and Pyeloenteric Fistula

▸ **Ureteroenteric fistula**

- **Commonest cause:** Crohn disease of the right ureter and terminal ileum, so most are right-sided.
- **Diagnosis:** retrograde pyelogram (RGP), CT or MR.
- **Treatment:** ureterolysis with stenting and possible bowel resection; ureteric resection may be unnecessary if the ureter separates from the bowel.

▸ **Pyeloenteric fistula**

- **Commonest sites:** duodenum on the right, descending colon on the left.
- **Presentation:** flank pain, fever, pyuria (60–70%), frequency, flank mass; iatrogenic ones cause minimal symptoms.
- **Diagnosis:** nephrostogram, urography, RGP, barium swallow or enema.

- **Non-functioning kidney:** like XGP, nephrectomy with GI reconstruction alongside a GI surgeon.
- **Other options:** internal urinary stenting, or a large nephrostomy tube positioned in the colon with bowel rest or enteric suction.

### Urethrorectal / Prostatorectal Fistula (RUF)

- **Further causes:** anorectal surgery, urethral instrumentation, ruptured prostatic abscess.
- **Presentation:** haematuria, nausea, vomiting, fever; rarely peritonitis.
- **Conservative (after open or laparoscopic prostatectomy):** catheter drainage, bowel rest, TPN.
- **Staged repair chosen for:** poor bowel preparation, large radiation fistulae, infection or abscess.
- **Transanal approach** (for example Latzko-type): possible, but exposure is poor.

### Urovascular Fistula

- **Types:** mostly ureteroiliac artery, then ureteroiliac vein; ureteroaortic is very rare.
- **Other causes:** penetrating trauma, pregnancy, prior ureterolithotomy.
- **Reno- or pyelovascular fistula after PCNL:** haematuria before or after nephrostomy removal; temporary control by replacing the nephrostomy tube.
- **Stable patient:** full radiological evaluation for diagnosis and reconstruction planning.
- **Definitive management:**
  - **Vascular side:** embolisation, endovascular stent, primary repair, or ligation with or without bypass.
  - **Urinary side:** ureteroureterostomy, transureteroureterostomy, cutaneous ureterostomy, or PCN with ureteric ligation.

## Ch 122 · Bladder and Female Urethral Diverticula

### Bladder Diverticulum

▸ **Histology and associations**

- **Wall:** mucosa, lamina propria, scattered thin muscle fibres and adventitia; the smooth muscle is disorganised and dysfunctional.
- **Syndromic association of congenital diverticula:** fetal alcohol syndrome.

▸ **Presentation**

- Congenital diverticula: enuresis, acute UTI, haematuria, pyelonephritis.

▸ **Imaging**

- **VCUG:** AP, lateral and oblique films.
- **CT or MRI:** needed especially when the neck is not patent or is blocked by tumour, so contrast does not fill the sac.
- **Pelvic ureter:** usually deviated medially, but lateral deviation can occur.

▸ **Endoscopy**

- Cellules and saccules are small outpouchings between hypertrophied muscle bands; no size criteria separate cellule, saccule and diverticulum, so the distinction is subjective and

arbitrary.

▸ **Malignancy**

- Stage at presentation is the most important prognostic factor; 5-year survival 85% for superficial and 45% for extradiverticular disease.

## Female Urethral Diverticulum (UD): Anatomy and Pathology

- **Position:** usually ventral (anterior vaginal wall), but can be dorsal or lateral.
- **Urethropelvic ligament:** formed by endopelvic and periurethral fascia, attached laterally to the tendinous arc of the obturator; the diverticulum lies within it, above the vaginal wall.
- **Lining:** urothelial, squamous, columnar or mixed, or absent leaving only fibrous tissue.
- About two-thirds show inflammatory change.
- **Size:** a few millimetres to several centimetres.

## UD: Presentation

- **Symptoms:** hesitancy, sense of incomplete voiding.
- **Signs:** suprapubic fullness or urinary retention.

## UD: Evaluation

▸ **Urine studies**

- **Culture:** commonest is E. coli; others are Neisseria gonorrhoeae, Chlamydia, Streptococcus, Staphylococcus.
- **Cytology:** if irritative symptoms are present.

▸ **Contrast studies**

- **MCUG:** both voiding and post-void films are needed.
- **Double-balloon positive-pressure urethrography:** uses the Trattner catheter; not widely performed.
- **IVP (intravenous pyelogram):** may also show the diverticulum.

▸ **Ultrasound and MRI**

- **Transvaginal ultrasound:** gives size and location; appears anechoic or hypoechoic.
- **MRI signal:** low on T1, high on T2.
- **MRI advantages:** no radiation, no dependence on voiding; surface-coil and endoluminal-coil MRI are both superior to VCUG and double-balloon urethrography.

## UD: Treatment

▸ **Indications and algorithm**

- Natural history is unknown, so no clear guidelines on indications for surgery.
- Surgery is also indicated for UD with SUI and when the patient asks for it.
- **UD with LUTS but no SUI:** treat other urodynamic findings or LUTS, then repair the diverticulum transvaginally.
- **Concomitant anti-incontinence surgery:** avoid synthetic materials (higher risk of erosion and infection).

▸ **Diverticulectomy technique**

- Urethra closed with vertical sutures, periurethral fascia with horizontal sutures.
- Most extravasation on the 14–21-day pericatheter VCUG resolves with weekly repeat studies.
- **Other (less used) procedures:** endoscopic unroofing, fulguration, incision with obliteration using oxidised cellulose, coagulation.

▸ **Complications**

- Female hypospadias.
- **Dyspareunia:** from vaginal scarring or narrowing.

**Other topics in this book**

## Female Urethral Occlusive Devices

- **Generations:** first, Reliance (no longer used); second, FemSoft insert.
- **Suited to:** mild SUI during sports and women unwilling to have SUI surgery.

## Intravaginal Continence Devices

- **Mechanism:** support the anterior vaginal wall and urethrovesical junction to improve urethral closure.
- **Options:** pessaries, contraceptive diaphragm, tampons.
- **Drawbacks:** mild side-effect profile but low efficacy; need regular cleaning; cannot address a fibrotic urethra or intrinsic sphincter deficiency (ISD); may dislodge or cause irritation.

# 7 Voiding Dysfunction

## Ch 103 · Physiology and Pharmacology of the Bladder and Urethra

### Functional classification of lower urinary tract (LUT) dysfunction

- **Failure to store, outlet causes:** urethral instability, intrinsic sphincter dysfunction, urethral hypermobility.
- **Failure to void, bladder causes:** severe infection, neurogenic disorders, overdistension, drugs.

### Bladder wall structure

▸ **Urothelium**

- About 7 layers: basal cells on the basal lamina, intermediate cells, apical umbrella cells, and a glycosaminoglycan (GAG) layer on the luminal surface.
- **Umbrella cells:** the largest urothelial cells; hexagonal, with microvilli.
- **Uroplakin plaques:** specialised areas of the apical membrane where uroplakins are assembled. Outer and inner leaflets differ in thickness, forming the asymmetric unit membrane (AUM).
- **Hinge region:** the symmetrical membrane between the plaques.

▸ **Lamina propria and vasculature**

- Coordinates the activities of the urothelium and detrusor.
- **Nerve plexus:** unmyelinated fibres.
- **Microvasculature:** mucosal and subepithelial plexuses.
- **Blood flow:** determined by intramural tension. In normal filling it adapts to the large increase in surface area until bladder pressure rises.

▸ **Stroma and matrix**

- Collagen and elastin in a proteoglycan matrix, filling the spaces among muscle bundles, vessels and nerves.
- Passive, low-metabolic tissue; with the detrusor, responsible for bladder viscoelasticity.
- **Non-fibrillar matrix:** proteoglycans and water.
- **Elastin:** amorphous, surrounded by a microfibrillar component containing glycoproteins. Found in all bladder layers. Mature elastic fibres are 90% elastin.
- Ageing causes a relative decrease of bladder smooth muscle.

▸ **Detrusor smooth muscle**

- Myofibrils form fascicles running in random directions, unlike the circular and longitudinal layers of gut or ureteric smooth muscle.
- Cells connect to form a functional syncytium.
- **Cells:** small, spindle-shaped, single central nucleus, no cross-striations, caveolae (flask-like membrane invaginations).
- **Interstitial cells (myofibroblasts):** proposed pace-making role in spontaneous bladder activity.

▸ **Contractile filaments**

| Filament | Diameter | Composition |
|---|---|---|
| Thick | 15 nm | Myosin II: two intertwined heavy chains, each with a rod, a head and a hinge. Each head carries light chains MLC20 and MLC17. |
| Thin | 6–8 nm | Actin with bound tropomyosin and caldesmon, which control the access points where myosin II heads form cross-bridges. |
| Intermediate | 10 nm | Desmin and vimentin. |

## Bladder mechanics: accommodation

- The lamina propria thins faster than the muscle wall.
- Wall thinning is due to:
  - muscle bundles rearranging from a "top to bottom" to a "side to side" configuration;
  - collagen coils becoming extended and taut, fibres oriented parallel to the lumen.

## Urothelium physiology

▸ **Permeability and transport**

- Modifies final urinary solute concentration depending on hydration status and protein.
- Permeable to water; expresses aquaporins.
- Sodium, potassium, chloride and urea all cross it, allowing their urinary concentration to be modified.
- **Sodium channels:** on the apical surface of umbrella cells and in cytoplasmic vesicles.
- **Electrical resistance:** high at the apical membrane; 10-fold lower at the basolateral membrane.
- Breakdown of apical cells raises water and urea permeability, producing the features of cystitis.

▸ **Tight junctions**

- **Proteins:** zona occludens-1, occludin, claudins 4, 8 and 12.
- Adapt to filling and voiding without affecting permeability.

▸ **GAG layer**

- Main GAG: chondroitin sulphate.
- Barrier role is controversial; may matter in bacterial antiadherence and in preventing damage by large macromolecules.

▸ **Signalling**

- Nociceptor and mechanoreceptor stimulation releases substance P, nitric oxide (NO), ATP and acetylcholine.
- Urothelial ATP may help regulate membrane trafficking.
- **Suburothelial interstitial cells (myofibroblasts):** in the lamina propria just below the basal layer; mediate interaction between urothelial cells and afferent nerves.

## Receptors and antimuscarinic mechanism

- Antimuscarinics are competitive antagonists at muscarinic receptors.

| Site | Muscarinic receptors |
|---|---|

| | |
|---|---|
| Bladder | M2 and M3, about 3:1 |
| Gastrointestinal tract | M2 more than M3 (about 4:1); stimulation increases motility |
| Eye | M1–M3; iris sphincter contraction |
| Brain | M1–M5 (M3 sparse); cognition |
| Salivary and parotid glands | M1 and M3 |

## Neural control

▸ **Efferent pathway (sympathetic)**

- Preganglionic fibres from the lumbar cord pass through the sympathetic chain ganglia.
- Postganglionic fibres relay via the inferior mesenteric (hypogastric) ganglia into the hypogastric nerves, then the pelvic ganglia.

▸ **Afferent fibres**

- **C fibres:** both volume (stretch) and pain sensors; more numerous at the bladder neck and urethra than in the rest of the bladder.
- **A-delta fibres:** evenly distributed through the detrusor, allowing a coordinated contraction.
- **Routes:** hypogastric nerve to the lumbar dorsal root ganglia; pelvic and pudendal nerves to the sacral dorsal root ganglia.

▸ **Modulators of afferent activity**

- **ATP:** released from urothelium on stretch; acts on P2X2 and P2X3 receptors to increase afferent signalling.
- **ATP:NO balance:** high NO with normal ATP tips towards detrusor underactivity; high ATP with normal NO gives detrusor overactivity.
- **TRP channels:** besides TRPV1 and TRPM8, TRPV2, TRPV4 and TRPA1 are expressed.
- **Stretch:** increases afferent impulses directly or via the neurokinin A–NK2 receptor complex.
- **Smooth muscle:** modulates afferents directly or by secreting nerve growth factor.
- **Cannabinoid receptors:** in urothelium and detrusor (CB1 more than CB2); modulate sensory signalling. Cannabis-based extracts improve urgency urinary incontinence and detrusor overactivity.

▸ **Efferent neurotransmitter drugs**

- **Duloxetine** (serotonin and noradrenaline reuptake inhibitor): increases external urethral sphincter (EUS) activity and reduces bladder activity; proposed for urge as well as stress incontinence.
- **Alpha-adrenoceptor blockade:** increases pelvic blood flow as well as improving voiding.

▸ **Central circuit**

- Limbic system and hypothalamus (emotion and safety) and prefrontal cortex (social aspect) feed into the periaqueductal grey (PAG) and pontine micturition centre (PMC).
- Ascending afferents reach the prefrontal cortex via the thalamus and insula.

# Ch 106 · Evaluation and Management of Males With Urinary Incontinence

## Continence mechanisms in males

- **Internal (lisso)sphincter:** bladder neck to distal verumontanum.

- The prostate itself adds to outlet resistance.
- **Rhabdosphincter support:** anteriorly by the pubourethral ligaments; posteriorly by musculofascial elements that condense and fuse at the perineal body.

## Ch 107 · Urodynamic and Videourodynamic Evaluation of the Lower Urinary Tract

### Indications and equipment

- **Indications (International Continence Society):** information on LUT function not expressed as a symptom or sign; predicting outcome of a planned treatment; confirming effects of an intervention; understanding why a previous treatment failed.
- **Intravesical catheters:** 4–10 Fr, dual- or triple-lumen; triple-lumen has channels for filling, Pves and urethral pressure profile.
- **Abdominal pressure catheters:** 8–12 Fr; usually rectal balloon, or a balloon in the vagina or a stoma.
- **Filling medium:** sterile water, saline, or radiographic contrast (if fluoroscopy is used).

### Normal values

- **Cystometric capacity:** 370–450 mL (plus or minus 100 mL).
- **Voiding detrusor pressure:** 40–60 cmH2O in men; 30–50 cmH2O in women.
- **Normal detrusor function:** voluntarily initiated continuous contraction emptying the bladder completely within a normal time span.

## Ch 109 · Neuromuscular Dysfunction of the Lower Urinary Tract

### Spinal cord anatomy

- 31 segments, each with paired ventral and dorsal roots that combine on each side to form a spinal nerve.
- **Vertebral to cord levels:** C1–C7 to C1–C8; T1–T8 to T1–T12; T9–T11 to L1–L5; T12–L1 to S1–S5.
- **Conus medullaris:** tapered distal cord at L1–L2.
- **Filum terminale:** pia mater extending to the coccyx; stabilises the cord.
- **Cauda equina:** L2–S5 and coccygeal roots.

### Classification

▸ **Lapides classification**

| Type | Cause | Features |
|---|---|---|
| Sensory | Diabetes, tabes dorsalis, pernicious anaemia; interrupted sensory fibres between bladder and cord, or afferent tracts to the brain | Sensation impaired early; late decompensation to a large-capacity atonic bladder |
| Motor paralytic | Extensive pelvic surgery or trauma destroying parasympathetic supply | Early painful retention or inability to initiate or maintain micturition; end stage decompensation |
| Uninhibited | Lesions in or above the brainstem: stroke, Parkinson disease, brain tumour | |

| | | |
|---|---|---|
| Reflex | Suprasacral spinal cord injury, transverse myelitis | |
| Autonomous | Lower motor neuron lesion destroying sacral cord, sacral roots or pelvic nerves | Complete motor and sensory separation of bladder from sacral cord |

▸ **Lesion level versus pattern**

- **Bladder sensation:** normal above the brainstem; absent with complete suprasacral, complete sacral and peripheral reflex arc lesions.
- The bladder neck is closed (competent) during spinal shock.

## Lesions at or above the brainstem

▸ **Stroke (CVA)**

- Uncommon findings after recovery from cerebral shock: loss of bladder sensation, persistent detrusor areflexia.

▸ **Parkinson disease**

- LUTS typically begin about 5 years after disease onset.
- Dopaminergic drugs have no impact on LUTS.

▸ **Other brain conditions**

- **Posterior fossa tumours:** urinary retention.
- **Acute traumatic brain injury:** retention from detrusor areflexia.
- **Cerebellar ataxia:** detrusor overactivity and incontinence common; spinal cord involvement adds detrusor sphincter dyssynergia (DSD) or areflexia.
- **Cerebral palsy:** non-progressive brain injury in the first year of life; most have normal storage and emptying.
- **Multiple system atrophy:** cell loss and gliosis in widespread brain areas.

## Spinal cord lesions

▸ **Multiple sclerosis**

- Most often involves the lateral corticospinal (pyramidal) and reticulospinal columns of the cervical cord, more than lumbar and sacral cord.

▸ **Spinal cord injury (SCI): epidemiology**

- **Causes:** road traffic accidents, falls, violence, sports; 70–80% male.
- **Leading causes of death:** pneumonia, septicaemia, pulmonary embolism, heart disease, accidents, suicide.

▸ **Spinal shock**

- **Definition:** decreased excitability of cord segments at and below the lesion, with absent somatic reflexes and flaccid paralysis below it.
- Short-lived in incomplete suprasacral injury.
- Striated sphincter has no voluntary control; guarding reflex absent.

▸ **Suprasacral and sacral SCI**

- **Suprasacral, musculoskeletal signs:** impaired superficial and deep sensation, abnormal plantar response.
- **Sacral:** Credé or Valsalva emptying may obstruct at the closed bladder neck or at the distal sphincter when its tone is fixed.

▸ **Urinary tract complications of SCI**

- **UTI:** incidence 57%. Choose an antimicrobial with little or no effect on normal flora where possible. Hydrophilic-coated catheters reduce infection by 21% in an institutional setting.
- **Vesicoureteric reflux (VUR):** more common with suprasacral lesions.
- **Bladder cancer:** earlier onset than in the general population; neither cystoscopy nor cytology is entirely reliable.
- **Other complications:** skin problems, depression or suicide.

▸ **Autonomic dysreflexia**

- **Incidence:** 20% of thoracic lesions, 60% of cervical lesions.
- **Prerequisite:** a viable distal spinal cord.
- **Triggers:** sexual activity, clot retention, electrocoagulation.
- **Heart rate:** bradycardia common, tachycardia rare.
- Symptoms resolve quickly once the stimulus is removed.
- **Procedural prophylaxis:** sublingual nifedipine 10 mg, 30 minutes before cystoscopy.
- **Less common drugs:** captopril, hydralazine, diazoxide.
- **Long-term prophylaxis:** terazosin 5 mg once daily for 3 months; careful monitoring still needed during procedures.
- **Drug-refractory cases:** neuroablative procedures include sacral neurectomy and dorsal root ganglionectomy.

## Lesions of the peripheral reflex arc

- Spinal stenosis causes storage and emptying dysfunction.

## Detrusor sphincter dyssynergia (DSD)

- **Mechanism:** interruption of spinobulbospinal pathways causes failure to inhibit the guarding reflex and excitation of Onuf's nucleus, so the EUS contracts with the detrusor.
- **Weld classification:** continuous versus intermittent.
- **Presentation:** mixed storage and voiding symptoms with neurological symptoms.
- **Complications:** upper tract deterioration, VUR, urolithiasis, hydronephrosis, sepsis, ureterovesical obstruction.
- **Examination:** perianal sensation, bulbocavernosus reflex, cremasteric reflex, anal tone on digital rectal examination.
- **EMG:** raised sphincter activity during bladder contraction without raised abdominal pressure (Valsalva), which separates it from straining.

## Non-neurogenic and functional voiding disorders

▸ **Bladder neck dysfunction**

- **Synonyms:** primary bladder neck obstruction, smooth sphincter dyssynergia, proximal urethral obstruction, dysfunctional bladder neck.
- Poorly understood non-neurogenic condition: incomplete bladder neck opening in voluntary and involuntary voiding; exact cause unknown.
- **Myogenic theory:** abnormal bladder neck muscle arrangement narrows the neck during detrusor contraction.
- **Neurogenic theory:** sympathetic hyperactivity in young, anxious people.

- Men report long-standing voiding and storage symptoms.

▸ **Bashful syndrome**

- Low-flow, low-pressure voiding in young men with marked hesitancy when others are present.
- Psychological: a tendency to overcontrol micturition.
- **Treatment:** behavioural therapy or drugs.

▸ **Hinman syndrome**

- **Synonyms:** occult voiding dysfunction, occult neuropathic bladder.

▸ **Fowler syndrome (retention in young women)**

- Typically women younger than 30 years.
- Painless retention: cannot void for one or more days without urgency; mild lower abdominal discomfort possible.
- **Diagnostic criteria:** capacity over 1 L without urgency, no neurological abnormality, normal MRI of brain and spinal cord.
- Associated with polycystic ovaries; may be a focal, hormonally dependent channelopathy.
- **Urodynamics:** detrusor acontractility, impaired sphincter relaxation, high urethral closure pressure.
- **Pathogenesis:** abnormal sphincter activity drives overactive urethral afferents, which inhibit bladder afferents and efferents, giving an acontractile bladder.
- Opioids make it worse.
- Apart from sacral neuromodulation, treatments (hormonal therapy, drugs, botulinum toxin) give poor outcomes.

## Ch 110 · Overactive Bladder

### Terminology

- Older terms: idiopathic detrusor overactivity was "detrusor instability"; neurogenic detrusor overactivity was "detrusor hyperreflexia".

### Epidemiology, aetiology and natural history

- More common in African-Americans than in white people.
- The female excess is greater in the young than in the old.
- "OAB dry" more common in men; "OAB wet" more common in women, owing to relative weakness of the female sphincter and bladder neck.
- **Neurological causes:** Parkinsonism, multiple sclerosis, stroke, probably via loss of cortical inhibition.
- May be related to fibromyalgia.
- Generally progressive, but can stay stable for long periods, and remission can occur.

### Pathophysiology

- **Neurogenic hypothesis:** generalised nerve-mediated detrusor excitation during filling, from loss of cerebral inhibition, re-emergence of primitive reflexes, new reflexes, or afferent sensitisation.

- Inappropriate sensory activity may come from abnormally numerous or pathologically sensitised afferent endings.
- Detrusor M2:M3 ratio is 3:1.

## Evaluation

▸ **OAB Symptom Score (OABSS, Homma et al)**

- Four questions on urinary condition during the last week.

| Question | Scoring |
|---|---|
| Q1 Daytime voids (waking to sleeping) | 7 or fewer: 0; 8–14: 1; 15 or more: 2 |
| Q2 Night-time voids (sleeping to waking) | 0: 0; 1: 1; 2 or more: 2 |
| Q3 Urgency (sudden desire difficult to defer) | Not at all 0; less than once a week 1; once a week or more 2; about once a day 3; 2–4 times a day 4; 5 or more times a day 5 |
| Q4 Urgency incontinence (leak because the desire cannot be deferred) | Same options, 0–5 |

▸ **Urodynamic findings**

- **Phasic DO:** the pattern in most idiopathic DO; contractions of increasing amplitude with filling; may or may not cause incontinence.
- **Terminal DO:** single involuntary contraction at cystometric capacity, often emptying the bladder completely.
- **Increased filling sensation:** early desire to void.
- On ambulatory urodynamics, DO appears in up to 60% of asymptomatic women.

## Treatment

▸ **Behavioural therapy**

- **Irritants to avoid:** caffeine, alcohol, tomato, chocolate, nuts.
- **Bladder training intervals:** half an hour, then 1 hour, 2 hours, then 3–4 hours.
- **Pelvic floor exercises:** 30–50 minutes daily for 8 weeks.
- Treat coexisting conditions such as diabetes and stones.

▸ **Refractory OAB**

- **Containment options:** appliances, catheters, urethral closure, urinary diversion.

# Ch 111 · Underactive Detrusor

## Epidemiology

- Population prevalence unknown, as there is no non-invasive marker.

## Pathophysiology and aetiology

- Can arise from combined myogenic and neurogenic pathophysiology.
- **Idiopathic:** attributed to ageing and, in men, unknown factors.
- **Neurogenic:** neurosyphilis, AIDS, pelvic fracture, pudendal nerve injury.
- **Iatrogenic:** radical pelvic surgery (including radical prostatectomy), detrusor myomectomy, pelvic radiotherapy.

- **Drugs:** antispasmodics.

## Diagnosis

- Current urodynamic methods neglect contraction velocity and sustainability.
- **Watts factor:** detrusor power per unit bladder surface area; lengthy, complex calculation; not validated.
- **Detrusor shortening velocity:** measurable, since a slow contraction may cause symptoms, but not part of the ICS definition of detrusor underactivity.
- **Occlusion tests** (voluntary stop, mechanical stop, continuous occlusion): measure isovolumetric pressure by obstructing outflow; impractical and uncomfortable or painful for some.
- **Contractility indices** such as the bladder contractility index (BCI): valued for simplicity.
- **Simple cut-offs** (for example Pdet at Qmax under 40 cmH2O, Qmax under 15 mL/s): easy, but no widely accepted diagnostic ranges.

## Management

- **Goal:** reduce complications such as UTI, bladder stones, VUR and skin damage from overflow incontinence.
- **Reducing outlet resistance:** alpha-blockers and muscle relaxants are routine.

### ▸ Parasympathomimetics (bethanechol, carbachol)

- Carbachol is a muscarinic agonist.
- Most likely to work with a reduced or absent contractile stimulus: reduced efferent input, impaired acetylcholine release, or increased acetylcholine breakdown.
- **Side effects:** include nausea, abdominal cramps, visual disturbance.

### ▸ Intravesical electrotherapy

- **Method:** saline-filled bladder; current through an electrode at the catheter tip, skin pad as return electrode.
- **Schedule:** daily sessions, usually 1 hour or more; 10–15 sessions as a trial.
- **Mechanism:** activates A-delta afferents, giving central reflex detrusor activation.
- **Drawbacks:** very time-consuming, resource-heavy, no definitive safety or application studies.
- **Results:** promising early results, mainly in children; a sham-controlled trial was negative.

# Ch 113 · Pharmacologic Management of Lower Urinary Tract Storage and Emptying Failure

## Drugs for stress urinary incontinence (SUI)

### ▸ Oestrogen

- Little effect on urodynamic stress incontinence, though two meta-analyses showed significant subjective improvement.
- **Sensitive tissues:** bladder, pelvic floor, urethra (epithelium, vasculature, connective tissue, muscle).

### ▸ Alpha-agonists and imipramine

- **Midodrine, methoxamine:** some alpha-1 selectivity; side effects limit use.

- **Ephedrine, phenylpropanolamine:** used, but no better than placebo.
- **Imipramine:** no RCTs in SUI.

▸ **Duloxetine**

- **Licence (Europe):** 40 mg twice daily for moderate to severe SUI (15 or more episodes per week).
- **Men:** SUI drugs poorly studied; duloxetine used off-label.
- **Other uses:** major depressive disorder, fibromyalgia, chronic musculoskeletal pain, generalised anxiety disorder.
- **Side effects:** include diarrhoea, fatigue, headache, somnolence.

## Drugs for storage failure

▸ **Antimuscarinics: pharmacology**

- Grouped by lipophilicity, charge and size into tertiary and quaternary amines.
- **Tertiary** (darifenacin, solifenacin, fesoterodine, oxybutynin, tolterodine, atropine): more lipophilic; more likely to cross the blood-brain barrier.
- **Quaternary** (propantheline, trospium): less lipophilic, charged, poorly absorbed, poor CNS entry, fewer CNS side effects.
- **Interactions:** enzyme inducers (phenytoin, phenobarbitone) lower levels; inhibitors (ketoconazole, cimetidine) raise them.
- **Cardiac adverse effects:** QT prolongation, raised heart rate, torsades de pointes.
- **Other adverse effects:** memory impairment, dizziness, fatigue, headache, photophobia.

▸ **Individual antimuscarinics**

| Drug | Key points |
|---|---|
| Oxybutynin | Half-life 2 hours. Three actions: antimuscarinic (higher affinity for M1 and M3), local anaesthetic, direct muscle relaxant (relevant intravesically). ER 5–10 mg OD. Transdermal form reduces desethyloxybutynin further but causes pruritus and erythema. Also topical gel, intrarectal and intravesical. |
| Darifenacin | Half-life 13–19 hours. |
| Tolterodine | IR 1 mg or 2 mg BD; ER 2 mg or 4 mg OD. |
| Trospium | Oral bioavailability below 10%; half-life about 20 hours; about 60% eliminated unchanged in urine; not metabolised by CYP450. |

▸ **Mirabegron (beta-3 agonist)**

- Beta1, beta2 and beta3 are all in detrusor and urothelium; beta3 predominates.
- Potassium channels, especially BKCa, may also be involved in (or matter more for) beta-mediated relaxation.
- Rapidly absorbed orally; liver metabolism by CYP450.
- Circulates unchanged and as glucuronide conjugates and other inactive metabolites; terminal half-life about 24 hours.
- Does not prolong QTc at therapeutic or supratherapeutic doses.

▸ **Flavoxate**

- Mixed actions, unknown mechanism: calcium antagonist, phosphodiesterase inhibitor, local anaesthetic.

- Used in detrusor overactivity or OAB; no RCTs.

▸ **Other agents**

- **Alpha-blockers:** silodosin and naftopidil also improve OAB symptoms.
- **COX inhibitors:** prostaglandin inhibition tried to aid storage; not a current option.

## Botulinum toxin

▸ **Preparations**

- Brands are wrapped by different proteins that modify potency: Botox (onabotulinumtoxinA), Dysport (abobotulinumtoxinA), Xeomin (incobotulinumtoxinA), Prosigne.
- Botox is about 3 times more potent than the others.
- Botox vials 100 U and 200 U; after reconstitution store at 2–8 degrees C and use within 24 hours.

▸ **Mechanism**

- Heavy and light chains of BoNT-A joined by a disulfide bond.
- Binds SV2 on the nerve terminal, enters by endocytosis; the light chain cleaves SNAP-25.
- **SNARE proteins:** synaptobrevin, SNAP-25, syntaxin.
- Also inhibits spinal release of glutamate, substance P and CGRP from sensory nerves.
- **Intravesical advantages:** high local concentration, minimal systemic effects.

▸ **Indications and contraindications**

- **Indications:** include refractory bladder outlet obstruction.
- **Contraindications:** hypersensitivity to BoNT; history of acute retention in a patient not on clean intermittent catheterisation (CIC).
- **Relative:** children, low pulmonary reserve, myasthenia gravis, aminoglycosides.
- **Commonest adverse effects:** UTI and bladder pain.

▸ **Injection technique**

- Cystoscopic, 4 mm ultrafine needle inserted about 2 mm into the wall.
- 30 injections 1 cm apart over the dome and posterior, right and left walls.
- Empty the bladder afterwards and observe for 30 minutes.

## Drugs to facilitate emptying

▸ **Bethanechol**

- Not hydrolysed by cholinesterase.
- **Dose:** 10–50 mg three or four times daily.
- **Side effects:** include abdominal cramps, sweating, headache, difficulty with visual accommodation.
- **Contraindications:** bowel obstruction, hyperthyroidism.

▸ **Prostaglandins**

- Only a paucity of literature for increasing bladder contractility.

▸ **Baclofen**

- GABA-B agonist (GABA is inhibitory); spinal inhibitory action controls micturition disorders and lowers outlet resistance.

- Not effective orally for voiding dysfunction, as it does not cross the blood-brain barrier.
- Intrathecal baclofen is an option to lower outlet resistance; few studies.

## Ch 115 · Electrical Stimulation and Neuromodulation in Storage and Emptying Failure

### Sacral neuromodulation (SNM)

▸ **Indications and contraindications**

- **Idiopathic non-obstructive retention:** Fowler syndrome, DSD, functional outlet resistance.
- Neurogenic bladder in multiple sclerosis.
- Faecal incontinence and constipation.
- **Relative contraindications:** pregnancy, significant spine abnormalities.

▸ **Staged trial**

- **Test stimulation:** 3–4 weeks for retention; 1–2 weeks for OAB.
- **Correct lead sensation:** between rectum and vagina in women; penoscrotal in men.

▸ **Mechanism**

- **OAB:** stimulation activates pudendal afferents, which inhibit bladder C-fibre afferents and their output to the PMC; DO is inhibited and sensation improves.
- A-delta afferents are not inhibited, so normal voiding is preserved.
- **Functional retention (DSD, Fowler):** abnormal pudendal afferent activity inhibits detrusor contraction; SNM restores normal activity so sensation reaches the PAG and PMC and parasympathetic efferents produce voiding.

▸ **Troubleshooting**

| Problem | Action |
|---|---|
| Pain at implantable pulse generator (IPG) site | Turn device off. If pain persists: pocket complication (erosion, infection, seroma, pocket too loose or tight); treat accordingly. If pain resolves: change unipolar to bipolar, or revise if due to current leakage. |
| Stimulation in wrong location | Unipolar or bipolar reprogramming; revise leads if this fails. |
| No stimulation felt | Exclude inadvertent switch-off and IPG end of life, then check impedances: all abnormal, revise; some abnormal, program with the normal electrodes; all equal and normal, suspect current leakage and revise. |
| Intermittent stimulation | Check impedances: normal suggests a positional problem; abnormal a mechanical problem. |

## Other topics in this book

### Urodynamic tests

▸ **Uroflowmetry: definitions**

- **Flow time:** time over which measurable flow occurs.
- **Average flow rate:** voided volume divided by flow time.
- **Time to maximum flow:** from flow onset to maximum flow.

▸ **Uroflowmetry: normal pattern**

- **Ideal voided volume:** 200–400 mL.

- Trace depends on voided volume and paper speed (0.25 cm/s).
- Interpret with age, sex and voided volume in mind.
- Maximum flow should be reached within 5 seconds of flow start.

| Age (years) | Minimum voided volume | Qmax male | Qmax female |
|---|---|---|---|
| 4–7 | 100 mL | 10 mL/s | 10 mL/s |
| 8–13 | 100 mL | 12 mL/s | 15 mL/s |
| 14–45 | 200 mL | 21 mL/s | 18 mL/s |
| 46–65 | 200 mL | 12 mL/s | 15 mL/s |
| 66–80 | 200 mL | 9 mL/s | 10 mL/s |

### ▸ Uroflowmetry: abnormal patterns

- **Detrusor overactivity:** very rapid rise to maximum in an abnormally short time (1–3 seconds).
- **Detrusor underactivity:** very variable time to maximum; maximum may fall in the second half of the trace.
- **Outlet obstruction:** compressive (BPH) or constrictive (stricture).
  - **Compressive:** first third may look fairly normal with reduced Qmax; later part elongated into a tail of falling flow.
  - **Constrictive:** little difference between Qmax and Qaverage; constant flow from fixed resistance.
- **Abdominal straining:** interrupted, variable traces; with or without obstruction or detrusor contraction.
- **Urethral overactivity:** interrupted flow that changes faster than with straining.
- **Poorly sustained or fluctuating contraction (hypotonic bladder):** low, undulating interrupted curve.

### ▸ Pressure-flow studies

- **Bladder outlet obstruction index** (PdetQmax minus 2 Qmax): also called the Abrams–Griffiths number.

### ▸ Detrusor leak point pressure (DLPP)

- Useful in upper motor neurone lesions with high storage pressure, non-neurogenic low-compliance bladders, and lower motor neurone lesions decentralising the bladder.

### ▸ Urethral pressure profile (UPP)

- **Urethral pressure:** fluid pressure needed to just open a closed urethra.
- **Method:** small catheter with lateral apertures infuses fluid while bladder and urethral pressures are measured as it is slowly withdrawn.
- **Urethral closure pressure profile:** difference between urethral and intravesical pressure along the urethra.
- **Maximum urethral pressure (MUP):** highest pressure on the profile.
- **Functional profile length:** length where urethral pressure exceeds intravesical pressure; about 3 cm in women.
- **Total profile length:** full length of the tracing (shown on the UPP diagram with MUCP, MUP and bladder pressure).

▸ **Videourodynamics (VUDS)**

- Children with abnormal voiding when invasive therapy is being considered.
- Impaired renal function where urethral overactivity may be the cause (for example with VUR).

▸ **Ambulatory urodynamics**

- **Three cycles:** resting (sitting on a chair), ambulant (moving around), exercising (specific incontinence-provoking acts).
- **Indication:** to decide whether DO or sphincter weakness is the main cause of incontinence.
- **Limitations:** time-consuming, technically challenging, no established normal and abnormal standards, more reproducibility data needed.

▸ **Coordination: non-relaxing sphincter**

- Classically in meningocele; a failure of urethral relaxation rather than active dyssynergia.

## Primary bladder neck obstruction (PBNO): treatment

- **Men:** alpha-blockers (partial relief), bladder neck incision (definitive relief), then TUIP or TURP.
- **Women:** pelvic floor or behavioural therapy and drugs first, as surgery carries a high risk of SUI.

## Storage failure at the outlet: additional therapies (Ch 87)

- **Options:** myoplasty, male urethral compression devices, female urethral occlusive devices, intravaginal continence devices, bladder outlet closure (functional or complete).

▸ **Myoplasty**

- Autologous gracilis transfer forming a neosphincter around urethra and bladder neck; large studies lacking.

▸ **Bladder outlet closure**

- **Indications:** failure of other surgery, or urethral or bladder neck erosion or destruction.
- **Functional:** autologous pubovaginal sling obstructs the bladder neck but allows a pop-off leak at higher pressures.
- **Complete:** when a functional outlet cannot be kept; multilayered bladder neck closure (transvaginal, transabdominal or combined) with an abdominal stoma for CIC.

## Augmentation cystoplasty

- **Poor compliance:** lowers maximum detrusor pressure by 65–82%.
- **Reduced capacity** (causing frequency and incontinence): increases cystometric capacity to over 500 mL.
- **Refractory DO** (after drugs, neuromodulation and botulinum toxin): especially in patients who fear incontinence.

## Continent catheterisable stomas

- **Further indications:** urethral or bladder neck destruction (erosion from chronic catheter, radiation, trauma); refractory urethral fistula or stricture.
- **Complications:** stomal prolapse and stricture, which may need revision.

## Emptying failure: catheters and manoeuvres

▸ **Long-term indwelling catheter**

- Generally for short-term drainage; for long-term (such as SCI) CIC is preferred.
- **Complications:** poor compliance (cause unknown), VUR, pyelonephritis, renal scars, calculi, periurethral abscess, blockage, haematuria.
- **Carcinoma risk:** 8–10% long term, from chronic inflammation and nitrosamines.

▸ **Suprapubic catheter**

- **Advantages:** preserves sexual function; low risk of urethral stricture and epididymitis.
- **Disadvantages:** invasive; incontinence may persist with a very weak sphincter or reflex DO.

▸ **Clean intermittent catheterisation (CIC)**

- **Prerequisites:** adequate urethral exposure, dexterity, cognition, compliance.
- More frequent with low bladder capacity.
- Women use a short (6-inch) catheter.
- Wash catheter with soap and water and air-dry before storage.
- **Advantages:** no stoma bag, preserved sexual function, no cancer risk, fewer complications than an indwelling catheter.

▸ **Credé and Valsalva**

- May succeed in an areflexic bladder with low outlet resistance (after surgery or botulinum toxin); these patients have stress incontinence and can empty when they choose.
- Credé can generate over 50 cmH2O intravesical pressure.
- **Credé methods:** open hand, closed fist, or rolled-up towel.
- **Valsalva:** sit with the abdomen protruding onto the thighs; hugging knees and legs may stop the abdomen bulging.

▸ **Triggering reflex contractions**

- In SCI with intact cord reflexes, rhythmic suprapubic pressure (seven or eight pushes every 3 seconds) summates on wall tension receptors and activates the reflex arc.
- **Other triggers:** pulling pubic, scrotal or thigh skin or hair; digital rectal stimulation; squeezing the clitoris.

## Collection devices and absorbent products

- No effective external collection devices for women; men can use a condom catheter, penile sheath or Texas catheter.
- **Absorbents:** pads, shields, drip collectors, guards, briefs, undergarments, underpads.
- **Incontinence-associated dermatitis:** surface skin inflammation with redness, oedema, sometimes bullae with clear exudate; mainly in skin folds; may develop candidiasis or bacterial infection.

## Diabetes and urology

▸ **FRIENDS mnemonic (urological complications)**

- **F:** Fournier gangrene and soft-tissue infection.
- **R:** Renal failure.
- **I:** Infections and UTI.

- **E:** Enhanced wound complications and emphysematous pyelonephritis (EPN).
- **N:** Necrosis (papillary).
- **D:** Diabetic cystopathy.
- **S:** Stones (low urine pH) and Sexual dysfunction.

▸ **Diabetic cystopathy: mechanism**

- 5–59% of diabetic patients report LUT dysfunction.
- **Neuropathy:** peripheral and autonomic; axonal degeneration with segmental demyelination.
- **Sequence:** impaired sensory afferents, impaired sensation, gradually longer intervals between voids, overdistension, impaired detrusor function, then decompensation (end-stage bladder).
- **"Temporal theory":** early hyperglycaemic osmotic diuresis overstretches the wall and raises pressure, causing compensatory hypertrophy with storage symptoms; later progression from accumulated toxic products of oxidative stress.
- Detrusor myocyte dysfunction may also contribute, by a poorly understood mechanism.

▸ **Diabetic cystopathy: evaluation**

- **Tests:** random or fasting blood sugar, HbA1c, urine routine, renal function, bladder diary.
- **Ultrasound:** high post-void residual, wall thickening, upper tract assessment.
- **Uroflowmetry:** impaired rate and pattern.
- **Urodynamics:** sphincters usually synergistic; pseudodyssynergia may appear on EMG.

▸ **Diabetic cystopathy: treatment**

- Start with glycaemic and infection control.
- **Early disease:** scheduled and double voiding.
- Reduce outlet resistance with drugs or surgery.
- **End-stage bladder:** CIC or urinary diversion.

# 8 Nocturia & Male Incontinence

## Ch 112 · Nocturia

### Definition and rationale for evaluation

- **Prevalence:** 28.4% at 30–80 years; commoner in women than men; rises with age.
- **Cost:** lost productivity to society.

▸ **Sleep measures**

- **Sleep latency:** time from fully awake to fully asleep.
- **Sleep efficiency:** actual sleep time divided by intended sleep time (in minutes); above 85% is normal.
- Nocturia affects both latency and efficiency.

### Physiology: arginine vasopressin (AVP, ADH)

- **Collecting duct:** water leaves across the basolateral membrane through aquaporin-3 and aquaporin-4. Aquaporin-2 sits in recycling vesicles that are inserted into the luminal membrane and retrieved by endocytosis.
- **Stimuli for release:** hyperosmolality, angiotensin II, reduced atrial receptor firing, sympathetic stimulation.

### Causes and nocturia index

- **Nocturia index (Ni) above 1:** night urine volume exceeds maximum bladder capacity, giving nocturia, or enuresis if the patient does not wake.
- **Low nocturnal or global bladder capacity:** ureteric or bladder calculi; carcinoma of bladder, prostate or urethra; nocturnal detrusor overactivity; anxiety disorders; learned voiding dysfunction; radiation; tuberculosis.

### Global polyuria: evaluation and treatment

▸ **Overnight water deprivation test**

- **First-morning urine osmolality above 800 mOsm/kg H2O:** normal; points to primary polydipsia (appropriate renal response and ADH secretion).
- **Below 800 mOsm/kg:** give desmopressin 40 µg intranasal or 0.4 mg orally; recheck osmolality 3–5 hours later.
  - 800 or more: central diabetes insipidus (DI).
  - Below 800: nephrogenic DI.

▸ **Diabetes insipidus**

- **Further central (pituitary or hypothalamic) causes:** metastasis, sarcoidosis, infarction, infection.
- **Extrarenal and other causes:** PGE2, ANP, lithium, tetracycline, hypercalcaemia, hypokalaemia.
- **Nephrogenic DI:** no specific treatment.

▸ **Primary polydipsia**

- **Types:** dipsogenic or psychogenic.

- **Dipsogenic:** history of central neurological abnormality or radiation; treat if a cause is found.
- **Psychogenic:** behavioural therapy or psychotherapy.

▸ **Diabetes mellitus**

- Treatment targets the glycosuria.

## Diminished global and nocturnal bladder capacity: treatment

- **Treating bladder outlet obstruction (BOO):** lowers PVR, raises functional bladder capacity and lowers frequency, so nocturia improves.
- **Other agents, none with conclusive evidence:** COX-2 inhibitors (block PGE2-mediated diuresis), Pygeum africanum, Cernilton (rye-derived).

## Nocturnal polyuria: drugs

▸ **Imipramine**

- **Effect:** variable in nocturnal polyuria.
- **Pharmacology:** complex. A tricyclic (TCA) that acts as an SNRI and non-selective antimuscarinic, and modulates AVP release and proximal tubular sodium and water reabsorption.
- **Black box warning:** increased suicidal tendency in depressed patients.
- **Cardiac effects:** prolonged PR, QRS and QTc; torsades de pointes; sudden death.
- **Other side effects:** dizziness, low blood pressure, blurred vision.

▸ **Desmopressin**

- **Dose, tablet:** 0.2 mg (1 tablet) late evening; raise to 0.4 mg after 10–14 days if needed.
- **Dose, melt:** 120 µg; raise to 240 µg after 10–14 days if needed.
- **Fluid limit:** 240 mL from 1 hour before until 8 hours after dosing, to prevent dilutional hyponatraemia.
- **Follow-up:** assess response at 1–2 weeks; if positive, continue for 3 months. After early checks, monitor sodium every 6 months.
- **Stop:** in severe vomiting and diarrhoea, fever or flu. Use with caution in hot weather or after strenuous exercise.
- **Side effect:** facial flushing.
- **Blood pressure:** vasopressin stimulates ACTH release but desmopressin does not (except in Cushing's disease), so it does not directly raise blood pressure.
- **Enuresis in children:** 2.2 fewer wet nights a week; 4.5 times more likely than placebo to sleep without disruption.
- **Nasal spray ban:** US regulators banned sprays for nocturnal enuresis after 2 deaths and 59 seizures from hyponatraemia. Tablets are still considered safe in otherwise healthy children.
- **Haematology:** raises von Willebrand factor and factor VIII 3- to 5-fold.
  - Used in von Willebrand disease, mild haemophilia A and thrombocytopenia.
  - Not effective in haemophilia B or severe haemophilia A.
  - Contraindicated in type IIb von Willebrand disease.

## Ch 123 · Surgical Procedures for Sphincteric Incontinence in the Male: Artificial Urinary Sphincter and Perineal Slings

### Risk factors and aetiology

- **Male incontinence:** residence in old-age (nursing) homes.
- **After prostatectomy:** preoperative bladder function is a risk factor.
- **Commonest cause after prostatectomy:** sphincter dysfunction or injury, in more than two-thirds.

▸ **Sphincter deficiency**

- **Internal (bladder neck):** pelvic surgery, bladder neck injury, sympathetic neuropathy, embryological disruption.
- **External:** radical prostatectomy (commonest), pelvic fracture, myelopathy, congenital (sacral agenesis, spinal dysraphism, exstrophy-epispadias complex).

### Evaluation

- **Pad test:** 24-hour is most accurate; 1-hour is most commonly used.

### Management algorithm

- **Significant OAB or reduced compliance** (on urodynamics or accurate voiding diary): antimuscarinics first. If OAB resolves, return to the sphincter pathway; if not, evaluate and treat bladder dysfunction before sphincter surgery.
- **Gravitational (continuous) incontinence:** AUS rather than sling; surgery need only be delayed about 6 months after prostatectomy.
- **Persistent incontinence after AUS revision options** (smaller cuff, transcorporal cuff, double cuff, bladder neck cuff): bladder neck closure or urinary diversion.

### History of continence devices

- **Evolution:** Kaufman I, II and III, then slings, then AMS 721, 742, 791 and 792, then AMS 800.

### Artificial urinary sphincter (AUS)

▸ **AUS versus sling**

- **AUS:** reproducible outcome; revision about 16% at 2 years and 28% at 5 years.
- **Sling:** an option for men who refuse an AUS for fear of its complications.

▸ **AMS 800 placement**

- **Position:** low lithotomy.
- **Pressure-regulating balloon:** prevesical space, slightly lateral, through a horizontal lower-quadrant abdominal incision.
- **Pump:** anterior scrotum through the same incision; tubing connections made in the abdominal incision.
- **Alternatives:** scrotal and perineal approaches.

▸ **Variations**

- **Transcorporal cuff:** dissection passes through the septum of the corpora cavernosa from side to side; reduces urethral injury during dissection.
- **Transscrotal AUS:** all components through one scrotal incision; continence results inferior.

▸ **Complications**

- **Retention:** rare. Suprapubic catheter for retention beyond 48 hours is placed under ultrasound guidance.
- **Infection:** scrotal pain, erythema, oedema, frank purulence. Non-eroded device: immediate reimplantation with antiseptic irrigation and device replacement; otherwise delayed.
- **Erosion:** confirm urethral healing by RGU or cystoscopy at 3–6 months before reimplantation.
- **Urethral atrophy:** from chronic cuff compression; commonest reason for revision.

## Male slings

- **Three types:** transobturator, quadratic-fixation, bone-anchored.
- **Contraindications:** detrusor overactivity, prior radiotherapy.
- **Complications:** perineal pain, urinary retention, infection, erosion, osteomyelitis (bone-anchored).

▸ **Transobturator bulbourethral sling**

- Fixed to the corpus spongiosum with 4–6 permanent or absorbable sutures.

▸ **Bone-anchored bulbourethral sling (InVance)**

- **Mechanism:** urethral compression.
- **Approach:** perineal incision exposing bulbospongiosus and the medial inferior pubic ramus.
- **Fixation:** sling lies over bulbospongiosus, fixed with 3 screws each side to the inferior pubic ramus.
- **Outcome:** cure or improvement 67–92%; failure 8–33%.

## Urethral bulking agents

- **Agents:** Deflux, Durasphere, Macroplastique.
- **Volume:** usually 2.5–5 mL.
- **Routes in men:** transurethral, periurethral, antegrade or retrograde.
- **Complication:** urethral inflammation leading to a "frozen" urethra.

**Other topics in this book**

## Male urethral compression devices

▸ **External devices**

- **Examples:** Hiesler metal clamp (no longer used), Cunningham clamp, C-3, U-Tex.
- Cause pressure necrosis, so release every 3–4 hours.
- Cannot be used with impaired penile sensation.

▸ **Internal devices**

- **Berry's implantable prosthesis:** no longer used.

- **Kaufman I and II:** compressed the bulbar urethra by cavernous–crural crossover (I) or approximation (II).
- **Kaufman III:** silicone gel prosthesis implanted perineally to compress the bulbar urethra.

# 9 Andrology & Sexual Medicine

## Ch 69 · Male Infertility

### Genetic disorders: classification

- **Three groups:** single-gene disorders, chromosomal disorders and non-Mendelian disorders. Non-Mendelian disorders account for most human disease.
- **Single-gene disorder:** caused by a mutant allele or a pair of alleles at a single genetic locus.
- **Chromosomal cause:** a truncated Y chromosome.

### Single-gene disorder: congenital bilateral absence of the vas deferens (CBAVD)

- Present in about 1% of infertile men.
- **CFTR protein:** maintains secretion viscosity through sodium and chloride channel balance.
- **Non-CF form ("Potter syndrome"):** abnormal differentiation of the mesonephric duct before 7 weeks of gestation causes CBAVD with renal agenesis or renal ectopia. Genetic screening is essential for diagnosis. (check)

### Chromosomal disorders

▸ **Klinefelter syndrome**

- Typically presents with infertility despite normal libido and erectile function.
- Higher risk of non-Hodgkin lymphoma.

▸ **46,XX male**

- SRY may be translocated to an autosome as well as to the X chromosome.

▸ **Y chromosome microdeletion**

- The Y is present as a single copy, so loss of any part of it has reproductive consequences.
- AZFa has a separate, distinct location on Yq; AZFb and AZFc overlap.

## Ch 70 · Surgical Management of Male Infertility

### Sperm retrieval techniques

- **Fresh retrieval:** sperm can be retrieved for immediate use when the female partner's IVF cycle is prepared in step with the retrieval, as an alternative to retrieving for cryopreservation.
- **Conventional TESE (testicular sperm extraction):** about 45% of couples achieve pregnancy; abortion (miscarriage) rate 19%.
- **TESA (testicular sperm aspiration) puncture site:** superomedial or superolateral aspect of the testis.
- **Post-mortem retrieval:** up to 24 million sperm with 70% grade I motility within 13 hours of death.

## Pre-requisites for IVF

▸ **Female partner**

- **Infection screen:** HBsAg, anti-HCV, HIV I and II, HTLV I and II, and syphilis by rapid plasma reagin (RPR) test.
- **Other tests:** Pap smear, blood group, Rh type and antibody screen.
- **Reproductive hormone assay:** FSH and oestrogen. FSH rises when the ovary cannot produce normal eggs; oestrogen influences FSH secretion and relates to egg quality.
- **Sonohysterogram:** assesses the endometrial cavity for polyps, scarring and fibroids.

▸ **Male partner**

- HIV, HBsAg, anti-HCV and RPR, with semen evaluation.

## Semen processing techniques

▸ **Simple wash**

- Single or double wash with culture medium, then centrifugation. Simplest method; simple and effective.
- Fails mainly because it generates reactive oxygen species (ROS).

▸ **Swim-up**

- Sperm wash medium is layered over liquefied semen; the tube is incubated at a 45 degree angle for 60 minutes.
- Viable spermatozoa swim up into the medium and are aspirated; non-viable sperm stay behind.

▸ **Swim-down**

- Semen is layered over bovine serum albumin medium and incubated at 37 degrees C for 1 hour. The most motile sperm move downward.

▸ **Density gradient centrifugation**

- **Principle:** each spermatozoon settles at the gradient level matching its own density. Morphologically normal mature spermatozoa have a density of 1.10 g/mL.
- **Method:** semen is layered over a lower-density and then a higher-density solution and spun for 20 minutes at 1600 rpm.
- **Layers, top to bottom:** seminal plasma; white cells and debris; abnormal and non-motile sperm; the interface; viable motile spermatozoa in the pellet.
- **Drawback:** in severe oligozoospermia, sperm are lost in the low-density fraction or at the interface.

▸ **Filtration**

- Glass wool, glass beads or cross-linked dextran gel; gravity helps pull motile spermatozoa through the column. Not used routinely.

## Sperm cryopreservation (sperm banking)

▸ **Definitions**

- **Cryopreservation:** the procedure that makes it possible to stabilise and store sperm at cryogenic temperature.
- **Sperm bank:** a place where semen is cryopreserved for later use in artificial insemination.

### ▸ Indications

- Autoimmune disorders or diabetes.
- Donor insemination programmes.
- To overcome the stress of producing a semen sample on the day of treatment.
- Before vasectomy, or at the time of the procedure.
- Exposure to potentially toxic agents that interfere with gametogenesis.

### ▸ Cryoprotectants

- Protect spermatozoa from freeze damage by ice crystallisation.
- Glycerol (most commonly used), ethylene glycol, dimethyl sulphoxide and 1,2-propanediol.

### ▸ Freezing techniques

- **Slow freezing:** manual or automated; 2 to 4 hours in total. Semen is cooled progressively while cryoprotectant is added.
  - Room temperature to 5 degrees C at 0.5 to 1 degree C per minute.
  - 5 to minus 80 degrees C at 1 to 10 degrees C per minute.
  - Store in liquid nitrogen at minus 196 degrees C.
- **Fast freezing:**
  - Mix the sample with an equal volume of cold cryoprotectant.
  - Load into straws and hold at 4 degrees C for 10 minutes.
  - Hold the straws 15 cm above liquid nitrogen (at minus 80 degrees C) for 15 minutes.
  - Immerse in liquid nitrogen at minus 196 degrees C.

### ▸ Cryo-injury to sperm

- **Mechanisms:** thermal shock; ice-crystal formation (impairs organelle function and survival); osmotic shock; cellular dehydration (toxic damage from high solute content).
- **Membranes:** plasma and mitochondrial membranes are injured, activating caspases 3, 8 and 9. This lowers ATP generation and so reduces motility.
- **DNA damage:** from oxidative stress, thermal denaturation and defects in DNA repair enzymes.

### ▸ Thawing

- Done gradually so cells recover normal biological activity without abrupt temperature change. Routine temperature: 37 degrees C.
- **Technique:** 10 minutes at room temperature, 10 minutes in a 37 degree C thermostat, 15 minutes at room temperature, then washing and centrifugation before use.

### ▸ Outcome

- Storage for up to 28 years.
- Long-term follow-up: no increase in birth defects or chromosomal anomalies in offspring compared with the general population.
- Better results with ICSI than with IVF or IUI, because ICSI needs only a small number of motile spermatozoa.
- Pregnancy rates vary with the source (testicular versus epididymal sperm) and the ART method used.

## Assisted reproductive technology (ART)

- **Definition:** methods that manipulate gametes and embryos outside the woman's body and return the result to the female genital tract to achieve pregnancy.
- **Components:** IUI, IVF (including ICSI), and third-party assisted ART (sperm donation, egg donation, surrogates or gestational carriers).

### ▸ Intrauterine insemination (IUI)

- Sperm placed in the uterus with a Wallace catheter.
- **Effective for:** women with a scarred or defective cervix; men with refractory erectile dysfunction.
- **Natural or stimulated:** stimulation uses human menopausal gonadotrophin (hMG), FSH, LH, hCG, clomiphene citrate or letrozole.
- **Success:** about 20% per monthly cycle; depends on the cause of infertility.

### ▸ In-vitro fertilisation (IVF)

- **Four steps:** stimulation of egg maturation (ovarian stimulation or induction), egg retrieval, fertilisation, embryo transfer.
- **Stimulation:** monitored by transvaginal ultrasound. Maturity judged by follicle size and oestrogen level; an hCG injection triggers ovulation; eggs retrieved 36 hours after it.
- **Egg retrieval:** mild sedation, about 30 minutes. Under transvaginal ultrasound a needle passes through the vaginal wall into the follicles and eggs are aspirated by suction.
- **Fertilisation:** sperm and egg placed together in a dish overnight in an incubator; fertilisation usually occurs on its own. Consider genetic testing if ICSI is needed.
- **Embryo transfer:** 1 to 6 days after retrieval, through the vagina with a long thin tube. Implantation 6 to 10 days after retrieval.
- **Assisted hatching:** a hole is made in the zona pellucida just before transfer to help the embryo hatch. Used after previous unsuccessful IVF.
- **GIFT (gamete intrafallopian transfer):** gametes placed laparoscopically into normal fallopian tubes.
- **ZIFT (zygote intrafallopian transfer):** the fertilised egg (zygote) placed into the fallopian tube.

### ▸ Intracytoplasmic sperm injection (ICSI)

- **Sperm selection:** mature sperm bind hyaluronan, the main constituent of the gel layer around the oocyte; a commercial device is available.
- **Procedure:** a holding micropipette steadies the mature oocyte with gentle suction from a micro-injector. A single sperm is immobilised by cutting its tail and drawn into an injection micropipette. The pipette pierces the oolemma and releases the sperm into the cytoplasm. The oocyte is cultured and watched for fertilisation before embryo transfer.
- **Safety:** may not be safe in HBsAg-positive mothers because of vertical transmission.
- **Follow-up:** regular prenatal care with aneuploidy screening based on maternal age, nuchal translucency scan and biomarkers.

### ▸ Third-party assisted ART

- **Donor sperm:** when the husband has a genetic disease.
- **Egg donation:** when the woman cannot produce eggs. Ova from another woman after ovarian stimulation are fertilised by IVF and the embryo placed in the wife's uterus.

    - **Indications:** ovarian insufficiency, after chemotherapy or radiotherapy, after oophorectomy, women born without ovaries, carriers of known genetic diseases, poor egg quality.
- **Surrogates and gestational carriers:** when the wife cannot carry a pregnancy to term.
    - **Surrogate:** inseminated with the male partner's sperm; the child is biologically related to the surrogate and the male partner.
    - **Gestational carrier:** receives an IVF embryo from the husband's sperm and the wife's ova; the child is not related to the carrier.

▸ **Risks of IVF/ART**

- Ovarian hyperstimulation (swollen, painful ovaries).
- Multiple pregnancy and preterm delivery.
- Ectopic pregnancy and bleeding.

## Ch 71 · Physiology of Penile Erection and Pathophysiology of Erectile Dysfunction

### Anatomic mechanism

- **Corpora cavernosa:** support the corpus spongiosum and glans.
- **Cavernous smooth muscle:** periarteriolar and perisinusoidal components, which together regulate blood flow into the sinusoids.
- **Ischiocavernosus:** pumps blood distally to hasten erection, and adds rigidity in the rigid erection phase.
- **Corpus spongiosum:** constricts the urethral lumen for forceful expulsion of semen.
- **Glans:** cushions the impact of the penis on the female organs; provides sensory input for erection and pleasure; its cone shape facilitates intromission.

### Haemodynamics

- **Onset of erection:** arterioles and arteries dilate, with increased flow in both systole and diastole.

### Neurogenic mechanism

- **Brain centres:** the forebrain controls sexual behaviour; the hypothalamus is responsible for sexual partner recognition. Both are involved in erection.
- **Afferent pathway:** free nerve endings and corpuscular receptors in the penis, dorsal nerve of the penis, pudendal nerve, S2 to S4 segments; then up the spinothalamic and spinoreticular pathways to the thalamus, hypothalamus and frontal cortex.
- **Inhibitory neurotransmitters:** include adrenaline as well as noradrenaline.
- **Renin-angiotensin system:** mixed effects, both pro-erectile and pro-detumescence, depending on receptor type and location.
- **Nocturnal (REM sleep) erections:** associated with increased activity in the pontine area, amygdalae and anterior cingulate gyrus.

## Evaluation

### ▸ Special vascular investigations

- Historical and investigational studies: penile brachial pressure index, penile plethysmography, radioisotope penography, penile MRI, penile near-infrared spectrophotometry, cavernous smooth muscle content on biopsy.

### ▸ Hormonal causes

- **Hyperprolactinaemia:** suppresses GnRH and so impairs LH secretion; also interferes with peripheral conversion of testosterone to DHT.
- **Hyperthyroidism:** increases aromatization of testosterone to oestrogen and increases alpha-adrenergic tone in cavernous smooth muscle.

### ▸ Neurological testing

- **Somatic:** biothesiometry, sacral evoked response (bulbocavernosus reflex latency), dorsal nerve conduction velocity, genitocerebral evoked potential.
- **Autonomic:** heart rate variability with sympathetic skin response, penile thermal sensory testing, corpus cavernosum EMG with single potential analysis of cavernous electrical activity.

### ▸ Psychophysiologic testing

- **Normal nocturnal penile tumescence and rigidity (NPTR):** mean erection duration more than 30 minutes; circumference increase more than 3 cm at the base and more than 2 cm at the tip.
- **Other tests:** audiovisual and vibratory stimulation; neuroimaging (functional MRI and PET).

## Hormonal therapy

### ▸ Testosterone replacement

- Start with short-acting preparations. Efficacy is best judged by clinical response.
- **Side effects:** urinary symptoms, gynaecomastia, acne, erythrocytosis, sleep apnoea, progression of prostate cancer.

| Preparation | Dose and use |
|---|---|
| Testosterone enanthate or cypionate | 200–250 mg every 2–3 weeks, or alternatively 100 mg weekly |
| Testosterone propionate | 200 mg every 2–3 days (short half-life) |
| Subcutaneous pellets (Testopel) | 75 mg each; 2–6 pellets every 3–6 months |
| Transdermal patches (Testoderm TTS, Androderm) | 5 mg daily on arm, back or buttock. Itching and chronic dermatitis, treated with cortisone cream |
| Gels: Androgel 1%, Testim 1% | Shoulder, upper arm or abdomen |
| Gel: Axiron 2% | Axilla (concealed site, high permeability) |
| Buccal (Striant) | 30 mg mucoadhesive tablet over the gum twice daily |
| Oral | Large doses (more than 200 mg/day) because first-pass liver metabolism inactivates most of it; not approved in the US (check) |

▸ **Alternative hormonal treatments**

- **DHT gel:** pure androgen with no peripheral aromatization, so no oestrogenic side effects.
- **DHEA and hCG:** only limited benefit on sexual function.

▸ **Hyperprolactinaemia**

- **Offending drugs to stop:** oestrogens and morphine.
- **Bromocriptine:** lowers prolactin, restores normal testosterone and shrinks the pituitary tumour.

## Oral therapy

▸ **PDE5 inhibitors**

- **Contraindications:** MI, stroke or arrhythmia within the past 6 months.
- **Caution:** left ventricular outflow obstruction, hypovolaemia, aortic stenosis, hypotension, QTc prolongation.
- **Visual side effects:** include photophobia. A causal link with NAION (nonarteritic anterior ischaemic optic neuropathy) has not been proven.

▸ **Oral phentolamine**

- Nonspecific alpha-1 and alpha-2 antagonist; relaxes corporal smooth muscle by postsynaptic alpha-1 blockade.
- About 40% response in mild ED; needs more study.

▸ **Yohimbine**

- Central alpha-2 antagonist. Best limited to psychogenic ED; does not enable intercourse in organic ED.

▸ **Apomorphine**

- Role in psychogenic ED. Sublingual 2, 4 or 6 mg; no erectile efficacy if swallowed.

▸ **Trazodone and melanocortin agonists**

- **Trazodone:** its active metabolite activates the pro-erectile 5-HT2C receptor and antagonises the anti-erectile 5-HT1A receptor.
- **Melanocortin receptor agonists:** early studies show efficacy in inducing erection.

▸ **Alternative therapies**

- Herbs, dietary supplements, ginkgo biloba, L-arginine (substrate for NO synthesis), Korean red ginseng.

## Intraurethral and topical therapy

▸ **Intraurethral alprostadil**

- **Route:** avoids needles; absorbed through the urethral mucosa into the corpus spongiosum, then through small vascular channels into the corpora cavernosa.
- **MUSE (Medicated Urethral System for Erection):** semisolid PGE1 pellets of 1 x 3 mm, inserted 3 cm from the meatus; optional adjustable constriction band improves local retention.
- **ALIBRA:** alprostadil plus prazosin.
- **Contraindications:** hypersensitivity to alprostadil, abnormal penile anatomy, increased priapism risk.

- **Problems:** minor urethral bleeding, hypotension, dizziness, priapism. Overall inferior to PDE5 inhibitors and intracavernosal injection.

▸ **Topical agents**

- **Alprostadil 0.3%:** significantly more successful intercourse than placebo.
- **Prostaglandin E1 ethyl ester:** a prodrug option.
- **Nitroglycerine 2%:** headache; only rarely a rigid penis.
- **Papaverine:** minimal absorption because of large molecular size.
- Topical and transdermal agents still need further trials.

## Intracavernosal injection

▸ **Alprostadil**

- 96% metabolised locally within 1 hour. Side effect: haematoma.

▸ **Papaverine**

- Derived from the opium poppy. Half-life 1–2 hours; liver metabolism.

▸ **Phentolamine**

- **Action:** erection by postsynaptic alpha-1 blockade, antagonised by its presynaptic alpha-2 inhibition.
- **Half-life:** 30 minutes.
- **Side effects:** include nasal congestion and GI upset.

▸ **Vasoactive intestinal peptide (VIP)**

- 28-amino-acid hormone; vasodilates by activating adenylate cyclase and raising cAMP. Not used as monotherapy.

▸ **Combinations**

- **Bimix:** alprostadil 20 µg/mL with phentolamine 0.5 mg/mL, or papaverine 30 mg/mL with phentolamine 0.5 mg/mL.
- **Invicorp:** VIP plus phentolamine; about 80% efficacy.

## Vacuum erection device

- **Problems:** local pain or discomfort; rarely penile pain, numbness, difficult ejaculation, ecchymosis, skin necrosis, urethral varicosities, Fournier gangrene.

# Ch 73 · Priapism

## Ischaemic priapism: histological changes

| Time | Histology |
|---|---|
| 12 hours | Interstitial oedema in the corpora |
| 24 hours | Sinusoidal endothelium destroyed, basement membrane exposed, platelets adhere |
| 48 hours | Thrombus in sinusoidal spaces; smooth muscle necrosis; transformation of smooth muscle into fibroblast-like cells |

## Aetiology

▸ **Haematological and thrombotic**

- **Besides sickle cell disease:** G6PD deficiency, hereditary spherocytosis, paroxysmal nocturnal haemoglobinuria (PNH), congenital dyserythropoietic anaemia, thalassaemia, granulocytic, myeloid and lymphocytic leukaemia, multiple myeloma, haemoglobin Olmsted variant, fat emboli with hyperalimentation.
- **Thrombotic:** asplenia, erythropoietin use, haemodialysis with heparin.

▸ **Intracavernosal and oral erectogenic drugs**

- After intracavernosal injection, prolonged erection is more common than true priapism.
- **PGE1 (Caverject, alprostadil):** prolonged erection 5%, priapism 1%.
- **Trimix (papaverine, phentolamine, alprostadil):** priapism in 5–35%.
- **Oral PDE5 inhibitors:** rare, usually in men with other risk factors.

▸ **Other drugs**

- **ADHD drugs:** atomoxetine, methylphenidate.
- **Antianxiety:** hydroxyzine.
- **Antidepressants and antipsychotics besides trazodone:** bupropion, fluoxetine, sertraline, lithium, clozapine, risperidone, olanzapine, chlorpromazine, thioridazine, phenothiazines.
- **Antihypertensives:** hydralazine, guanethidine, propranolol.
- **Hormonal:** gonadotropin-releasing hormone, testosterone.
- **Recreational:** alcohol, crack cocaine, marijuana.

▸ **Other causes**

- **Genitourinary:** arteriovenous or arteriocavernous bypass surgery, urinary retention.
- **Infectious and toxins:** rabies, malaria, scorpion sting, spider bite. The spider is Phoneutria nigriventer, whose neurotoxin causes paralysis, asphyxia and priapism.
- **Metabolic:** amyloidosis, Fabry disease, gout.
- **Neoplastic infiltration:** from urethra, testis, lung and kidney as well as prostate, bladder and rectum.
- **Neurogenic:** syphilis, spinal cord injury, cauda equina compression, autonomic neuropathy, lumbar disc herniation, spinal stenosis, cerebrovascular accident, brain tumour, spinal anaesthesia, cauda equina syndrome.
- Idiopathic.

▸ **High-flow (non-ischaemic, arterial) priapism: iatrogenic**

- Optical internal urethrotomy (OIU), Nesbit corporoplasty, penile revascularisation.

## Diagnosis

- **Examination:** abdomen and genitalia for malignancy.
- **Aspirate in ischaemic priapism:** first dark like crankcase oil, later bright red.
- **Persisting erection after treatment:** always consider colour Doppler. Differential: resolved ischaemia with penile oedema, persistent ischaemia, or conversion to high flow.
- **Penile arteriography:** too invasive for routine diagnosis; blood gas data after arteriography are confusing.
- **MRI, three reported roles:**

- Imaging a well-established arteriovenous fistula.
- Imaging corporal malignancy or metastases.
- Showing presence and extent of tissue thrombosis or corporal smooth muscle infarction.

## Management of ischaemic priapism

▸ **Aspiration**

- Repeat until fresh bright red blood is obtained.

▸ **Sympathomimetic drugs**

- **Pure alpha-agonist:** metaraminol.
- **Mixed alpha- and beta-agonists:** etilefrine, phenylephrine, epinephrine.
- **Pure beta-agonist:** terbutaline.
- **Alpha-agonists:** contract cavernous smooth muscle, so sinusoidal blood leaves through the subtunical veins.
- **Beta-agonists:** relax cavernous smooth muscle and dilate the cavernous artery and arterioles, improving oxygenation.

▸ **Phenylephrine**

- **Technique:** aspirate between injections while tightly pinching the shaft at the penoscrotal junction, just below the needle.
- **Side effects:** include dizziness and tachycardia. Haemorrhagic stroke with doses of 2 mg or more.

▸ **Shunts**

- **Objective:** re-oxygenate cavernous smooth muscle by relieving venous outflow obstruction and re-establishing corporal inflow.
- **Indications:** also when aspiration and sympathomimetics have caused significant cardiovascular problems. Shunt early (rather than prolonged sympathomimetic dosing) in poorly controlled hypertension and in men taking MAO inhibitors.
- **Ebbehoj:** straight incision into glans and corpus cavernosum with a No. 11 blade.
- **T-shunt:** No. 10 blade inserted as for Ebbehoj, rotated 90 degrees outward and withdrawn.
- **After a percutaneous distal shunt:** deoxygenated blood is milked out; close incisions once bright red blood comes out.
- **Al-Ghorab:** excises a 5 mm circular core of distal tunica albuginea. The corporal snake is done after excising a 5 mm tunica core.
- **Quackels/Sacher (proximal):** bilateral, transscrotal or transperineal.
- **Complication:** urethral stricture.
- **Patency assessment:** bright red corporal aspirate, corporal blood gas, colour Doppler, fall in intracavernous pressure, penile squeeze-and-release (penis should detumesce and refill).
- **Preventing obstruction:** avoid compressive dressings, periodic squeeze and release, anticoagulation.

▸ **Penile implant after priapism**

- Exact timing of early implantation is not known.
- Semirigid (malleable) implant preferred with corporal fibrosis.

## Management of high-flow priapism

- **Conservative:** ice to the perineum and site-specific compression; spontaneous resolution is due to fistula closure. Expectant management recommended especially in children.
- **Active intervention:** selective internal pudendal catheterisation, angiography and embolisation, for patients wanting immediate relief.
- **Embolisation side effects:** penile gangrene, gluteal ischaemia, purulent cavernositis, perineal abscess. Commonest notable effect of bilateral embolisation: erectile dysfunction.
- **Erections after embolisation:** normally preserved in 75–85%.
- **Surgery, indications:** unwilling for conservative care and poor embolisation candidate; refuses embolisation; embolisation unavailable or failed.
- **Surgical ligation:** transcorporeal approach with intraoperative Doppler ultrasound. Done weeks to months after trauma so a pseudocapsule forms over the fistula; earlier surgery risks ligating the cavernous artery instead of the fistula alone.

## Stuttering priapism

- **Cause:** past history of ischaemic priapism.
- **Pattern:** waking with an erection lasting up to 4 hours, progressively painful from ischaemia, then remitting.
- **Education:** parents and emergency staff should be educated.
- **Hormonal prophylaxis:** includes diethylstilbestrol (DES, by feedback inhibition) and androgen synthesis inhibitors such as ketoconazole.

▸ **Baclofen**

- Inhibits erection and ejaculation through GABA receptor activity.
- Intrathecal pumps risk withdrawal syndrome if the pump fails.

▸ **PDE5 inhibitor prophylaxis**

- Long-term use apart from erection-stimulating conditions reduces recurrences without affecting normal erectile capacity.
- Effect appears after a week or more.
- **Regimen:** sildenafil 25 mg once daily, increased to 50 mg once daily; then tadalafil 5 or 10 mg three times weekly.

## Sickle cell disease and priapism

- **Mutation:** valine for glutamic acid at position 6 of the beta-globin chain.
- **Haemolysis and reduced nitric oxide:** pulmonary hypertension, priapism, leg ulcers, stroke.
- **Increased viscosity:** painful crises, osteonecrosis, acute chest syndrome.
- **Precipitants of full-blown priapism:** stuttering nocturnal or early-morning erections, dehydration, fever, cold.
- **ASPEN syndrome:** sickle cell disease, priapism, exchange transfusion and neurological events.
- **Hydroxyurea:** part of haematological therapy; used for sickle cell crisis.

## Physiology of ejaculation

- **Supraspinal pathway:** sensory cortex to hypothalamus (which also receives thalamic input) and brainstem, down to the lumbosacral cord, with pudendal and sympathetic nerves to the penis. Tactile penile input ascends as somatosensory input; post-ejaculation afferent input returns to the thalamus.

## Premature ejaculation (PE)

### ▸ Epidemiology and classification

- **Prevalence:** about 20%; exact figure not known.
- **Lifelong PE:** about 25% of cases. Ejaculation within 30–60 seconds in the majority (80%), and between 1 and 2 minutes in the rest (20%).
- **Acquired PE:** about 75%; often situational.
- **Subjective PE:** also called premature-like ejaculatory dysfunction.

### ▸ Evaluation

- Medical history should include the partner.
- **Optional diagnostic questions, five areas:**
  - **Lifelong versus acquired:** when first experienced; whether since the first sexual experience on every or almost every attempt and with every partner.
  - **Erectile function:** erection hard enough to penetrate; difficulty maintaining it until ejaculation; rushing intercourse to avoid losing it.
  - **Relationship impact:** how upset the partner is; whether the partner avoids intercourse; effect on the overall relationship.
  - **Previous treatment.**
  - **Quality of life:** avoiding intercourse from embarrassment; feeling anxious, depressed or embarrassed.
- **Full IIEF:** normal erectile function domain score is 26 or more.

### ▸ Pharmacology basis

- **5-HT2C stimulation:** inhibits ejaculation but facilitates erection.
- **SSRIs:** block 5-HT reuptake by serotonin transporters, raising synaptic 5-HT; 5-HT1A receptors downregulate, which increases the intravaginal ejaculatory latency time (IELT).

### ▸ Dapoxetine (Priligy)

- **Efficacy:** IELT increased 2.5–3 times; better ejaculatory control, less distress, more satisfaction.
- **Side effects:** nausea, diarrhoea, headache, dizziness, dry mouth, insomnia.
- No suicidal tendencies, no withdrawal symptoms, negligible effect on sexual function.
- **Contraindications:** severe hepatic impairment; CYP3A4 inhibitors such as ketoconazole; heart failure or significant ischaemic heart disease.

### ▸ Other drugs

- **Paroxetine:** greatest IELT increase, about 8.8-fold.

- **Other SSRIs and TCAs, adverse effects:** nausea, diarrhoea, perspiration, yawning, agitation, weight gain, increased risk of type 2 diabetes.
- **Topical lignocaine/prilocaine:** main problem is penile hypoaesthesia.
- **Intracavernosal alprostadil:** used, but very low level of evidence.

▸ **Behavioural therapy and surgery**

- **Physical techniques:** stop–start, squeeze, pelvic floor exercises.
- **Behavioural and psychological therapy:** success 45–60%.
- **Surgery:** dorsal penile nerve division reduces sensation and increases IELT.

## Delayed ejaculation (DE), anejaculation and anorgasmia

- **Investigations:** urine routine examination, urine culture, urine cytology, PSA, ultrasound, testosterone.
- **Recommended questions:**
  - How often he can ejaculate during intercourse.
  - How long after penetration he ejaculates or stops intercourse.
  - How often he feels close to ejaculation when he cannot ejaculate.
  - Why he stops intercourse if he cannot ejaculate.
  - Whether he feels he has ejaculated but fails to release semen.
  - Whether he feels bothered, annoyed or frustrated.
  - How often he can ejaculate during masturbation alone or with his partner.
- **Optional questions:** the same five areas as for PE (lifelong versus acquired, erectile function, relationship impact, previous treatment, quality of life).
- **Psychoeducation or psychosexual therapy:** involve both patient and partner.
- **Lifestyle:** more time together, minimal alcohol, making love when not tired.
- **Technique:** maximise penile stimulation, for example pelvic floor training.

## Retrograde ejaculation (RE)

- **Share of infertility:** less than 2% of cases.
- **After TUIP (transurethral incision of the prostate):** 5–45%. Rate depends on unilateral versus bilateral incision, and incision of the bladder neck only versus extension to the verumontanum.
- **Other causes:** idiopathic; congenital bladder neck abnormality.
- **Treatment:** drug therapy (alpha-adrenergic agents) is the most promising approach.
- **Urinary sperm retrieval:** urine can be diluted by oral fluid intake as an alternative to alkalinising with oral sodium bicarbonate.

# Ch 75 · Surgery for Erectile Dysfunction

## Penile revascularization

- **Further arterial options:** inferior epigastric artery to deep dorsal vein with venous ligation (arterialization with venous reconstruction), or anastomosis directly to the corpus cavernosum.
- **Best results:** young men under 30 years with isolated arterial stenosis after trauma.

## Penile prosthesis: types

- **Semirigid:** does not increase penile length.
- **Malleable examples:** AMS 600; Shah penile prosthesis with a hinge.
- **Positional (mechanical):** central metal cable with polyethylene discs, holding upward and downward positions better. Example: Dura-II.
- **Two-piece inflatable:** no separate reservoir; inflates fully with few pump squeezes. But the small pump is hard to manipulate and a larger tunical incision is needed. Examples: Coloplast Excel, AMS Ambicor.

## Penile prosthesis: indications and contraindications

- **Indication:** psychological impotence after failure of all other treatments.
- **Contraindications:** situational ED; ED from relationship conflict; poor hygiene and skin cleanliness; non-compliance with concurrent medication (for example for hypertension or diabetes).

## Penile prosthesis: procedure

- **Preparation:** chlorhexidine soap wash for 3 days before surgery.
- **Corporotomy:** longitudinal, about 2 cm, on the tunica albuginea.
- **Dilatation and measurement:** can use the dilators supplied with the prosthesis (Dilamezinsert dilators).

### ▸ Reservoir placement

- **Space of Retzius:** a defect is created between the crus of the penis medially and the spermatic cord laterally, with a 0.5 cm defect in the floor of the inguinal canal.
- **Avoid the space of Retzius after:** previous hernia surgery, abdominoperineal resection, pelvic fracture with pelvic surgery.
- **Flat submuscular reservoir:** placed above the transversalis fascia.

### ▸ Postoperative care

- No heavy lifting or sitting on the scrotum in the first week.
- Sutures removed at the 2-week review; follow-up at 3 months, 6 months, then annually.
- No sexual activity for 1 month; resume after cycling, once comfortable with the device.

## Penile prosthesis: complications

### ▸ Intra-operative

- Damage to the device.
- **Corporal perforation:** one option is to delay the prosthesis to 3 months instead of repairing and proceeding.

### ▸ Infection

- **Risk factors:** patient (spinal cord injury, immunosuppression); intra-operative (surgical time over 2 hours, repeated device-skin contact); post-operative (prolonged hospitalisation).
- **Organisms:** skin organisms that form biofilm (Staphylococcus epidermidis, Staphylococcus aureus, Candida); others (Pseudomonas, E. coli, Enterococcus).
- **Precautions:** open the device at the last moment, minimal instruments, control theatre traffic, laminar flow ventilation.

- **Coloplast Titan coating:** hydrophilic polyvinylpyrrolidone, which absorbs and elutes any antibiotic solution.
- **Conservative antibiotics:** reasonable when infection appears within 2 weeks, the patient is improving and the device is not infected.
- **Aggressive salvage:** for an infected device, clinical deterioration or rising total leucocyte count, before systemic symptoms develop.
- **Immediate salvage replacement contraindicated in:** Enterococcus infection, tissue necrosis, sepsis, diabetic ketoacidosis, cylinder erosion into the urethra.
- **Antibiotic spacer after explantation:** synthetic high-purity calcium sulphate cast placed in the corpora; delivers antibiotic for 30–60 days as it is absorbed and prevents fibrosis; the device is then reimplanted.

▸ **Mechanical failure**

- Cracks in silicone tubing; leaks at tubing-pump connections.
- Malfunction within a few months of implantation: replace only the defective component.

▸ **Other complications**

- Scrotal haematoma.
- **Poor glans support (supersonic transporter deformity):** from undersized cylinders. Insert longer cylinders, or coronal hitch or plication.
- **S-shaped deformity:** from overlong cylinders; needs device replacement.

## Special situations

▸ **Implant after radical prostatectomy**

- Retropubic fibrosis puts bladder, vena cava and bowel at risk during reservoir placement.
- A two-piece device (no reservoir) is an option.
- If a separate reservoir incision is used (scrotal incision for the prosthesis), prefer the right side to avoid the sigmoid colon.

▸ **Prosthesis with straightening manoeuvres**

- An inflatable prosthesis gives superior correction of curvature with improved girth.

## Outcome

- Satisfaction higher with newer devices.
- Satisfaction lower in Peyronie disease with ED, after radical prostatectomy, and with BMI over 30.

# Ch 76 · Diagnosis and Management of Peyronie Disease

## Definition and associated conditions

- **Definition:** a wound-healing disorder of the tunica albuginea forming an exuberant scar, presumably after penile injury.
- **Diabetes:** 8% to 20% of diabetics have PD; risk relates to duration of diabetes and poor glucose control.
- **ED:** 38% to 58% of men with PD, partly from comorbidities.

- **Psychological:** anxiety, low self-esteem, shame, loss of confidence and depression, from penile shortening and inability to have sex.
- **Radical prostatectomy:** possible risk factor, through perioperative penile trauma, neurological consequences and local cytokine release.
- **Hypogonadism:** increases severity.
- **Tympanosclerosis:** associated.

## Aetiopathogenesis

- **Trigger:** repetitive microtrauma or a single event in a susceptible man, in the erect or even flaccid state.
- **Plaque formation:** microfractures (chronic) or delamination (acute) of the tunical layers (outer longitudinal, inner circular) create clot-filled spaces that heal abnormally.
- **Predisposition:** genetic, autoimmune and infective factors probably responsible.
- **ROS and oxidative stress:** injury with blood extravasation leads to thrombus; fibronectin and fibrin deposit and inflammatory cells accumulate. ROS production follows, causing overexpression of fibrogenic cytokines and increased collagen synthesis.
- **Myofibroblasts:** normally undergo apoptosis when tissue tension or trauma falls. Persistent tension during rigid erections causes failure of apoptosis; myofibroblasts, activated by TGF-beta1, deposit excess extracellular matrix and collagen, giving inappropriate healing and fibrosis.
- **Nitric oxide:** antifibrotic. Wound cells (monocytes, macrophages, fibroblasts) make NO, which reacts with ROS and lowers ROS levels, so fibrosis decreases.

▸ **TGF-beta1**

- Promotes ROS production.
- Directly inhibits collagenase and promotes collagen synthesis.
- Fibrogenic growth factor: strong activator of myofibroblasts; stimulates extracellular matrix deposition.
- Decreases matrix-degrading proteases and increases their inhibitors.
- Modulates integrin expression.
- Upregulates myostatin, an inducer of fibrosis.

▸ **Plaque gene expression**

- **Overexpressed:** PTN (OSF-1, recruits fibroblasts and osteoblasts); MCP-1 (inflammatory response and ossification); MMPs, especially MMP-2 and MMP-9 (break down collagen); myofibroblast-formation genes.
- **Downregulated:** Id-2 (inhibitor of myofibroblast differentiation); apoptotic genes.

## Evaluation

- **History:** include ejaculatory problems.
- **Peyronie's Disease Questionnaire:** 15 questions in 3 domains; higher score means greater negative impact.
- **Examination:** femoral pulses; hands and feet (Dupuytren and Ledderhose disease); penis flaccid and erect if possible, including pain on palpation and nodules.
- **Penile duplex ultrasonography:** also identifies corporal fibrosis.

## Non-surgical treatment

- **Reassurance alone:** enough if the patient is not bothered, as PD carries no cancer risk.

▸ **Oral agents**

| Agent | Mechanism | Results | Adverse effects |
|---|---|---|---|
| Potaba (potassium aminobenzoate) | Increases MAO activity and lowers serotonin, increasing tissue antifibrotic properties | Decreased plaque size; no other benefit | Acute hepatitis |
| Vitamin E | Antioxidant limiting oxidative stress | Most frequently used oral agent | — |
| Tamoxifen | Raises TGF-beta, causing macrophage deactivation and T-lymphocyte suppression, preventing fibrogenesis | — | Thromboembolism |
| Colchicine | Microtubule depolymerisation; inhibits transcellular collagen movement; stimulates collagenase | — | GI upset |
| Carnitine | Increases mitochondrial respiration; decreases free radicals | — | GI upset, seizures |
| Pentoxifylline | Blocks TGF-beta1-mediated inflammation; prevents type I collagen deposition; decreases platelet-activating factor; non-specific PDE5 inhibitor | Decreased curvature (by 23 degrees) | GI upset, headache, flushing, dizziness |
| PDE5 inhibitors | Antifibrotic; septal scar remodelling | Also treat ED | Like pentoxifylline (hypotension) |

▸ **Intralesional verapamil**

- **Mechanism:** inhibits incorporation of proline into extracellular matrix protein.
- **Results:** reduces plaque-associated narrowing; improves erection quality.
- **Adverse effects:** nausea, lightheadedness, penile pain, ecchymosis.

▸ **Intralesional nicardipine**

- Dihydropyridine calcium channel blocker; reduces glycosaminoglycan (GAG) biosynthesis and extracellular matrix production. Rated better than verapamil.
- Reduced pain and plaque size improve erections; no severe side effects.

▸ **Intralesional interferon alfa-2b**

- **Adverse effects:** flu-like symptoms, sinusitis, ecchymosis, penile swelling.

▸ **Electromotive drug administration (EMDA)**

- Bypasses hepatic metabolism and raises drug concentration in target tissue.
- Verapamil alone: no benefit. Verapamil plus dexamethasone: decreased plaque volume and curvature.
- **Adverse effect:** temporary erythema at the electrode site.

▸ **Extracorporeal shock wave therapy (ESWT)**

- **Mechanism:** direct plaque damage induces inflammation and increased vascularity, leading to plaque lysis.
- Improves quality of life and erections. Can worsen curvature; can cause local ecchymosis.

▸ **Penile traction**

- **Mechanism:** mechanotransduction; decreases alpha-smooth muscle actin and increases matrix metalloproteinases.
- Increases girth and shrinks the plaque. Overall satisfaction 85%. Adverse effect: erythema.

▸ **Vacuum therapy**

- Mechanism unknown, probably mechanical like traction. Reduces curvature.
- **Adverse effects:** development of PD, ecchymosis, skin necrosis, urethral bleeding.

▸ **Radiotherapy**

- Anti-inflammatory effect; no clinical benefit. Risks: malignancy, and ED in the elderly.

▸ **Topical drugs and combination therapy**

- **Topical drugs tried:** beta-aminopropionitrile, superoxide dismutase, verapamil, trifluoperazine. None effective.
- **"Three-armed approach":** daily pentoxifylline plus L-arginine, verapamil injections twice weekly, daily traction. Best opportunity for deformity correction.

## Surgery

- **Further indications:** desire for the most rapid and reliable result; pain for patient or partner during intercourse; patient anxiety.
- **Consent:** decreased sexual sensation rarely compromises orgasm or ejaculation and resolves in 1-6 months.
- **Algorithm:** with inadequate erection, pre-prosthesis plication may be followed by a penile prosthesis.

▸ **Tunical shortening (plication)**

- **Levine modification of Duckett-Baskin tunica albuginea plication (TAP):** two partial-thickness transverse incisions 0.5-1 cm apart on the most convex part, involving only the longitudinal fibres. The fibres between them are reduced to remove bulk, and the defect is sutured transversely.
- **Advantages:** short operation, good cosmesis, minimal effect on rigidity.
- **Results:** straightening in 80-100%; satisfaction 65-100%.
- **Problems:** plaque left in situ; pain from knots and suture granulomas; new ED in 0-38%; haematoma, urethral injury, phimosis.

▸ **Tunical lengthening (plaque incision or partial excision with grafting)**

- **Indications (complex disease):** include shaft narrowing and extensive plaque calcification.
- **Results:** satisfactory straightening in 75-100%.
- **Commonest grafts:** Tutoplast (processed human and bovine pericardium) and porcine SIS.
- **Ideal graft:** strength and elasticity close to normal tunica albuginea; minimal morbidity and tissue reaction; readily available and inexpensive; not too thick; pliable; easy to size and suture; infection resistant; preserves erectile capacity.
- **Postoperative care:**
  - Dressing removed after 3 days.
  - Massage and stretch from 2 weeks (5 minutes twice daily for 2-4 weeks).
  - PDE5 inhibitors to improve vascularity and reduce ED.

- Traction 3 or more hours daily for 3 months, starting 3-4 weeks after surgery.

**Other topics in this book**

## Testicular biopsy

▸ **Indications and principles**

- **Conditions for diagnostic biopsy:** normal testes, palpable vasa, normal FSH, negative antisperm antibody assay.
- **Both sides:** biopsy both whatever the size difference; a small firm testis can have good spermatogenesis and a large healthy one may show maturation arrest.
- **Fresh unfixed tissue:** examined for sperm with tails; several samples if none found at first.

▸ **Open (microsurgical) technique**

- The wound should be dry before incising the tunica albuginea, so blood does not contaminate the sample.
- **Touch prep:** blot the cut testis surface several times on a slide with a drop of saline; examine under high power on a light microscope.
- **Wet squash prep:** a second specimen on a slide with saline, crushed under a coverslip.
- If the squash prep shows no sperm, biopsy another area through the same skin incision.

▸ **Micro-biopsy and complications**

- **Micro-biopsy:** 26-gauge needle taking a single seminiferous tubule (SST).
- Inadvertent epididymal biopsy does no harm if that tubule contains no sperm.

▸ **Interpretation**

- **Normal:** several stages of spermatogenesis in one tubular section (patchwork pattern).
- **Obstruction:** azoospermia with normal biopsy and dilated tubules.
- **Maturation arrest:** normal testicular architecture.
- **End-stage testis:** germ cells and Sertoli cells both absent; thickened basement membrane with tubular and peritubular sclerosis.

## Unconsummated marriage

- **Definition:** to consummate a marriage is to complete the married relationship by sexual intercourse.
- **Frequency:** common; many sexologists report about every third patient has this problem.
- **Effects:** severe stress and frustration, hostility, even divorce, ruined lives.
- Treat as a semi-emergency, because delay builds frustration and hostility that hinder treatment and understanding.
- Cases with deep-rooted psychopathology are difficult and most often fail.

▸ **Aetiology**

- **Ignorance:** many young couples, though educated, are ignorant about sexual practice; false expectations or wrong technique lead to failure.
- **Fear of pain:** a preconditioned notion in many women that sex is very painful leads to severe vaginismus, making intercourse impossible.
- **Performance anxiety with erectile failure:** may stem from myths such as masturbation guilt and Dhat syndrome.

    - **Dhat syndrome:** culture-bound; the man complains of premature ejaculation or impotence and believes he passes semen in his urine, with no organic cause.
- **Personality disorders:** phobias, fear and depression affect the couple's sexual life.
- **Physical defects:** women, local inflammatory pathology and rarely a tough hymen; men, severe penile curvature or veno-occlusive disease causing inadequate erections.
- **Social factors:** family pressure, lack of privacy and job stress can make intimacy impossible.

▸ **Treatment**

- **Basic sex counselling:** correct misconceptions, allay fears, teach the right technique.
- **Social counselling:** correct excessive in-law interference, create privacy, minimise stress, improve communication between wife and husband.
- **Vaginal self-dilatation:** when fear of pain is the main problem.
- **Surgical hymenectomy:** for a tough hymen.
- **Temporary medical support:** sildenafil or intrapenile injections for performance anxiety; after a few successful intercourses men gain confidence and usually no longer need it.
- **Correction of physical defects:** severe chordee, circumcision for phimosis, penile revascularisation, penile prosthesis.

# 10 Testis, Penis & Urethra

## Ch 79 · Neoplasms of the Testis

### Epidemiology and Age Groups

▸ **Incidence**

- **Annual incidence:** 5.4 per 100,000 (1 lakh) men.
- **Geography:** highest in Scandinavia and Germany; lowest in Africa and Asia.
- **Side:** more common on the right than the left.
- In men aged 15-19 years, second commonest malignancy after leukaemia.

▸ **Age peaks**

- Infancy: yolk sac tumour.
- 30-34 years: seminoma.
- About 60 years: malignant lymphoma.

### Risk Factors

▸ **Cryptorchidism**

- In unilateral cryptorchidism, the contralateral normally descended testis is also at slightly increased risk (1.5-2 times).

▸ **Family history**

- 10-fold increased risk; affected relatives present 2-3 years earlier.

▸ **Infertility**

- 3-fold increased risk.

▸ **Exposures**

- **Environmental:** organochlorines and polyvinyl chloride.
- **Prenatal:** hormone therapy (diethylstilbestrol) or radiation.

### Genetics and Pathogenesis

▸ **CCND2 (cyclin D2) and 12p**

- CCND2 overexpression is seen in most germ cell tumours (GCTs).
- CCND2 overexpression plus an extra copy of 12p activate CDK4 and CDK6, which protect against apoptosis and lead to genomic instability.

▸ **Associated syndromes**

- Down syndrome, persistent Mullerian duct syndrome and cutaneous ichthyosis.

▸ **Germ cell transformation**

- Primordial germ cell, then gonocyte (fetal), then ITGCN (before puberty), then after puberty either seminoma or embryonal carcinoma.
- Embryonal carcinoma can give rise to choriocarcinoma, yolk sac tumour and teratoma.
- **Seminoma:** DNA hypomethylation and hyperdiploidy (hypertriploidy) (check).
- **Embryonal carcinoma lineage** (choriocarcinoma, yolk sac tumour, teratoma): extraembryonic differentiation and hypermethylation.

## ITGCN (Intratubular Germ Cell Neoplasia)

▸ **Definition and incidence**

- Also called carcinoma in situ (CIS) of the testis.
- **Incidence:** 0.8%.

▸ **Models of origin**

- **Fetal gonocyte model:** primordial gonocytes persist abnormally beyond the neonatal period without differentiating into spermatozoa; invasive growth is then driven by postnatal and pubertal gonadotropin stimulation.
- **Second model:** zygotene-pachytene spermatocytes transform to a precursor germ cell phenotype.

▸ **Diagnosis**

- Found on testicular biopsy for infertility, on contralateral biopsy in GCT patients, or in the affected testis during testis-sparing surgery.

▸ **Treatment and prognosis**

- Choice also depends on the patient's wish to avoid testosterone replacement therapy.
- **Low-dose radiotherapy:** local control similar to radical orchidectomy; Leydig cell function is relatively preserved because Leydig cells are radioresistant compared with germ cells.
- **Radical orchidectomy:** the most definitive treatment.
- ITGCN in the orchidectomy specimen of a man with testis cancer has no prognostic implication for relapse risk.

## Pathology

▸ **Seminoma**

- **Gross:** soft, tan to white, diffuse or multinodular mass, with or without focal necrosis (necrosis is more prominent in other GCTs).
- **Microscopy:** polygonal nuclei.
- Syncytiotrophoblastic cells have no clear prognostic significance.
- Immunohistochemistry has a limited role in diagnosing GCTs.

▸ **Spermatocytic seminoma**

- Not bilateral; does not stain for glycogen.
- **Three cell types:** small lymphocyte-like cells; medium-sized cells with eosinophilic cytoplasm and round nuclei; large mononucleated or multinucleated cells.

▸ **Embryonal carcinoma**

- The most undifferentiated GCT type.
- **Gross:** tan to yellow with haemorrhage and necrosis.
- **Microscopy:** variable; solid sheets, papillary, tubular or glandular-alveolar patterns.
- **Immunohistochemistry:** PLAP positive.
- **Markers:** modest hCG rise when syncytiotrophoblastic cells are present; LDH concentration is prognostic in metastatic disease.

▸ **Yolk sac tumour**

- **Other names:** endodermal sinus tumour, orchidoblastoma, juvenile or infantile embryonal carcinoma.
- Commonest infantile testicular tumour; second commonest prepubertal testicular tumour after teratoma.
- **Schiller-Duval bodies:** represent endodermal sinuses.
- **Hyaline globules:** cytoplasmic and extracellular eosinophilic globules in 84%.

▸ **Teratoma**

- **Immature:** incompletely differentiated.
- **Mature:** well differentiated; may contain hair, teeth, bone and cartilage.

## Extragonadal GCT

▸ **Site and behaviour**

- Arises more often in the mediastinum than the retroperitoneum; seminoma is the commonest type.
- **Retroperitoneal GCT:** behaves like testicular GCT in biology and prognosis.
- **Mediastinal seminoma:** same prognosis as testicular seminoma.
- **Mediastinal mature teratoma:** low malignant potential; generally cured surgically.

▸ **Mediastinal NSGCT**

- Less chemosensitive than testicular or retroperitoneal NSGCT; 5-year overall survival 45%.
- More likely to contain a yolk sac component, so AFP is raised.
- Associated with haematological malignancies carrying extra 12p.

▸ **Diagnosis when suspected**

- Extragonadal mass plus raised AFP and/or hCG plus normal testis examination: enough to start chemotherapy.
- Normal markers and normal testis examination: biopsy the mass to confirm.

## Clinical Features

▸ **Local**

- Heaviness or a pressure-like sensation.
- Acute pain from haemorrhage, or rapid expansion with infarction, is much more common in NSGCT than seminoma.
- hCG-secreting tumours can cause hyperprolactinaemia.

▸ **Metastatic**

| Site | Presentation |
|---|---|
| Bulky retroperitoneal | Palpable mass, abdominal pain, ureteric obstruction, lower-limb oedema, GI symptoms (retroduodenal metastasis); back pain from psoas or nerve-root involvement |
| Pulmonary | Cough, dyspnoea, haemoptysis |
| Bone | Skeletal pain |
| Brain | Neurological symptoms (choriocarcinoma) |
| Hepatic | Jaundice, ascites, nausea |

## Diagnosis

### ▸ Scrotal ultrasound

- **Probe:** 5-10 MHz.
- **NSGCT:** haemorrhage, calcification, necrosis and heterogeneous echotexture.
- Spontaneous regression (burned-out primary) is common in GCTs, most commonly in seminoma.

### ▸ AFP (alpha-fetoprotein)

- **Normal:** below 10 ng/mL.
- Raised in 100% of yolk sac tumours.
- **Other cancers:** hepatocellular carcinoma; biliary tract, pancreas, lung and stomach cancers.
- **Non-malignant causes:** alcohol abuse, ataxia telangiectasia.

### ▸ hCG (human chorionic gonadotropin)

- Beta subunit is unique to hCG; normally produced by the placenta.
- Raised in 40-60% of embryonal carcinomas.
- **Other cancers:** gastric, hepatic, pancreatic, lung, breast, biliary, kidney and bladder.
- Also raised in pregnancy.

### ▸ LDH (lactate dehydrogenase)

- Five isoenzymes; LDH-1 is the one mostly raised in GCT.
- Non-specific: also from cardiac, smooth and skeletal muscle.
- Not a good indicator of recurrence.

### ▸ Other markers

- **Serum PLAP (placental alkaline phosphatase):** raised in advanced seminoma.
- **GGT (gamma-glutamyl transpeptidase):** raised in 30% of seminomas.

## Staging

### ▸ AJCC 8th edition

- pT1 pure seminoma: pT1a below 3 cm, pT1b 3 cm or more.
- Stage 0: pTis, N0, M0, S0.

### ▸ CT

- With oral and IV contrast.
- **Node cut-off:** above 10 mm commonly used, though size criteria are not clearly defined.
- Nodes of 5-9 mm in the primary landing zone are especially suspicious when anterior to the great vessels.

### ▸ Other staging tools

- **MRI:** alternative to CT but longer and costlier.
- **Chest X-ray:** routine in all patients before management decisions.
- Mediastinal or hilar nodes without retroperitoneal disease suggest a non-GCT cause (lymphoma, sarcoidosis); biopsy.
- **Laparoscopic RPLND** (retroperitoneal lymph node dissection): in some European countries, done in CS I and IIA NSGCT for staging without curative intent, to identify who needs chemotherapy.

## Sperm Cryopreservation

- At diagnosis, 52% of GCT patients are oligospermic and 10% azoospermic.
- Spermatogenesis recovers 2-3 years later after radiotherapy than after chemotherapy.

## Treatment of Seminoma

### ▸ Stage I

- **Carboplatin:** less ototoxicity, neurotoxicity and nephrotoxicity than cisplatin; follow-up after adjuvant carboplatin still needs CT.
- **Prophylactic radiotherapy:** in-field recurrence below 1%, so routine follow-up CT of the treated field is not required.

### ▸ Stage II

- **IIA:** 30 Gy dog-leg as 15 fractions of 2 Gy to para-aortic and ipsilateral iliac nodes.
- **IIB:** boost of 2 Gy x 3 = 6 Gy to enlarged nodes, on top of 30 Gy to the para-aortic and ipsilateral iliac field.
- **Disease-free survival** with radiotherapy or chemotherapy: about 90%.

### ▸ Advanced (stage IIC and III)

- Induction chemotherapy: response 70-90%, relapse 15-20%.
- 4 cycles of VIP is an alternative to 4 cycles of BEP for intermediate prognosis.

### ▸ Post-chemotherapy residual mass

- 60-80% have a radiologically detectable residual lesion; 50-60% of these resolve spontaneously within 13-18 months.
- **Malignancy risk:** smaller than 3 cm, 0-4%; larger than 3 cm, 27-38%.
- Radiotherapy has no role.
- **Surgery for PET-positive residual:** viable malignancy, give 2 further cycles of chemotherapy; necrosis, follow up.

### ▸ Relapse

- Relapse after single-agent carboplatin: treat as chemotherapy-naive with first-line cisplatin-based chemotherapy.
- **Early relapse after chemotherapy:** poor prognosis, long-term survival 30-50%; biopsy the relapse site before second-line VeIP, because it may contain teratoma.
- **Late relapse after chemotherapy:** pure seminoma accounts for less than 80% (check).

## Treatment of NSGCT

### ▸ Clinical stage I

- **Low risk** (no embryonal carcinoma predominance, no lymphovascular invasion): adjuvant BEP (1 or 2 cycles) if unwilling to have surveillance; nerve-sparing RPLND if unwilling to have either.
- **High risk, unwilling to have chemotherapy:** nerve-sparing RPLND or surveillance.
- **Relapse:** most present with a node mass larger than 3 cm, distant metastases or raised markers; treat with 3 or 4 cycles of BEP or VIP, then resect any residual tumour.

### ▸ Clinical stage IIA

- **Marker positive (S1):** 3 cycles of BEP, then resect any residual tumour.

- **Marker negative:** follow up and reimage at 6 weeks.
  - Progressive, markers positive: 3 cycles of BEP with or without residual tumour resection.
  - Progressive, markers negative: nerve-sparing RPLND or chemotherapy.
  - Unchanged: nerve-sparing RPLND.
  - Regressing: follow up.
- **Survival in CS IIA/IIB** with chemotherapy or RPLND: over 90%.

▸ **Clinical stage IIB**

- Persistently elevated markers after primary chemotherapy: second-line chemotherapy.

▸ **VIP regimen**

- Etoposide (VP-16), ifosfamide and cisplatin.

▸ **Stage IIC/III: response and relapse**

- Elevated markers after induction (partial remission) or progressive disease: second-line chemotherapy.
- **Early relapse (within 2 years):** can also be treated with high-dose chemotherapy with bone marrow (stem cell) transplant.
- **Late relapse (after 2 years):** poor survival; resected tissue may show viable tumour, teratoma or adenocarcinoma transformation.

▸ **Desperation surgery**

- For progression after first- and second-line chemotherapy, which has a dismal prognosis; evidence only from limited literature and small studies.
- The single resectable site is usually the retroperitoneum.

## Choriocarcinoma Syndrome and Brain Metastasis

▸ **Choriocarcinoma syndrome**

- **Features:** very high beta-hCG (above 50,000 IU/L), mediastinal, abdominal or testicular mass, gynaecomastia, hyperthyroidism; can cause tumour haemorrhage.
- Medical emergency needing immediate chemotherapy.
- **Sequence:** first course EP for 3 days with beta-blockers, then BEP or VIP; radical orchidectomy between cycles.

▸ **Brain metastases**

- Highly vascular; tend to bleed during chemotherapy, with 4-10% death from intracranial bleeding.
- **5-year survival:** 33% in disseminated NSGCT, 57% in seminoma.
- **Options:** chemotherapy, surgical resection, whole-brain radiation, stereotactic radiosurgery and multimodality treatment; most receive multimodality treatment.
- Brain relapse after first-line chemotherapy: second-line chemotherapy, then radiotherapy or surgery.

# Ch 82 · Tumours of the Penis

## Aetiology and Epidemiology

- Incidence is decreasing.

- Peaks in the 6th decade, though also reported in younger patients.
- African American patients have higher risk and poorer survival.
- More lifetime sexual partners is a risk factor.

## Carcinoma in Situ

▸ **Erythroplasia of Queyrat**

- Well marginated; may ulcerate or discharge.

▸ **Bowen disease**

- Same histology as erythroplasia of Queyrat; well-marginated plaques of scaly erythema; can ulcerate.
- **Differential diagnosis:** bowenoid papulosis, superficial basal cell carcinoma, nummular eczema, psoriasis.

▸ **Behaviour and treatment**

- Metastasis is extremely rare; no association with internal malignant tumours.
- Treat only after histopathological confirmation.
- **Laser:** KTP is an alternative to Nd:YAG.
- **Radiotherapy:** for CIS resistant to topical therapy in men who are not surgical candidates.

## Evaluation of the Primary

▸ **Presentation and examination**

- Usually a mass or discharge; bleeding, pain and urinary retention are rare; systemic symptoms may occur.
- DRE and bimanual examination assess perineal body involvement and any pelvic mass.

▸ **Primary sites**

- Glans 48%, prepuce 20%, glans and prepuce 10%, coronal sulcus 6%, shaft less than 2%.

▸ **Growth patterns**

- Superficial spreading, flat, and ulcerated with vertical growth.

▸ **Biopsy**

- Biopsy then definitive surgery later, or frozen section then surgery at the same sitting.

▸ **Differential diagnosis**

- Condyloma acuminatum, Buschke–Löwenstein tumour, lichen sclerosus (BXO).
- **Infectious:** chancre, chancroid, herpes, granuloma inguinale, tuberculosis.

## Histology and Grading

- **Typical SCC:** keratinisation, epithelial pearl formation, varying mitotic activity.
- **Broders classification:** grade I well differentiated, II moderately differentiated, III poorly differentiated, IV anaplastic.
- **Basaloid carcinoma:** HPV-related in about 80%.

## Investigations

▸ **Laboratory**

- Anaemia, leukocytosis, hypoalbuminaemia.

- **Hypercalcaemia treatment:** aggressive saline hydration (promotes sodium and calcium excretion), diuretics if volume overloaded, bisphosphonates, calcitonin.

▸ **Imaging of the primary**

- **Ultrasound:** 7.5 MHz linear probe; hypoechoic lesion; cannot reliably tell subepithelial glans invasion from corpus spongiosum involvement.
- **MRI:** artificial erection induced with PGE1.

▸ **Lymph nodes and metastatic work-up**

- A palpable inguinal node appearing during follow-up after treatment of the primary is metastatic in essentially 100%.
- Metastatic work-up only when nodes are metastatic: CT chest and abdomen or chest X-ray, liver function tests and bone scan, or PET-CT.
- **Stage 0:** Tis or Ta.

## Treatment of the Primary

▸ **Penile preservation**

- **Goals:** keep glans sensation where possible; maximise shaft length.
- About 75% of men resume sexual activity with satisfaction.
- **Mohs surgery:** an organ-preserving option, but not commonly done.
- **Limited excision with frozen section:** local recurrence under 10%.

▸ **Glans resurfacing**

- Preserves sensation.
- **Recurrence:** about 20% after total stripping, about 50% after partial stripping.

▸ **Laser**

- **Options:** CO2, Nd:YAG, KTP and argon.
- Recurrence can be managed by repeat ablation in selected cases.

▸ **Partial penectomy**

- Done when excision with a clear margin leaves at least 4 cm of unstretched stump, allowing standing micturition and possibly vaginal penetration.

▸ **Radiotherapy**

- Circumcision beforehand also lets any infection resolve.
- **EBRT (external beam radiotherapy):** widely available and easy to give but needs a 45-day course; penile preservation at 5 years about 50%.
- **Brachytherapy:** needs expertise; not widely available.
- **Positioning:** supine with a wax or Perspex block around the penis, or prone with the penis suspended in a small water-bath container.
- **Problems:** moist desquamation, urethral fistula, urethral stricture, penile necrosis, pain, oedema.

## Metastatic Disease

- Cisplatin plus 5-FU is an alternative to TIP.
- Bleomycin regimens are avoided because of lung toxicity, which can be fatal.

- If disease responds to chemotherapy, surgery for resectable disease improves survival, to about 32 months.

## Non-Squamous Malignancies

▸ **Basal cell carcinoma**

- Slow growing; diagnosed by biopsy; local excision is curative.

▸ **Melanoma**

- **Presentation:** blue-black or reddish-brown pigmented papule or ulcer on the glans.
- **Biopsy:** atypical junctional cell activity with pigmented cells displaced into the dermis.
- **Treatment by site:** prepuce, circumcision; glans, glansectomy or partial penectomy; shaft, partial or total penectomy.

▸ **Sarcoma**

- Benign sarcomas occur more often distally.
- **Malignant, by frequency:** vascular, then neural, then myogenic, then fibrous.
- **Presentation:** mass, pain, penile enlargement, priapism, urinary retention.

▸ **Extramammary Paget disease**

- **Presentation:** pruritus, discharge, discomfort.
- **Paget cells:** large, oval or round, clear-staining hydropic cells with hypochromatic nuclei.
- Slow growing but can become invasive.
- **Treatment:** positive margins need further excision; invasive tumours need more extensive resection with reconstruction.
- **Palpable nodes:** radical node dissection, but poor prognosis; neoadjuvant chemotherapy such as docetaxel may be added.

▸ **Adenosquamous carcinoma**

- Large firm mass; CEA positive; probably best treated by local excision and surveillance.

▸ **Lymphoreticular malignancy**

- Primary disease is rare; penile involvement may be metastatic; leukaemia can cause priapism.
- Low-dose radiotherapy has been reported to treat primary penile lymphoma successfully.

▸ **Metastases to the penis**

- **Sources:** rarely gastrointestinal tract, testis or kidney.
- **Routes:** direct extension, retrograde venous, lymphatic, arterial embolism.
- **Presentation:** penile swelling, nodularity or ulceration, and priapism.
- **Commonest histology:** replacement of the corpora cavernosa by the metastatic lesion.
- Penectomy may be considered if the primary can be excised or controlled.

# Ch 84 · Inguinal Node Dissection

## Lymph Node Management

▸ **Before planning groin management**

- The primary should be completely resected, its histopathology available, and the groins examined.

▸ **Bulky and progressive nodes**

- A mobile node mass of 4 cm or more carries a risk of extranodal extension; manage as metastatic nodal disease.
- **Progression on induction chemotherapy:** clinical trial, palliative chemotherapy or radiotherapy to the affected side; hospice referral on further progression.

## Dynamic Sentinel Node Biopsy (DSNB)

▸ **Setting**

- Limited to larger centres with experienced surgeons.

▸ **Technique**

- **Tracer:** 0.3–0.4 mL technetium-99m sulphur colloid around the tumour, 4 hours or 1 day before surgery.
- **Blue dye:** 1 mL intradermally around the circumference of the penile shaft, under anaesthesia.
- Small inguinal incision where the gamma camera directs; remove nodes found by gamma signal and blue dye; groin radioactivity should fall sharply afterwards.

▸ **Pathology and next step**

- Assess on permanent sections, not frozen section.
- Embed the entire node, cut at 2 mm intervals, analyse with staining and immunohistochemistry.
- Positive node: inguinal lymph node dissection within 4 weeks.

# Ch 85 · Benign Disorders of the Penis and Urethra

## Flaps and Grafts in Substitution Urethroplasty

▸ **Flaps**

- **Axial flap example:** groin flap, based on the superficial circumflex iliac artery.
- **By elevation technique:**
  - Peninsular: keeps vascular and cutaneous continuity at its base.
  - Island: keeps vascular continuity, divides cutaneous continuity.
  - Free: divides both; vascular continuity re-established at the recipient site.

▸ **Graft take**

- Three phases: imbibition, inosculation, revascularisation.
- Appendix is listed as a urethral substitution material.

## Oral Mucosa

▸ **Histology**

- **Four layers:** oral epithelium, superficial lamina propria, deep lamina propria, muscularis with minor salivary glands.
- The buccal graft consists of oral epithelium and superficial lamina propria.

▸ **Buccal mucosa**

- **Extent:** anterior tonsillar pillar (posteriorly) to outer commissure of the lips (anteriorly); limited superiorly and inferiorly by the maxillary and mandibular vestibular folds.
- **Nerves:** branches of the maxillary and mandibular nerves.
- **Artery:** buccal artery, a branch of the maxillary artery.
- **Advantages:** concealed donor site (cosmetic); high regenerative power.
- **Limitation:** not enough tissue to tubularise into long tubes.

▸ **Lingual mucosa**

- Non-keratinised stratified squamous epithelium of the ventral tongue; the dorsum is keratinised.
- **Harvest:** posterolateral aspect of the posterior tongue; a graft 7–8 cm long can be obtained.
- **Risk:** overzealous harvest can injure the submandibular duct at the base of the tongue.
- Especially useful with oral submucous fibrosis or unhealthy buccal mucosa.

## Other Grafts

▸ **Labial (lip) mucosa**

- Limited tissue; may cause lip disfigurement.

▸ **Bladder mucosa**

- **Pros:** harvested under regional anaesthesia; gives long grafts.
- **Cons:** too invasive to harvest; complications of bladder surgery; prone to encrustation.

▸ **Tunica vaginalis**

- **Pros:** easily and universally available; no general anaesthesia; no extensive surgery as it is a local tissue transfer.
- **Cons:** less reliable than oral mucosa; may be unavailable after previous scrotal (hydrocele) surgery.

# Ch 86 · Urethral Stricture Disease

## Diagnosis and Evaluation

- Stricture depth and density are hard to measure objectively; judged from physical examination, urethral appearance on radiography, and ultrasound or MRI.

## RGU (Retrograde Urethrogram)

▸ **Indication**

- Suspected urethral foreign body.

▸ **Preparation**

- Examine the genitalia to exclude tight phimosis or local infection, which may make RGU difficult or impossible.
- Ask the patient to pass urine first.

▸ **Method**

- Retract the foreskin; clean glans and meatus with povidone-iodine (Betadine) or another antiseptic.

- Instil a small amount of 2% lidocaine jelly before contrast.
- Take a plain (scout) film supine before contrast.
- Contrast can go through a feeding tube just inside the meatus, as an alternative to a catheter balloon in the fossa navicularis.
- **Films:** 30-degree left and 30-degree right anterior oblique during injection.
- **Oblique position:** patient turned about 45 degrees, lower (dependent) leg flexed 90 degrees at the knee, upper leg straight.

▸ **Complications**

- Contrast extravasation or urethral rupture, contrast reactions, urethral trauma, urinary tract infection.

## Internal Urethrotomy (OIU)

- **Success:** about 75% moderately long-term for bulbar stricture under 1.5 cm with minimal deep spongiofibrosis.
- **Erectile dysfunction:** due to cavernosal veno-occlusive dysfunction.
- **Colchicine:** blocks tubulin, inhibiting wound contraction, to prevent recurrence.
- **Mitomycin C:** injected submucosally; antifibroblast and anticollagen activity.

## Open Urethroplasty

▸ **Flaps**

- Hockey-stick penile skin flaps and Q-flaps for anterior urethral stricture.

▸ **Two-stage urethroplasty**

- **Indications:** urethral plate too poor or absent; necrosed or stenosed bulbar urethra.
- **Stage I:** excise the whole scarred plate; place the graft on dartos fascia, tunica albuginea or corpora cavernosa.

▸ **Dorsal onlay BMG (buccal mucosal graft) technique**

- Place a throat pack before harvest.
- Infiltrate the donor site submucosally with 2% lignocaine (Xylocaine) with adrenaline.
- Outline the graft with two parallel incisions about 1.5 cm apart, from the angle of the mouth to the anterior tonsillar pillar.
- Keep the graft in saline, with or without gentamicin, before defatting.
- Dorsal urethrotomy opens the whole stricture length plus 0.5 cm on either side.
- Lay the graft on the corpora cavernosa, mucosa facing the lumen; suture one edge to one cut urethral edge, place quilting sutures, then suture the other edge to the other margin over a catheter.

▸ **BMG outcomes**

- **Barbagli et al (50 cases):** ventral, dorsal or lateral graft placement showed no significant difference in success.
- **Restricture:** around 5%; other perineal complications rare.
- **Donor site:** about 15% have long-term oral numbness and mouth tightness.

## Pelvic Fracture Urethral Injuries (PFUI)

▸ **Approach**

- **Transpubic:** very rarely needed; for marked prostatic displacement, retropubic fixation of the prostate, and fistulas.
- **Abdominoperineal:** may be needed with severe fibrosis.

▸ **Aftercare and follow-up**

- At catheter removal (21 or 28 days), flexible endoscopy and a pericatheter study are alternatives to voiding cystourethrogram.
- Remove the suprapubic catheter once voiding well for 5–7 days.
- Flexible endoscopy at 6 months and again at 1 year.

### Other topics in this book

## PET-CT

- **Seminoma residual mass after chemotherapy:** raised beta-hCG leads directly to salvage chemotherapy, without PET.
- **Penile cancer:** no role in assessing the primary lesion.

# 11 Stones, Obstruction & Trauma

## Ch 93 · Management of Upper Urinary Tract Obstruction

### Retroperitoneal fibrosis (RPF)

▸ **Epidemiology**

- Typical age 40–60 years, but cases under 18 years have been reported.

▸ **Presentation**

- The plaque is centred on the distal aorta at L4–L5 and may extend distally to involve the iliac vessels.

▸ **Medical management**

- **Hormonal agents:** progesterone, medroxyprogesterone acetate and tamoxifen.

▸ **Open ureterolysis**

- Midline transperitoneal incision.
- Deep biopsies of the mass go for both frozen section and permanent (paraffin) section.

▸ **Laparoscopic ureterolysis**

- Four-port transperitoneal approach.
- Same success and complication rates as open ureterolysis.

### Retrocaval ureter (pre-ureteral vena cava): treatment

- Repair is done over a double-J stent, removed after 4 to 6 weeks.

### Stone types

▸ **CT and Hounsfield units**

- Hounsfield unit values correlate with composition, but many stone types overlap.
- Uric acid stones can be told apart from calcium stones; calcium stones usually cannot be told apart from struvite stones.

▸ **Matrix stones**

- Matrix is the non-crystalline component of stones; predominantly matrix stones are rare.
- Pure matrix stones may be up to 65% protein.
- **Matrix proteins:** osteopontin and calprotectin.
- Beta-2-microglobulin (microfibrillar protein) matrix stones occur in chronic renal failure patients on dialysis.
- **Imaging:** typically radiolucent; may be mistaken for a tumour or a uric acid stone.
- **Treatment:** PCNL is the best choice, but ureteroscopy can also be used.

▸ **Ammonium acid urate stones**

- **Prime risk factor:** conditions causing metabolic acidosis.
- **Further associations:** inflammatory bowel disease, morbid obesity and recurrent uric acid stones.
- **Treatment by cause:**
  - Bowel disease: standard fluid intake, oral calcium, alkalinisation and oxalate reduction.

- History of uric acid calculi: more fluid, protein and salt restriction, potassium citrate alkalinisation and allopurinol.
- Laxative abuse: a healthier bowel regimen.
- Chronic infection: as for struvite calculi.

▸ **Infection (struvite) stones**

- Calcium, phosphate and carbonate concentrations allow precipitation of hydroxyapatite as well as carbonate apatite.
- Most urea-splitting organisms belong to the Enterobacteriaceae family. Some Mycoplasma species and some yeasts also produce urease.
- Bacterial infection (E. coli, Proteus) alters urokinase and sialidase activity, which promotes stone formation.

▸ **Cystine stones**

- Cystine is a dimer of two cysteine molecules joined by a disulfide bond.

▸ **Drug-related stones**

- Phosphate-binding antacids indirectly promote stone formation.

## Metabolic abnormalities

▸ **Dietary hyperoxaluria**

- Potatoes are an oxalate-rich food that contributes.
- Prolonged antibiotic use reduces or eliminates Oxalobacter formigenes, so less oxalate is degraded in the gut and more is absorbed.

▸ **Idiopathic hyperoxaluria**

- In the Arabian Peninsula population, hyperoxaluria with almost no hypercalciuria leads to idiopathic calcium oxalate stones.
- **Proposed mechanism:** abnormal oxalate transport.

▸ **Hypocitraturia**

- Citrate enhances the inhibitory effect of Tamm-Horsfall protein on crystallisation.
- Metabolic acidosis lowers urinary citrate by enhanced tubular reabsorption and by decreased citrate synthesis in renal tubular cells.
- Strenuous exercise causes hypocitraturia through lactic acidosis.

▸ **Hypomagnesiuria**

- **Cause:** poor dietary magnesium intake.
- Promotes stones by reducing magnesium complexation of calcium and oxalate salts, and also by decreasing urinary citrate.

▸ **Hyperuricosuria (uric acid stones)**

- **Most common cause:** dietary purine intake.
- **Myeloproliferative examples:** multiple myeloma and polycythaemia.
- Haemoglobinopathies, including thalassaemia.

## Renal trauma

### ▸ Mechanism

- **Blunt causes:** road traffic accidents, falls from height and assault.
- **Penetrating injuries:** 86% gunshot, 14% stab.
- Penetrating trauma at or anterior to the anterior axillary line more often injures the hilum and pedicle; at the posterior axillary line it more often causes parenchymal injury.
- A rib fracture carries a threefold increase in the incidence of renal trauma.

### ▸ Haematuria and indications for imaging

- Microscopic haematuria is detected by dipstick or urine microscopy of the first aliquot; any gross haematuria is significant.
- Image all paediatric trauma patients with more than 5 red blood cells (RBCs) per high-power field, without waiting for shock.

### ▸ AAST grading (revised 2011)

- Only 5% of renal injuries are grade II to grade V.

### ▸ Indications for surgery and angioembolisation

- Delayed diagnosis of a renal arterial injury is a relative indication for surgery, as delayed nephrectomy.
- In the acute setting, angioembolisation can stop significant bleeding and avoid nephrectomy; evidence is from case reports only.

### ▸ Delayed bleeding

- Occurs in 9% of renal injuries.
- **Management:** bed rest and hydration, or angioembolisation.

### ▸ Post-traumatic hypertension

- Caused by vascular stenosis, parenchymal compression or post-traumatic arteriovenous fistula; mediated by the renin-angiotensin-aldosterone system.

## Ureteric injury

### ▸ Causes and incidence

- Occurs in less than 4% of penetrating trauma and less than 1% of blunt trauma.
- Gunshot ureteric injuries have associated bladder injury in 5%.
- **Surgical causes after hysterectomy, in descending order:** colorectal, ovarian, then vascular surgery.
- 0.2% of ureteroscopic perforations require open surgery.
- **Perforation risk by intracorporeal lithotripter:** electrohydraulic lithotripsy highest, then Nd:YAG laser, then holmium:YAG laser.

### ▸ CT diagnosis

- CT may miss an upper ureteric injury because Gerota's fascia can contain the extravasation.
- Circumferential contrast extravasation is a feature of ureteric injury.

▸ **Intraoperative recognition**

- Recognition at operation can be as low as 0% during laparoscopic surgery.
- The wound trajectory is a guide.
- Inject 1–2 mL of methylene blue into the renal pelvis and look for leakage.

▸ **Damage control**

- Do nothing to the ureter at the first operation and plan re-operation at 24 hours.

▸ **Failed initial management**

- If stenting (or, for a delayed-recognised injury, stenting and nephrostomy) fails, consider endoscopic options before open reconstruction: balloon dilation, laser incision and Acucise (cutting-balloon) endoureterotomy.

▸ **Principles of repair**

- Anastomosis with an absorbable monofilament suture such as 5-0 PDS.

## Controversies in fluid and dietary therapy for stones

▸ **Fluids**

- The amount of fluid per day matters, not the type.
- **Caffeine:** increases stone risk by raising urinary calcium/creatinine, magnesium/creatinine and urine flow, but not the oxalate/creatinine ratio.
- **Possibly decrease risk:** lemonade, citrus juice, grapefruit juice, apple juice, tea, beer and wine (check).

▸ **Diet**

- Low-carbohydrate, high-protein weight-loss diets (Atkins, South Beach, Sugar Busters) give a marked acid load to the kidney and increase stone risk.

▸ **Calcium, vitamins and bisphosphonates**

- **Calcium:** restriction is an old concept; a moderate supplement, preferably calcium citrate, is advised.
- **Vitamin D:** repletion does not change urinary calcium, but urinary calcium should be monitored.
- **Bisphosphonates with thiazides:** may reduce hypercalciuria and protect bone.
- **Vitamin C:** limit to less than 2 g/day.

# Ch 95 · Urinary Lithiasis: Etiology, Epidemiology and Pathogenesis

## Stone types and analysis

- **Brushite:** calcium hydrogen phosphate dihydrate ($CaHPO_4 \cdot 2H_2O$).
- **Whitlockite:** mineral name of tricalcium phosphate.
- **Dual-energy CT (DECT):** distinguishes uric acid from non-uric-acid stones with 92-100% accuracy.
- "Unclassified calcium nephrolithiasis" is a category in the metabolic classification of nephrolithiasis.

## Epidemiology of urolithiasis

▸ **Climate and occupation**

- More stone episodes in summer and in hot, dry climates.
- Heat-exposed occupations, such as cooks, cause dehydration and predispose to stones.

▸ **Obesity, diabetes and metabolic syndrome**

- Stone risk is especially marked in women.
- Obesity increases urinary oxalate, uric acid, sodium and phosphate.
- Obesity predisposes to calcium oxalate stones (increased promoters) and uric acid stones (low urine pH).

▸ **Water hardness and race**

- Whether hard or soft water affects stone risk is controversial.
- In the United States, White people form more stones than Hispanic, Asian and Asian-American people.

## Physical chemistry of stone formation

▸ **State of saturation**

- Concentration product (CP) example: for sodium chloride, CP equals [Na+] times [Cl-].

▸ **Nucleation**

- Nuclei are the earliest crystals that will not dissolve. Small nuclei are unstable and dissolve; large nuclei persist.

▸ **Crystal retention**

- Oxalate-induced tubular epithelial cell injury promotes adherence of calcium oxalate crystals.
- Calcium oxalate monohydrate causes oxidative stress to tubular epithelial cells.
- Retention in the tubular lumen without extrusion by epithelial erosion gives nephrocalcinosis; retention with subepithelial location and epithelial erosion gives urolithiasis.

▸ **Inhibitors**

- **Citrate:** reduces crystal growth, much more for calcium phosphate than for calcium oxalate.
- **Osteopontin:** also called uropontin.
- Urinary prothrombin fragment 1.
- **Polyanionic macromolecules:** glycosaminoglycans (chondroitin sulfate, heparan sulfate), acid mucopolysaccharide and RNA.

▸ **Promoters**

- Oxidative stress.

## Mineral metabolism

▸ **Calcium**

- 30-40% of dietary calcium is absorbed: 90% in the small intestine and 10% in the colon.
- Absorbed in its ionic state.
- Phosphate, citrate, sulfate, oxalate and fatty acids complex with calcium in the gut and reduce absorption.
- About 98% of filtered calcium is reabsorbed by the kidney.

▸ **Oxalate**

- Absorbed dietary oxalate is taken up 50% in the small intestine and 50% in the colon.

▸ **Magnesium**

- Absorbed in the small intestine (mostly the ileum) and the large intestine.
- Mostly passive diffusion, with an active transport component.
- Hormonally regulated by vitamin D.

▸ **Phosphorus**

- About 60% of dietary phosphorus is absorbed.
- Absorption is active: transcellular at low concentrations, paracellular at high concentrations.
- Vitamin D increases absorption. Low luminal pH decreases it and high luminal pH increases it.
- About 80-90% of filtered phosphate is reabsorbed in the renal tubule.

## Hypercalciuria

▸ **Definition and renal calcium handling**

- **Alternative definition:** urinary calcium above 7 mmol/day in men or above 6 mmol/day in women.
- The normal kidney filters about 270 mmol of calcium daily and reabsorbs all but about 4 mmol (about 98%).

▸ **Absorptive hypercalciuria**

- **Definition:** urinary calcium above 0.2 mg per mg creatinine after an oral calcium load.
- **Causes:** vitamin D-dependent and vitamin D-independent processes, including upregulation of the vitamin D receptor, and hypersensitivity to vitamin D.
- **Type 3:** renal phosphate wasting (renal phosphate leak).

▸ **Resorptive hypercalciuria**

- **Thiazide challenge:** a thiazide increases renal calcium reabsorption even when serum calcium is already high, so in primary hyperparathyroidism it worsens the hypercalcaemia.

▸ **Hypercalciuria from other diseases**

- Glucocorticoids inhibit intestinal calcium absorption, so they treat the hypercalciuria of sarcoidosis, although glucocorticoid excess itself causes resorptive hypercalciuria.
- Tuberculosis and thyrotoxicosis.

▸ **Distal renal tubular acidosis (RTA)**

- The hypercalciuria results from metabolic acidosis with a secondary rise in parathyroid hormone (PTH).

▸ **Fast and calcium load test**

- Restricted diet for at least 7 days beforehand.
- Drink 300 ml of water at 12 hours and again at 9 hours before the calcium load.
- Two hours before the load, empty the bladder and drink 600 ml of water.
- Collect urine for 2 hours: the fasting urine.
- Give a 1 g oral calcium load as 250 ml of a liquid synthetic diet (Calcitest).
- Collect urine for the next 4 hours: the post-load urine.

- Measure calcium and creatinine in both samples.
- **Normal post-load urinary calcium:** less than 0.2 mg per mg creatinine.

## Primary hyperoxaluria (PH)

▸ **Oxalate metabolism in the liver**

- Hydroxyproline becomes 4-hydroxy-2-oxoglutarate, which HOG aldolase (HOGA, deficient in PH3) splits into glyoxylate and pyruvate.
- **Fates of glyoxylate:**
  - Reduced to glycolate by GRHPR.
  - Transaminated to glycine by AGT (alanine donates the amino group and becomes pyruvate; vitamin B6 is the cofactor; deficient in PH1).
  - Oxidised to oxalate by lactate dehydrogenase (LDH).
- Hydroxypyruvate is converted to D-glycerate by GRHPR (deficient in PH2). Without GRHPR, LDH converts it to L-glycerate instead, which is why L-glycerate is the PH2 urinary marker.

▸ **Renal damage**

- Interstitial nephrocalcinosis causes interstitial fibrosis, and obstructing stones cause parenchymal inflammation; together they cause progressive damage.

▸ **PH1**

- Untreated, it leads to ESRD by 15 years of age in 50% of children; overall 30% of patients die.

▸ **PH2**

- Thirty GRHPR gene mutations have been identified.

▸ **PH3**

- Mechanism of hyperoxaluria unknown.
- Associated with severe hypercalciuria and recurrent calcium oxalate stones; stone formation may stop later in life.
- No reported cases of ESRD, and no kidney transplants reported.

## PUJ obstruction and stones

- UPJ obstruction is associated with stones in 20–30%.
- **Two situations:** a primary PUJO with secondary calculi, or a renal pelvic/UPJ stone causing PUJ oedema that looks like PUJO.
- **If unclear:** treat the stones first and repeat CT or ultrasound after 4–6 weeks. If hydronephrosis persists, do a DTPA scan; operate if it confirms PUJO, follow up if it does not.

# Ch 97 · Surgical Management of Ureteral Calculi

## Ureteric calculi

▸ **Natural history and management**

- Alpha-blockers increase passage of proximal/mid ureteric stones under 5 mm by 29%.
- Associated ureteral pathology is an indication for PCNL (antegrade approach) for a proximal/mid ureteric stone.

▸ **Stone in a megaureter**

- **Non-obstructed:** medical expulsive therapy (MET), URS or ESWL.
- **Obstructed:** any of
  - retropulsion to the kidney plus PCNL plus ureteroneocystostomy;
  - laparoscopic/robotic ureteroneocystostomy with stone retrieval;
  - URS with endoureterotomy (segment under 3 cm);
  - open ureteroneocystostomy with stone retrieval.

▸ **Stricture below a stone**

- Consider the cause of a pathological stricture. Options: endoureterotomy with double-J stent and URS at the same or a second sitting, or open/laparoscopic/robotic ureterotomy with definitive repair.

▸ **Ureteral avulsion**

- Transureteroureterostomy is contraindicated in stone formers.

## Special situations

▸ **Stones threatening renal function**

- **Causes:** staghorn calculus, recurrent pyelonephritis or abscess, XGPN and chronic obstruction from stone.
- Cortical atrophy or thinned parenchyma on ultrasound or CT should prompt a diuretic renogram before choosing nephrectomy, drainage with reassessment, or stone surgery.

▸ **Obstructed solitary kidney**

- Either drain first (PCN or double-J stent) with later definitive surgery, or do primary URSL with a double-J stent, depending on clinical stability, UTI and electrolyte abnormality.

▸ **Urinary diversion**

- **Retrograde access possible:** PCNL for a large stone burden, flexible URS for a small one.
- **No retrograde access:** PCNL or antegrade URS.
- Locate the ureteric orifice by loopogram, pouchogram or intravenous indigo carmine.

▸ **Transplant kidney**

- **Presentation:** vague symptoms, graft-site discomfort, fever, oliguria and raised creatinine.
- **Ureteric stone under 1.5 cm:** SWL or antegrade/retrograde URS.
- **Over 1.5 cm, or failed URS/SWL:** mini-PCNL.
- Tract dilatation may be tough because of the fibrotic capsule.

## Intracorporeal lithotripters

▸ **Electrohydraulic lithotripsy (EHL)**

- Fibre tip projected 2–5 mm beyond the ureteroscope tip, about 1 mm from the stone surface.
- **Safety (probe more than 1 mm from tissue):** under 500 mJ causes no injury; over 1000 mJ causes ureteric perforation.
- Flexible probe can reach the lower pole.
- Causes stone retropulsion and produces larger fragments than other sources.
- **Success rate:** about 90%.

▸ **Pneumatic (ballistic) lithotripsy**

- First ballistic lithotrite: the Swiss Lithoclast.
- **Probe sizes:** 2.4, 3, 4.8, 6 and 10.5 Fr.
- **Tissue effect:** focal erosion and haemorrhage; no thermal injury.
- Little maintenance; probes have a long life; easily available.
- Least efficient of EHL, pneumatic and laser.
- **Goal:** fragments smaller than 2 mm.
- **Success rate:** 73–100%.

▸ **Ultrasonic lithotripsy**

- A piezoceramic plate resonates at 23,000–25,000 Hz, making the stone resonate and break. The unit has a generator, a foot pedal and suction.
- **Probes:** 2.5 Fr to 12 Fr; needs a rigid scope with a large working channel for the 4.5 Fr hollow probe.
- Minimal heat, so minimal soft-tissue damage.
- Particularly helpful for large stones; less steinstrasse.
- **Disadvantages:** risk of ureteral injury; stone can block the hollow suction probe.
- **Success rate:** 70–100%.

▸ **Lasers**

- **Ruby laser:** continuous-wave, produces heat; not appropriate for clinical lithotripsy.
- **Pulsed dye laser:** does not fragment hard stones, and the eye protection hampers vision of the stone; no longer used.
- **Holmium fibres:** 200, 365, 550 and 1000 µm; 200 and 365 µm are used for lithotripsy.
- **Holmium power:** raise frequency rather than pulse energy, because higher pulse energy narrows the safety margin.
- **Holmium stone-free rate:** 85–100%.
- **Holmium disadvantages:** high initial cost and fibre cost; can cut guidewires and baskets.

## Shock wave lithotripsy (SWL/ESWL)

▸ **Generators**

| Generator | Features |
|---|---|
| Electrohydraulic | High-voltage spark causes explosive vaporisation of water, giving a spherically expanding shock wave. |
| Electromagnetic | Plane or cylindrical shock waves, focused by an acoustic lens or reflector. Less pain, because energy enters over a larger skin area. |
| Piezoelectric | Barium titanate piezoceramic elements; needs 300–3000 crystals for enough shock pressure. Focuses accurately. |

▸ **Imaging systems**

- Modern lithotripters use a rotatable C-arm for fluoroscopy, in place of the HM3's two fixed X-ray units at 90°.

| Targeting | Advantages | Disadvantages |
|---|---|---|

| | | |
|---|---|---|
| Fluoroscopy | Familiar to most urologists; shows radio-opaque stones; shows radiolucent stones as filling defects with contrast. | Radiation exposure; difficulty with radiolucent stones; high maintenance. |
| Ultrasound | Inexpensive; no radiation; detects non-opaque stones. | Needs a highly trained operator; stones hard to see in the middle third of the ureter or with a stent in situ. |

### ▸ Mechanism of comminution

- **Direct shock wave:** a plane wave front.
- **Cavitation (indirect):** with no boundary, the bubble collapses symmetrically and gives a spherical wave front; at a boundary such as the stone surface, asymmetrical collapse forms microjets that apply a local compressive force progressing spherically into the stone.
- Together these cause compression-induced tensile cracks and spalling.
- Modern lithotripters have small focal zones, so make little use of squeezing-splitting (circumferential compression).

### ▸ Bioeffects

- **Acute vascular:** damage and rupture of veins and small arteries, rupture of glomerular and peritubular capillaries, venous thrombi, intraparenchymal haemorrhage.
- **Acute tubular:** mild tubular dilatation with cast formation, mild tubular necrosis.
- **Acute glomerular:** breaks in Bowman's capsule; damage to mesangial cells and podocytes.
- **Chronic:** nephron loss, dilated veins, streaky fibrosis, diffuse interstitial fibrosis, calcium and haemosiderin deposits, hyalinised acellular scars from cortex to medulla.
- **Proposed chronic effects:** more stone recurrence and induction of brushite stone disease.
- **Risk factors for injury:** first-generation lithotripters (more than second and third generation); juvenile kidney (more vulnerable than adult); pre-existing renal impairment.
- **Ramping:** start with 100–500 low-energy shocks, then raise to standard treatment energy.

### ▸ Factors affecting fragmentation

- **Composition:** matrix stones resist SWL; apatite breaks less well than calcium oxalate dihydrate.
- **Location:** upper ureter and iliac ureter have lower stone-free rates; middle calyx, pelvic ureter, renal pelvis and upper calyx are among the best.
- **Habitus:** obesity makes imaging and focusing difficult and increases skin-to-stone distance; BMI 20–28 gives the best outcome.
- **Coupling:** an air pocket over just 2% of the coupling area reduces breakage by 20–40%.
- **Power:** higher settings narrow the focal zone, so breathing moves the stone in and out of it and breakage falls.

### ▸ Success and residual fragments

- **Success:** stone-free 3 months after treatment. Residual fragments larger than 5 mm are generally a failed session.
- **Residual fragments:** all fragments remaining in the kidney 3 months after the last session; seen in about 24–36% (about 30%).
- **Risk factors:** multiple stones and congenital anomalies.
- **Assessment:** plain KUB overestimates stone-free rates by about 35%; ultrasound has interobserver variability.

- Whether clinically insignificant residual fragments (CIRF) need treatment is controversial, because residual fragments carry a significant recurrence risk.
- Alkali citrate after ESWL decreases stone growth and recurrence from residual fragments.

▸ **Steinstrasse**

- **Site:** distal ureter about 64% (65%), proximal about 29% (30%), mid about 8% (5%).
- **Presentation:** includes bladder irritation.
- **Causes:** include meatal stenosis and ureteric stricture.

## Complications of PCNL

▸ **Clavien-Dindo classification**

- **Rates:** none (grade 0) 77%, grade I 11%, grade II 7%, grade III 4.1%, grade IV 0.6%.
- **Grade I:** allows antiemetics, antipyretics, analgesics, diuretics, electrolytes and physiotherapy.
- **Grade II:** includes total parenteral nutrition and blood transfusion.
- **Grade IIIa / IIIb:** intervention without / under general anaesthesia.
- **Grade IVa / IVb:** single-organ dysfunction (including dialysis) / multi-organ dysfunction.

▸ **Acute haemorrhage**

- **Further factors:** patient characteristics, supracostal access, tract dilatation, renal pelvic perforation, excessive torque and misuse of instruments.
- Parenchymal tract bleeding is mostly not significant; the access sheath gives haemostasis during the operation.
- If a vein or artery in the collecting system is injured, stop if vision is lost.
- **Venous:** nephrostomy tube or Councill catheter, with or without gelatin matrix.
- **Arterial in the collecting system:** fulgurate under direct vision.

▸ **Delayed haemorrhage**

- Occurs in about 1%.
- Pseudoaneurysm is far commoner than arteriovenous fistula; a fistula usually bleeds continuously rather than intermittently.
- Any bright red blood in the urine after PCNL needs admission and angiography.
- **Alternatives to embolisation:** endovascular stent, or ultrasound-guided puncture of the pseudoaneurysm with thrombin or fibrin tissue adhesive.

▸ **Collecting system injury**

- Small infundibular tears are common and harmless without haemorrhage.
- Ureteral injury is rare and heals with a stent.
- **Renal pelvic injury:** during initial access or dilatation, or from pushing too hard on a pelvic stone during lithotripsy.
- **Recognition:** usually intraoperative, by collapse of the previously distended collecting system.
- **Action:** abort, or finish quickly with low irrigation pressure if nearly complete.
- **Drainage:** nephroureteral stent, or double-J stent plus nephrostomy; nephrostogram and tube removal at 2 to 7 days.
- **Missed perforation:** postoperative abdominal pain and ileus, with or without fever.

- **Urinoma:** percutaneous drainage if stent and nephrostomy are already in place; with no tubes, try a double-J stent or nephrostomy, with or without a drain.

▸ **Colon injury**

- Occurs in less than 1%, matching the incidence of retrorenal colon. Left outnumbers right 2:1.
- **Risk factors:** advanced age, dilated colon, previous surgery, thin body habitus.
- **Missed injury:** unexplained fever, prolonged ileus, unexplained leucocytosis, rectal bleeding, peritoneal inflammation, faecaluria or pneumaturia.
- **Conservative management:** add broad-spectrum antibiotics and nil by mouth for a few days, then clear liquids and a gradually increased diet. Remove the colostomy tube once contrast shows no communication.
- Intraperitoneal perforation needs open repair.

▸ **Small bowel and other viscera**

- **Small bowel:** rare; detected by peritonitis or a nephroenteric fistula on nephrostogram.
- **Duodenum:** conservative with a percutaneous intraduodenal catheter or nasoduodenal drainage, plus upper tract drainage, fasting and parenteral feeding; otherwise open surgery.
- **Spleen and liver:** rare unless access is above the 10th rib; very rare between the 10th and 11th ribs.
- **Gallbladder or biliary tree:** can cause biliary peritonitis.
- **Pancreatitis:** can occur without evidence of pancreatic trauma.

▸ **Pleural injury**

- Occurs in 24% of accesses above the 11th rib and 4.6% above the 12th rib.
- Urinothorax is one form.
- Every supracostal access needs formal chest radiography, during or after the operation.
- **Hydrothorax:** found intraoperatively, intercostal tube drain; minimal postoperatively, conservative; large or affecting breathing, intercostal drain.
- In a urinoma, rule out distal obstruction.

▸ **Metabolic and physiological complications**

- **Saline extravasation:** can cause congestive cardiac failure and respiratory distress; rare in uncomplicated cases.
- **Venous gas embolism:** hypoxaemia, pulmonary oedema, increased airway pressure, hypotension, jugular venous distension, facial plethora, dysrhythmias, and right heart strain with wide QRS on ECG.

▸ **Venous thromboembolism**

- Deep vein thrombosis risk is low; thromboprophylaxis is not mentioned in the AUA best practice statements.
- **Prevention:** early ambulation is best; stockings or intermittent compression pumps during the operation.

▸ **Fever and sepsis**

- **Additional risk factors:** paraplegia, indwelling stents, previous PCNL, hydronephrosis.
- With appropriate prophylaxis, most patients with fever do not have an infection.

- If bacteriological cure is not possible beforehand, suppress the bacterial count as much as possible.
- A negative preoperative urine culture does not exclude sepsis.

▸ **Positioning injuries**

- Facial nerve injury, nose injury, stroke, peripheral nerve injury and rhabdomyolysis.

▸ **Tube dislodgement**

- Prevent with proper skin fixation.
- **Retention strength:** balloon, then Cope loop, then Malecot.
- Tissue ingrowth can make removal difficult, especially with a Malecot.

▸ **Collecting system obstruction**

- Transient obstruction from clot or oedema: conservative. Late strictures: mostly endoscopic.
- Infundibular stenosis occurs in 2%.

▸ **Renal function and death**

- Function falls slightly at first, with a nadir at 48 hours; long-term loss is negligible.
- When function is reduced by stone obstruction, PCNL may improve it.
- Death is extremely rare, usually from underlying cardiovascular disease.

## Ch 98 · Surgical Management of Upper Urinary Tract Calculi

### Natural history of renal stones

- **Staghorn stones:** named for their likeness to deer or stag antlers on imaging. Untreated, complete loss of renal function can occur in 50% after 2 years.
- **Non-staghorn stones:** only about 15% pass spontaneously, mainly those of 5 mm or less.
- **Asymptomatic stones:** 10-20% (about 15%) will require surgery at 3-4 years.

### Choice of treatment

▸ **Non-staghorn stones by size**

- **1 cm or less:** failed URS and SWL is an indication for PCNL.
- **Over 2 cm:** URS is the alternative to PCNL for patients too frail for PCNL, who refuse blood transfusion, or with coagulopathy.

▸ **Miniaturised PCNL**

- **Super-mini PCNL:** 10F, 12F or 14F sheaths with active suction.
- **Micro-PCNL:** 4.85 Fr instrument.

▸ **Staghorn stones**

- "Sandwich therapy" (PCNL combined with ESWL) is an option.

▸ **Lower pole stones**

- Asymptomatic stone under 1 cm: observation is also an option.
- **Unfavourable anatomy:** infundibular length more than 3 cm; infundibular width less than 5 mm.

### Ectopic (pelvic) kidney with stone

- **Problems:** high ureteral insertion, UPJO and malrotation.

- **Laparoscopic pyelolithotomy:** with or without pyeloplasty, is an option.
- **SWL:** needs the prone position; exclude impaired drainage first (check).

## Combined ultrasonic–pneumatic devices

- **Lithoclast Ultra:** pneumatic and ultrasonic probes as two separate handpieces joined together. The "Small Stone" pedal activates only the ultrasonic inner probe, for finer control; the "Large Stone" pedal activates both.
- **CyberWand:** dual ultrasonic probe design.
- Both modalities appear safe, but minimise contact time with dual-modality settings.

## Colon injury in PCNL

- **Detected intraoperatively:** easily managed by separate drainage of the colon and the urinary tract.
- Most perforations are extraperitoneal and managed conservatively: pull the nephrostomy tube back into the colon as a colostomy, and drain the kidney with a double-J stent or another nephrostomy.

## Evaluation of outcome: residual fragments

### ▸ Imaging

| Method | Features |
|---|---|
| Plain KUB X-ray | Sensitivity about 60%; detects radio-opaque stones as small as 2 mm. |
| Nephrotomography | Better than plain X-ray; obsolete in many centres. |
| Ultrasound | Sensitivity 50–70%; also detects hydronephrosis. |
| CT | Preferred after PCNL (100% sensitivity); radiation exposure and high cost. |

### ▸ Fate of residual stones

- May pass spontaneously, cause symptoms if they impact and obstruct, act as a nidus for new stones, or be a source of persistent infection.

### ▸ Clinically insignificant residual fragments (CIRF)

- Fragments of 4 mm or less, non-obstructive and non-infectious, with sterile urine, in an asymptomatic patient.

### ▸ Reporting outcomes

- Stone-free rate; success rate (stone-free rate plus CIRF); effectiveness coefficient.
- **Effectiveness coefficient** (for comparing modalities): % stone free divided by (100% + % retreatment + % auxiliary procedures), multiplied by 100.

# Ch 99 · Prevention of Urinary Stone Disease

## Metabolic evaluation

### ▸ Indications

- Osteoporosis, pathological fracture, renal insufficiency, and infirm health (unable to tolerate repeated stone episodes).

▸ **Low-risk single stone former (abbreviated protocol)**

- Further evaluation is optional and should be discussed with the patient.
- **Multichannel blood screen:** basic metabolic panel (sodium, potassium, chloride, CO2, BUN, creatinine), calcium, uric acid and intact PTH.

▸ **High-risk or recurrent formers**

- **Extensive evaluation:** time-consuming and difficult, needs a restricted diet (older protocols) and two 24-hour urine collections; high diagnostic yield.
- **Simplified evaluation:** omits the fast and calcium load test; may not need strict dietary adherence.
- **Blood tests (extensive):** complete blood count and comprehensive metabolic panel (14 tests covering kidney and liver function, glucose, parathyroid, electrolytes and fluid balance), plus calcium, PTH, uric acid and creatinine.

▸ **24-hour urine**

- A single specimen may lead to misdiagnosis.
- Collected on ice or with an added preservative.
- Urinary sulfate is one of the parameters.

▸ **Imaging**

- **Digital tomosynthesis:** newer technique; plain KUB plus a single tomographic sweep of the X-ray emitter, reconstructed by digital software.
- NCCT KUB is usually for the acute episode; for metabolic evaluation, plain KUB and IVU document current stones and their density.

## Conservative measures

▸ **Fluids**

- Work by mechanical diuresis (preventing stasis) and by preventing supersaturation. Amount matters more than type.
- **Hard water:** a minor concern; raises urinary calcium, magnesium and citrate, with no significant change in pH, oxalate or uric acid.
- **Carbonated drinks:** may protect by raising urinary citrate; lemonade or citric-acid-flavoured soda helps.
- **Phosphoric-acid-flavoured soda:** increases stone risk.

▸ **Protein and sodium**

- Protein restriction: 0.8-1 g/kg/day.
- Atkins diet (high protein, low carbohydrate) increases stone formation.
- **Controversy:** some studies suggest hypocitraturic calcium oxalate formers may benefit from sodium supplementation.

▸ **Calcium**

- Daily requirement 1000 mg. Moderate dietary calcium from dairy, or supplements with meals, lowers stone risk.
- Calcium citrate is stone-friendly because of citrate's inhibitory action.

▸ **Vitamins and bisphosphonates**

- **Vitamin C:** avoid very high doses; limit to less than 2 g/day, because it may increase urinary oxalate.
- **Vitamin D:** supplementation is controversial as it may increase stones; repletion is likely safe, but monitor 24-hour urinary calcium.
- **Bisphosphonates:** alone or with thiazides, improve bone density and decrease hypercalciuria, reducing stone formation.

## Treatment of hypercalciuria

▸ **Thiazides**

- Do not decrease intestinal calcium absorption.
- **Side effects:** muscle cramps, lassitude, sleepiness, decreased libido.
- Less effective long term, so a drug holiday may be needed.
- **Renal hypercalciuria:** correct the renal calcium leak, normalising secondary hyperparathyroidism, vitamin D and calcium.

▸ **Sodium cellulose phosphate**

- Non-absorbable resin that binds calcium and inhibits its absorption; no longer used.

▸ **Orthophosphate**

- Sodium or potassium salt of phosphorus; may have a role when other treatments fail.
- **Action:** inhibits vitamin D synthesis and directly alters renal tubular calcium reabsorption.
- **Problems:** parathyroid stimulation, soft-tissue calcification, marked rise in urinary phosphorus, increased urinary brushite saturation, GI distress, hypomagnesaemia and hyperoxaluria (check).
- May be used in mild to moderate resorptive hypercalciuria.

▸ **Absorptive hypercalciuria type II**

- May need no specific drug, as the defect is not severe. Manage with high fluid intake, moderate calcium intake and avoiding a high-sodium diet.

▸ **Resorptive hypercalciuria**

- No established drug therapy; orthophosphate in mild to moderate cases.
- Oestrogen may be used in postmenopausal women when surgery is not possible.

## Enteric hyperoxaluria

| Agent | Role |
|---|---|
| Calcium citrate | Calcium binds oxalate in the gut and reduces its absorption; citrate also helps prevent stones. |
| Magnesium gluconate | Better tolerated than magnesium oxide or hydroxide. |
| Cholestyramine | Binds bile salts, leaving calcium free to bind oxalate as calcium oxalate, reducing oxalate absorption. |
| Probiotics | Oxalate-degrading Oxalobacter formigenes or lactic acid bacteria. |
| Pyridoxine | Alternative in refractory cases. |

## Hypomagnesiuric calcium stones

- Typically also low urinary citrate and low urine volume.
- **Drugs:** magnesium oxide and magnesium hydroxide; potassium-magnesium citrate is newer.
- **Effect of magnesium:** raises urinary citrate, lowers urinary oxalate, and gives a favourable urinary calcium-to-magnesium ratio.

## Hyperuricosuric calcium stones

- **Diet:** reduce purine intake.
- **Allopurinol:** with moderate sodium restriction (150 mEq/day).
- **Potassium citrate:** 30-60 mEq/day in divided doses with high fluid intake; lowers supersaturation of both uric acid and calcium oxalate.
- Evidence that higher-dose potassium citrate causes calcium phosphate stones by over-alkalinisation is conflicting.

## Hypocitraturic calcium oxalate stones

- **Chronic diarrhoea:** slow-release potassium citrate is poorly absorbed because of rapid transit; use a liquid form or more frequent dosing.
- **Distal RTA:** potassium citrate corrects metabolic acidosis and hypokalaemia as well as urinary citrate. Urine pH rises little, because it is already raised.

## Cystine stones

- **Alkalinisation:** difficult, as the target pH is often unrealistic.
- **Tiopronin (alpha-mercaptopropionylglycine):** side effects include mental status changes.
- **Captopril:** acts through its free -SH (sulfhydryl) group; not often used. Side effects: hypotension, hyperkalaemia, chronic cough, bad taste.

## Ammonium acid urate stones

- Alkalinisation plus the common therapies: fluids, moderate calcium intake and a low-oxalate diet.
- Correct the cause: encourage healthy bowel habits in laxative abuse; rectify bowel disease or chronic diarrhoea if possible.

## Infection stones

- **Acetohydroxamic acid side effects:** DVT, tremor, headache, palpitations, GI upset, oedema, rash, alopecia, anaemia, abdominal pain and loss of taste; these prevent its widespread use.
- **Renacidin (hemiacidrin) irrigation:** significant side effects, even death.

# Ch 125 · Genital and Lower Urinary Tract Trauma

## Penile fracture

▸ **Tunica albuginea**

- Made of collagen and elastin.
- The outer longitudinal layer sets its strength and thickness, which vary along the shaft.
- Resists rupture until intracavernous pressure exceeds 1500 mmHg.

### ▸ Mechanism and site

- In vigorous intercourse the rigid penis slips out of the vagina and strikes the perineum or pubic bone, causing a buckling injury.
- In the US and Europe, intercourse-related fractures are commoner in stressful situations such as extramarital sex.
- Most are distal to the suspensory ligament, though rupture can occur anywhere on the shaft.
- The tear is usually transverse and 1-2 cm long.

### ▸ Diagnosis

- Fear and embarrassment may delay presentation greatly.
- **MRI:** useful but expensive, of limited availability and time-consuming.
- **Cavernosography:** discouraged; time-consuming and unfamiliar to most urologists.
- **Retrograde urethrography:** time-consuming and can be inaccurate, so intraoperative flexible cystoscopy just before catheter placement is now often routine.

### ▸ Surgical approach

- A distal circumcision (degloving) incision exposes all three compartments.
- In uncircumcised men it puts the distal prepuce at risk of ischaemia, so strongly consider limited circumcision at the end.

### ▸ Outcomes

| Management | Outcome |
|---|---|
| Surgical reconstruction | Faster recovery, lower morbidity, fewer complications; long-term curvature in fewer than 5%. |
| Conservative | Curvature in more than 10%; abscess or debilitating plaques in 25-30%; longer hospital stay and recovery. |

## Anterior urethral injury

- **Causes:** blunt trauma, sexual intercourse, penetrating trauma, constricting bands and iatrogenic injury.
- **Complete disruption with suprapubic catheter:** delayed reconstruction usually after 3 months, preceded by MCU/RGU.

## Posterior urethral injury

### ▸ Causes

- Usually part of multisystem trauma from road traffic accidents, falls or industrial accidents.
- In children the tear more often extends to the bladder neck, because the prostate is rudimentary.

### ▸ Immediate repair

- Some authors advocate immediate anastomotic reconstruction for a high-riding bladder, as well as for bladder neck injury.

### ▸ Female urethral injury

- Needs immediate primary repair, or at least realignment over a catheter, to prevent later urethrovaginal fistula or urethral obliteration.
- Suture vaginal lacerations to prevent vaginal stenosis.

- Delayed reconstruction is very difficult, because the small urethra is embedded in scar and hard to mobilise.

▸ **Primary realignment**

- **Technique:** pass a Coudé catheter antegrade through the bladder out to the meatus, tie it to a second catheter, and draw that back into the bladder.
- Avoid prolonged attempts, which risk infecting the pelvic haematoma.
- May not prevent symptomatic stenosis, but may allow endoscopic management of the stricture, and eases open urethroplasty by aligning prostate and urethra and shortening the stricture.

▸ **Delayed posterior urethroplasty**

- **End point (open perineal approach):** scar fully excised when a dilator of at least 28 Fr passes into the bladder without resistance. Some advise dilating up to 32 Fr.

▸ **Delayed endoscopic treatment**

- A stricture recurring after optical internal urethrotomy (OIU) or cut-to-the-light needs 3 months of urethral rest before definitive surgery.

▸ **Complications**

- **Anejaculation:** under 4%, the same rate as incontinence.
- Impotent men are more likely to restenose after urethroplasty, because the bulbar urethra is ischaemic.

## Ch 62 · Paediatric Genitourinary Trauma

### Ureteral injuries: causes

▸ **External trauma**

- **Incidence:** under 4% of penetrating trauma and under 1% of blunt trauma.
- **Gunshot wounds:** associated bladder injury in about 5%.

▸ **Iatrogenic (surgical)**

- **Hysterectomy:** commonest cause, about 54% of iatrogenic injuries.
- **Order after hysterectomy:** colorectal surgery, then ovarian surgery, then vascular surgery.

▸ **Endourological**

- About 0.2% of ureteroscopic injuries need open surgery.
- **Risk by intracorporeal lithotripsy energy:** electrohydraulic (EHL) greater than Nd:YAG laser, greater than Ho:YAG laser.

### Diagnosis

▸ **Helical CT**

- Needs delayed images at 5–15 minutes.
- **Sign:** circumferential contrast extravasation.
- **Pitfall:** can miss an upper ureteric injury, because Gerota's fascia can contain the extravasation.

▸ **Retrograde pyelography (RGP)**

- Some centres use it as the primary diagnostic test.

▸ **Intravenous pyelography (IVP)**

- Use intraoperatively when injury is suspected but preoperative imaging was negative.

## Intraoperative recognition

- The wound trajectory guides the search.
- **Methylene blue:** inject 1–2 mL and watch the pelvis for leakage.
- **Negative single-shot IVP but still suspicious:** watch for delayed presentation, with fever, leucocytosis and peritoneal irritation.
- **Recognition rate:** as low as 0% at laparoscopy, compared with about one-third at open surgery.

## Algorithm after external trauma

- **Unstable, blunt trauma:** a normal intraoperative one-shot IVP should prompt a search for other sources of haematuria (bladder, kidney, urethra).
- **Unstable, penetrating trauma with haematuria:** do an intraoperative one-shot IVP. If it is normal, or if there is no haematuria, explore the ureter when the bullet or knife track passes near it.

## Management

▸ **Damage control**

- One option: do nothing to the ureter at the first operation and plan re-operation at 24 hours.

▸ **Failed stented repair**

- **Endoscopic options:** Acucise endoureterotomy, balloon dilation, laser incision.

▸ **Major injury recognised in the postoperative period**

- If retrograde or antegrade stenting fails, place a percutaneous nephrostomy (PCN) and wait about 6 weeks before definitive repair.

▸ **Autotransplantation**

- Indicated when several attempts at ureteral repair have failed.

▸ **General principles of repair**

- **Debridement:** minimal in general, but wider in gunshot injuries.
- **Suture:** absorbable monofilament such as 5-0 PDS for the anastomosis.

## Ureteral strictures

▸ **Aetiology**

- **Malignant:** transitional cell carcinoma, cervical cancer.
- **Periureteral fibrosis:** may be secondary to aortic aneurysm or endometriosis.
- Some remain idiopathic.

▸ **Indications for intervention**

- Recurrent pyelonephritis, need to exclude malignancy, ongoing obstruction or pain.

▸ **Ureteral stent**

- Effective acutely for most strictures, especially intrinsic ones; about 88% success.

▸ **Balloon dilatation**

- **Approach:** most commonly retrograde; antegrade when retrograde access is not possible.
- **Balloon:** high-pressure, 4 cm long, 5-8 mm diameter ("Endoburst").
- **Technique:** under fluoroscopy; keep inflated as a tamponade for about 10 minutes, after which the waist usually disappears.
- **Alternative:** a low-profile balloon through the ureteroscope, dilating under direct vision.
- **Follow-up:** diuretic renography at 1, 6 and 12 months, continuing to 2 years.
- **Overall success:** 50-75%.

▸ **Endoureterotomy**

- **Tools:** cold knife, cutting electrode or holmium laser.
- **Holmium laser success:** 65-80%.
- **Afterwards:** larger double-J stent (8-12 Fr) for 6 weeks; diuretic renogram at 1, 6, 12 and 24 months.

▸ **Psoas hitch**

- May be fixed to the psoas major muscle as an alternative to the psoas minor tendon.

▸ **Boari flap**

- **Incisions:** Pfannenstiel, midline or Gibson.

▸ **Davis intubated ureterotomy**

- Used for very long strictures; a modern modification adds a buccal mucosa graft.

▸ **Ileal ureter**

- **Variation:** ileocalycostomy, used when the renal pelvis is scarred or intrarenal.

▸ **Ureteroenteric anastomotic stricture**

- **Assess length and renal function with:** CT urography, IVU, loopogram, antegrade nephrostogram, DTPA renogram.

**Other topics in this book**

## Stones in calyceal diverticula

▸ **Diagnostic and access points**

- A flattened renal papilla means an obstructed calyx rather than a diverticulum.
- Retrograde contrast may not fill the diverticulum because of a narrow infundibulum; target it with ultrasound-guided puncture or CT-guided direct contrast injection.
- Puncture is difficult in small and upper-pole diverticula.
- In an anterior diverticulum, dilating or incising the neck is difficult because of unfavourable angles.
- Passing a guidewire from the diverticulum into the pelvis is almost impossible.
- **If direct puncture fails:** puncture a neighbouring calyx and enter the diverticulum indirectly, by perforating its wall or retrogradely through its neck.

▸ **Percutaneous nephrolithotomy (PCNL)**

- First-line for most diverticular stones; stone-free rate 70–100%.
- **Diverticular ablation rate:** over 70%.

- **Ablation:** a resectoscope with a roller-ball electrode allows easy fulguration of the lining; dilatation and trauma can also cause ablation.

▸ **Ureteroscopy (URS)**

- **First-line for:** stones under 2 cm in the upper or middle calyx with a short, identifiable neck.
- **Stone-free rate:** 50–90%; diverticular obliteration only about 20%; a staged procedure may be needed.
- The ostium can be identified in only 75%.

▸ **Shock-wave lithotripsy (SWL)**

- **Stone-free rate:** 4–58%. It neither ablates the lining nor dilates the neck.

▸ **Laparoscopic or robotic**

- **Stone-free rate:** about 100%.

## Lasers

▸ **Laser physics**

- Emission begins when an external energy source excites an atom's electrons; they release the excess energy as photons, which form highly concentrated energy.

▸ **CO2 laser**

- Infrared gas laser emitting a continuous wave.
- **Delivery:** rigid articulated arm rather than a fibre.
- **Tissue effects:** vaporisation and coagulation.
- **Other indications:** skin resurfacing, vocal cord lesions.

▸ **Holmium versus erbium:YAG for lithotripsy**

- Er:YAG is the second laser (with holmium) used for lithotripsy.
- **Er:YAG wavelength:** 2940 nm.

| Feature | Holmium | Er:YAG |
|---|---|---|
| Mechanism | Photothermal vaporisation | Optical (photomechanical) energy |
| Shock wave | Weak | Strong |
| Vapour bubble | Pear-shaped, more heat loss | Torpedo-shaped |
| Soft-tissue penetration | Greater | Smaller, safer |
| Fragmentation | Slow, fine fragments | Fast, large fragments |
| Pulse duration | Longer | Shorter |
| Standard silica quartz fibres | Compatible | Cannot carry the energy |

▸ **Problems with Er:YAG fibres**

- **Main barrier:** no fibres with adequate mechanical, chemical and thermal reliability.
- **Sapphire:** too brittle and too thick.
- **ZBLAN fluoride glass:** low melting point, inadequate durability.
- **Hybrid** (fluoride core, sapphire tip, circumferential cladding): under evaluation.

## Nanobacteria and fetuin in stone formation

### ▸ Nanobacteria

- Proposed as cell-walled microorganisms smaller than the accepted lower size limit for life (about 200 nm, for bacteria such as mycoplasma).
- **Properties:** bacteria-like, pleomorphic, infectious; form a biofilm.
- **Proposed sequence:** Nanobacterium sanguineum is engulfed by phagocytosis, provokes only a weak inflammatory response (low virulence), and produces a biofilm that host defences cannot penetrate. Within it, calcium and phosphate fixation forms calcium carbonate and hydroxyapatite, giving calcification and stones.
- **Evidence:** form calcium stones in modified Loeffler's medium; in rats, stone formation after inoculation was dose dependent.

### ▸ Fetuin

- Plasma protein secreted by the liver; binds calcium and phosphorus and so inhibits biomineralisation.
- When calcium and phosphorus reach saturation, its inhibition is overcome and it precipitates as mineraloprotein complexes that grow into a nidus for stones.
- **Factors overcoming inhibition:** fetuin gene polymorphism, saturation levels of calcium and phosphorus, altered microenvironment such as pH.

## PCNL variants

### ▸ Supine PCNL

- **Advantages:** less risk of hypothermia and sepsis than prone; less surgeon fatigue, as the surgeon can sit.
- **Drawbacks:** posterior calyces not always accessible; less familiar to most urologists.
- **Positioning:** patient at the table edge, with a support such as a 3 L saline bag under the ipsilateral flank.

### ▸ Tubeless PCNL

- **Indication:** low-volume stone disease (less than 3 cm).
- Faster return to work.
- A ureteric stent may not drain the kidney as well as a nephrostomy tube.

## Renal tubular acidosis (RTA)

### ▸ Plasma bicarbonate

- **Type 1:** very low, below 12 mEq/L.
- **Type 4:** mildly low, above 17 mEq/L.

### ▸ Type 1 (distal)

- **Associated causes:** autoimmune thyroiditis, toxic nephropathy, lithium, renal transplantation, sickle cell anaemia.
- **Sodium bicarbonate:** alkalinises the urine and lowers aldosterone, helping correct sodium and potassium.

### ▸ Type 2 (proximal)

- **Primary:** idiopathic or familial.

- **Acquired:** amphotericin B, 6-mercaptopurine.

▸ **Type 4**

- Often associated azotaemia and hypertension.

## Stone analysis

▸ **Methods**

- MR spectrometry, chromatography, ultraviolet-visible spectroscopy, photomicroscopy, fluorescence.

▸ **Fourier transform infrared spectroscopy (FTIR)**

- Obtains an infrared spectrum of a solid, liquid or gas by measuring the coherence of a radiation source.
- The Fourier transform mathematically converts the raw data (interferogram) into the actual spectrum.
- **Pathway:** infrared source, interferometer, stone powder on the spectrophotometer surface, interferogram, Fourier-transform computer, spectrum, result.
- Fast; needs less than 2 mg of fragments.
- Gives core and surface information separately.
- The ATR (attenuated total reflectance) technique increases sensitivity.

▸ **FTIR procedure**

- Receive the stone in a sterile container; clean, dry and store in an air-conditioned environment.
- Fracture large stones so core and surface can be analysed separately.
- Pulverise with pestle and mortar to a homogeneous powder.
- Place about 2 mg on the flat surface of the spectrophotometer.
- Collect spectra with software and compare with synthetic kidney-stone spectra in a reference library; then issue the report.

## Stones in pregnancy

▸ **Epidemiology**

- More common in multiparous women, but after adjusting for age the incidence is the same as in primiparous women.

▸ **Aetiology and presentation**

- Increased renal blood flow and GFR cause hyperuricosuria as well as hypercalciuria.
- The rise in urine pH is most marked in the second trimester.
- The gravid uterus may mask signs and symptoms, making diagnosis difficult.
- **Presentation:** commonly flank pain; microscopic or macroscopic haematuria in most; UTI in some; nausea and vomiting.

▸ **Ultrasound**

- Colour Doppler tells a dilated ureter from the iliac artery.
- A dilated ureter below the iliac vessels (infra-iliac) suggests a distal stone.
- **Resistive index (RI):** unchanged in normal pregnancy, so a raised RI points to obstruction.

▸ **Other imaging**

- **Limited IVP:** scout film, 30-second film, 20-minute film.
- **Nuclear renography:** limited radiation and can diagnose obstruction, but the bladder acts as a reservoir exposing the fetus; reduce this with extra water intake, frequent voiding and possibly a catheter.
- **HASTE MRI** (half-Fourier single-shot turbo spin echo): may improve diagnosis.
- **Low-dose CT:** reliability for calculi in pregnancy is controversial.

▸ **Radiation**

- The fetus is most sensitive in the first trimester, during organogenesis.
- **Effects:** teratogenesis, carcinogenesis, mutagenesis; depend on dose and stage of pregnancy.
- In the second and third trimesters teratogenic risk falls, but a risk of childhood malignancy remains.
- No threshold dose has been defined below which harm cannot occur.

▸ **Treatment**

- **DJ stent:** can be placed under local anaesthesia.
- **Ureteroscopy:** reduce fetal X-ray exposure with a below-table X-ray source as well as a lead apron under the patient.
- **Nephrostomy drawbacks:** morbidity of carrying a tube, blockage, need for flushing and replacement, source of infection.
- **First-trimester upper ureteric stone:** ureteroscopy with laser is best. Nephrostomy or DJ stent with surgery after delivery is not a good option, because the drain stays long and needs many changes.

## Complications of ureteroscopy (URS)

▸ **Overall**

- **Overall rate:** 3.5%.
- **Fever:** the most frequent, at 1.8%.

▸ **Perforation**

- Rates have decreased with improved technique and instruments.
- **Causes:** splitting of the ureter after balloon dilatation, basket and grasper injury, pressurised irrigation.
- **Management:** stop, stent and give antibiotics; image after stent removal at 4 weeks.

▸ **Stricture**

- **Risk factor:** previous ureteral surgery.
- **Impacted stone:** one past which a guidewire cannot be passed.

▸ **Avulsion**

- The most catastrophic complication.
- **Mechanism:** withdrawing a large instrument that was forcefully advanced.
- **Commonest presentation:** a portion of ureter withdrawn with the instrument.
- **Repair:** delayed preferred, but immediate is also an option.
- Drain any urinoma percutaneously.

▸ **Submucosal and lost stone**

- **Stone extrusion:** occurs in 2% of procedures.
- **Submucosal migration:** difficult to remove; treat with laser excision and double-J stenting; late stricture is a complication.
- **Lost stone** (complete extrusion outside the ureter): can be left, as retrieval increases injury and extravasation. Stop, stent and give antibiotics to prevent urinoma or abscess; late stricture can follow.

▸ **Other complications**

- Fluid extravasation causing congestive cardiac failure and pulmonary oedema.
- Deep vein thrombosis.
- Musculoskeletal injuries.

## Ureteral access sheath

▸ **History and design**

- First reported by Takayasu and Aso in 1974; popularised two decades later.
- Hydrophilic coating gives a low-friction surface; strengthened to resist kinking.
- **Manufacturers:** Cook, Boston Scientific, Rocamed, Coloplast, Olympus; various lengths and diameters.

▸ **Advantages**

- Decreases operative time.
- No need to drain the bladder separately.
- Higher stone-free rates reported.
- Facilitates active extraction of fragments with minimal ureteric trauma.

# 12 Bladder Cancer & Diversion

## Ch 129 · Tumours of the Urinary Bladder

### Benign bladder tumours

▸ **Inverted papilloma**

- **Associations:** chronic inflammation and bladder outlet obstruction.
- **Histology:** anastomosing islands of normal urothelial cells that invaginate from the surface into the lamina propria, but not into the muscularis propria.
- **Site:** commonest at the trigone, but can occur anywhere in the bladder.
- **Types:** Type I (benign) and Type II (malignant).

▸ **Urothelial papilloma**

- About 75% carry FGFR3 mutations.

▸ **Nephrogenic adenoma**

- **Origin (two theories):** metaplasia from chronic irritation, or nests of displaced mesonephric tissue in the urothelium activated by mucosal injury.
- **Causes of chronic irritation:** trauma, previous surgery, renal transplantation, intravesical chemotherapy, stones, catheters and infection.
- **Histology:** glandular tubules resembling renal tubules, involving the mucosa and submucosa.
- **Presentation:** a vascular lesion; gross haematuria in most cases.
- **Treatment:** transurethral resection plus elimination of the source of chronic irritation.

▸ **Leiomyoma**

- Commonest non-epithelial benign tumour of the bladder.
- **Cystoscopy:** smooth indentation of the bladder covered by normal mucosa.

### Bladder cancer: epidemiology

- **Race:** commoner in whites than in African Americans, the opposite of most other tumours.
- **Adolescents and young adults:** tumours are usually well differentiated and non-invasive, the opposite of most other tumours.
- **Mortality:** urothelial carcinoma is the second commonest cause of death among genitourinary tumours.

### Etiology: genetic factors

▸ **NAT-2**

- Slow polymorphism: odds ratio 1.4 compared with the fast polymorphism.

▸ **GSTM1 (glutathione-S-transferase M1)**

- **Role:** conjugates reactive chemicals such as arylamines and nitrosamines.
- Null polymorphism increases risk 1.5 times.

▸ **Slow NAT-2 with null GSTM1**

- High levels of 3-aminobiphenyl and a higher risk of bladder cancer.

## Etiology: external risk factors

▸ **Aromatic amines**

- **Sources:** benzidine, beta-naphthylamine, polycyclic aromatic hydrocarbons, diesel exhaust and paint substances.
- **Route:** inhalation or through the skin.
- **Latent period:** 10-20 years.

▸ **Smoking**

- Risk is linearly related to duration and intensity.
- **After stopping:** relative risk is still 2.6 at 1-3 years, falling to 1.1 after 15 years.

▸ **HPV**

- Codes two oncoproteins, E6 and E7. E6 interacts with TP53, which has a role in bladder cancer formation and progression.

▸ **BK virus**

- Causes severe haemorrhagic cystitis in bone marrow transplant patients, but no consistent link with bladder carcinoma has been shown.

▸ **Chronic urinary tract infection**

- Relative risk 1.4-1.6, possibly because bacteria produce carcinogens such as nitrosamines.

▸ **Radiation**

- Evidence includes atomic bomb survivors.

▸ **Green tea (protective)**

- Antioxidant polyphenolic compounds, especially epigallocatechin-3-gallate.

## Urothelium, precursor lesions and molecular pathways

▸ **Umbrella cells**

- Membrane is rigid and made of uroplakin proteins.
- Part of the urine-bladder barrier that stops urinary toxins from transforming urothelial cells.

▸ **Chromosome 9**

- Deletion of part or all of chromosome 9 is the earliest mutation in low malignant potential NMIBC.
- Its tumour suppressor gene (9q21 site) is a negative regulator of RB.
- **Hyperplasia:** may show loss of parts of chromosome 9, the most frequent deletion in hyperplasia.
- **Dysplasia:** allelic loss of chromosome 9 and occasional TP53 abnormalities.

▸ **Pathways of development**

- Three pathways are proposed. The third runs from normal urothelium through hyperplasia or dysplasia to high-grade papillary carcinoma, then muscle-invasive disease.

▸ **Proliferation markers**

- PCNA and MIB-1 are found in high-grade tumours and indicate poor prognosis.
- PUNLMP has the same proliferation rate as low-grade urothelial carcinoma.

▸ **Primary CIS**

- **Definition:** CIS not associated with a previous or concurrent papillary or invasive lesion, and with no previous CIS.

## NMIBC: grade and stage distribution

- **Grade:** about 25% PUNLMP, 50% low grade, 25% high grade.
- **Stage at presentation:** about 70% Ta, 20% T1, 10% CIS.

## Recurrence, dissemination and prognosis

▸ **Field change**

- Multifocality and rapidly recurring tumours strongly predict histologically abnormal urothelium in a normal-appearing bladder.

▸ **Angiolymphatic invasion**

- Poor prognostic sign with a 40% risk of nodal disease.
- Independent predictor of overall and cancer-specific survival.
- Use immunohistochemistry with CD31 and CD34 monoclonal antibodies to identify blood vessels accurately.

▸ **Pagetoid spread**

- **Definition:** spread of cancer cells underneath a layer of normal-appearing surface urothelium, seen mainly in urothelial CIS.
- **Frequency:** 15% of bladders with CIS and 11% of high-grade papillary NMIBC.
- Can extend into the prostatic urethra and distal ureters.

## Presentation: haematuria

- Microscopic haematuria is present in all patients with bladder cancer.
- Haematuria can be intermittent, so evaluate any episode even if a later urinalysis is negative.
- **Gross haematuria:** 50% have a demonstrable cause and 12% have a bladder tumour.
- **Microscopic haematuria:** 4.1% risk of bladder malignancy.
- Irritative voiding symptoms with microscopic haematuria double the risk, especially of CIS.

## Cystoscopy and optical imaging

▸ **White-light cystoscopy**

- **Accuracy:** sensitivity 87%, specificity 85%.
- Cannot distinguish low-grade from high-grade tumour, or CIS from inflammation.

▸ **Blue-light cystoscopy (photodynamic diagnosis)**

- **Method:** 5-ALA or HAL instilled into the bladder 1 hour beforehand; blue light of 375–440 nm.
- **Cysview:** under 2 ounces delivered 1 hour before cystoscopy; normal white-light cystoscopy is followed by blue-light cystoscopy.
- **Cysview side effects:** allergic reaction, bladder pain, cystitis, dysuria and haematuria.
- Does not distinguish high-grade from low-grade tumour.
- **False positives:** after BCG, after a previous biopsy or TURBT, and with inflammation.

▸ **Narrow-band imaging (NBI)**

- Depth of light penetration into the bladder wall increases with increasing wavelength.
- Vessels appear green or dark brown against a pink or white mucosal background.
- **Accuracy:** sensitivity 100%, specificity 82%.
- Does not distinguish high-grade from low-grade tumour.

▸ **Confocal laser endomicroscopy**

- **Principle:** a low-power laser illuminates tissue and the emitted fluorescent light is detected, using topical or intravenous fluorescein, to a depth of about 120 µm.
- **Strength:** distinguishes high-grade from low-grade tumours.
- **Limits:** restricted view, needs contrast, cannot detect muscle invasion.

▸ **Raman shift spectroscopy**

- Named after the Indian physicist C.V. Raman.
- **Principle:** measures the energy photons gain or lose from intramolecular bonds in tissue. Emitted wavelengths are studied as wave numbers; the Raman shift is the change in photon energy.
- Can assess invasion and distinguish low-grade from high-grade tumour, but is still in trials.

▸ **UV autofluorescence**

- Ultraviolet excitation of tissue; detects invasion and distinguishes low-grade from high-grade tumours.
- **Drawback:** toxicity.

▸ **Multiphoton microscopy**

- **Principle:** absorption of two infrared photons (fluorescence imaging) to a depth of about 0.5 mm.
- Distinguishes benign from malignant tissue and allows cytological grading, but does not detect muscle invasion.

▸ **Scanning fibre endoscopy**

- Small flexible endoscope with laser-based imaging giving a panoramic view of the urothelium; still in trials.

▸ **Molecular imaging**

- **Definition:** visualises and characterises biological processes at the molecular and cellular level.
- **In the bladder:** fluorescently labelled anti-CD47 antibody is instilled, the bladder is emptied, then optical imaging with confocal laser endomicroscopy is done. Anti-CD47 binds only tumour cells, so fluorescent tissue is biopsied.

## Random bladder biopsies and urine cytology

▸ **Random biopsies**

- **Detection rate:** 2.5% overall in specimens with known or suspected tumour; 23% with a concurrent bladder tumour.
- Reasonable in high-risk patients who have had intravesical therapy.

▸ **Urine cytology**

- Performance depends on the cytopathologist, number of samples, and tumour stage and grade.
- Instrumented (bladder wash) cytology improves sensitivity and specificity but needs an invasive procedure.

## Urinary molecular markers

- **FDA-approved:** NMP-22, FISH (UroVysion), BTA stat, BTA TRAK and ImmunoCyt/uCyt+.

▸ **NMP-22 (BladderChek)**

- **Cut-off:** 10 units/mL. Lowering it to 5 improves sensitivity but significantly worsens specificity.
- Urine concentration is 20 times higher in bladder cancer than in non-cancer patients.
- **Sensitivity by stage:** Ta 35%, T1 65%, T2 88%.
- Results in 30 minutes; costs less than cytology.
- **False positives:** active urinary tract infection or significant haematuria.
- Combined with cystoscopy it gives a higher detection rate.

▸ **FISH (UroVysion)**

- Useful for high-grade disease.
- May be anticipatory, turning positive before a new tumour forms.

▸ **BTA**

- **Basis:** complement factor H and factor H-related proteins in bladder cancer cells inhibit the complement cascade to prevent cell lysis.
- **False positives:** urolithiasis, inflammation, recent instrumentation, other genitourinary malignancies and intravesical BCG therapy.

▸ **ImmunoCyt**

- **Antibodies (three fluorescent monoclonals):**
  - LDQ10 and M344: fluorescein-labelled, against mucin.
  - 19A211: Texas Red, against a CEA antigen.
- Needs 500 negative cells on the slide for a negative report.
- **False positives:** BPH, infection and urolithiasis.

▸ **FGFR-3 (CertNDx)**

- High in low-grade pTa tumours; marker of favourable disease with low recurrence and improved survival.
- Not affected by the degree of haematuria or other urinary tract diseases.

▸ **Microsatellite analysis**

- PCR amplification detects tumour-associated loss of heterozygosity.
- **Accuracy:** sensitivity 72-97%, specificity 80-100%; described as the most promising marker.

▸ **DNA methylation (CpG islands)**

- Sensitivity 75%, but methylated CpG islands can occur in normal urothelial cells of older patients.

▸ **Lewis blood group antigen X**

- No commercially available test.
- Usually absent from adult urothelial cells except occasional umbrella cells.

▸ **CK20 and CYFRA 21-1**

- Cytoskeletal proteins detectable in urine by protein or mRNA assays.
- CYFRA 21-1 is not a useful marker in its current form.

▸ **Survivin**

- Anti-apoptotic protein highly expressed in urothelial cancer cells.
- May predict response to intravesical therapy, but is relatively poor at detecting advanced-stage or high-grade tumours.

▸ **Other markers**

- **Hyaluronic acid:** role in cell proliferation; urinary amount correlates with stage.
- **Telomerase:** activity detected in 80% of urine samples from bladder cancer patients.
- **BLCA-4:** nuclear transcription factor marker with high specificity and sensitivity.

## TNM staging and survival

▸ **Lymph nodes**

- **Regional (N1/N2), true pelvis:** hypogastric, obturator, external iliac and presacral nodes.
- **Distant:** all nodes above the aortic bifurcation.

▸ **Five-year survival by stage**

| Stage | Five-year survival |
|---|---|
| 0a / 0is | Over 85% |
| I | 65-75% |
| II | 55% |
| III | About 30% |
| IV | 15-25% |
| M1 | 5-20% |

# Ch 130 · Management Strategies for Non–Muscle-Invasive Bladder Cancer

## WHO grading and outcome by grade

▸ **WHO 1973 to WHO 2004**

| WHO 1973 | WHO 2004 |
|---|---|
| Grade 1 | PUNLMP or low-grade carcinoma |
| Grade 2 | Low-grade or high-grade carcinoma |
| Grade 3 | All high-grade carcinoma |

▸ **Outcome by lesion**

| Lesion | Recurrence | Grade progression | Stage progression | Survival |
|---|---|---|---|---|
| Urothelial papilloma | 0-8% | 2% | 0% | 100% |

| | | | | |
|---|---|---|---|---|
| PUNLMP | | 11% | | 93-100% |
| Low-grade papillary carcinoma | | 7% | | 82-96% |
| High-grade carcinoma (papillary and CIS) | 55-58% | | | 74-90% |

## Risk stratification (EAU 2018)

▸ **Intermediate risk**

- **Definition:** a residual category, all tumours not defined as low or high risk.
- **Intravesical chemotherapy:** maximum 1 year; optimal schedule not known.
- **One-year full-dose BCG:** induction plus three weekly instillations at 3, 6 and 12 months.
- **Follow-up:** consider upper tract imaging, especially at recurrence; image the upper tracts if haematuria occurs.

▸ **High and highest risk**

- TaG1G2/low-grade tumours are high risk only when multiple, recurrent and large (more than 3 cm), with all three present.
- **Highest-risk subgroup:** multiple large T1 high-grade tumours, and T1 high-grade tumour with CIS in the prostatic urethra.
- **Management:** radical cystectomy, with full-dose BCG for 1–3 years in those who refuse.

## Transurethral resection of bladder tumour (TURBT)

▸ **Technique**

- A systematic stepwise procedure: insert the resectoscope under vision and inspect the whole urethra.
- **Final steps:** record findings in the operation record and describe the specimen precisely for pathology.
- **Quality marker:** detrusor muscle in the specimen, required in every TURBT except TaG1/low-grade tumours.

▸ **Random biopsies of normal-looking mucosa**

- **Sites:** trigone, dome, and right, left, anterior and posterior walls.
- Also recommended when a high-risk exophytic tumour of non-papillary appearance is expected.

▸ **Prostatic urethral biopsy**

- **Indication:** visible abnormalities of the prostatic urethra.
- If not done at the initial TURBT, complete it at the second resection.

▸ **Second (repeat) TURBT**

- **Causes of incomplete first resection:** extensive tumour volume, anatomical inaccessibility, medical instability requiring premature termination, risk of perforation.
- **Indication:** no muscle in the initial specimen, except Ta low-grade tumours and primary CIS.
- BCG appears more effective at preventing progression in high-grade papillary tumours and CIS when restaging TURBT precedes it.

## Laser and office-based management

### ▸ Laser

- **Best probe:** end-fire non-contact fibre with 5–15 degree divergence, giving penetration up to 5 mm.
- White tissue means protein denaturation; stop at that point.
- **Drawbacks:** higher cost and no tissue for biopsy.

### ▸ Office-based fulguration

- Xylocaine (lidocaine) jelly with a 15–30 minute dwell time gives mucosal analgesia.

## Single immediate (perioperative) instillation

- **Rationale:** primary tumours most often arise on the floor and side walls, but early recurrences appear at the dome where floating tumour cells implant.
- **Agent in Europe:** epirubicin.
- **Benefit:** reduces recurrence by up to 12%, or by 50% if therapy is optimised.
- No effect on high-grade tumours.
- Maintenance after it gives no further reduction in recurrence.
- **Problems:** cost, complexity, and it can make later cystectomy more difficult.

## Intravesical chemotherapy

- No risk of BCG infectious complications, so it is the favoured adjuvant approach in many European countries.

### ▸ Mitomycin C

- **Action:** inhibits DNA synthesis.
- **Toxicity:** rash in 8-19%.

### ▸ Doxorubicin (Adriamycin)

- **Action:** anthracycline antibiotic; inhibits topoisomerase II and protein synthesis.
- **Use:** can be used perioperatively; low- to intermediate-risk disease.
- **Toxicity:** cystitis in 20-40%; also fever, allergy, and contracted bladder in 5%.
- **SWOG trial versus BCG (Lamm et al, 1991):** progression 15% with BCG versus 37% with chemotherapy.

### ▸ Valrubicin

- Semisynthetic analogue of doxorubicin.

### ▸ Epirubicin

- **Use:** low- to intermediate-risk disease.
- **Toxicity:** cystitis in 10-30%; contracted bladder is rare.

### ▸ Thiotepa

- **Use:** low- to intermediate-risk disease.
- **Toxicity:** cystitis in 10-30%; myelosuppression in 8-19%.
- Low molecular weight causes serious systemic side effects, especially myelosuppression.

### ▸ Gemcitabine

- Salvage agent; other toxicity is occasional nausea.

## Intravesical BCG

### ▸ History

- Morales's original regimen also used the percutaneous route; Brosman showed similar success with intravesical BCG alone.

### ▸ Mechanism

- **Cytokines:** GM-CSF, IL-1, IL-6, TNF-α, IFN-γ, IL-2, IL-8 and IL-12.
- Stimulates an anti-angiogenic response as well as a cell-based immune response.

### ▸ Schedule and dose

- **Dose:** 120 mg in 50 mL, reduced if side effects develop.
- **Induction:** 120 mg weekly for 6 weeks.
- **Alternative maintenance:** 1 dose per month for 10 months.

### ▸ Practical points

- **Before instillation:** restrict fluids, diuretics and caffeine to limit dilution.
- **Afterwards:** the patient should clean the toilet with bleach.
- **No data:** prosthetic materials (theoretical risk of more infections) and vesicoureteric reflux.
- Can effectively treat residual papillary tumour, but is not a substitute for surgical resection.

## BCG toxicity

- Cystitis occurs in 60-80%.
- Do not use antituberculous monotherapy.
- Do not use cycloserine: it causes severe psychiatric symptoms.

### ▸ Grade 1

- Topical antispasmodics such as phenazopyridine, plus anticholinergics, analgesics and anti-inflammatory drugs.

### ▸ Grade 2

- Consult immediately with a physician experienced in mycobacterial infections and their complications.
- Oral isoniazid 300 mg/day and rifampin 600 mg/day until symptoms resolve.

### ▸ Grade 3

- **Allergic reaction (joint pain, rash):** isoniazid 300 mg/day and rifampin 600 mg/day for 3–6 months depending on response.
- **Solid-organ involvement:** isoniazid 300 mg/day, rifampin 600 mg/day and ethambutol 15 mg/kg/day as a single daily dose for 3–6 months.
- **Organs involved:** bone and joints, epididymis, liver, lung, kidney and prostate.
- **Prednisone 40 mg/day:** consider for inadequate response or septic shock, never without effective antimycobacterial therapy.

## Interferon and other immunotherapy

### ▸ IFN-α

- The most extensively studied intravesical interferon.
- As a single agent, more expensive and less effective than BCG or intravesical chemotherapy.

- **Use:** salvage agent; not used perioperatively.
- **Toxicity:** cystitis in under 5%; flu-like symptoms in 20%.

▸ **Other investigational agents**

- IL-2 and a (mycobacterial) cell wall DNA extract (check).

## Carcinoma in situ

▸ **Classification**

- **Secondary CIS:** found during follow-up of a patient whose previous tumour was not CIS.
- **Concurrent CIS:** found in the presence of another urothelial tumour in the bladder.

▸ **Natural history and prognosis**

- At cystectomy, CIS is associated with upstaging in 55%, versus 5% without CIS.
- **Response to BCG** is an important prognostic factor:
  - Progression to muscle invasion: 10–20% of complete responders versus 66% of non-responders.
  - Progression at 5 years: 19% of initial responders versus 95% of non-responders.
  - Over 10 years, about 30% remain free of recurrence or progression.

▸ **Treatment**

- Cannot be treated by endoscopic procedures alone.
- **Isolated CIS:** intravesical chemotherapy gives complete response in about 50% and reduces recurrence only; BCG reduces both recurrence and progression.
- **With low-grade Ta/T1 tumour:** BCG or intravesical chemotherapy.
- **With high-grade Ta/T1 tumour:** re-TURBT first, then BCG or chemotherapy.
- **Assessment at 3 months:**
  - Responders: continue maintenance.
  - Non-responders: TURBT plus a second course (BCG preferred).
  - Still not responding: partial or radical cystectomy.
- **Upper tract CIS:** treat according to UTUC guidelines.

## BCG failure and recurrence after BCG

▸ **Definitions**

- **BCG-refractory:** disease that does not improve or worsens during BCG. Includes:
  - CIS without concomitant papillary tumour present at both 3 and 6 months.
  - A high-grade tumour that appears during BCG therapy.
- **BCG failure:** muscle-invasive disease detected at any time during follow-up.
- **Not failure:** low-grade recurrence during or after BCG.
- **Timing:** declaring failure may take up to 6 months, because the response of high-grade disease rose from 57% to 80% between 3 and 6 months.

▸ **Management**

- More than 2 BCG courses are not recommended: they fail 80% of the time.
- **High-grade recurrence after previous intravesical chemotherapy:** BCG gives better disease-free survival.

- **BCG-intolerant high-grade disease:** salvage chemotherapy carries a high rate of progression.
- **Non-high-grade recurrence after BCG for a primary intermediate-risk tumour:** repeat BCG or intravesical chemotherapy, then radical cystectomy.

## Photodynamic therapy

- **Role:** an alternative for refractory NMIBC.
- **Mechanism:** a photosensitiser such as hexaminolevulinate (HAL) is instilled; red laser light with molecular oxygen causes free-radical cytotoxic injury.
- **Agents:** porfimer sodium can be given systemically instead of intravesical HAL. Emerging agents are hypericin and radachlorin.
- **Light:** red laser at 630 nm for 12–20 minutes, 2–3 days after the photosensitiser, once it has cleared from normal tissue.
- **Side effects (limit use):** bladder contracture or irritability in 50%; dermal photosensitivity in 19%.

## Radiation therapy

- **Indication:** patients who refuse radical cystectomy after failed intravesical therapy, or who are unfit for major surgery.
- TUR plus radiation gives complete response in 50–75% of NMIBC patients.
- No significant effect on CIS.

# Ch 131 · Management of Muscle-Invasive and Metastatic Bladder Cancer

## Natural history

- Untreated, 85% of patients with MIBC die within 2 years.

## Clinical staging

▸ **Prostatic urethral loop biopsy**

- May start at the bladder neck in shorter prostates and runs to the mid-to-distal verumontanum.
- The 5 and 7 o'clock area beside the verumontanum has the most prostatic ducts, so CIS is most likely there.

▸ **Bimanual examination**

- **Technique:** dominant hand on the mass suprapubically; one or two fingers of the other hand in the rectum or vagina. Bladder empty, but no Foley catheter left in.
- **Staging scheme** (T2a non-palpable to T4b fixed to sidewall): attributed to Marshall.

▸ **CT**

- If done after TUR, delay about 7 days, because inflammatory artefact may be mistaken for T3 disease.
- **Local staging:** sensitivity about 90%, specificity about 95%.
- **Nodal metastasis:** specificity 70–100%.

▸ **MRI**

- May be better than CT at predicting grade and chemosensitivity, but poorer at detecting lymph node metastasis.

## Radical cystectomy and organ preservation

- With bilateral PLND, local control is excellent: about 4% local recurrence in node-negative disease.

▸ **Prostate sparing**

- 15–75% have prostate involvement on the final specimen; if sparing is considered, take TUR samples of the bladder neck and prostatic urethra.
- **Techniques:** apical sparing, posterior sparing (posterior prostate and seminal vesicles) and total prostate sparing.
- All give better preservation of continence and erection.

## Pelvic lymph node dissection (PLND)

▸ **Lymphatic zones**

- **Primary ("HOPE"):** hypogastric, obturator, external iliac and presacral nodes.
- **Secondary:** common iliac, paracaval, para-aortic and interaortocaval nodes.

▸ **Templates**

- **Standard (level I):** up to the common iliac bifurcation. Inferior (distal) boundary is Cooper's ligament or the circumflex iliac vein.
- **Extended (level II):** up to the aortic bifurcation.

▸ **Sentinel node biopsy**

- No role, because bladder cancer has no skip metastases.

▸ **Lymph node density**

- Positive nodes as a percentage of nodes removed. Below 20% is associated with better survival, though this is controversial.

## Intraoperative decision making

▸ **Extrapelvic nodes**

- Grossly positive nodes outside the pelvis on imaging: confirm by CT-guided biopsy. If positive, give neoadjuvant chemotherapy and then radical cystectomy.

▸ **Reasons not to proceed**

- Extensive periureteral disease is a further reason not to proceed with cystectomy.

▸ **Upper tract**

- **Risk factors for recurrence:** bladder CIS, tumour in the distal ureter, and high-grade pTa–T1 disease.
- **Ureteric frozen section:** controversial, as it is debatable whether a negative CIS margin reduces recurrence.

▸ **Male urethra**

- **Risk factor for recurrence:** non-muscle-invasive disease on final pathology.
- Diffuse CIS of the prostatic urethra or ducts: consider urethrectomy.

- Pre-operative prostatic urethral biopsy has only moderate sensitivity and specificity, so some advise urethrectomy only if the apical urethral margin of the specimen is positive.

▸ **Female urethra**

- Inguinal lymphadenopathy increases the risk of urethral tumour.
- Complete urethrectomy for bladder-neck tumours is controversial: 60% of these women have no urethral involvement on final pathology.

## Neoadjuvant and adjuvant chemotherapy

▸ **Neoadjuvant**

- Lets the clinician assess each patient's response to therapy.
- **NCCN:** strongly consider cisplatin-based neoadjuvant chemotherapy for cT2 N0 M0; recommended for cT3–T4a N0 M0.

▸ **Adjuvant**

- Allows immediate cystectomy without delay.
- Often difficult after cystectomy because of surgical deconditioning, deteriorating renal function and peri-operative complications.

## Bladder preservation

- **Suboptimal candidates:** obvious T3 disease on imaging, multifocal tumours.

▸ **TUR monotherapy**

- **Only if:** negative restaging TUR, no hydronephrosis, no adenopathy, tumour under 3 cm and not multifocal.
- Counsel the patient about recurrence.

▸ **Partial cystectomy**

- **Recurrence:** NMIBC in 12–50%, MIBC in 20–60%, metastatic in 15–50%.
- May be combined with neoadjuvant chemoradiation or adjuvant chemotherapy (small studies).

▸ **Radiation monotherapy**

- 1.5–2 Gy per fraction after debulking TUR; local failure in more than 30%.

## Metastatic bladder cancer

- Ultimate 5-year overall survival is 5–20%.

▸ **First line**

- **Carboplatin:** better tolerated but less effective than cisplatin.
- **PGC (paclitaxel, gemcitabine, cisplatin):** another option for cisplatin-fit patients.

▸ **Second line**

- **Nivolumab:** antibody blocking the PD-1 receptor.
- Chosen regardless of the time to progression after first line.
- **Performance status 2 or more:** immunotherapy, clinical trials or best supportive care.

▸ **Performance status 2 or more with GFR under 60 mL/min**

- No combination chemotherapy data; give pembrolizumab or atezolizumab monotherapy, or best supportive care.

▸ **Other drugs tried**

- **Vinflunine:** antitubulin vinca alkaloid, approved in Europe as a second-line agent.
- **Taxanes:** paclitaxel, docetaxel, larotaxel.
- **Antifolates:** pemetrexed and piritrexim (synthetic).
- **VEGF inhibitors (alone or in combination):** bevacizumab, sunitinib, sorafenib.
- **EGFR-directed:** cetuximab with or without paclitaxel or GC, gefitinib, lapatinib, erlotinib.
- **Others:** trastuzumab (anti-HER2), ipilimumab (CTLA-4 inhibitor), cabozantinib (anti-angiogenic).

## Radical cystectomy: preparation and positioning

- **Preparation:** counselling by an enterostomal therapist and stoma site marking.
- **Anaesthesia:** general or spinal.
- **Men:** supine, table flexed 15 degrees at the level of the anterior superior iliac spine (ASIS).
- **Women:** supine with spreader bars for vaginal access, or low lithotomy.
- **Prep and drape:** xiphoid to upper thighs; povidone-iodine, not chlorhexidine, on the genitals.
- **Open incision:** from 2 cm above the umbilicus to just above the pubis.

## Open radical cystectomy: technique

▸ **Steps**

- Open the peritoneum lateral to each medial umbilical ligament down to the internal inguinal rings; divide the vas deferens or round ligament there.
- On the right, incise the white line around the caecum to mobilise the root of the small bowel mesentery.
- Divide the superior vesical artery on each side for maximal ureteric length, then divide the ureter between ligatures.
- Pelvic lymphadenectomy can be done at this stage or after the bladder is removed.

▸ **In women**

- Historically the bladder, urethra, anterior vagina, uterus and cervix were removed, because women present with more advanced disease than men.
- The urethral meatus is incised from the pelvis or from the vaginal introitus.

## Robotic radical cystectomy: set-up and ports

- **Theatre layout:**
  - Anaesthetist at the head, with a monitor on each side of the head end.
  - A right and a left assistant, one at each side.
  - Scrub nurse with a third monitor at the lower left.
  - Da Vinci system docked between the legs; surgeon at the console away from the table.
- **Position:** fixed to the table in dorsal lithotomy with 30 degree head-down tilt.
- **Ports:**
  - 12 mm camera port 5 cm above the umbilicus.

- 8 mm robotic arm ports at about umbilical level, including the fourth arm laterally on one side.
- 12 mm assistant port laterally on the other side, plus an additional 12 mm assistant port.
- 15 mm lower midline port for the Endo GIA stapler if intracorporeal diversion is planned.

## Robotic radical cystectomy: procedure

▸ **Pedicles**

- **Superior vesical artery (anterior pedicle):** at the most posterior part of the plane between the external iliac vessels and the lateral bladder wall.
- The bedside assistant divides the posterior pedicles with LigaSure or staplers; then the endopelvic fascia on each side and the prostatic pedicles are divided.

▸ **Anterior dissection**

- Drop the bladder anteriorly: divide the urachus near the umbilicus and take down wide peritoneal wings before fully dissecting the Retzius space.

▸ **Extended lymph node dissection**

- Takes around 45 to 60 minutes; split-and-roll technique.

▸ **Specimen**

- Placed in a 15 cm Endo Catch bag.
- **Extraction:** Pfannenstiel incision with intracorporeal diversion; vertical midline infraumbilical incision with extracorporeal diversion.

▸ **In women**

- Ligate the uterine as well as the ovarian vessels.
- Transect the round ligaments and suture the uterus, tubes and ovaries to the obliterated umbilical ligaments to help visualise the posterior dissection.

# Ch 132 · Surgical Management of Bladder Cancer: Transurethral, Open and Robotic

## Endoscopic management of NMIBC (TURBT)

▸ **Equipment**

- **Lens:** 12–30 degrees, because this deflection lets the surgeon see the loop where it is placed.
- **Video TUR advantages:** magnification, resident teaching, documentation and recording, and less exposure of the surgeon to body fluids.

▸ **Obturator reflex**

- Obturator nerve block is one way to avoid it.

▸ **Complications**

- Uncontrolled haematuria or bladder perforation in fewer than 5%; minor perforation is common.
- Intraperitoneal perforation carries a low risk of tumour seeding.

▸ **Ureteric orifice obstruction**

- **Detection after resection near the orifice:** intravenous indigo carmine or methylene blue, or a retrograde pyelogram.
- **Treatment:** balloon dilatation, endoscopic incision or ureteric reimplantation.

## Urinary diversion: indications and types

- **Before renal transplantation:** indicated when the bladder cannot receive the transplant ureter.
- **Incontinent forms:** cutaneous ureterostomy, and the ileal and other conduits.

## Principles of bowel segment use

### ▸ Stomach

- The right gastroepiploic artery is much larger than the left.
- A gastric antral segment is mobilised on the right gastroepiploic artery.

### ▸ Jejunum and ileum

- **Mesentery:** jejunum has a thin mesentery with large vessels in its arcade; ileum has a thick mesentery with multiple arcades of small vessels.
- **Division:** the isolated segment's arcade should be supplied by a palpable artery arising from the base of the mesentery.
- **After pelvic radiation:** do not use the last 5 cm of terminal ileum or the bowel lying in the pelvis ("5 feet bowel 6 feet distal to ligament of Treitz"), as they may show radiation effect.

### ▸ Colon

- **Ileocolic artery:** the direct continuation of the superior mesenteric artery (SMA). It also supplies the terminal 15 cm of ileum, which is usually left behind for B12 absorption, so dissect the mesentery accordingly.
- **Weak point:** the midpoint between the right colic and middle colic arteries is a further weak point of the colonic blood supply. Avoid anastomosis there.

## Bowel preparation

### ▸ Mechanical preparation

- The concentration of organisms per gram of faeces is unchanged, so any spillage gives the same inoculum as unprepared bowel.
- **Conventional preparation:** 3-day diet, whole-gut irrigation, elemental diet and 10% mannitol. It left the patient exhausted and nutritionally depleted.

### ▸ Whole-gut irrigation

- **Method:** Ringer lactate or normal saline through a nasogastric tube over several hours.
- **Contraindications:** unstable cardiovascular disease, congestive cardiac failure, cirrhosis, severe renal disease (fluid overload) and bowel obstruction.
- **Advantages:** dietary freedom, short preparation time, no enema.

### ▸ Polyethylene glycol (PEG)–electrolyte regimen

- **Diet:** low-residue diet on day 2 and clear liquids on day 1 before surgery.
- **Adults:** PEG solution 1–1.5 L per hour for 3 hours, up to 4 L or until the rectal effluent is clear.
- **Children:** 20–40 mL/kg per hour.
- **Nausea:** metoclopramide 10 mg may be given.

▸ **Oral cathartics**

- **Doses:** sodium phosphate 45 mL or magnesium citrate 148 mL.
- Typically well tolerated, and smaller volumes are needed.
- **Sodium phosphate caution:** potassium, calcium and phosphate derangements in frail patients; use with caution in renal failure.

▸ **Antibiotics**

- **Kanamycin (best oral preparation):** 1 g orally every hour for 4 doses, then 4 times a day on day 1 of the regimen; then 1 g orally 4 times a day on each of the next 2 days.
- **Intravenous prophylaxis:** a third-generation cephalosporin is suggested.

## Diarrhoea and pseudomembranous enterocolitis

▸ **Pathogenesis**

- Antibiotics suppress normal bowel flora, so Clostridium difficile grows and releases toxin.
- The toxin causes diffuse inflammation with cream-coloured plaques, erythema, oedema, polymorphonuclear infiltrate and mucosal sloughing.
- Can be a pseudomembranous colitis, an enteritis, or both.

▸ **Features**

- Abdominal pain and diarrhoea, usually without fever or chills, though systemic toxicity may supervene.
- May progress to toxic megacolon, which carries a mortality of 15–20%.

▸ **Diagnosis**

- Symptoms and endoscopy, confirmed by culture or by toxin identification (quick result).

▸ **Treatment**

- Vancomycin or metronidazole, with withdrawal of the offending antibiotic.
- Subtotal colectomy if toxic megacolon develops.

## Intestinal anastomosis: types

▸ **Sutured**

- Two-layer enteroenterostomy, single-layer enteroenterostomy, and end-to-side sutured ileocolic anastomosis. Usable with or without discrepant bowel sizes.

▸ **Stapled**

- **Staplers:** linear, anastomotic and circular.
- **Theoretical benefits:** better blood supply to the healing margin, less tissue handling, minimal oedema from uniformly placed staples, wider lumen, ease and speed, less postoperative ileus.

▸ **Laparoscopic and robotic**

- Bowel anastomosis uses endoscopic stapling devices; the ureteroileal anastomosis is sutured.
- A purely laparoscopic intestinal anastomosis at radical cystectomy has a much higher complication rate (intestinal and ureteric anastomotic leaks) than an exteriorised approach through a mini-laparotomy (extracorporeal surgery).

▸ **Compression anastomosis**

- The biofragmentable ring has proved safe, effective and time-efficient.

## Postoperative care

▸ **Return of bowel activity**

- Small bowel within a few hours of surgery; stomach as early as 24 hours.

▸ **Nutrition**

- **Parenteral nutrition indications:** impaired preoperative nutrition, a postoperative complication delaying oral feeding, or paralytic ileus persisting on postoperative day 5.
- Some surgeons recommend jejunal feeding tubes for early enteral feeding.

▸ **Stomach tube**

- **Gastrostomy decompression:** helps patients with severe pulmonary disease by easing pulmonary toilet and improving comfort.
- **Stress ulcer prophylaxis:** in significant comorbidity, an H2 blocker or antacid through the stomach tube every 2 hours to keep gastric pH above 5.

## Complications of intestinal anastomosis

▸ **Leak and sepsis**

- Faecal or urinary fistula with sepsis, wound infection, pelvic abscess and wound dehiscence.

▸ **Obstruction**

- Recurrent cancer is the second most common cause, after adhesions.
- **Stenosis:** oedema if immediate; ischaemia if delayed.
- **Pseudo-obstruction:** usually within the first 3 days, in patients with multiple comorbidities.

▸ **Haemorrhage**

- Commonest when stomach is used.

▸ **Isolated intestinal segment**

- **Stricture:** lymphoid depletion, bacterial infection, then submucosal oedema with fibrosis.
- **Elongation:** usually due to distal obstruction.

## Ureterointestinal anastomosis: principles and renal outcome

▸ **Refluxing anastomosis**

- An advantage when the remaining urothelium is prone to malignant change, because the upper tracts can be imaged by instilling contrast into the conduit.

▸ **Technique**

- Clean the ureter of adventitia for only 2–3 mm at the distal end where the anastomosis is made.
- **Non-refluxing submucosal tunnel:** inject saline submucosally with a 25-gauge needle to raise the mucosa and ease dissection.
- **Stricture prevention:** soft silastic stents and better suture material.

▸ **Renal outcome**

- About 12.5% develop renal failure after ureteroileal diversion, from lack of ureteric motility, infection, stones and obstruction at the anastomosis.
- Renal deterioration does not depend much on the type of urinary diversion.

## Ureterocolic anastomosis

▸ **Leadbetter–Clarke combined technique**

- Similar to the extravesical technique used in transplantation.

▸ **Strickler**

- Non-refluxing ureterocolic tunnel; the ureter enters from the lateral side.

▸ **Pagano**

- Both ureters through a single incision into two submucosal tunnels.
- Lowest stricture rate among ureterocolonic anastomoses.

▸ **Cordonnier and Nesbit**

- Refluxing; not preferred for ureterosigmoidostomy.

## Ureteroileal anastomosis: techniques

▸ **Wallace**

- Not recommended with extensive carcinoma in situ (CIS) or a high likelihood of recurrent tumour, because a single tumour can block both ureters.

▸ **Tunnelled small-bowel antireflux anastomosis**

- Not widely used.

▸ **Split-nipple (Griffiths)**

- The spatulated ureter is turned back on itself and the nipple is implanted into the bowel.

▸ **Le Duc**

- The ureter is laid onto the inner bowel wall and becomes a submucosal tunnel when the mucosa re-epithelialises over it.

▸ **Hammock**

- The conjoined ureters are sutured to the intestinal mucosa, with longitudinal seromuscular incisions cross-hatched by multiple incisions.

▸ **Ureteral dipping**

- The ureter is dipped into the bowel lumen and secured to the seromuscular layer without a mucosal anastomosis.

▸ **U-SB anastomosis**

- Antireflux mechanism: serosal compression of an extramural ureter.

▸ **Intestinal antireflux valves**

- Nipple valve, intussuscepted ileocaecal valves and intussuscepted ileal valves.

## Complications of ureteroileal anastomosis

- **Urinary fistula:** usually within 7–10 days.
- **Ureterointestinal stricture, endoscopic treatment:** balloon dilatation, cold-knife incision, laser incision.
- **Pyelonephritis (early or long-term):** considerable morbidity and significant mortality.

## Ileal conduit

▸ **Positioning**

- The ureteroileal anastomoses can also be done after the ileoileostomy is made.
- Keep the base of the conduit mesentery as wide as possible.
- The conduit lies posterior and caudal to the ileoileostomy.
- Mobilising the caecum, small bowel and colon cephalad makes it easier to bring both ureters into the right lower quadrant.

▸ **Bricker anastomosis**

- **Ideal:** the spatulated ureters pass under the conduit without tension or kink.
- **Type:** single-layer, full-thickness, interrupted, 4-0 or 5-0 Vicryl.
- **Steps:** incise conduit serosa and muscularis, excise a small disc of mucosa and submucosa, then sew the spatulated ureter to the opening.

▸ **Stents and fixation**

- Ureteric stents are brought out through the conduit, with distal ends outside the stoma.
- The conduit may be sutured to the lateral abdominal wall to prevent an internal hernia.

## Ileostomy (rosebud stoma)

- Excise a circular plug of skin at the marked site.
- Spread a Kelly clamp to make a hiatus through the rectus muscle and posterior rectus sheath.
- Suture the anterior rectus sheath to the conduit serosa.
- **Eversion:** four three-point sutures, each through the bowel mucosal edge, then bowel serosa and muscularis, then out through the dermis (not the epidermis), so the stoma pouts.

## Abdominal stomas: complications

- **Early:** bowel necrosis, bleeding, prolapse, obstruction, stomal retraction, parastomal hernia and stenosis.
- **Peristomal dermatitis:** irritative, pseudoverrucous, or erythematous with erosive lesions.

## Rectal-bladder urinary diversion

▸ **Types and variations**

- **Types:** ureterosigmoidostomy, folded rectosigmoid bladder, augmented valved rectum, hemi-Kock and T-Kock procedures, sigma-rectum pouch (Mainz II).
- **Variations of ureterosigmoidostomy:** ileocaecal sigmoidostomy, the rectal bladder, and the sigmoid hemi-Kock with proximal colonic intussusception. Urine is passed by evacuation.

▸ **Patient selection**

- **Contraindications:** dilated ureters, extensive pelvic radiation, existing renal insufficiency, incompetent anal sphincter, pre-existing colorectal disease.
- **Anal sphincter test:** instil 400–500 ml of a thin oatmeal and water mixture into the rectum; the patient retains it for 1 hour upright.

▸ **Folded rectosigmoid bladder**

- Ureters implanted through serosa-lined tunnels.

▸ **Augmented valved rectum**

- A proximal sigmoid intussusception confines urine to a smaller surface area.

▸ **Hemi-Kock and T-Kock**

- The ileal patch is anastomosed to the rectum.

▸ **Mainz II (sigma-rectum pouch)**

- Simple and reproducible. Partial reconfiguration of the rectosigmoid junction creates a low-pressure reservoir with increased capacity.

## Continent catheterising pouches

▸ **Stoma site**

- Alternative to the umbilicus: the lower abdominal quadrant, through the rectus bulge and below the bikini line.

▸ **Continence mechanisms**

- **Appendiceal tunnelling:** the simplest of all to perform, for right colon pouches.
- **Intussuscepted nipple valve:** the most technically demanding. The flap valve avoids the need for intussusception.
- **Hydraulic valve (fourth principle):** as in the Benchekroun nipple, now largely abandoned.

▸ **Kock pouch (continent ileal reservoir)**

- 15 cm of terminal ileum with an intussuscepted nipple valve.
- Preserves the ileocaecal valve.
- High complication rate; abandoned.

▸ **Double T-pouch**

- 70 cm of terminal ileum in a W-shaped configuration.
- **Continence:** an antiperistaltic 12–15 cm segment of distal ileum.
- **Antireflux:** the proximal 10 cm.
- Preserves the ileocaecal valve; fewer complications than the Kock pouch.

▸ **Mainz I pouch**

- W configuration from limbs of about 10–15 cm, 10–15 cm, 10–15 cm and 20–25 cm (caecum/ascending colon and ileum).
- **Nipple valve:** intussusception of ileum through the ileocaecal valve.
- High complication rate (31%).

▸ **Indiana pouch**

- The entire right colon is reconfigured in a Heineke–Mikulicz fashion; appendicectomy is done.
- Buttressing of the ileocaecal valve alone is probably enough.
- The remaining ileal neourethra is tapered to a 12–14 Fr lumen and brought out.
- One of the most reliable reservoirs, easy to construct, low complication rate.
- **Variations:** Florida pouch and Miami pouch.

▸ **Penn pouch**

- Ileocaecal pouch with 10–12 cm of ileum; the first diversion to employ the Mitrofanoff principle.

- The appendix is excised with a button of caecum and reversed on itself before tunnelled reimplantation.
- **Alternative:** bury the appendix in the taenia with the tip brought out as the stoma.
- Easy to construct; suits patients without a previous appendicectomy.

▸ **Gastric pouch**

- A wedge-shaped segment of stomach from the greater curvature, based on the right gastroepiploic artery (also recorded as the left gastroepiploic artery).
- **Continence:** the contralateral distal ureter (after transureteroureterostomy) tunnelled into the reservoir, like an appendix.
- **Alternative:** a Mitrofanoff procedure, with a composite gastric reservoir and ileum segment and an appendix channel.
- **Possible advantages:** fewer infections and fewer metabolic complications; usable when the bowel is irradiated or short bowel syndrome is anticipated.
- May be especially useful in paediatric patients, but experience is limited at present.

## Complications of urinary diversion

▸ **Classification**

- **Intra-operative:** haemorrhage, trauma, ureteric stripping.
- **Post-operative:** metabolic, intestinal-anastomotic, conduit or pouch and stoma related, and general (chest and wound problems).

▸ **Conduit**

- Conduit infarction, volvulus, conduit–enteric fistula, excessive conduit length, conduit stenosis, GI bleeding.

▸ **Bowel anastomosis**

- Internal hernia.

▸ **Continent pouch**

- **Pouchitis (a reservoir complication):** needs at least 10 days of antibiotics.
- **Retention:** most common with a nipple valve.

## Complications of intestinal segments: electrolyte disturbances

- **Classification:** metabolic, neuromechanical and technical-surgical.

▸ **Stomach: hypochloraemic metabolic alkalosis**

- Normally the kidney compensates for gastric H+ loss by excreting bicarbonate, so alkalosis becomes a problem in renal failure.
- Failure to empty the gastric reservoir distends it and releases gastrin, which increases HCl production and worsens the alkalosis.
- **Cycle:** renal compensation (H+ retention with K+ loss) causes hypokalaemia; vomiting then causes water and sodium loss, dehydration with raised aldosterone, acute renal failure and possibly death.
- **Severe alkalosis:** respiratory insufficiency, lethargy, seizures, ventricular arrhythmias.

▸ **Jejunum: hyponatraemic, hypochloraemic, hyperkalaemic acidosis**

- About 25% of patients with a jejunal conduit may develop it.

- **Features:** nausea, vomiting, dehydration, muscle weakness, elevated temperature, azotaemia.
- Hyperalimentation solutions worsen the electrolyte abnormalities.

▸ **Ileum and colon: hyperchloraemic metabolic acidosis**

- Usually minor; severe disturbances are much less common.
- **Symptoms:** easy fatigability, anorexia, lethargy, weight loss, polydipsia; worsening diarrhoea in ureterosigmoidostomy.

▸ **Hypokalaemia**

- Caused by renal potassium wasting (renal damage and osmotic diuresis); gut loss plays only a minor role.
- Ileum absorbs some potassium from urine, colon absorbs none, so colonic diversions and ureterosigmoidostomy are more likely to cause potassium loss.

## Treatment of hyperchloraemic acidosis

▸ **Oral sodium bicarbonate**

- Produces considerable intestinal gas, which may not be tolerated, and tastes bad.

▸ **Citrate solutions**

- Sodium citrate, potassium citrate and citric acid (for example Bicitra or tricitrate preparations). Used when the sodium load is a problem, as in cardiac or renal patients.

▸ **Chlorpromazine or nicotinic acid**

- Used in persistent acidosis, combined with low-dose alkalinisation.
- **Chlorpromazine side effect:** tardive dyskinesia.
- **Nicotinic acid:** can worsen liver dysfunction or peptic ulcer disease; causes headache, double vision, flushing and dermatitis.

## Renal function assessment and altered sensorium

▸ **Intestinal mucosa exposed to urine**

- Histological changes develop over time; less marked in colon.

▸ **Magnesium deficiency**

- Causes altered sensorium; results from nutritional depletion and increased renal loss.

▸ **Ammoniagenic encephalopathy**

- Empty the reservoir continuously with a Foley catheter or rectal tube.
- **Lactulose:** combines with ammonium in the gut and reduces its absorption.
- **Arginine glutamate:** 50 g in 1 L of 5% dextrose; complexes with ammonia to form glutamine.

## Abnormal drug absorption

- **Drugs at risk:** those absorbed from the gut and excreted unchanged by the kidney, because they are re-exposed to the intestinal segment and reabsorbed to toxic levels.
- **Chemotherapy:** usually well tolerated, though methotrexate toxicity has been reported.

## Osteomalacia

- **Bone:** mineralised bone is reduced and the osteoid component increases.
- **Mechanism:** in persistent acidosis, bone hydroxyapatite buffers H+ and releases calcium, which is lost in the urine. An altered calcium to PTH ratio contributes.
- **Symptoms:** lethargy, joint pain, proximal myopathy.
- **Investigations:** low serum calcium, low phosphate, raised alkaline phosphatase.

## Growth and development

- Children with intestinal diversions are susceptible to fractures and to complications after orthopaedic procedures.
- Growth problems occur when diversions were made in childhood and kept for more than 10 years.

## Stones

- **Risk factor:** pre-existing pyelonephritis.

## Infection

- Increased bacteriuria, bacteraemia and septic episodes; high incidence in continent diversions.
- About 13% have septic or major infectious complications.
- **Proposed reasons:** intestine cannot inhibit bacterial proliferation, makes urine less bacteriostatic, allows bacterial translocation when distended, and has an altered immunological mucosal barrier.

## Intestinal motility, short bowel and nutritional problems

### ▸ Loss of the ileocaecal valve

- Fat reaching the colon causes fatty diarrhoea and deficiency of fat-soluble vitamins A, D, E and K.
- **Osteoporosis:** vitamin D deficiency and reduced calcium absorption, because calcium binds fat to form soaps.

### ▸ Jejunal loss

- Significant loss is rare in urology; causes malabsorption of fat, calcium and folic acid.

### ▸ Colonic loss

- Reduced fluid, electrolyte and bicarbonate reabsorption, causing dehydration.

### ▸ Any major bowel loss

- Nocturnal bowel movements, faecal urgency, faecal incontinence, diarrhoea, nutritional deficiencies.

## Cancer

- **Ileal segments:** adenomatous polyps and anaplastic carcinoma reported.
- Exposure of gut epithelium to urine alone is probably not the cause of secondary cancer.

## Enhanced recovery after surgery (ERAS)

- **Aim:** standardise and improve perioperative management.
- **Goal:** faster, more efficient recovery using evidence-based practices.
- **Outcomes:** faster recovery, shorter stay, cost-effectiveness, improved quality of life.
- **Audit:** needed to ensure successful implementation and to assess clinical and financial outcomes.

## Pre-op ERAS elements

### ▸ Lifestyle

- Stop smoking 8 weeks before surgery; avoid excessive alcohol.

### ▸ Nutrition

- **Screening:** nutritional risk score.
- **Immunonutrition:** arginine, fish oil and nucleotides.
- **Carbohydrate loading:** oral or IV; safe in diabetics; up to 2 hours before surgery.

### ▸ Alvimopan

- Concern about cardiovascular events.

### ▸ Premedication

- Short-acting benzodiazepines preferred to reduce anxiety; long-acting ones are associated with cognitive impairment.

### ▸ Antibiotic prophylaxis

- **EAU:** continue for up to 24 hours; extend to 72 hours with specific infection risk factors or a prolonged operation (more than 3 hours).

### ▸ Anaesthesia

- No prospective studies of a specific anaesthetic protocol for urological surgery.

## Intra-op ERAS elements

- **Fluids:** excess fluid and hypervolaemia cause splanchnic hypoperfusion, leading to ileus.
- **Urinary catheter:** optimal duration unknown, but early removal appears beneficial.

## Post-op ERAS elements

### ▸ Postoperative nausea and vomiting (PONV)

- **Prevention:** multimodal antiemetic prophylaxis (ondansetron) and dexamethasone, intraoperative fluid optimisation, and stenting of the ureteroileal anastomoses.
- **Benefit:** better patient satisfaction, shorter hospital stay, fewer readmissions.

### ▸ Ileus and nutrition

- Prophylactic oral laxatives for ileus.
- Total parenteral nutrition (TPN) not routine unless enteral nutrition is delayed for more than 5–7 days.

### ▸ Analgesia

- Opioid-sparing multimodal analgesia with regional or local anaesthesia, using oral or IV paracetamol with or without NSAIDs.

- Enables early mobilisation, which reduces DVT, ileus and chest infection.

▸ **Discharge criteria**

- Resumed oral intake, normal bowel function, effective oral pain control.

## Ch 134 · Continent Urinary Diversion

### Principles

- **Orthotopic diversion, the "3 Cs":** capacity, compliance, continence.
- **Creating a reservoir:** splitting the bowel on its antimesenteric wall interrupts the coordinated activity fronts, but this activity is regained over time.

### Patient selection

▸ **Tumour-related factors**

- The patient must agree to indefinite follow-up before a neobladder is offered.
- In men with prostatic mucosal, ductal or stromal invasion, urethrectomy is an option when neoadjuvant chemotherapy is not possible.

▸ **Patient-related factors**

- **Obesity:** not a contraindication; in obese patients stoma creation itself can be difficult.

▸ **Choice of bowel segment**

- **Stomach:** acid secretion reduces bacterial contamination of the urine.

### Orthotopic reservoirs

- **Ileal:** Camey II, Hemi-Kock, Ghoneim serous-lined extramural tunnel, Hautmann, Studer, T pouch.
- **Ileocolic:** Mainz, Le Bag.
- **Colonic:** right colon pouch, sigmoid pouch.

▸ **Camey II**

- Detubularised ileum folded in a transverse U orientation.
- **Barre modification:** ileum in a Z configuration.

▸ **Ghoneim (serous-lined extramural tunnel)**

- The ureters pass into a 40 cm W-shaped ileal reservoir through extramural serous-lined tunnels, made with continuous seromuscular sutures over the ileum, which back the ureters.

▸ **Hautmann pouch**

- **Ureters:** originally a Le Duc antireflux anastomosis, later replaced by a freely refluxing end-to-side anastomosis. Implanted from inside the neobladder over stents.
- Limbs joined with running absorbable sutures, then the rest of the anterior wall is closed.
- **Modification:** only 40 cm of ileum.
- **Short ureters:** one or both ends of the W can be left long (as chimneys) for the ureteroileal anastomosis.

▸ **Studer pouch**

- The two 22 cm limbs are opened from the apex towards the afferent limb opening.
- On folding and closing, the distal one fingerbreadth is left open for the urethral anastomosis.

- The afferent limb can be combined with a flap-valve or modified T-pouch antireflux technique.

▸ **T pouch**

- A modification of Ghoneim's tunnel, designed to prevent its long-term complications while keeping an antireflux mechanism. Can also be used with a Studer neobladder.
- **Antireflux:** the distal 3–4 cm of the 15 cm ileal segment is anchored into the serosa-lined trough formed by the bases of the two adjacent 22 cm ileal segments.

▸ **Mainz ileocolonic neobladder**

- W-shaped segment opened fully; the posterior wall is made by joining the three opposing limbs; closed side to side.
- **Urethral anastomosis:** button-hole incision in the base of the caecum.

▸ **Le Bag**

- Ileocolic: ileum, caecum and colon, about 20 cm of colon and 20 cm of ileum, opened on the antimesenteric border and sutured to form the posterior wall.
- Urethra joined end-to-side to the caecum, ureters anastomosed, neobladder closed anteriorly.

▸ **Right colon pouch**

- Uses the entire right colon and caecum; the ileal stump is closed.
- Opened along the anterior taenia, leaving the proximal 2–3 inches of caecum intact.
- Antirefluxing ureterocolonic anastomosis and a urethrocolonic anastomosis; folded in a Heineke–Mikulicz manner and closed.

▸ **Sigmoid (Reddy) pouch**

- Based on the inferior mesenteric artery.
- The medial taenia is incised down to just short of the urethral anastomosis site, then the posterior wall is sutured.

## Advantages and outcome of neobladder

- **Advantage:** kidneys protected from infections.
- **Mortality:** 0–6%.
- **Complications:** early 5–30%, late 10–35%.

## Complications

▸ **Classification**

- **Not directly related to the diversion:**
  - Early: bleeding, thrombotic events, infection, cardiovascular and pulmonary complications.
  - Late: bowel obstruction, ventral hernia, thrombotic and cardiovascular events.
- **Directly related:** includes afferent limb obstruction and ureteroileal obstruction.

▸ **Perforation**

- Rare, because outlet resistance is low.

▸ **Night-time incontinence**

- Decreased sphincter tone at night and reversal of the normal night-time ADH effect.

▸ **Retention**

- **Risk factors:** excessive intestinal length (more than 60 cm); nerve-sparing or prostate-sparing procedure.
- **Options:** clean intermittent catheterisation, biofeedback, pelvic floor muscle training.

## Follow-up

- **Examination:** includes pelvic and rectal examination.
- **PSA:** yearly if prostate cancer was found on cystectomy pathology.
- **Imaging, pT2 or lower:** CT and chest radiograph at 4 months, 12 months, then yearly to 5 years, then every 2 years. Ultrasound may replace CT after 3 years.
- **Imaging, pT3, node-positive or higher:** CT abdomen and pelvis with chest CT or chest radiograph at each visit.

# Ch 135 · Minimally Invasive Urinary Diversion

## Contraindications to augmentation

- Bladder pathology that would preclude use of the native bladder.

## Operative technique: incision and native bladder

- **Incision:** midline or lower transverse.
- **Native bladder:** formerly the supratrigonal bladder was removed; now it is usually preserved as long as it is widely opened.
- **Bivalving:** sagittal incision from several centimetres above the bladder neck anteriorly to just above the trigone posteriorly. Eases the bowel anastomosis and keeps the native bladder's contribution to capacity.
- Transverse incisions into the two halves give a greater circumference.

## Ileocystoplasty and caecocystoplasty

- **Bowel length:** depends on native bladder volume, urine output and desired final capacity.
- Irrigate the isolated ileum with 0.25% neomycin before opening it on the antimesenteric border.
- **Reconfiguration:** U, W or S shape, with full-thickness sutures inverting the mucosa.
- **Anastomosis to bladder:** start posteriorly (easier), complete anteriorly; absorbable sutures in one or two layers.
- Caecocystoplasty alone is no longer done.

## Ileocaecocystoplasty (Mainz)

- Mobilise the right colon up to the hepatic flexure.
- **Segment lengths:** in a typical ileocaecal augmentation, colon equals ileum; in the Mainz, ileum is twice the length of the caecum.
- Alternatively, the ileum can be opened and used as a patch on the caecal segment.
- **Suprapubic tube (Mainz):** through the native bladder.
- **Appendix:** removed if not used; for a continent stoma, kept in situ with its base tunnelled beneath a taenia.

- **Short ureters:** bridged with ileum with an imbricated ileocaecal valve, which acts as an antireflux mechanism.

## Sigmoid cystoplasty

- **Detubularisation:** open along the antimesenteric border and close the two ends, or U- or S-shaped reconfiguration.
- Keep a large suprapubic tube.

## Gastrocystoplasty

- Rarely done nowadays.
- **Technique:** gastric body or antrum. The body technique takes a 10–20 cm segment (more acid-producing cells) on the right gastroepiploic artery; a bladder drainage tube is kept.
- **Advantage:** less bacterial colonisation.
- **Disadvantages:** haematuria–dysuria syndrome in about 70%, peptic ulceration, raised gastrin.

## Postoperative management

- **Nasogastric tube:** until bowel function returns, though some surgeons do not use one.
- **Extravesical drains:** removed after several days if dry.
- **At 3 weeks:** cystogram, then the suprapubic catheter is clamped; clean intermittent catheterisation (CIC) once or twice at night.
- Check post-void residual; continue CIC if high.

## Complications

▸ **Gastrointestinal**

- Bowel obstruction; early satiety after use of stomach.

▸ **Bowel anastomosis**

- Leak, peritonitis, obstruction, stricture, internal hernias.

▸ **Persistent rhythmic bowel contractions**

- May cause persistent incontinence, vesicoureteric reflux (VUR), hydronephrosis or delayed perforation.

▸ **Delayed spontaneous perforation**

- **Cause:** catheter-induced injury or transmural infection.
- Patients are very sick.

## Pregnancy after augmentation

- The expanding uterus is probably not detrimental to the reconstruction pedicle.
- **Caesarean section:** take care of the augmentation pedicle and any stoma.

## Alternatives to GI cystoplasty

- Ureterocystoplasty, autoaugmentation and synthetic material; none carries a tumour risk.

▸ **Ureterocystoplasty**

- Capacity up to 470 mL, but applicable only in limited cases.

▸ **Autoaugmentation**

- **Technique:** the detrusor over the dome is split or excised and its edges sutured to the psoas, so the mucosa bulges out.
- Increases capacity by up to 40%.
- **Why unpopular:** collagen deposition over the mucosa causes contraction and about a 50% reduction in capacity, so long-term results are poor when capacity is the concern.
- **Indication:** only reasonable capacity with poor compliance due to uninhibited contractions.

▸ **Seromuscular enterocystoplasty**

- Good early results, technically more difficult, long-term results unknown.

▸ **Bioengineered bladder**

- Harvesting the patient's own bladder cells is invasive and carries anaesthetic risk.
- **Urine cell culture:** cells isolated from urine are cultured in vitro instead.

▸ **Urinary buttons**

- Alternative to a conduit or suprapubic catheter for drainage.
- **Design:** internal part like a Foley tip with a balloon; external flat button on the skin.
- A one-way valve maintains continence; no stoma bag needed.

**Other topics in this book**

## Bladder diverticulum: management

▸ **Endoscopic**

- Transurethral resection of the diverticular neck with fulguration of the entire urothelial lining of the sac.
- **Indications:** not a candidate for open repair, or undergoing TURP with an associated poorly draining diverticulum.

▸ **Open diverticulectomy**

- **Approaches:** transvesical (commonest, including a transvesical submucosal technique), extravesical, or combined.
- **Combined approach:** for large diverticula and/or considerable peridiverticular inflammation.

▸ **Laparoscopic and robotic**

- Smaller incision, less pain, shorter hospital stay, shorter convalescence.

## Complications of diverticulectomy

- **Ureteric injury:** the most troublesome. Partial injury recognised intraoperatively is repaired and stented; complete transection is reimplanted.
- **Others:** bleeding, vascular injury, rectal injury.

## Cystitis glandularis

▸ **Features**

- Usually near the trigone; cobblestone appearance at cystoscopy.
- **Types:** typical (superficial); intestinal (less common, resembles neoplasia).

- Rarely causes ureteric obstruction with hydronephrosis.

▸ **Management**

- Eliminate chronic irritation, including replacing an indwelling (per-urethral) catheter with clean intermittent catheterisation.
- Manage distal obstruction such as prostatic enlargement or urethral stricture.
- Bladder augmentation or cystectomy rarely reported for intractable disease or bladder contracture.

## Late complications of neobladder

- **Influencing factors (seven):** tumour recurrence, adjuvant or salvage chemotherapy, radiation, type of bowel segment, technique of diversion, general condition (nutrition), associated comorbidities.
- **Classification:** not directly related, or directly related to the diversion.
- **Directly related:** incontinence, perforation/rupture, retention, afferent limb obstruction, urethral stricture, vaginal fistula, stones.

▸ **Afferent limb obstruction**

- Highest risk with the antireflux valve in the hemi-Kock pouch and the T-pouch.

▸ **Pouch stones**

- Rare in Studer and Hautmann neobladders.

▸ **Pouch–vaginal fistula**

- **Prevention:** careful watertight closure of the vaginal cuff and an omental flap.

# 13 Kidney & UTUC

## Ch 136 · Benign Renal Tumours

### Classification of renal tumours

- **Malignant masses:** leukaemia is included.
- **Inflammatory masses:** infected renal cyst and rheumatic granuloma are included.

### Oncocytoma

▸ **Epidemiology and genetics**

- 3 to 7% of kidney tumours; female-to-male ratio 2:1; peak in the 7th decade.
- Commonest genetic abnormality: loss of heterozygosity of chromosome 1 and/or chromosome 14.

▸ **Pathology**

- **Gross:** mahogany or tan, homogeneous, well circumscribed, with a pseudocapsule.
- **Microscopy:** large, uniform, round or polygonal, highly eosinophilic cells in a nested pattern.
- **Differential diagnosis:** clear cell RCC with eosinophilic characteristics, and chromophobe RCC.

▸ **Diagnosis**

- **Angiography:** spoke-wheel pattern, but of low predictive value.
- **CT enhancement:** peak HU in the nephrographic phase; RCC peaks in the corticomedullary phase.
- **Core biopsy immunohistochemistry:** optimal panel of 3 markers (vimentin, ECAM, glutathione-S-transferase alpha) separates RCC from oncocytoma with 100% sensitivity and 100% specificity.
- **Frozen section:** not sensitive for oncocytoma.

### Angiomyolipoma (AML)

- Less than 10% of renal tumours.
- 70–80% sporadic; 20–30% genetic (tuberous sclerosis).
- Strongly expresses oestrogen receptor beta, progesterone receptor and androgen receptor.
- **CT fat criteria:** more than 20 pixels below −20 HU, or more than 5 pixels below −30 HU, gives a 100% positive predictive value.
- **CT signs:** indentation of the renal parenchyma is characteristic; aneurysmal dilatations on CT angiography in 50%.
- **Liposarcoma mimic:** HMB-45 staining is positive in AML and confirms the diagnosis.
- **Selective embolisation:** often needs repeat procedures.

### Renal cysts

- Commonest benign renal lesion; 70% of asymptomatic renal masses.
- **Risk factors:** increasing age, male gender, hypertension, renal insufficiency.
- **Bosniak on MRI:** usable, but MRI tends to exaggerate some findings.

- **Parapelvic cyst:** laparoscopic approach is better than percutaneous because of nearby vital structures.
- **Complications:** pain from haemorrhage, infection and rupture.

## Mixed epithelial and stromal tumour (MEST) and cystic nephroma

▸ **Pathogenesis (both)**

- Sex hormones play a role, suggested by female predilection, a history of hormonal ablation therapy in affected males, and frequent ER/PR expression.

▸ **Cystic nephroma**

- **Imaging:** solitary, central lesion with curvilinear calcification and septal enhancement; cannot be told apart from cystic RCC.
- **Histology:** fibrous pseudocapsule; cysts lined by flattened, cuboidal or hobnail epithelium; ovarian-like stroma.
- **Staining:** epithelium cytokeratin positive; stroma ER, PR, CD10 and calretinin positive.

▸ **MEST**

- Peaks in the 5th decade, usually in women with a history of oestrogen replacement therapy.
- Appears as a Bosniak III/IV cyst, indistinguishable from cystic RCC; consider it pre-operatively in a woman on HRT.
- **Treatment:** partial nephrectomy if feasible, otherwise radical nephrectomy.

## Metanephric adenoma and papillary adenoma

▸ **Metanephric adenoma**

- Radiologically indistinguishable from RCC; shares some features with Wilms tumour.
- About 10% have polycythaemia.
- Peripheral or central calcification; enhances less than the surrounding cortex and medulla.
- Usually confirmed after excision: WT1 and S100 positive, and BRAF V600E mutation testing.

▸ **Papillary adenoma**

- May be premalignant for papillary RCC; many believe all solid epithelial renal masses are potentially malignant and need treatment.
- Incidence rises with age, male sex and acquired renal cystic disease.

## Tuberous sclerosis complex (TSC)

▸ **Genetics and pathogenesis**

- Classically called epiloia: triad of epilepsy, low intelligence and adenoma sebaceum.
- TSC1 and TSC2 are tumour suppressor genes regulating cell proliferation and differentiation.
- **Pathway:** loss of TSC1/TSC2 leads to Rheb accumulation, then FKBP38 accumulation, which activates mTORC1 (mTOR complex 1). mTOR inhibitors block this step.

▸ **Urological manifestations**

- Renal disease is the second commonest cause of morbidity and mortality, after neurological disease.
- Renal cysts are the second most common renal manifestation, after AML.
- Oncocytoma and perirenal cysts also occur.

- Periodic RCC screening with ultrasound or CT is needed.

▸ **Diagnostic criteria**

- **Major** (besides renal AML, facial angiofibromas and cortical tubers): ungual or periungual fibroma, 3 or more hypomelanotic macules, shagreen patch, multiple retinal hamartomas, subependymal nodule, subependymal (giant cell) astrocytoma, cardiac rhabdomyoma, lymphangioleiomyomatosis.
- **Minor:** multiple renal cysts, non-renal hamartoma, hamartomatous rectal polyps, retinal achromic patch, bone cysts, gingival fibromas, confetti skin lesions, dental enamel pits, cerebral white matter radial migration lines.

▸ **Management of AML in TSC**

- **Before surgery:** fully evaluate neurological status and optimise because of seizure risk; involve a neurophysician.
- **mTOR inhibitors:** sirolimus, as well as everolimus, has been used in trials.
- **Acute haemorrhage:** embolisation preferred, because emergency surgery almost always ends in nephrectomy.
- **Ablation:** consider for comorbid patients or multiple bilateral tumours.

## Ch 137 · Malignant Renal Tumours

### RCC: general aspects and epidemiology

- **Incidence:** about 12 per lakh (100,000) per year.
- **Children:** Wilms tumour is commoner in younger children, but RCC equals Wilms in the second decade.
- Childhood RCC is more often symptomatic, locally advanced, high grade and of unfavourable histology than adult RCC.

### Aetiology

▸ **Cell of origin**

- **Proximal convoluted tubule:** papillary and clear cell RCC.
- **Collecting duct cells:** medullary carcinoma, collecting duct carcinoma, chromophobe RCC.

▸ **Tobacco**

- All forms implicated; risk begins to fall after stopping.

▸ **Obesity**

- Relative risk 1.07 for each additional unit of BMI.
- **Proposed mechanisms:** raised IGF-1, raised oestrogen, increased arteriolar nephrosclerosis, local inflammation.

▸ **Hypertension**

- Thought to act through renal injury and inflammation, and metabolic and functional changes in the tubules.

## Pathological subtypes

### ▸ Fuhrman nuclear grade

| Grade | Nuclear size | Features |
|---|---|---|
| I | About 10 µm | Round, uniform; nucleoli absent or inconspicuous |
| II | About 15 µm | Irregular; small nucleoli visible at 400x |
| III | About 20 µm | Irregular; prominent nucleoli |
| IV | 20 µm or more | Bizarre, multilobed; prominent nucleoli, heavy chromatin clumps |

### ▸ Clear cell RCC

- Can result from VHL gene hypermethylation as well as mutation.
- Loss of chromosomes 8p and 14q, in addition to 9p.
- **Gross:** well circumscribed, lobulated, golden yellow; arises from the proximal convoluted tubule.

### ▸ Papillary RCC

- Fleshy, with a fibrous pseudocapsule; necrosis and haemorrhage common; arises from the proximal convoluted tubule.
- **Type I:** basophilic cells, low-grade nuclei, better prognosis.
- **Type II:** eosinophilic cells, high-grade nuclei.

### ▸ Chromophobe RCC

- Well circumscribed, homogeneous, tan or light-brown cut surface; Hale's colloidal iron positive.

### ▸ Collecting duct carcinoma

- Less than 1% of RCC; firm, with high-grade nuclei and frequent mitoses.

### ▸ Renal medullary carcinoma

- Many are locally advanced or metastatic at diagnosis.
- Same histology as collecting duct carcinoma.

### ▸ Rare subtypes

- Post-neuroblastoma RCC is a separate pathological subtype.

## Familial RCC syndromes

### ▸ Succinate dehydrogenase (SDH)-deficient RCC

- **Genes:** SDHB on chromosome 1; SDHD on chromosome 11.
- **Renal tumours:** chromophobe RCC, clear cell RCC, type 2 papillary RCC, oncocytoma.
- **Extrarenal:** paraganglioma and papillary thyroid carcinoma.

### ▸ Birt–Hogg–Dubé syndrome

- Chromophobe and hybrid oncocytic tumours, and occasionally clear cell and papillary RCC.

### ▸ Von Hippel–Lindau (VHL) syndrome

- **VHL protein:** alpha-domain binds the elongin B/elongin C–cullin-2–RBX1 complex; there is also a beta-domain. Alpha-domain mutation disrupts the VHL complex.
- **Haemangioblastoma risk:** high in type 1, 2A and 2B; type 2C has neither RCC nor haemangioblastoma.

- **Screening:** relatives who test negative genetically need only less stringent follow-up.

▸ **Nephron-sparing surgery in familial RCC**

- Tumours are generally low stage, though multiple, bilateral and more often recurrent.
- SDH-RCC, like HLRCC, is an exception to the 3 cm rule because it is more aggressive.

## Clinical presentation

▸ **IVC involvement**

- **Suspect tumour thrombus with:** leg oedema, non-reducing right varicocele, dilated superficial anterior abdominal wall veins, proteinuria, deteriorating or non-functioning ipsilateral kidney, pulmonary embolism, right atrial (right-sided cardiac) mass.

▸ **Metastatic disease**

- Symptoms also include cervical lymphadenopathy and jaundice. Common sites are lung, liver, lymph nodes and bone.

▸ **Paraneoplastic syndromes**

- Also anaemia, pyrexia, cachexia with weight loss, and amyloidosis.
- **Hypercalcaemia:** management includes focused radiotherapy to metastatic bone lesions.

▸ **Stauffer syndrome**

- **Clinical:** fever, weight loss, signs of liver dysfunction such as oedema and bleeding tendency.
- **Laboratory:** raised ALP, raised prothrombin time, low albumin, raised SGOT/SGPT, low platelets, neutropenia.
- **Liver biopsy:** non-specific hepatitis with prominent lymphocytic infiltrate.

## Diagnosis

▸ **Intravenous pyelography (IVP)**

- No longer used for renal masses: poor sensitivity and specificity; misses small anterior or posterior lesions that do not distort the collecting system or contour.
- **Features of malignancy:** calcification in the mass, increased tissue density, irregular margin, collecting system distortion.

▸ **CT**

- 10–20% of renal masses are indeterminate; consider renal mass biopsy.
- **Renal pseudotumour:** enhancing segment isodense with the rest of the kidney; biopsy if uncertain.

▸ **MRI and nephrogenic systemic fibrosis**

- **Risk reduction:** avoid gadolinium in CKD, dialyse after the procedure, or use CT with periprocedural hydration instead.

▸ **Renal arteriography**

- Now a limited role.
- **Indications:** renal mass with renal artery disease; equivocal cases where neovascularity may help; showing vascularity of IVC tumour thrombus (seen in 35–40%).

▸ **Inferior venacavography**

- Can assess IVC thrombus but is rarely done.

▸ **Renal mass biopsy**

- **Technique:** under sedation; an 18 G, 16 cm needle is ideal; co-axial technique (multiple cores through one cannula) avoids seeding; take cores away from necrotic areas.
- **Pneumothorax:** less than 1%.
- **Tract seeding:** a concern mainly for central infiltrative masses and poorly differentiated urothelial cancer.

▸ **Metastatic evaluation**

- Every case: liver function tests, chest X-ray, and abdominopelvic CT or MRI.
- **FDG PET-CT:** good specificity, suboptimal sensitivity; best for equivocal findings on conventional imaging.

## Staging

▸ **TNM changes**

- **2002:** T1 split into T1a and T1b, reflecting different prognosis.
- **T2:** later subdivided into T2a/T2b, supporting the 10 cm break-point.
- **2009:** isolated renal vein thrombosis downgraded from T3b to T3a (more favourable prognosis).
- **Nodes:** subdivision by number of nodes was abolished, as it had no prognostic significance.

▸ **Robson staging**

| Stage | Extent |
|---|---|
| I | Within the renal capsule |
| II | Into perinephric fat, confined within Gerota's fascia |
| III | Regional lymph nodes and/or renal vein or IVC |
| IV | Adjacent organs or distant metastasis |

## Prognostic factors

▸ **Clinical, anatomical and histological**

- **Clinical (adverse):** symptomatic rather than incidental presentation, systemic symptoms, raised ESR, CRP and alkaline phosphatase.
- **Anatomical:** adrenal involvement, metastatic burden.
- **Histological:** microvascular invasion, surgical margin status.

▸ **Molecular**

- **CA-IX:** also a marker of response to systemic therapy, so useful in advanced disease.
- **B7-H1 (PD-L1):** T-cell co-regulatory molecule; strong independent predictor of progression, even after other molecular, clinical and pathological predictors.
- **Ki-67:** correlates with reduced survival in clear cell RCC; independent factor; used in predictive algorithms.

## Management of localised RCC

▸ **Zonal NePhRO score**

- Four components, each scored 1, 2 or 3 by zone 1, 2 or 3.

| Component | 1 point | 2 points | 3 points |
|---|---|---|---|
| Nearness | Touches cortex | Touches medulla | Touches collecting system or crosses into renal sinus |
| Physical zone | Lower pole, below collecting system | Lateral to, not touching, pelvicalyceal system | Upper pole, or touching pelvicalyceal system |
| Radius | Under 2.5 cm | 2.5–4 cm | More than 4 cm |
| Organisation | More than 50% exophytic | 50–75% endophytic | More than 75% endophytic |

- **Total:** 4–6 low, 7–9 intermediate, 10–12 high risk.

▸ **Active surveillance**

- **Suitable:** small, solid, enhancing, homogeneous masses. **Not suitable:** poorly defined margins or heterogeneous masses.

▸ **Partial nephrectomy: local recurrence**

- **Options:** repeat partial nephrectomy, radical nephrectomy, thermal ablation, active surveillance.

▸ **Radical nephrectomy**

- **Adrenalectomy indications:** tumour more than 10 cm, direct involvement, locally advanced disease, upper-pole location, diffuse involvement.
- **Lymphadenectomy:** extended dissection from the diaphragmatic crus to the aortic bifurcation is controversial; enlarged hilar and adjacent para-aortic or paracaval nodes are removed.

▸ **Lymph node dissection templates**

- **Right tumour:** paracaval, precaval, retrocaval and interaortocaval nodes, from the right crus to the IVC bifurcation.
- **Left tumour:** from the superior mesenteric artery to the aortic bifurcation; nodes anterior and lateral to the aorta and retroaortic. Interaortocaval nodes only if palpable or radiologically enlarged.

## Locally advanced RCC

▸ **IVC tumour thrombus surgery**

- With complete IVC occlusion, no grafting or reconstruction is needed after resection, because collaterals have developed.

▸ **Locally invasive RCC**

- Usually presents with pain from invasion of posterior abdominal wall, nerve roots and paraspinal muscles.
- **Neoadjuvant options:** systemic therapy (no defined protocols), preoperative embolisation.

▸ **Local recurrence after RN or NSS**

- Complete resection may carry significant morbidity.

## Surveillance after partial or radical nephrectomy

- **Low risk:** pT1 N0 Mx. **Moderate-to-high risk:** pT2–4 N0 Mx, or pT any N1 M0.

- **All patients:** history and examination; renal function tests, urinalysis (CUE), eGFR, complete blood count, LDH, liver function tests, alkaline phosphatase, calcium.

| Group | Abdominal imaging | Chest X-ray |
|---|---|---|
| Low risk, after PN | CT or MRI at 3–6 months, then ultrasound, CT or MRI yearly for 3 years | Yearly for 3 years, then as clinically indicated |
| Low risk, after RN | Ultrasound, CT or MRI at 3–6 months; after 12 months at clinician's discretion | Yearly for 3 years, then as clinically indicated |
| Moderate-to-high risk | CT or MRI at 3–6 months, then ultrasound, CT or MRI 6-monthly for 3 years, then yearly to year 5; beyond 5 years at clinician's discretion; site-specific imaging as warranted | Same as low risk |

## Ch 138 · Urothelial Tumours of the Upper Urinary Tract and Ureter

### Epidemiology

- **Western incidence:** about 2 per 100,000.
- **Balkan countries with highest incidence (South Eastern Europe):** Albania, Bulgaria, Bosnia and Herzegovina, Croatia, Greece, Romania.
- **Taiwan:** sex ratio reversed, male:female 1:2.
- **Race:** commoner in whites than African-Americans.
- **Bilateral disease:** synchronous 1.6%; metachronous contralateral 2–6% after a UTUC.
- Metachronous occurrence of 80% after bladder cancer; this conflicts with the 2–4% figure for UTUC after bladder cancer elsewhere (check).
- **Survival:** overall 5-year disease-specific survival 75%; worse with increasing age, male gender, black non-Hispanic race and advanced stage.

### Aetiology

▸ **Lynch syndrome (HNPCC)**

- **Cancers:** colon, uterus, pancreas, breast, ovary, stomach.
- Associated UTUC presents younger (mean age 55) and is more likely in females.

▸ **Aristolochic acid nephropathy**

- **Source plants:** Aristolochia fangchi and Aristolochia clematitis.
- Mutates codon 139 of p53, seen in both Balkan endemic and Chinese herb nephropathy.
- Explains the very high UTUC incidence in Taiwan and China (herbal medicine).

▸ **Balkan endemic nephropathy**

- Degenerative interstitial nephropathy; much higher UTUC incidence, but not bladder cancer.
- Tumours are usually low grade, multiple and bilateral.

▸ **Chinese herb nephropathy**

- Caused by a herb containing A. fangchi; leads to renal failure and UTUC.

▸ **Smoking and coffee**

- Smoking risk is dose related and falls after stopping.
- Smoking causes more ureteric than renal pelvic cancers.

- Coffee has no proven association.

▸ **Phenacetin (analgesics)**

- Used in combination with caffeine, codeine, acetaminophen and aspirin.
- Replaced by paracetamol, which is non-toxigenic.
- Long latent period of 20 years.

▸ **Arsenic**

- Well water in Taiwan: blackfoot disease, dry gangrene and increased UTUC.
- Ureteric outnumber renal pelvic tumours 2:1; male:female reversed at 1:2.

▸ **Occupation**

- Aromatic hydrocarbons in chemical, petrol and plastic industries.
- Coal, asphalt and tar.
- Chlorinated solvents in printing, by skin contact and vapour inhalation.

▸ **Iatrogenic**

- **Ifosfamide and cyclophosphamide:** raise risk through the metabolite acrolein.
- **Laxatives:** habitual anthranoid and chemical laxative use for more than 1 year raises risk 9-fold.

## Pathology and natural history

▸ **Renal pelvic tumours**

- Invasive in about 50% of papillary and about 80% of sessile tumours, similar to the bladder.

▸ **UTUC after bladder cancer**

- Develops 17–170 months later; rates up to 25% reported.
- **Risk factors:** NMIBC, low grade, multiple and multifocal tumours, previous UTUC, positive ureteral margin, N0 status, involvement of the male prostatic urethra or female urethra, vesicoureteral reflux, tumours near the ureteral orifice.

▸ **Bladder recurrence after UTUC**

- Explained by downstream seeding, longer carcinogen exposure and the greater number of urothelial cells exposed to carcinogenic events.

▸ **Metastasis at presentation**

- 20% of UTUC.

▸ **Carcinoma in situ (CIS) of the upper tract**

- More bilateral and multifocal disease: bilateral in about 25%, compared with 3–5% normally.
- Higher risk of later pan-urothelial disease.
- **Pan-urothelial disease:** in a female, both upper tracts; in a male, both upper tracts or one upper tract plus the prostatic urethra.

## Diagnosis

▸ **CT urography (CTU)**

- With 3D reconstruction, detects filling defects under 5 mm.
- Drawback: higher radiation dose.
- UTUC attenuation 10 to 70 HU (average 46 HU).

▸ **Radiolucent filling defect: differential**

- UTUC, blood clot, stone, sloughed papilla, overlying bowel gas, fungus ball, external compression.

▸ **Cytology**

- Sensitivity for grade II tumours: 45%.
- Ureteral washings are more accurate, as irrigation releases loosely adherent cells.
- Ionic high-osmolar contrast (as in retrograde pyelography) worsens cytological abnormality, so sample before using it.

▸ **Brush biopsy**

- Through a retrograde catheter or ureteroscope; sensitivity and specificity about 90%.
- **Complications:** massive haemorrhage, perforation, extravasation.

▸ **Ureteroscopic biopsy**

- By forceps or brushings.
- **Risks:** pyelovenous and pyelolymphatic tumour cell migration, extravasation, dissemination.

▸ **Antegrade endoscopy**

- Risks tumour implantation in the retroperitoneum and along the nephrostomy tract.

## Prognostic factors

▸ **Broders grade**

| Grade | Description |
|---|---|
| I | Papilloma |
| II | Well differentiated |
| III | Moderately differentiated |
| IV | Anaplastic or poorly differentiated |

▸ **Clinical and pathological factors**

- **Hydronephrosis:** independent predictor of advanced stage and survival.
- **Size:** above 3–4 cm may mean worse survival and more bladder recurrence.
- **Advanced age:** predicts recurrence-free, disease-specific and overall survival.
- **Race:** Black non-Hispanic patients have higher mortality than White non-Hispanic patients.
- **Multifocality:** tumour at two or more urothelial sites; independently predicts poor outcome.
- **Tumour necrosis:** correlates with stage, grade, lymphovascular invasion, CIS and nodal metastasis.
- **Lymph node density:** 30% or more predicts poor outcome.
- **Lymphovascular invasion:** independent factor for disease-specific survival; correlates with higher stage, grade and recurrence.

▸ **Molecular markers**

- None yet validated for clinical use.
- **Cell cycle:** TP53 mutation; CDKN1B (formerly p27).
- **Apoptosis:** Bcl-2 and survivin, associated with advanced disease.
- **Migration and invasion:** E-cadherin and MMP (matrix metalloproteinase) expression, poor prognosis.

- **Angiogenesis:** HIF-1alpha, poor prognosis.
- **Proliferation:** overexpression of Ki-67, EGFR, NF-kB, HER-2.
- **Differentiation:** uroplakin III, associated with lower stage.
- **Mitosis:** Aurora-A, poor prognosis.
- **MET proto-oncogene family:** cMET linked to worse prognosis; RON (with cMET) not correlated with outcome.
- **COX-2:** poor prognosis.

## Treatment

▸ **Risk stratification**

- **Size cut-off:** older cut-off 1 cm (low risk below, high risk above); current cut-off 2 cm.

▸ **Site-based algorithm**

- **Low-risk calyceal tumour:** URS first, RNU second.
- **High-risk distal ureteric tumour:** distal ureterectomy with or without LND is an alternative to RNU.

▸ **Metastatic UTUC**

- Platinum-based chemotherapy, with or without surgical resection; no regimen is defined as standard.
- **Other regimens:** PCG (paclitaxel, carboplatin, gemcitabine), CG (carboplatin–gemcitabine), cabozantinib (MET and VEGF pathway inhibitor).
- **Radiotherapy:** no role, alone or with chemotherapy.
- **RNU:** no survival benefit; palliation only.

# Ch 139 · Surgical Management of Upper Urinary Tract Urothelial Tumours

## Kidney-sparing surgery

▸ **Open**

- **Renal pelvic options:** pyelotomy with excision and base fulguration; partial nephrectomy guided by intraoperative ultrasound.
- **Recurrence:** 7–60%, with inferior oncological outcomes and the drawback of open surgery.

▸ **Endoscopic (URS) and lasers**

- Ureteroscopic management can be done as an outpatient.
- **Holmium:YAG:** 0.6–1.2 J at 8–10 Hz; efficient ablation and precise cutting.
- **Neodymium:YAG:** 20–30 W; coagulative necrosis; excellent haemostasis; only a narrow safety margin in the ureter.

▸ **Distal ureterectomy**

- A non-refluxing reimplant also limits ascending infection.

## Radical nephroureterectomy (RNU) with bladder cuff

▸ **Indications**

- Low-grade, non-invasive renal pelvic and upper ureteric tumours that are large, multifocal and rapidly recurring.

▸ **Distal ureter and bladder cuff**

- Cuff excised is about 1 cm.
- **Five techniques:**
  - Traditional open: intravesical, extravesical or combined.
  - Transvesical ligation and detachment (Collins knife and endoloop).
  - Transurethral resection of the ureteric orifice ("pluck"): abandoned because of tumour seeding into the extravesical space.
  - Intussusception ("stripping"): contraindicated when ureteric tumours are present.
  - Total laparoscopic: contraindicated by distal ureteric tumours.

▸ **Adjuvant intravesical instillation**

- Strong EAU 2018 recommendation (for example mitomycin C) after RNU to prevent bladder recurrence.

▸ **Lymphadenectomy**

- EAU 2018: only a weak recommendation for template or extended lymphadenectomy; more studies needed.
- Limited (regional) lymphadenectomy with RNU has limited morbidity.
- Nodal involvement rises with stage and grade.
- **Problems:** no uniform templates in the literature; less consistent spread than bladder cancer.

## Follow-up

- **After open surgery or once endoscopically tumour free:** examination, cystoscopy and urine cytology (high grade only) 3-monthly for 1 year, 6-monthly in years 2 and 3, then yearly.
- **Contralateral upper tract:** yearly IVU or RGP.
- **After organ-sparing therapy:** ipsilateral endoscopy 6-monthly for the first several years, then yearly.
- **Metastatic evaluation (significant progression risk):** examination, chest X-ray and comprehensive metabolic panel with liver enzymes 3-monthly in year 1, 6-monthly in years 2 and 3, yearly in years 4 and 5. After 5 years, urothelium only.
- **CT or MRI abdomen and pelvis:** 6-monthly in years 1 and 2, yearly in years 3–5.
- **Bone scan:** only if alkaline phosphatase is raised or there is bone pain.

# Ch 141 · Contemporary Open Surgery of the Kidney

## Laparoscopic renal surgery: approaches and ports

- **Five approaches:** transperitoneal, retroperitoneal, hand-assisted, robotic, LESS/NOTES.

▸ **Transperitoneal**

- **Umbilical port:** camera.
- **Anterior axillary line port** (at or below the umbilicus): instruments, and passing sutures, bulldog clamps or staplers for the hilar vessels.
- **Third port:** 2 cm below the xiphoid; on the right, an extra port for liver retraction.
- All ports shift laterally in obese patients.

▸ **Retroperitoneal**

- **First port:** blunt-tip trocar after balloon dilatation, through a 15 mm transverse incision in the posterior axillary line.
- 5 mm trocar just off the tip of the 12th rib; 12 mm trocar posterior and superior to the camera port.

▸ **Hand-assisted**

- Surgeon's left hand goes in the hand port.

▸ **Robotic**

- **3-arm:** 4-5 ports (8 mm robotic ports, 12 mm camera port, 12 mm assistant trocar, plus an extra assistant trocar).
- **4-arm:** 5-6 ports in the same layout, robotic trocars shifted to avoid arm clashing.

▸ **LESS (laparoendoscopic single-site surgery)**

- All ports in one incision, often hidden in the umbilicus; virtually all renal operations have been done this way.
- Modified or full flank position; multiple low-profile trocars or purpose-built access devices; flexible scopes and instruments may be used.
- Most difficult of the minimally invasive techniques.

▸ **NOTES (natural orifice transluminal endoscopic surgery)**

- Limited human experience; hybrid NOTES with laparoscopic assistance, using the vagina for access and extraction at nephrectomy, has been described.
- Transvesical nephrectomy recently done in animals.

## Simple and radical laparoscopic nephrectomy

- Ureter identified but not divided early; the lower pole may be dissected.
- Renal artery divided with multiple titanium clips or a stapler.
- Intact removal through an extended trocar site or Pfannenstiel incision.

## Partial nephrectomy: ischaemia and hilar control

- **Hypothermia (cold ischaemia):** consider for complex tumours.
- **Warm ischaemia:** historical safe limit 30 minutes, but some data suggest even 90 minutes may be reasonable.
- **Selective arterial clamping:** needs extensive hilar dissection; longer operative time and more transfusion.
- **Artery-only clamping:** theoretically allows retrograde venous flow and some oxygen supply; controversial.
- **Parenchymal compression:** not feasible for central, hilar or large tumours, or in pure laparoscopy without special instruments.
- **Outcomes:** long-term overall survival similar for laparoscopic and open; 5-year cancer-specific survival after laparoscopic PN up to 100%.

## Open incisions

▸ **Midline**

- Xiphisternum to symphysis pubis.

- **Layers:** skin, subcutaneous tissue, linea alba, preperitoneal fat, peritoneum.
- **Indications:** include renovascular operations.
- **Advantage:** coexisting intraperitoneal pathology can be dealt with.

▸ **Anterior subcostal**

- Supine, right or left.
- Two fingerbreadths below the costal margin, from the midline to the tip of the 11th rib.
- **Layers:** skin, subcutaneous fascia, external oblique, internal oblique, transversus abdominis, transversalis fascia, retroperitoneum.
- Good exposure of the hilum.

▸ **Chevron**

- From the 11th rib on one side, across the midline, to the 11th rib on the other.

▸ **Subcostal flank**

- One fingerbreadth below and parallel to the 12th rib, from the lateral border of sacrospinalis to the lateral border of rectus.
- **Layers:** skin, subcutaneous tissue, latissimus dorsi and external oblique, serratus posterior inferior, lumbodorsal fascia, internal oblique and transversus abdominis, peritoneum.
- Sweeping the peritoneum medially gives a retroperitoneal approach; incising it gives a transperitoneal approach.
- Poor choice for radical nephrectomy of a large mass because of poor hilar access.

▸ **Supracostal**

- Same skin landmarks as the subcostal flank incision.
- **Divided:** posteriorly latissimus dorsi, serratus posterior inferior and intercostals; anteriorly external and internal obliques; then lumbodorsal fascia and transversus abdominis; peritoneum swept medially.
- Extrapleural and extraperitoneal; less morbidity than transcostal.

▸ **Dorsal lumbotomy**

- **Incision:** vertical, from the inferior border of the 12th rib to the iliac crest, along the lateral line of sacrospinalis.
- **Steps:** incise subcutaneous tissue; incise posterior layer of lumbodorsal fascia; retract sacrospinalis medially; divide costovertebral ligament; divide fused middle and anterior layers of lumbodorsal fascia; retract quadratus lumborum medially; incise transversalis fascia; enter paranephric space.
- **Suited to:** also thin adults needing bilateral renal surgery.
- **Disadvantages:** poor exposure, especially of the hilum; prone turning after induction, with its complications.
- **Poorly tolerated by:** older adults, cardiopulmonary patients, obese patients.

▸ **Thoracoabdominal**

- **Ideal for:** large or upper-pole masses, locally advanced tumours, tumours with IVC thrombus.
- **Position:** semioblique, bean bag supporting the flank.
- **Incision:** 8th, 9th, 10th or 11th intercostal space by tumour; from the lateral sacrospinalis through the intercostal space, extendable to the contralateral rectus abdominis.

- **Disadvantages:** pleural opening needing a chest tube; risk of phrenic nerve injury.

## Ch 142 · Laparoscopic and Robotic Surgery of the Kidney

### Laparoscopic and robotic partial nephrectomy

- **Robotic advantage:** eases tumour resection and renorrhaphy; helps surgeons without advanced laparoscopic skills.
- **Bedside assistant (robotic):** clamps the hilum, suctions and retracts, delivers and cuts sutures, places clips.
- **Alternative haemostasis:** lasers, microwave, ultrasonic and radiofrequency energy.
- **Collecting system repair:** longer operating time.
- **Hypothermia:** rarely used, as it is complex and most tumours do not need long warm ischaemia.

### Complications of laparoscopic renal surgery

▸ **Types**

- **Intraoperative:** tumour rupture.
- **Postoperative:** seroma, wound dehiscence, internal hernia, pulmonary complications, urinary retention or UTI, renal insufficiency, testicular infarction or ischaemia, epididymitis.

▸ **Prevention**

- After multiple previous surgeries, place trocars away from suspected adhesions; a retroperitoneal approach may be needed.
- Review imaging just before surgery and keep it in the operating room.

▸ **Bowel injury**

- If recognised at surgery: debride and close in two layers; consider a drain; nil by mouth until bowel function returns.

▸ **Spleen, liver and gallbladder injury**

- **Spleen:** topical haemostatic agents or argon beam coagulation; splenectomy rarely needed.
- **Liver:** haemostatic agents. **Gallbladder:** best treated by cholecystectomy.

▸ **Pulmonary complications**

- **Prevention:** stop smoking, spirometry. **Treatment:** spirometry, deep breathing, early ambulation.

▸ **Wound infection and hernia**

- **Infection:** swab culture, appropriate antibiotics, drainage of any abscess, dressings.
- **Incisional hernia:** under 2 cm may be closed anatomically; over 2 cm may need mesh.

## Ch 143 · Focal Therapy for Renal Tumours

### Focal therapy: indications and advantages

- **Indications:** include local recurrence after nephron-sparing surgery, and hereditary RCC with many multifocal lesions where repeated partial nephrectomy is cumbersome.
- **Advantages over NSS:** technically easier, less time-consuming, can be open, laparoscopic or percutaneous.

- **Future therapies needing evaluation:** laser interstitial thermal therapy, targeted embolisation with ablation.

## Choice of approach and outcomes

- **Laparoscopic preferred:** anterior tumours or those next to bowel, tumours near the collecting system, tumours without suitable imaging access.
- **Percutaneous ideal:** posterior tumour at least 0.5 cm from the renal pelvis or pelvicalyceal system.
- **Local recurrence-free outcome:** 80–90% overall after thermal ablation; 80–86% after cryoablation; 88–92% after RFA.
- **Repeat biopsy after ablation:** new enhancement, progressive enlargement, new nodules, satellite or port-site lesions.

## Ablative modalities

### ▸ Cryoablation

- Cooper and Lee built the first modern cryoprobe with pressurised liquid nitrogen (−196 °C); it could not monitor the extent of the cryolesion.
- Argon-gas probes reach down to −185.7 °C.

### ▸ Radiofrequency ablation (RFA)

- **Frequency:** typically 460 kHz.
- **Probes:** single-electrode monopolar; Christmas-tree-shaped device; LeVeen insulated monopolar probe with 12 deployable tines (RF antennas).
- Bipolar probes produce higher temperatures than monopolar.
- **Impedance-based systems:** start at 40–80 W, increase by 10 W per minute to a maximum of 130–200 W, until impedance reaches 200–500 ohms.
- Selective embolisation can help prevent the heat-sink effect.

### ▸ Microwave ablation (MWA)

- **Spectrum:** ultrasound about 10^6 Hz, microwave about 10^9 Hz, laser about 3 x 10^14 Hz.
- Energy carried by an insulated coaxial cable and delivered by an antenna or emitter.
- **Further advantages:** more uniform heating zone, less dependence on tissue properties, no need for tissue contact.

### ▸ High-intensity focused ultrasound (HIFU)

- Renal HIFU is extracorporeal, using ultrasound from 20 kHz to 20 MHz.
- **Transducers:** parabolic, non-invasive or minimally invasive; focusing geometric or electronic.
- **Focal point:** 3 by 3 by 11 mm, usually 3–4 cm from the transducer.
- **Advantages:** precision (primary); minimal or no risk of seeding, haemorrhage or urinary extravasation.
- **Disadvantages:** skin burns; poor targeting, improved by laparoscopic HIFU.
- **Uses:** destroying solid tumours of bone, kidney, prostate, liver, pancreas and brain; palliating bone metastases; ablating breast fibroadenoma; neuropathic pain, Parkinson disease, kidney stones, prostate cancer (Sonablate and Ablatherm devices).

▸ **Irreversible electroporation (IRE)**

- Single or multiple electrodes placed laparoscopically or percutaneously under CT or ultrasound.
- **Pathology:** no cell viability immediately; diffuse necrosis by day 7; chronic inflammation or fibrosis by day 14.

## Complications of renal ablation

- **Most common:** pain and paraesthesia at the probe insertion site (8%).
- **Post-ablation syndrome:** low-grade fever, pain, malaise, myalgias; self-limiting or settles with NSAIDs.
- Mild haematuria or clot colic.
- **Thermal injury:** genitofemoral nerve gives chronic groin pain; psoas gives impaired hip flexion; adrenal gives hypertensive crisis.

# Ch 144 · Treatment of Advanced Renal Cell Carcinoma

## Surgery in metastatic RCC (mRCC)

▸ **Cytoreductive nephrectomy (CN)**

- A third proposed reason for benefit: systemic agents, especially cytokines, fail to produce a meaningful response in the primary tumour in most patients.
- Role unclear in non-clear cell histology.
- CN for palliation of local symptoms does not always give clinical benefit.

▸ **CARMENA trial**

- **Design:** phase III, prospective, multicentre, open-label randomised; primary end point overall survival.
- **Arms:** 226 CN plus sunitinib and 224 sunitinib alone; 40 in the CN arm and 11 in the sunitinib arm did not receive sunitinib.
- **Safety populations:** 186 (CN) and 213 (sunitinib); deaths 165 and 161.
- **Follow-up:** 50.9 months.
- **Results:** no significant difference in response rate or progression-free survival.

▸ **Palliative resection of metastases**

- **Indications:** metastases in weight-bearing joints and bones, vertebral metastases with impending cord compression, solitary brain metastases.

## Immunotherapy and transplant

▸ **Interferon-alpha (IFN-alpha)**

- Earliest cytokine evaluated in mRCC.
- Combined with IL-2: high toxicity, no survival advantage.

▸ **Interleukin-2 (IL-2)**

- Given intravenously or subcutaneously; high dose is US FDA approved for mRCC.
- **Additional side effects:** third-space fluid retention, respiratory compromise.

▸ **Allogeneic haematopoietic stem cell transplant**

- Replaces the recipient's immune and haematopoietic systems with those of a healthy HLA-compatible donor; acts by a graft-versus-tumour (allogeneic antitumour) effect.

## Targeted therapy

▸ **Molecular mechanism**

- The VHL complex (cullin-2, elongin B, elongin C, Rbx1 and an E2 enzyme) binds the beta-domain of the VHL protein; a mutant alpha-domain disrupts the complex, so HIF accumulates.
- HIF-driven TGF-alpha acts on the EGF receptor (EGFR).

▸ **Bevacizumab**

- Humanised antibody (more than 90% human) blocking angiogenesis.
- **Dose:** intravenous every 14 days.
- **Side effects:** bleeding (main); also fatigue, infection, bowel perforation, headache, nose bleed, allergy, nasal septal perforation.
- **Other uses:** eye diseases; cancers of lung, colon and rectum, breast, brain, ovary.
- **AVOREN:** no benefit of bevacizumab plus IFN-alpha in MSKCC poor-risk patients.
- Bevacizumab plus IFN-alpha can be used after failed standard first-line therapy, and in good- and intermediate-risk patients only.

▸ **Sorafenib**

- **Side effects:** also hypertension, fatigue, diarrhoea.

▸ **Sunitinib**

- **Action:** inhibits vascular endothelial development and proliferation, and vascular pericyte function.
- **Side effects:** also diarrhoea, rash, asthenia, bone marrow suppression.
- **Alternative schedule:** 2 weeks on, 1 week off.
- **EFFECT and STAR trials:** better quality of life than IFN-alpha; mostly mild-to-moderate side effects manageable with close monitoring and proactive management; superior PFS and OS compared with IFN-alpha.
- **Second line after cytokines:** 30-40% partial response.

▸ **Pazopanib**

- **Dose:** 800 mg orally once daily.

▸ **Axitinib**

- Second-generation inhibitor. **Dose:** 5 mg twice daily orally, increased to 7 mg twice daily.

▸ **Other VEGF-pathway agents**

- Nintedanib, dovitinib.

▸ **mTOR inhibitors**

- mTOR is encoded by FRAP-1 and regulates translation and stability of HIF-1alpha, driving growth, multiplication and angiogenesis.
- **Temsirolimus side effects:** also fatigue, rash, hypophosphataemia.

## Sequencing, chemotherapy and non-clear cell RCC

- **After progression on mTOR inhibition:** VEGF-targeted therapy.

- **After both VEGF and mTOR inhibition:** sorafenib.
- **Sarcomatoid or rapidly progressive mRCC:** gemcitabine plus doxorubicin.
- **Papillary tumours:** sunitinib is better than everolimus.

## Other topics in this book

## Autosomal dominant polycystic kidney disease (ADPKD)

### ▸ Clinical features

- **Kidneys:** bilaterally enlarged, reniform, diffusely cystic through cortex and medulla.
- **GI symptoms:** from renomegaly or colonic diverticula.
- **Anaemia:** unusual in its renal failure, because erythropoietin is raised.
- **Tumours:** RCC is not commoner, but renal adenoma incidence is high.
- **Hepatic cysts:** usually asymptomatic; commoner in females (oestrogen effect).
- **Berry aneurysm:** 9% of those with subarachnoid haemorrhage die.
- **Risk factors for earlier ESRD:** also first haematuria before 30 years, hypertension before 35 years, hyperlipidaemia, sickle cell trait.

### ▸ Evaluation

- A 3-generation family history is a must.
- **Ultrasound criteria:** 100% sensitive at age 30 or more and in younger PKD1 individuals; only 67% sensitive in PKD2 individuals under 30.
- CT and MRI are superior to ultrasound.
- **Presumptive diagnosis without family history:** bilateral renal cysts plus at least 2 of bilateral renal enlargement, 3 or more hepatic cysts, cerebral artery aneurysm, or a solitary cyst of the arachnoid, pineal gland or spleen.

### ▸ Aneurysm screening

- Routine MRA or CTA for asymptomatic aneurysms is not recommended.
- **Indications:** family history of aneurysm or SAH, previous rupture, preparation for major surgery, high-risk occupation (for example airline pilot), marked patient anxiety.

### ▸ Treatment and emerging therapy

- No cure; aims are to lessen complications and delay ESRD.
- **UTI:** treat effectively, especially in women; chloramphenicol is another lipophilic antibiotic that penetrates cyst walls.
- **Somatostatin:** like tolvaptan, lowers vasopressin-induced cAMP.
- **mTOR inhibitors:** reduce cyst burden.
- **Emerging:** EGF receptor (ErbB1 and ErbB2) inhibitors, Src kinase inhibitors, CDK inhibitors.

## PET-CT in RCC

### ▸ FDG PET-CT

- Indicated in patients at high risk of metastasis with equivocal CT or MRI; good specificity but suboptimal sensitivity for staging.

▸ **Immuno-PET/CT with 124I-cG250**

- cG250 is the antibody against CA-IX; iodine-124 is a positron-emitting radionuclide. The complex binds CA-IX on clear cell RCC and is detected by PET/CT.
- Promising for diagnosis and staging of clear cell RCC: sensitivity 96%, specificity 100%.

## UTUC: additional points

▸ **Positive upper tract cytology or CIS**

- If cytology stays positive for more than 1 year after adequate bladder treatment, evaluate outside the bladder.
- Upper-tract CIS is often multifocal, so segmental resection fails. Keep nephroureterectomy for radiographic or endoscopic evidence of more than surface disease; re-evaluate frequently.

▸ **Adjuvant therapy**

- **PCN-tract brachytherapy** after percutaneous organ-sparing therapy: uses iridium wire; risk of cutaneous fistula, which then needs RNU.
- **Adjuvant radiotherapy:** after complete excision, reduces local relapse after radical surgery for T3, T4 or N+ disease; without chemotherapy it does not protect against distant failure.

▸ **Lymph node involvement**

- 12–25%, rising with stage and grade.

▸ **Follow-up**

- Contralateral imaging yearly by IVU or retrograde pyelogram (RGP); metastatic and cross-sectional schedule as in Ch 139.

# 14 Adrenal & Retroperitoneum

## Ch 145 · Surgical and Radiographic Anatomy of the Adrenals

### Gross anatomy and relations

- **Colour:** the medulla is brown; the cortex is yellow-orange.
- **Left adrenal:** the greater curvature of the stomach is an anterior relation.

### Embryology

▸ **Cortex**

- Over the 2 weeks after condensation, coelomic epithelial cells proliferate and migrate to the cranial end of the mesonephros to form the adrenogenital ridge (intermediate mesoderm).
  - Cephalic portion: adrenal cortex.
  - Caudal portion: gonads.
- At 8–10 weeks the cortex differentiates morphologically into a fetal zone and a definitive zone; the definitive zone replaces the fetal zone after birth.
- Zona glomerulosa and zona fasciculata develop within the definitive cortex over the first year, under the influence of ACTH.
- Zona reticularis reaches adult volume only at 6–8 years of age. This explains the relatively late importance of adrenal sex steroid production.

▸ **Medulla**

- Neural crest cells migrate into the medial aspect of the fetal adrenal cortex by the 9th week and reach a central location by the 18th week.

▸ **Neonatal adrenal**

- Twice the adult weight at birth, but not completely developed.

### Anomalies

- **Unilateral adrenal agenesis:** rare.
- **Adrenal rests, surgical importance:** compensatory hypertrophy after adrenalectomy, inadvertent excision during unrelated surgery, and neoplasia.

### Histology

| Zone | Cells | Product |
|---|---|---|
| Zona glomerulosa | Small polyhedral cells, vacuolated appearance (less cytoplasm, more lipid), in ovoid clusters | Aldosterone, synthesised in mitochondria and smooth endoplasmic reticulum, 100–150 µg/day |
| Zona fasciculata | Large polyhedral cells in cords | |
| Zona reticularis | Smaller cells in anastomosing cords | More than 20 mg/day of DHEA, DHEA sulphate and androstenedione |

## Vasculature, lymphatics and nerve supply

▸ **Blood supply**

- Arterial branches penetrate the capsule to form a subcapsular plexus.
- **Medulla, dual supply:** arterial blood directly from medullary arterioles, and venous blood from the cortical sinusoidal capillaries.
- Catecholamines from the medulla reach the circulation through fenestrated capillaries.
- The overlapping arterial and venous anatomy complicates partial adrenalectomy.

▸ **Lymphatics**

- **Right:** paracaval nodes. **Left:** para-aortic nodes.

▸ **Nerve supply**

- **Medulla:** preganglionic sympathetic fibres form a plexus at the capsule and traverse the cortex to reach the medulla.
- **Cortex:** postganglionic fibres from the splanchnic ganglia.
- **Parasympathetic** innervation of cortex and medulla is not well defined.

## Radiological points

▸ **CT**

- Most widely used adrenal imaging. Perinephric fat allows clear visualisation with outstanding resolution.
- Normal limbs are usually thinner than the adjacent diaphragmatic crura.

▸ **MRI**

- Slightly inferior to CT in spatial resolution.
- **Technique:** transverse plane most commonly used, slice thickness 3 mm or less.
- **T1:** uniform intermediate signal, slightly lower than liver and renal cortex.

▸ **Ultrasound**

- Normal adrenal is hypoechoic but hard to distinguish from surrounding structures because of retroperitoneal fat.
- Right-sided lesions are visualised better.

# Ch 146 · Pathophysiology, Evaluation and Medical Management of Adrenal Disorders

## Adrenal physiology

▸ **Aldosterone**

- **Synthesis:** aldosterone synthase converts corticosterone to 18-hydroxycorticosterone; 18-oxidase converts this to aldosterone.
- **Regulators:** angiotensin II and potassium; ACTH is an additional stimulator.
- **Main inhibitor:** atrial natriuretic peptide.
- **Distal nephron action:** sodium and chloride reabsorption, hydrogen and potassium excretion, and free water reabsorption. Sodium concentration stays the same while total body water volume changes.

▸ **Glucocorticoid actions**

- Enhance skeletal and cardiac muscle contraction.

- Anti-inflammatory and immunosuppressive.
- Inhibit bone formation and collagen synthesis; cause protein catabolism.
- Maintain normal GFR.
- Increase vascular contractility and decrease vascular permeability.

▸ **Sex steroids**

- Androstenedione is converted to testosterone (then DHT) and to oestrone. Testosterone is converted to oestradiol.

▸ **Catecholamine metabolism**

- **PNMT:** unique to the adrenal medulla, although the brain and the organ of Zuckerkandl also express it.
- **COMT:** in the adrenal medulla, liver and kidney. Medullary COMT: 90% (metanephrine) and 20% (normetanephrine).
- Normetanephrine and metanephrine are conjugated to sulphated normetanephrine and metanephrine sulphate; together these make up total metanephrines.

## Adrenal insufficiency (Addison disease)

▸ **Prevalence and causes**

- **Prevalence:** 39–60 per million.
- **Commonest primary cause:** autoimmune adrenalitis in the Western world; tuberculosis in developing countries.
- **Other primary causes:**
  - Autoimmune polyendocrine syndrome.
  - Cytomegalovirus (CMV) adrenalitis in HIV.
  - Fungal infection.
  - Infiltration: amyloidosis, sarcoidosis, haemochromatosis.
- **Other secondary causes:** pituitary or hypothalamic radiation, autoimmune disease, trauma.

▸ **Symptoms and signs**

- Features are mostly non-specific and constitutional.
- **Symptoms:** dizziness, fatigue and lack of energy, anorexia and weight loss, gastric pain, nausea and vomiting, joint pain and myalgias, dry itchy skin, loss of libido in women.
- **Signs:** fever, low blood pressure, postural hypotension.
- **Laboratory:** hypoglycaemia, hypercalcaemia, anaemia, lymphocytosis, eosinophilia.
- Women may lose axillary or pubic hair; children may show absent adrenarche or pubarche.
- **Pigmentation (primary only):** melanocortin receptor stimulation by high serum ACTH.

▸ **Adrenal crisis**

- Life-threatening; often preceded by hypotension unresponsive to fluids.
- **Presentation:** abdominal pain, nausea, vomiting, fever, hypotension.
- Children can have hypoglycaemic seizures.

▸ **Diagnosis**

- May be made clinically, then secured by tests.
- Abnormal aldosterone and renin levels occur in primary insufficiency only.
- **Confirmatory test:** corticotropin (ACTH stimulation) test.

- Secondary causes need detailed hypothalamic–pituitary–adrenal axis testing by an endocrinologist.

▸ **Treatment**

- **Cortisone acetate:** alternative to hydrocortisone; the remainder of the daily dose is given in the evening to follow the circadian rhythm.
- **Adrenal androgen replacement:** usually limited to patients with constitutional symptoms despite glucocorticoid and mineralocorticoid replacement.

▸ **Peri-operative steroid dosing**

- The usual steroid dose is sufficient, but surgeons continue to give stress-dose steroids (controversial).
- **Major surgery:** usual glucocorticoid dose within 2 hours before surgery.

▸ **Prognosis**

- **Primary:** effect on life expectancy not established; significant impact on quality of life.
- **Secondary:** established cause of premature death.

## Primary aldosteronism (Conn syndrome)

▸ **History and epidemiology**

- First described by Dr Jerome Conn in 1955.
- Most common form of secondary hypertension.

▸ **Mineralocorticoid escape**

- Escape from the sodium-retaining effects of excess aldosterone.
- Mediated by pressure natriuresis and atrial natriuretic peptide (ANP).
- Limits volume expansion to 1.5 kg or less.

▸ **Subtypes**

- **Bilateral (idiopathic) hyperplasia:** mostly micronodular.
- Hypertension and hypokalaemia are more severe with an aldosterone-producing adenoma than with bilateral hyperplasia.

▸ **Familial hyperaldosteronism (FH)**

| Type | Features |
|---|---|
| FH type I | Autosomal dominant. Aldosterone circadian rhythm governed by ACTH instead of the RAAS. Family history of early-onset hypertension or cerebrovascular accident (stroke). |
| FH type II | Autosomal dominant. Adenoma or hyperplasia indistinguishable from sporadic cases. |
| FH type III | Bilateral adrenal hyperplasia, refractory hypertension, severe hypokalaemia, overproduction of hybrid steroids. |

▸ **Clinical features**

- Usually diagnosed in the third to sixth decade, during evaluation for refractory hypertension.
- Only a minority have hypokalaemic symptoms: headache, palpitations, weakness.
- **Cardiac:** left ventricular hypertrophy (LVH), prolonged QT interval.
- **Vascular risk:** higher risk of myocardial infarction, atrial fibrillation and stroke; appropriate treatment decreases it.

- **Renal:** proteinuria.
- **Metabolic:** higher rate of type 2 diabetes mellitus.

▸ **Screening**

- **Further indications:** when other secondary causes of hypertension (such as phaeochromocytoma or renovascular hypertension) are being considered; unexplained hypokalaemia, spontaneous or diuretic-induced.
- **Aldosterone-to-renin ratio (ARR):** sensitivity 66–100%, specificity 61–96%.
- Plasma renin activity is time-consuming to measure; plasma renin concentration has been proposed as an alternative.

▸ **Confirmatory testing**

- Mandatory, because aldosterone and plasma renin activity vary with day-to-day oscillation, posture, diet and antihypertensive drugs.
- Only 50–87% of screen-positive patients have a positive confirmatory test.
- **Fludrocortisone suppression test:** once the gold standard, now out of favour.
  - Fludrocortisone 0.1 mg 6-hourly with NaCl 2 g 8-hourly for 4 days.
  - Plasma aldosterone concentration (PAC) then measured upright; PAC above 6 ng/dL is diagnostic.
- **Oral sodium loading test:** supplemental NaCl to ensure at least 12.8 g of sodium per day; measure 24-hour urine aldosterone, sodium and creatinine. Contraindicated in cardiac and renal disease.
- **IV saline infusion test:** done in the morning after an overnight fast.
- **Furosemide upright test and ACTH stimulation test:** not widely used.

▸ **Subtype differentiation: CT and genetics**

- **CT appearance of an aldosterone-producing adenoma:** unilateral, low density, non-enhancing (below 10 HU), average size 1.6–1.8 cm, normal-appearing opposite gland.
- CT alone is acceptable below age 40, because incidental non-functioning adenomas are rare in this group.
- **Genetic screening:** not done for all because familial hyperaldosteronism is rare. Considered with a family history of primary aldosteronism, onset before 20 years, or a family history of stroke at a young age.

▸ **Adrenal venous sampling (AVS)**

- **Preparation:** morning, after an overnight fast and 1 hour of recumbency.
- **Technique:** percutaneous femoral vein access; catheter tips sample the right adrenal vein, left adrenal vein and IVC.
- **Cut-offs:** selectivity (adrenal vein to IVC cortisol) above 1.1:1 (lower) or above 5:1 (higher), and lateralisation 2:1 to 4:1, depending on whether cosyntropin (ACTH) stimulation is used.
- **Non-selective sample:** discard the data and consider repeat AVS, with ACTH stimulation if not used initially.

▸ **Inconclusive AVS: alternatives**

- **Options:** nuclear scintigraphy (functional and anatomical information), postural stimulation testing, measurement of cortisol metabolites.

▸ **NP-59 scintigraphy**

- **Tracer:** I-131 6β-iodomethyl-norcholesterol, a cholesterol-based cortisol analogue that labels adrenocortical cells.
- **Thyroid protection:** saturated potassium iodide (Lugol's solution) before injection, against uptake of free I-131.
- **ACTH suppression:** dexamethasone 1 mg 6-hourly for 7 days.
- **Timing:** imaging begins on day 4 after injection and may continue through day 10.
- **Interpretation:** unilateral uptake within 5 days suggests adenoma; bilateral uptake suggests idiopathic hyperplasia.
- Detection is proportional to adenoma size and function; SPECT-CT improves sensitivity for smaller adenomas.

▸ **Posture stimulation test**

- Aldosterone measured after overnight recumbency and after 4 hours upright.
- **Normal:** upright position raises angiotensin and aldosterone 2–4 times above the recumbent level.
- **Idiopathic hyperplasia:** aldosterone rises by at least 33% over baseline on standing.
- **Aldosterone-producing adenoma or FH type I:** no change with posture.
- **Limits:** no information on location; 30–50% of adenomas show a rise in PAC, and 20% of idiopathic hyperplasias show no rise.

▸ **18-hydroxycorticosterone (18-OHB)**

- Immediate precursor of aldosterone.
- Above 100 ng/dL suggests adenoma; below 100 ng/dL suggests hyperplasia.
- Accuracy below 80%; no information on side (right versus left).

▸ **Treatment and outcome**

- **Predictors of persistent hypertension after adrenalectomy** also include: 2 or more antihypertensives before surgery, a first-degree relative with hypertension, and the degree of cardiovascular remodelling and renal insufficiency.
- **Spironolactone:** can cause menstrual disturbances.
- Even on mineralocorticoid receptor antagonists, patients still need other oral antihypertensives, regular exercise with weight loss, and a low-sodium diet.
- **FH type I:** add a mineralocorticoid receptor antagonist if BP is not controlled with glucocorticoid alone or if iatrogenic Cushing syndrome develops.

## Cushing syndrome

▸ **Definition and HPA axis**

- **Incidence:** rare, 2–5 per million per year.
- Stress and circadian regulation drive the hypothalamus to release CRH; oxytocin and vasopressin also stimulate the pituitary but play a lesser role.
- The pituitary makes ACTH (corticotropin) by cleaving POMC.
- Cortisol feeds back negatively on both pituitary and hypothalamus.

▸ **Cushing disease (pituitary)**

- Named after Harvey Cushing, a neurosurgeon, who described it in 1932.

- **Commonest cause:** ACTH-producing microadenoma.
- **Rare causes:** macroadenoma, pituitary carcinoma, hyperplasia of ACTH-producing cells.
- Two thirds of patients are female.

▸ **Ectopic ACTH and CRH**

- **Ectopic ACTH tumours:** mostly malignant. Include islet cell tumours, thymic tumours, medullary carcinoma of the thyroid, bronchial carcinoid, small cell lung cancer, phaeochromocytoma.
- **Ectopic CRH:** rare ACTH-dependent cause (less than 1%), most often bronchial carcinoma.

▸ **ACTH-independent syndromes**

- **AIMAH:** less than 1% of Cushing syndrome.
- **McCune–Albright syndrome:** AIMAH, fibrous dysplasia, skin problems, other endocrine abnormalities.
- **Carney complex:** autosomal dominant; PPNAD, spotty skin and mucous membrane lesions, multiple tumours including Sertoli cell tumours.

▸ **Clinical features by frequency**

| Frequency | Features |
|---|---|
| 90–100% | Decreased libido |
| 70–90% | Menstrual disturbance |
| 50–70% | Muscle weakness, buffalo hump, easy bruising, diabetes or glucose intolerance, osteoporosis, depression |
| 20–50% | Headache, backache, limb oedema, recurrent infections, nephrolithiasis |
| 0–20% | Acne, alopecia |

- **Stones:** definitive treatment lowers the stone risk but not to that of the general population.
- **Subclinical Cushing syndrome, other names:** pre-Cushing, non-Cushingoid Cushing, preclinical Cushing syndrome.

▸ **Cortisol excess without Cushing syndrome**

- Other causes: pregnancy, glucocorticoid resistance, poorly controlled diabetes, malnutrition, anorexia, intense chronic exercise.

▸ **Establishing the diagnosis and cause**

- **Second-line tests:** 2-day low-dose DST (less practical), midnight plasma cortisol.
- **No longer recommended:** random serum cortisol, plasma ACTH level, urinary 17-ketosteroids, insulin tolerance test, loperamide test.
- **Low ACTH, normal abdominal imaging:** consider PPNAD or exogenous steroids.
- **Low ACTH, bilateral adrenal lesions:** adrenal vein sampling to lateralise.
- **ACTH-dependent:** imaging struggles to localise pituitary microadenomas and ectopic tumours, so IPSS is needed.

▸ **Exogenous Cushing syndrome**

- Taper the steroid gradually when possible.
- **Steroid withdrawal syndrome:** the patient cannot tolerate dose reduction even though HPA axis tests appear normal.

▸ **Trans-sphenoidal surgery**

- **Relapse:** about 25%, so long-term follow-up is needed.
- **Macroadenoma:** 15% or fewer cured.
- Severe Addisonian state is common; glucocorticoid replacement needed for the year after surgery.
- **Hypopituitarism:** under 5% to 50%, a wide range.
- **Re-operation:** challenging, variable success, higher risk of hypopituitarism.
- **Salvage pituitary radiation:** ACTH falls slowly; hypopituitarism may follow.

▸ **Bilateral adrenalectomy**

- **Advantages:** no risk of hypopituitarism; very high success with rapid resolution of hypercortisolism.
- Residual functioning adrenal tissue rarely remains.
- **Nelson syndrome:** can rarely cause raised intracranial pressure. Prophylactic pituitary radiotherapy to prevent it is controversial.

▸ **Ectopic ACTH syndrome**

- Tumour resection is the single best treatment but is possible in only about 10%.
- In about 35% the primary is unresectable or not found; bilateral adrenalectomy with lifelong replacement is then best.

▸ **ACTH-independent disease and medical treatment**

- **Cortisol-secreting adrenal mass:** adrenalectomy or partial adrenalectomy.
- **Medical treatment:** bridge to surgery or when surgery is not possible; best left to the endocrinologist.
- **Aminoglutethimide:** another steroid synthesis blocker.
- **Abiraterone** (17-alpha-hydroxylase inhibitor): possible future treatment.

▸ **Prognosis**

- Life expectancy is shorter than normal; treatment reduces but does not remove this risk.

## Adrenal incidentaloma

▸ **Frequency**

- Rises with age: less than 0.5% in the 20s to up to 7% in the 70s.
- **Median diameter:** 3 cm.

▸ **Ultrasonography**

- Suboptimal for detecting and characterising adrenal masses.
- **Right:** IVC and liver give a better window.
- **Left:** less sensitive; lesions are overlooked or mistaken for splenic, pancreatic, para-aortic lymphatic or gastric anatomy.

▸ **CT**

- **Unenhanced below 10 HU:** sensitivity 71%, specificity 98% for adenoma.
- **Enhanced CT:** about 1 minute after bolus, 5 mm images. An enhancing rim with irregular margins suggests malignancy.
- **15-minute delayed CT:** rapid washout in lipid-poor and also lipid-rich adenomas.

▸ **MRI**

- **Chemical-shift MRI:** uses interference between fat and water signals. Signal loss on out-of-phase (opposed-phase) versus in-phase images shows intracellular lipid and identifies an adenoma.
- Gadolinium-enhanced washout studies are not used.

▸ **Functional imaging**

- Not used routinely, because cross-sectional imaging plus biochemistry already gives high accuracy.
- **NP-59** (radiolabelled cholesterol analogue): taken up by adrenocortical cells; distinguishes adenomas from other lesions, but time-consuming and not currently approved.
- **11C-metomidate:** inhibits 11β-hydroxylase and aldosterone synthase; preferentially taken up by adrenocortical cells; cannot separate benign lesions from adrenocortical carcinoma.
- **131I-MIBG:** limited role.
- **Alternatives to MIBG:** 18F-FDG PET, 18F-dopamine PET, 111In-octreotide. Routine dopamine PET and octreotide use not evaluated.
- **18F-FDG PET:** sensitivity 93–100%, specificity 80–100% for benign versus malignant; reserved for inconclusive imaging and clinical data.

▸ **Follow-up of a non-functional incidentaloma**

- Re-image at 6, 12 and 24 months; repeat functional work-up yearly for 4 years.
- **Indications for adrenalectomy:** growth by more than 1 cm or a positive repeat functional work-up.

## Hormone hypersecretion testing

▸ **Cortisol: low-dose dexamethasone suppression test (LD-DST)**

- The three first-line tests (1-mg DST, late-night salivary cortisol, 24-h UFC) have roughly equivalent accuracy.
- **Cut-off below 1.8 µg/dL:** specificity about 80%.
- **Other CYP3A4 inducers** (faster dexamethasone metabolism, false-positive DST): phenobarbitone, primidone, ethosuximide, pioglitazone, rifapentine.
- **Other CYP3A4 inhibitors** (slower dexamethasone metabolism): cimetidine, fluoxetine, aprepitant, fosaprepitant.
- **Mitotane:** like oestrogen, raises cortisol-binding globulin and total cortisol, giving a false-positive DST.

▸ **Late-night salivary cortisol**

- Tests loss of diurnal variation (inability to suppress cortisol at night); abnormal even in mild Cushing syndrome.
- Reflects serum cortisol in near real time.
- **Normal:** below 145 ng/dL by liquid chromatography–tandem mass spectrometry.
- **Accuracy:** sensitivity and specificity above 90%.
- **False positives:** high in depression, altered sleep patterns, chronic illness.
- **Patient instructions:** no tobacco on the day; no eating or tooth-brushing within 30 minutes of collection.

### ▸ 24-hour urinary free cortisol (UFC)

- Endocrine Society guideline (2008): low-dose DST or late-night salivary cortisol has supplanted 24-h UFC for initial screening.
- **Normal cut-off:** 50–100 µg per 24 hours.
- **Pregnancy (second and third trimesters):** cut-offs 2 to 3 times the normal threshold.
- **Falsely increased by:** carbamazepine, fenofibrate, carbenoxolone.

### ▸ Catecholamines

- **Plasma free metanephrines, upper limits:** metanephrine 61 ng/L, normetanephrine 112 ng/L.
- **24-h urinary fractionated metanephrines, positive:** metanephrine above 1531 nmol/day, normetanephrine above 4001 nmol/day, or total above 1563 nmol/day. Specificity about 91%.

## Adrenal biopsy

- **Commonest complication:** bleeding, which can complicate or prevent later laparoscopic resection.
- **Others:** pneumothorax, haemothorax.

## Benign adrenal lesions

### ▸ Adrenal adenoma

- **Gross:** yellow, from lipid content. Usually smooth, homogeneous, smaller than 4 cm.
- About 90% are benign and about 90% non-functional. Only about 2% of non-functioning adenomas become active on follow-up.
- **Histology:** hard to tell from malignancy; Weiss criteria or immunohistochemistry used.
- **PET:** no significant FDG uptake.
- **Cortisol-secreting:** normalising cortisol reduces but does not eliminate cardiovascular risk.
- **Aldosterone-secreting:** aldosterone usually normalises within 3 months of removal; baseline essential hypertension may still need medication.

### ▸ Adrenal oncocytoma

- Extremely rare, about 50 cases reported; 2 reported in pregnancy.
- Oncocytomas also arise in kidney, salivary glands and other endocrine organs.
- **Ratios:** female to male 2.5:1; left to right 3.5:1.
- **Function:** roughly 10% are functional, secreting cortisol or sex steroids.
- **Size:** up to 20 cm.
- About 23% are of indeterminate malignant potential.
- Nearly always diagnosed on histopathology after resection; evaluated like other adrenal masses.
- Unlike renal oncocytoma, a central stellate scar is not regularly reported.
- **Prognosis:** uncertain but probably benign; little evidence.

### ▸ Myelolipoma

- **Incidence:** below 0.1%. Men and women equally.
- **Site:** 85% adrenal, about 15% extra-adrenal (presacral, thoracic, retroperitoneal, pelvic, renal, hepatic, gastric).

- **Origin:** clonal stem-cell proliferation. One hypothesis is an adrenal stress response; other theories unproven.
- Haematopoietic elements are relatively normal trilineage cells but do not reach the circulation, because there are no capillary venous sinuses.
- Sometimes associated with a functional adenoma, most commonly cortisol-secreting.
- **CT:** higher-density myeloid components enhance. About a quarter calcify; many show haemorrhage.
- **MRI:** macroscopic fat can also be identified.
- **Differential:** upper-pole renal angiomyolipoma, liposarcoma.
- Some experts biopsy if the diagnosis is in doubt.
- **Growth:** variable; some decrease in size.
- **Bleeding:** spontaneous haemorrhage extremely rare; rupture reported in lesions larger than 10 cm.

▸ **Ganglioneuroma**

- Benign neuroectodermal neoplasm of ganglion cells and Schwann cells.
- Malignant transformation is rare. Can grow extremely large.
- VIP secretion with diarrhoea reported.
- **CT:** lower attenuation than muscle on plain and contrast CT. A metabolically silent mass below 40 HU on non-contrast CT with stippled calcification suggests ganglioneuroma.
- Natural history when unresected is not known.

▸ **Adrenal cysts**

- **Frequency:** 1–22% of incidental adrenal lesions.
- **Age and sex:** any age, even prenatal; commoner in women, peak 3rd to 6th decades.
- **Bilateral:** 8–10%.
- **Associations:** polycystic kidney disease, Beckwith–Wiedemann syndrome, Klippel–Trenaunay–Weber syndrome.
- **Types:**
  - Pseudocyst: no cellular lining; from previous haemorrhage or infarction.
  - Endothelial: no proliferating epithelium; lymphangiomatous or angiomatous.
  - Epithelial: true epithelial lining; glandular cysts, embryonal cysts, cystic adenomas.
  - Parasitic: rare; Echinococcus.
- A benign cyst is hard to distinguish from a cystic neoplasm.
- **CT:** well-defined, sharply marginated, fluid attenuation, no enhancement. Intracystic debris, rim-like calcification and haemorrhage raise attenuation.
- **MRI:** T1 hypointense, T2 hyperintense.
- **Management:** imaging cannot rule out malignancy, so excise or aspirate. Aspiration can be diagnostic or therapeutic.
- **Aspirate tests:** Gram stain, cytology, cortisol, amylase (for left-sided lesions).
- **Surveillance:** with caution, as malignancy is possible. Small, asymptomatic, thin-walled cysts with homogeneous near-water attenuation can be observed.
- **Prognosis:** depends on histology.

## Adrenocortical carcinoma (ACC)

### ▸ Malignant adrenal lesions

- Adrenocortical carcinoma, metastasis, malignant phaeochromocytoma, neuroblastoma.

### ▸ Epidemiology and genetics

- Sporadic ACC is usually unilateral; 2–6% are bilateral and hereditary.
- McCune–Albright syndrome is also a syndromic association.
- **TP53 loss** (as in Li–Fraumeni): 20–33% of sporadic ACC.
- **IGF-2 overexpression** (as in Beckwith–Wiedemann): about 90%.
- **Wnt/β-catenin constitutive activation:** about 30%.

### ▸ Clinical presentation

- A single ACC commonly secretes several hormones; functional status and output may change as it grows.
- **Virilisation alone:** excess 17-ketosteroids and DHEA; male-pattern baldness, hirsutism, dysmenorrhoea.
- **Feminisation in men:** testicular atrophy, gynaecomastia, from peripheral conversion of testosterone to oestrogen.
- **Hyperaldosteronism:** rarest functional presentation; hypertension with hypokalaemia.
- **Non-functional (mass-effect or systemic):** back pain, nausea and vomiting, abdominal fullness, constitutional symptoms.

### ▸ Imaging

- **CT:** also irregular enhancement and necrotic areas with cystic degeneration.
- Local extension and lymphadenopathy are ominous, suggesting locally advanced or metastatic disease.
- **Commonest metastatic sites:** lung, liver.
- **MRI:** T1 isointense to liver and spleen; T2 intermediate to increased; marked gadolinium uptake.
- **18F-FDG PET:** avid, but rarely needed.

### ▸ Weiss criteria

- High nuclear grade corresponds to Fuhrman grade 3 or 4.
- **Score 3 or more:** sensitivity 100%, specificity 96% for malignancy.
- Use with caution, with alternative criteria, in paediatric ACC and oncocytic tumours.

### ▸ Management of stage IV disease

- **Unresectable:** confirm by needle biopsy, start mitotane monotherapy, follow up every 2 months. Stable or regressing disease continues mitotane; progression moves to EDP chemotherapy.
- **Resectable (oligometastatic):** debulking surgery or metastasectomy plus mitotane.

### ▸ Surgery and follow-up

- Local and systemic adjuvant therapy is often given despite no clear survival evidence.
- Cytoreduction or metastasectomy may not improve survival but reduces hormonal symptoms and facilitates further therapy.
- **Follow-up imaging:** CT; PET has no established role.

▸ **Radiotherapy**

- Limited role for the primary.
- **Adjuvant tumour-bed radiotherapy:** local recurrence 14% versus 80% without; no significant difference in disease-free or overall survival.
- Best for positive or indeterminate margins without metastasis.

▸ **Mitotane**

- **Action:** direct cytotoxicity to the adrenal cortex, oxidative damage through free radicals, inhibition of steroid-synthesis enzymes.
- **Other side effects:** lethargy, depression, feminisation, rash, transaminitis, dyslipidaemia, thrombocytopenia.
- **Response rate alone:** 15–35%.
- Controls hormonal symptoms only after a delay, so adrenostatic drugs lower cortisol in symptomatic patients: aminoglutethimide, ketoconazole, metyrapone, etomidate.

▸ **Prognosis**

- **Additional adverse factors:** tumour necrosis, atypical mitotic figures.

## Adrenal metastasis

- **Commonest primaries:** melanoma, lung cancer.
- **Other primaries:** RCC, breast, MCT, contralateral ACC, gastrointestinal, prostate, cervical, basal cell, pancreatic, cholangiocarcinoma, urothelial, squamous cell carcinoma, seminoma, thymoma, CML.
- Bilateral metastases larger than 4 cm can cause adrenal insufficiency.
- **CT:** well circumscribed, homogeneous, above 10 HU on non-contrast CT, no significant washout.

## Phaeochromocytoma

▸ **Epidemiology and paraganglia**

- Only about 0.5% of all hypertension.
- **Paraganglia:** chromaffin-producing neural crest tissue paralleling the sympathetic and parasympathetic ganglia, in the head, neck, thorax, abdomen and pelvis.

▸ **Pathophysiology**

- Tumours differ in enzyme content and in ability to metabolise their own catecholamines. This gives great variability in amount and ratio of catecholamines, explaining their heterogeneous behaviour.
- **Malignancy:** pathology, local invasion and immunohistochemistry (including Ki-67) are of limited value; only clinical metastasis defines it.

▸ **Clinical characteristics**

- **Rule of 10s:** also includes 10% bilateral and 10% paediatric.
- **Malignancy:** under 5% of adrenal tumours; about one-third of extra-adrenal tumours.
- **Age:** familial tumours present younger; sporadic tumours in the 4th to 5th decade (30s and 40s).
- **Right-sided tumours:** larger, more common, recur more often.
- **Episodic hypertension** (classic sign): only 30–50%.

- **Other symptoms:** tiredness, nausea, flushing, weight loss, anxiety and panic attacks, orthostatic hypotension, in addition to sustained or paroxysmal hypertension.

▸ **Metastatic disease**

- **Commonest sites:** bone, lung, liver, lymph nodes.
- Present at diagnosis or appear during surveillance after excision; most found within 5 years.

▸ **Biochemical evaluation**

- Catecholamines and metabolites circulate free or conjugated with a sulfate moiety.
- **Catecholamines (urine or blood):** sensitivity about 85%, specificity about 85%, because release is often paroxysmal. Urinary catecholamines are still recommended together with urinary fractionated metanephrines.
- Whether plasma free metanephrines or 24-hour urinary fractionated metanephrines should be the initial screen is controversial.
- **Vanillylmandelic acid (VMA):** primary end metabolite; sensitivity below 65%, specificity 99% (especially non-familial cases).

▸ **Clonidine suppression test**

- Clonidine (alpha-2 agonist) suppresses sympathetic catecholamine (especially norepinephrine) production but not that from a phaeochromocytoma.
- Normetanephrine is compared before and after clonidine.
- **Main problem:** significant hypotension.

▸ **Chromogranin A**

- A granin, found in secretory vesicles of neuroendocrine and nervous tissue.
- **Sensitivity:** about 85%.
- **Use:** when free plasma metanephrines are mildly to moderately raised (less than 4-fold).
- Renally cleared, so specificity falls when GFR is below 80 mL/min.

▸ **Cross-sectional imaging**

- **CT:** well circumscribed; high attenuation reflects low lipid content and rich vascularity.
- **MRI:** no signal drop-out on out-of-phase (chemical-shift) sequences, unlike a lipid-rich adenoma.

▸ **Functional imaging**

- **MIBG scintigraphy:** sensitivity 83–100%, specificity 95–100%.
- **FDG-PET:** avidity independent of biochemical activity or phenotype. More accurate than MIBG for metastatic disease in nearly all patients, except MEN-2 mutations, where sensitivity is below 50%.
- Routine functional imaging in every mass with positive biochemistry is controversial.

▸ **Hereditary phaeochromocytoma**

- **Gene testing by phenotype** (after informed consent):
  - Family history or age under 50 years: VHL, RET, SDHB, SDHD.
  - Multiple tumours: VHL, SDHB, SDHD.
  - Malignant tumour: VHL, SDHB.
  - Bilateral tumours: VHL, RET, SDHD.
- A positive result is followed by screening of first-degree relatives.
- **MEN-2A:** also cutaneous lichen amyloidosis.

- **SDHB (PGL-4):** head and neck paragangliomas (carotid body, vagal, jugular, tympanic) as well as abdominal and thoracic paragangliomas.

▸ **Preoperative preparation**

| Drug | Dose and notes |
|---|---|
| Phenoxybenzamine | Start 10 mg orally twice daily; increase by 10–20 mg every 2–3 days |
| Selective alpha-1 blockers | Evidence on efficacy contradictory |
| Beta-1 blockers | Atenolol 12.5–25 mg twice or three times daily; metoprolol 25–50 mg orally three or four times daily |
| Metyrosine (catecholamine synthesis blockade) | Use depends on institutional preference. Increase by 250–500 mg every 2–3 days. About 3 days to full effect; 500 mg orally the night before surgery. Other CNS side effects: galactorrhoea, mood depression |
| Calcium channel blockers (added if BP stays poorly controlled) | Amlodipine 10–20 mg daily; nicardipine 60–90 mg daily; nifedipine 30–90 mg daily; verapamil 180–540 mg daily (avoids reflex tachycardia) |

- **Admission:** 1–2 days before surgery for BP and heart-rate monitoring and intravascular volume restoration.
- **Option 2:** calcium channel blocker then catecholamine synthesis inhibitor as outpatient for 7–14 days, then 1–2 days inpatient for BP and heart-rate monitoring and IV fluid restoration.

▸ **Treatment of malignant phaeochromocytoma**

- Largely palliative. Metastasectomy of resectable disease is standard of care, but survival benefit uncertain.
- Ablation and embolisation described for local control.
- **131I-MIBG therapy:** catecholamines fall in 40%, tumours shrink in 30%. High-dose therapy is controversial because of significant toxicity.
- **CVD chemotherapy:** short-lived response (under 2 years); radiographic shrinkage 50%, biochemical response 75%. Given mainly after MIBG failure or with no MIBG uptake initially.

▸ **Follow-up and prognosis**

- Repeat metabolic testing 2 weeks after adrenalectomy.
- Post-operative cross-sectional imaging is reasonable to document resection and bed healing; later imaging guided by biochemistry.
- **Recurrence risk factors:** extra-adrenal, hereditary, bilateral, large tumour.
- Up to 50% of recurrences are malignant.
- Bone metastases appear to carry the most benign prognosis.

## Ch 147 · Surgery of the Adrenal Glands

### Laparoscopic adrenalectomy: contraindications

- **Absolute:** severe cardiopulmonary disease; local recurrence of a previously resected adrenal mass.
- **Relative:** malignant phaeochromocytoma, morbid obesity, significant adhesions, history of recurrent pyelonephritis, virilising tumour (presume ACC).

## Preoperative preparation

▸ **Conn syndrome**

- Replete magnesium along with potassium.
- Normalise intravascular fluid status.
- Stress dose of cortisol.

▸ **Incidentaloma**

- Same anaesthetic precautions as for phaeochromocytoma, because about 5% of phaeochromocytomas have normal diagnostic studies.

▸ **ACC**

- Consent for removal of adjacent organs.
- Look for vena cava involvement preoperatively; missing it is a recognised pitfall.

▸ **Cushing syndrome**

- Control diabetes, do a cardiopulmonary work-up, give preoperative antibiotics.

## Surgical approach

- **Transperitoneal:** better access to the renal hilum and great vessels.
- **Retroperitoneal:** shorter hospital stay; better for morbid obesity; smaller operative field.

## Oncological principles of resection for ACC

- Preserve intact anterior peritoneum if there is no invasion through it.
- Exclude the rest of the peritoneal cavity with barriers such as mops.
- Minimise bleeding and fluid spillage into the peritoneal cavity.
- Change gloves, gowns and instruments after tumour removal and before closure.

## Complications of adrenalectomy

▸ **Intraoperative**

- **Access-related:** abdominal wall bleeding, cutaneous nerve injury.
- **Haemorrhage:** lumbar vein, hepatic vein, remnant gland after partial adrenalectomy.
- **Ischaemic:** inadvertent ligation of renal artery or vein, or superior mesenteric artery or vein.

▸ **After surgery for primary hyperaldosteronism**

- **Hypokalaemia:** continued potassium loss in the immediate postoperative period.
- **Hyperkalaemia:** failure (suppression) of the contralateral adrenal.

▸ **After surgery for Cushing syndrome**

- Hyperglycaemia, poor wound healing, increased infection risk, osteoporotic fractures.

▸ **After surgery for phaeochromocytoma**

- **Hypoglycaemia:** high catecholamines inhibit insulin release through alpha-2 stimulation; after tumour removal rebound hyperinsulinaemia follows.

▸ **General**

- Pancreatitis or pancreatic fistula, pneumonia, prolonged ileus, collections.

## Retroperitoneal tumours: classification

- Retroperitoneal sarcomas arise de novo. Benign tumours are much less common and do not become malignant.

| Tissue | Benign | Malignant |
|---|---|---|
| Adipose | Lipoma, angiolipoma, lipoblastoma, lipomatosis | |
| Fibrous | Fibroma, angiofibroma, elastofibroma | Fibrosarcoma |
| Fibrohistiocytic | Fibrous histiocytoma, xanthoma | Malignant fibrous histiocytoma |
| Muscle | Leiomyoma, angiomyoma, rhabdomyoma | Leiomyosarcoma, angiomyosarcoma, rhabdomyosarcoma |
| Neural | Phaeochromocytoma (also listed: schwannoma, neurofibroma, ganglioneuroma) | Malignant schwannoma, malignant phaeochromocytoma (also listed: MPNST, neuroblastoma) |
| Synovial | Synovioma | Synovial carcinoma/sarcoma |
| Vascular | Haemangioma | Angiosarcoma, Kaposi sarcoma |
| Lymphatic | Lymphangioma | Lymphangiosarcoma |

## Retroperitoneal sarcoma

### ▸ Clinical presentation and work-up

- **Features:** abdominal lump, pain, constitutional symptoms, early satiety, fatigue, leg oedema.
- **Further indications for image-guided core biopsy:** suspected metastasis from a primary elsewhere, surgically unresectable mass, suspected sarcoma with metastases on imaging.
- **CT:** a higher-density mass without fat favours biopsy, as it suggests lymphoma, metastasis or germ cell tumour rather than liposarcoma.

### ▸ Staging

- **Spread:** haematogenous. Commonest site lung, then liver, bone and brain.
- **T (size):** T1 under 5 cm, T2 5 to 10 cm, T3 10 to 15 cm, T4 over 15 cm.
- **N and M:** N1 must be histologically proven (pN1); M0 or M1.
- **Residual tumour (R):**
  - R0: entirely resected, no microscopic or macroscopic residual.
  - R1: microscopic residual.
  - R2: macroscopic residual.
  - R3: tumour spillage and dissemination during resection.

### ▸ Radiotherapy

- **Pre-operative:** partial tumour regression and less tumour cell implantation. Dose 45–50 Gy total at 1.8 Gy per fraction.
- **Adjuvant:** considered for high-grade tumours, positive margins or tumours larger than 5 cm. It is not compensation for incomplete or poor surgery.

### ▸ Chemotherapy

- **MAID:** mesna, doxorubicin (Adriamycin), ifosfamide, dacarbazine.

▸ **Surgery**

- Lymph node dissection only for bulky or grossly involved nodes.

▸ **Unresectable or metastatic disease**

- If still unresectable after neoadjuvant therapy, palliative options are chemotherapy, radiotherapy, surgery for symptom control, supportive care, and observation if asymptomatic.

▸ **Follow-up**

- CT or MRI after resection; consider chest imaging.

# 15 Renal, Nephrology & Transplant

## Ch 88 · Surgical, Radiologic and Endoscopic Anatomy of the Kidney and Ureter

### Lumbodorsal fascia

- Deep investing membrane over most of the posterior thorax and abdomen; only a thin fibrous lamina in the thorax.
- **Insertions:** anterior and middle layers onto the vertebral transverse processes; posterior layer onto the spinous processes.
- **Laterally:** bridges the aponeuroses of internal oblique and transversus abdominis, the costal angles and the iliac crest to the medial vertebral column and sacrum.
- **Relations:**
  - Superficial to posterior layer: serratus posterior inferior and latissimus dorsi.
  - Between posterior and middle layers: erector spinae.
  - Between middle and anterior layers: quadratus lumborum.
  - Anterior to anterior layer: psoas major.

## Ch 90 · Fundamentals of Nephrology

### Haemodialysis access

▸ **Venous catheters**

- **Preferred site:** right internal jugular vein, a short straight path to the right atrium.
- **Left internal jugular:** used if the right internal jugular and right subclavian veins are contraindicated.
- Cuffed catheters migrate less than uncuffed ones.
- **Designs:** double-D or side-by-side ports.
- May be impregnated with antiseptic or silver-based coatings to reduce bacterial growth.

▸ **AV fistula: pre-operative evaluation**

- **Vein dilatation test:** internal vein diameter should rise by more than 50% on proximal occlusion.
- **Arteriolar dilatation test:** the normal triphasic pulse contour becomes biphasic after the fist is clenched for 2 minutes and then opened.
- **Arteriography indications:** absent or reduced pulse at the desired site, or mean arterial pressure differing by more than 20 mmHg between the arms.

▸ **AV fistula: sites**

- **Snuffbox fistula:** the most distal site, between the tendons of extensor pollicis longus and extensor pollicis brevis; a variant of the radiocephalic fistula.
- **Uncommon conventional:** ulnar artery to forearm basilic vein.
- **Unconventional upper limb:** forearm basilic vein to radial artery at the wrist; loops of forearm basilic or cephalic vein to brachial artery.
- **Konner modification of the Gracz fistula:** perforating veins of the proximal forearm to the proximal radial artery.

- **Elbow fistula indications:** radial or ulnar artery at the wrist under 2 mm, calcified or segmentally stenosed; or cephalic vein under 2.5 mm.
- **Leg:** saphenous vein to popliteal artery; superficial femoral artery to femoral vein.

▸ **AV fistula: technique and maturation**

- **End-to-side:** avoids venous hypertension in the hand.
- **Side-to-side:** sometimes causes red hand syndrome.
- **Modified "piggyback" Slot technique:** reduces anastomotic stenosis and torsion of the anastomotic vein.
- **Immediate flow:** should be 200–300 mL/min; under 120 mL/min predicts failure or closure.
- Hand exercises are advised afterwards.
- A mature fistula also needs a straight cannulation segment of at least 6 cm.

▸ **AV fistula: complications**

- **Thrombosis:** the commonest.
- Red hand syndrome.
- **Steal:** arterial flow diverts into the fistula; with distal small-vessel arteriopathy this causes hand ischaemia.

▸ **AV grafts**

- **Shapes:** straight, looped or curved.
- **Sites:** brachial artery to axillary vein; brachial artery to basilic vein; radial artery to basilic vein at the wrist; axillary artery to axillary vein (same side, or opposite-side artery as a "necklace graft").
- **Advantages:** large surface area for needles; easy cannulation and surgical handling.

## Haemodialysis prescription

- **Dialysate:** bicarbonate 25 mmol/L, calcium 1.5 mmol/L; flow 500 mL/min; temperature 35–36 °C.
- **Fluid removal:** typically 2.2 L per session.

## Peritoneal dialysis (PD)

▸ **Membrane and fluid**

- **Three-pore model:** ultrasmall pores (water channels); small pores (major pathway, water and small solutes); large pores (large molecules).
- **Fluid:** lactate buffer in a balanced salt solution devoid of potassium.

▸ **Modalities and technique**

- **Automated PD:** continuous cycling PD (CCPD), intermittent PD (IPD) and nocturnal intermittent PD; more suitable for children.
- **Each exchange:** drain over 10–15 minutes; instil 2–2.5 L over 10–15 minutes.
- **Catheters:** straight single-cuff, straight double-cuff, double-cuff coiled, swan-neck and Toronto Western.
- **Further absolute contraindications:** large diaphragmatic defect; acute ischaemic or infectious bowel disease.

▸ **Adequacy**

- Also includes no oedema or hypertension, potassium below 5 mmol/L, bicarbonate above 24 mEq/L, BMI 20–30 kg/m², albumin above 3.5 and a stable mid-arm circumference.
- **Large-solute clearance:** albumin clearance below 0.15 mL/min.

▸ **Outcome and complications**

- Lower risk of death than haemodialysis (HD) in the first 3 years.
- Patients eventually need HD or a transplant.
- **Catheter inflow failure:** fibrin plug, omental wrapping or kinking.
- Thickening of the peritoneal membrane; metabolic acidosis.

## Hyperfiltration injury

▸ **Definition and adaptation**

- Variably defined as a rise in whole-kidney GFR, in filtration fraction, or in single-nephron GFR (SNGFR).
- Even after losing 90% of renal function, most kidneys keep excretory and secretory function compatible with life, through compensatory physiological and morphological changes.
- **Kidney enlargement after nephron loss:** 10–25% hyperplasia, the rest hypertrophy.

▸ **Causes**

- **Physiological:** high-protein meal; pregnancy.
- **Others:** FSGS, ADPKD, sickle cell disease, thalassaemia, cancer, obesity.

▸ **Pathogenesis**

- Raised glomerular pressure activates the renin–angiotensin–aldosterone system, causing hypertension and driving tubulointerstitial inflammation.
- Increased glomerular permeability leaks protein, which injures tubular cells; cytokine-mediated tubulointerstitial inflammation then leads to tubulointerstitial fibrosis. In parallel, podocyte injury leads to glomerulosclerosis.

▸ **Treatment**

- **ACE inhibitors and ARBs:** can promote regression of glomerulosclerosis, retarding diabetic nephropathy.
- **Pro-insulin C-peptide:** potential renoprotective agent; in diabetics it constricts afferent arterioles and prevents hyperfiltration.
- **Rosiglitazone:** improves endothelial dysfunction, reduces glomerular hyperfiltration, lowers microalbuminuria and progression to ESRD.
- Avoid obesity and nephrotoxic drugs.

## Proteinuria

▸ **Causes**

- **Pre-renal:** haemoglobinuria, myoglobinuria, multiple myeloma.
- **Renal, functional** (mild, transient): fever, exercise, epilepsy.
- **Renal, glomerular:** SLE, IgA nephropathy, Alport syndrome, post-streptococcal glomerulonephritis, diabetes, amyloidosis.
- **Renal, tubular:** sarcoidosis, Balkan nephropathy.

- **Post-renal (surgical):** protein entering urine after it reaches the renal pelvis.
  - Urinary: UTI, neoplasm, calculus.
  - Extraurinary: genital infections such as prostatitis.

▸ **Albumin dipstick**

- Qualitative and semi-quantitative.
- **Principle:** protein-induced pH change alters the indicator colour; the darker the green, the more protein.
- Alkaline urine gives a false positive.
- Threshold of 20–30 mg/dL is too high to detect early kidney disease.

▸ **Quantitative and specific tests**

- **24-hour urinary protein:** gold standard; quantifies total protein by chemical assay or dye binding.
- **Protein electrophoresis:** separates glomerular from tubular proteinuria. In tubular proteinuria only 10–20% is albumin and immunoglobulin predominates.
- **Immunoassay:** identifies specific proteins.
- **Microalbuminuria:** higher risk of diabetic nephropathy and CKD; detected by 24-hour collection or immunoturbidimetry.

▸ **Algorithm for a positive dipstick**

- **By timing:** transient, persistent or intermittent. Intermittent proteinuria unrelated to posture is quantified like persistent; posture-related is orthostatic.
- **Glomerular** (more than 2 g/24 h, mainly albumin):
  - With dysmorphic red cells and red cell casts: glomerular haematuria work-up.
  - With no or little haematuria: look for specific diseases such as diabetes and amyloidosis.
- **300–2000 mg/24 h, mainly globulin:**
  - Normal proteins: tubular proteinuria; evaluate the cause.
  - Abnormal proteins: overflow proteinuria; evaluate the specific protein.

## Renal artery stenosis (RAS)

▸ **Manifestations and causes**

- May be asymptomatic and incidental.
- Accelerated cardiovascular and cerebrovascular disease is one of the four manifestations.
- **Renovascular hypertension:** with 70% stenosis of one or both arteries, or with 50% stenosis when post-stenotic dilatation is present.
- **Further causes of renovascular hypertension and ischaemic nephropathy:** arterial thrombosis; arteriovenous malformation; endovascular or iatrogenic injury (for example after aortic aneurysm repair).

▸ **Fibromuscular dysplasia (FMD)**

- Non-atherosclerotic, non-inflammatory disease with abnormal growth within the arterial wall; most often renal and carotid arteries.
- Usually bilateral.

| Type | Share | Who | Features |
|---|---|---|---|

| | | | |
|---|---|---|---|
| Intimal fibroplasia | 10% | Children and younger adults | Collagen within the intima; usually proximal renal artery; may be complicated by dissection or wall haematoma; progresses with loss of renal function if untreated |
| Medial fibroplasia | 70-80% | Women aged 25-50 years | Fibrotic stenotic segments alternate with thinned-out media in aneurysmal segments |
| Perimedial fibroplasia | 10-20% | Younger women (5-15 years) | Mid renal artery; widespread collagen over the outer half of the media; also string-of-beads appearance; progresses with loss of renal function unless treated |
| Medial hyperplasia | Rare | | No fibrosis |

### ▸ Atherosclerotic RAS

- Bilateral in 20-40%.
- 30% non-ostial, usually 1-3 cm distal to the ostium.

### ▸ Ischaemic nephropathy: pathogenesis

- **Four phases:** cellular activation and injury, fibrogenic signalling, fibrogenic, destructive.
- Reduced renal blood flow triggers pro-inflammatory mediators and inflammatory cell attraction, leading to fibrosis.
- **Angiotensin II:** profibrogenic peptide and powerful vasoconstrictor; acts through NF-kB and TGF-beta.
- **NF-kB:** drives fibroblast proliferation and differentiation, and makes tubular cells secrete profibrotic cytokines (matrix) and chemoattractant adhesion proteins (inflammation); all converge on fibrosis.

### ▸ Clinical clues to renovascular disease

- Severe or refractory hypertension with grade III-IV hypertensive retinopathy.
- Paradoxical worsening of hypertension with diuretics.
- Systolic-diastolic abdominal bruit radiating to both flanks.
- Diffuse vascular disease and/or cholesterol embolisation.

### ▸ Investigations

- **Gold standard:** intra-arterial digital subtraction angiography (DSA), as well as conventional renal angiography.
- **Angiography indication:** high clinical suspicion with a negative screening test, since screening can miss distal or intrarenal disease.

| Test | Accuracy | Points |
|---|---|---|
| MR angiography (MRA) | Sensitivity 100%, specificity 70-95% | No radiation; also assesses renal blood flow and GFR; may miss distal, intrarenal and accessory artery lesions; gadolinium carries a risk of nephrogenic systemic fibrosis (NSF) when GFR is below 30 mL/min |
| Spiral CT angiography | Sensitivity 98%, specificity 94% | Both decline when serum creatinine is above 1.7 mg/dL |
| Duplex Doppler | | Accurate in renal failure; time-consuming; intravascular or even main-artery lesions may be poorly seen in obesity or with bowel gas |

### ▸ Medical therapy

- Controls BP in more than 90%, usually with several antihypertensives.

- **Monitor:** renal size and cortical blood flow, as well as serum creatinine and GFR or creatinine clearance.
- **ACE inhibitors in bilateral RAS:** acute reversible renal failure in 10-20%, particularly with volume contraction.

▸ **Angioplasty with or without stenting**

- **Indications:** intolerance of medical therapy; haemodynamically significant RAS with accelerated or malignant hypertension.
- **Procedural success:** 85-100%.
- **Hypertension:** cured or improved in about 90%.
- **Renal function:** improves in 40%, stabilises in another 40%.
- **Restenosis:** PTRA alone 30% (non-ostial) and 50% (ostial); PTRA with stent 15-25%.
- **Trials:** STAR, ASTRAL and CORAL showed no significant difference in hypertension between medical treatment and angioplasty.
- **Complications:** 5-15%; major in less than 5%.
  - Minor: renal artery spasm or renal infarction, reversed with intra-arterial nitroglycerin.
  - Stenting: stent malpositioning or dislocation.
- **Predictors of improved or stabilised function after revascularisation:** pyelogram seen on IVP; kidney length more than 9 cm; collaterals filling the distal vasculature beyond a total occlusion on angiography; biopsy showing viable glomeruli with minimal arteriolar glomerulosclerosis.

▸ **Surgery**

- **Indication:** malignant or accelerated hypertension not responding to, or not tolerating, medical therapy when angioplasty is technically impossible.
- **Iliorenal bypass:** an option when the aorta is diseased.
- **Supraceliac or lower thoracic aorta inflow:** when the infrarenal aorta is diseased, as it is less involved by atherosclerosis.
- **Ablative:** laparoscopic or open nephrectomy.

## Ch 91 · Renal Transplantation

### Laparoscopic donor nephrectomy (LDN)

▸ **Advantages**

- No increase in complications compared with open donor nephrectomy.
- Fewer wound problems and incisional hernias.

▸ **Risks and complications**

- **Intraoperative:** bleeding, visceral injury, ureteric stripping or short ureter.
- **Postoperative, non-life-threatening:** ileus, port-site infection, deep vein thrombosis, lymphatic leak.
- **Causing donor death:** haemorrhage and pulmonary embolism.
- **Specific to the laparoscopic route:** bowel injury; internal hernia.
- **Donor nephrectomy in general:** avoid hypotension during surgery to prevent graft dysfunction; donor should avoid nephrotoxic drugs afterwards to protect the remaining kidney from CKD.

▸ **Hand-assisted LDN**

- **Advantages:** less traction on the kidney; faster removal and less warm ischaemia; shorter learning curve.
- **Disadvantages:** higher cost; awkward surgeon position; higher risk of traumatic injury from manual manipulation.

## Renal preservation

- **Purpose (deceased donors):** time for cross-matching, sharing organs between centres and recipient preparation.

▸ **Harmful effects of ischaemia and hypothermia**

- Activate lysosomal enzymes in the renal cell.
- Loss of cellular potassium and magnesium, with a rise in cellular calcium, leads to ATN and cell swelling.

▸ **Solutions**

- The original Collins solution is a preservation solution in its own right, separate from Euro-Collins.
- **UW solution:**
  - Dexamethasone, with MgSO4, is a membrane stabiliser.
  - KH2PO4 is a potassium source.
  - Disadvantage: particles become caught in the capillary bed and prevent reperfusion.

▸ **Methods**

- **Cold storage:** kidney kept in a bag so crushed ice does not touch its surface.
- **Machine pulsatile perfusion:** maintains viability for up to 72 hours.

## Rejection

▸ **Pathogenesis**

- Acute rejection can occur in all transplants without immunosuppression, except between identical twins.
- **Antibody** from plasma cells damages the graft by antibody-dependent cell-mediated cytotoxicity and complement-mediated damage.
- **CD4+ T-helper cytokines** activate macrophages and NK cells, producing delayed-type hypersensitivity.

▸ **Banff classification**

- T-cell-mediated rejection includes a chronic category (chronic T-cell-mediated rejection) besides acute and borderline.

▸ **Acute rejection: presentation and investigations**

- Hypertension is a presenting feature.
- **Urinalysis:** may show proteinuria, pyuria or haematuria.
- **Imaging:** may show decreased cortical blood flow and decreased tubular function.
- **Adequate biopsy:** at least 10 glomeruli and 2 small arteries; stained with H&E, PAS or silver, and trichrome. Take two separate cores.

▸ **ATN versus acute rejection**

- **Post-transplant ATN timing:** commonly within the first week.
- **Frequency:** cadaveric donor 25%, live donor 5%.
- **Course:** delayed graft function usually recovers within 10–14 days; conservative treatment, dialysis may be needed.
- **Doppler resistive index:** about 0.9 in ATN, about 1.0 in acute rejection.

▸ **Treatment of cell-mediated rejection**

| Drug | Dose |
|---|---|
| Methylprednisolone | 3–5 mg/kg (250–500 mg/day) for 3–5 days |
| Muromonab (OKT3) | 5 mg IV for 7–14 days |
| ATG | 1–1.5 mg/kg IV |
| Alemtuzumab (anti-CD52) | 30–60 mg on day 0 and day 2 |

- Anti-T-cell antibodies restore kidney function and prevent graft loss better than corticosteroids, but have more adverse effects.

▸ **Treatment of antibody-mediated rejection**

| Treatment | Regimen |
|---|---|
| Plasma exchange | 2–5 treatments, daily or alternate days, with IVIG |
| IVIG | 100–200 mg/kg after plasma exchange |
| Rituximab | 375 mg/m2 with plasma exchange and IVIG |
| Bortezomib | 1.3 mg/m2 for 4 doses over 1–2 weeks, usually with plasma exchange and IVIG |
| Eculizumab (prevention in cross-match-positive transplant) | 600–1200 mg weekly for 4 doses, then every two weeks until successful antibody reduction |

## Immunosuppression

▸ **Classification: antibodies and others**

- **Muromonab (OKT3):** monoclonal anti-CD3.
- **Daclizumab:** like basiliximab, a monoclonal anti-CD25 (IL-2 receptor) antibody and induction agent.
- **Leflunomide:** inhibits pyrimidine synthesis.

▸ **Corticosteroids**

- **Targets:** T cells, B cells and macrophages; mechanism complex and not fully understood.
- **Pathway:** steroid binds the cytoplasmic receptor/heat shock protein complex; the heat shock protein is released; the steroid–receptor complex enters the nucleus, binds promoter regions of target genes and induces or suppresses cytokine genes.
- At higher doses, also a receptor-independent effect.
- **Suppress:** IL-2, TNF-alpha, chemokines and MHC class II; inhibit leukotrienes and prostaglandins; regulate NF-kB.
- **Side effects:**
  - Renal: hypokalaemia, diuretic resistance.
  - Gastrointestinal: peptic ulcer, pancreatitis, colonic perforation.

- Haematological: leucocytosis, polycythaemia.
- Neuromuscular and psychiatric: myopathy, psychosis, insomnia.
- Cosmetic: cushingoid features, hirsutism, acne, buffalo hump.
- Pregnancy: fetal adrenal suppression.

▸ **Calcineurin inhibitors**

- Increase TGF-beta, which adds to immunosuppression but causes tissue fibrosis.
- Cause hypophosphataemia and type 4 renal tubular acidosis.
- **Tacrolimus more than ciclosporin:** myocardial hypertrophy.
- **Ciclosporin more than tacrolimus:** coarse facial features and oedema.

▸ **Mycophenolate mofetil**

- **Diarrhoea:** inhibits gastrointestinal epithelial replication, disrupting fluid absorption.
- **Pregnancy:** wait 1 year after the graft stabilises and switch to azathioprine.

▸ **Azathioprine**

- Converted to 6-mercaptopurine, which inhibits purine synthesis.
- Inhibits T-cell proliferation much more than B-cell, so mainly inhibits cell-mediated immunity.
- **Side effects:** hepatotoxicity, alopecia.

▸ **mTOR inhibitors**

- **Side effects:** leucopenia, thrombocytopenia, diarrhoea.
- Hyperlipidaemia is more common with sirolimus than everolimus.

▸ **Antithymocyte globulin (ATG)**

- **Forms:** equine and rabbit.
- Also used for aplastic anaemia.
- **Action:** binds multiple T-cell surface epitopes; rapid lymphocytopenia by phagocytosis, apoptosis and complement-dependent cytolysis. Potent; depresses T and B cells for up to 24 hours after a dose.
- **Side effects:** thrombocytopenia, serum sickness-like syndrome, rarely ARDS.

▸ **Rituximab**

- **Action:** depletes B cells by antibody-dependent cell-mediated cytotoxicity, complement-mediated lysis and apoptosis.
- **Infusion reactions** within 2 hours: rash, itching, low blood pressure, shortness of breath; also headache, fever, chills, dizziness.
- **Other risks:** hepatitis B reactivation, progressive multifocal leukoencephalopathy, toxic epidermal necrolysis.

▸ **Intravenous immunoglobulin (IVIG)**

- **Action** (not fully understood): inhibits antibody production, increases catabolism of circulating antibodies, inhibits cytokines and complement-mediated injury. Also complement downregulation, Fc receptor interaction, inhibition of B- and T-cell proliferation, inhibition of CD8 T-cell toxicity and increased B-cell apoptosis.
- **Side effects:** thrombotic events, flushing, chills, headache, myalgia, anaphylaxis, meningitis.
- **Serious:** neutropenia, thrombotic events, neurotoxicity.
- **High dose in ABO-incompatible transplant:** haemolysis and increased risk of antibody-mediated rejection.

▸ **Induction**

| Drug | Induction dose |
|---|---|
| Methylprednisolone | 500 mg IV intraoperatively, tapered over 1–5 days |
| Rabbit ATG (rATG) | 1–1.5 mg/kg IV for 4–14 days |
| OKT3 | 5 mg IV for 7–14 days |
| Daclizumab, basiliximab | Anti-CD25 agents |

▸ **Regimens by number of drugs**

- **Monotherapy:** ciclosporin, tacrolimus or an mTOR inhibitor.
- **Dual:** one of these plus a steroid, azathioprine or MMF.
- **Triple:** steroid plus an antiproliferative plus ciclosporin, tacrolimus or an mTOR inhibitor.
- **Quadruple:** ATG induction plus triple therapy.

## ABO-incompatible and swap transplantation

▸ **Background**

- **Recipient blood types:** A 45%, O 40%, B 10%, AB 5%.
- About 35% of medically suitable donor-recipient pairs are ABO incompatible.
- ABO-incompatible transplantation reduces the deceased-donor waiting list and waiting time.

▸ **Problems**

- Significantly higher infection rate.
- Adverse events from antibody depletion therapy.
- Needs intensified immunosuppression.
- High cost.
- Lack of experience in many centres of developing countries.
- No RCTs compare desensitisation protocols, so no single ideal protocol exists.

▸ **Antibody titres and monitoring**

- **Anti-A/anti-B measurement:** tube technique, gel technique or flow cytometry.
- High pre-operative titre: poor graft survival.
- High post-operative titres: increased risk of antibody-mediated rejection.
- **Monitor:** anti-ABO IgG titre and B-cell count.

▸ **Desensitisation methods**

- **Rituximab:** 375 mg/m2 for B-cell depletion.
- **Plasmapheresis:** plasma replaced by human albumin, colloid and/or fresh frozen plasma. Disadvantages: removes coagulation factors and antiviral and antibacterial antibodies, increasing infections.
- **Double filtration plasmapheresis:** selectively removes the desired immunoglobulin; needs less fluid substitution.
- **Immunoadsorption:** safe and more effective selective removal of anti-A or anti-B antibodies; no volume replacement needed.
- **Johns Hopkins protocol:** plasma exchange and rituximab (500 mg) daily or alternate days from 2 weeks before transplant; tacrolimus and mycophenolate mofetil from 1 week before; ATG with methylprednisolone the day before (check).

▸ **Swap and donor chains**

- **Domino paired exchange:** a non-directed (altruistic, Good Samaritan) donor gives to the first recipient, whose incompatible donor gives to the next recipient, completing the exchange.
- **Ongoing chain:** continued, or ended by giving the last living donor's kidney to a patient on the deceased-donor list.

## Post-transplant infections

▸ **Vaccination**

- Any live vaccine at least 4 weeks before transplant.
- **Before transplant:** HPV, hepatitis B, pneumococcus.
- **Barred after (live):** polio, MMR.
- **Safe after:** hepatitis A and B, diphtheria, tetanus.

▸ **Timeline**

- **First month:**
  - Bacterial: UTI, wound infection, aspiration pneumonia, intra-abdominal infections, catheter and drain-site infections. Multidrug-resistant bacteria are an important cause of mortality and morbidity.
  - Viral: mainly HSV; others uncommon.
  - Fungal: Candida, including azole-resistant Candida.
- **1 to 6 months (opportunistic, from immunosuppression):**
  - Viral: CMV, HHV-6, HSV, VZV, EBV, HBV, HCV, BK virus.
  - Fungal: Aspergillus, Cryptococcus, mucormycosis.
  - Bacterial: Nocardia, TB, Legionella.
  - Parasitic: Pneumocystis jirovecii, Toxoplasma, Leishmaniasis.
- **After 6 months:**
  - Good graft function, no chronic viral infection, low-dose immunosuppression: mainly community-acquired infections.
  - Chronic viral infection: progressive liver disease (HBV/HCV), BK nephropathy or PTLD (EBV).
  - Multiple rejection episodes: opportunistic bacterial, viral and fungal infections.

▸ **Prophylaxis**

| Target | Prophylaxis |
|---|---|
| UTI | TMP-SMX for 3–6 months, or nitrofurantoin |
| Pneumocystis pneumonia | TMP-SMX, or inhaled pentamidine |
| Oral candidiasis | Nystatin suspension for 1–3 months, or clotrimazole lozenges (check) |
| Vaginal candidiasis | Intravaginal nystatin or clotrimazole (check) |
| HSV/CMV | Acyclovir or valacyclovir for 3 months |
| Peptic ulcer | H2-receptor blocker, plus a second agent if symptomatic |

▸ **Cytomegalovirus (CMV)**

- **Timing:** mainly 1–6 months after transplant.

- **Patterns:** primary infection (recipient negative, donor positive); reactivation of latent virus (recipient positive, donor positive or negative); superinfection with a new virus in a seropositive recipient (both positive).
- **Effects:** any organ (for example hepatitis, retinitis); lowers platelet and total leucocyte counts; raises the risk of opportunistic infection and PTLD.
- **Prevention:** acyclovir or valganciclovir.
- **Diagnosis:** CMV PCR assay.
- **Treatment:** IV ganciclovir; ganciclovir-resistant disease with cidofovir or foscarnet.

**Other topics in this book**

## Urinalysis: proteinuria

▸ **Dipstick**

- **Grades:** trace 5–20 mg/dL; 1+ 30 mg/dL; 2+ 100 mg/dL; 3+ 300 mg/dL; 4+ more than 2000 mg/dL.
- **Tetrabromophenol blue:** protein shows as shades of green.
- Alkaline urine is a cause of a false-negative result.

▸ **Other points**

- **3% sulfosalicylic acid precipitation:** detects protein from 15 mg/dL.
- **Orthostatic proteinuria:** attributed to pressure on the renal vein while standing.

## Nanobacteria in other urological diseases

▸ **ADPKD**

- Proposed to target the tubule cells that are the primary source of cysts.
- Higher urinary nanobacterial antigen levels.

▸ **Chronic prostatitis/CPPS**

- Prostatic calcifications are typically calcium apatite, said to be the hallmark of nanobacterial action.
- Higher serum nanobacterial antigen levels by ELISA.

## Acid-base regulation

- **Three systems:** kidney, lungs and buffers.
- **Kidney:** distal H+ secretion alone would drop urine pH below 2.5, toxic to tubular cells, so the acid is buffered with ammonium in collecting duct cells.
- **Proximal tubule:** luminal H+ (secreted in exchange for Na+) combines with filtered $HCO_3$ to form $H_2CO_3$; carbonic anhydrase converts it to $CO_2$ and water, which enter the cell; carbonic anhydrase regenerates H+ and $HCO_3$, and the $HCO_3$ passes to blood.
- **Lungs:** exhale acidic $CO_2$; a rise in blood $CO_2$ drives hyperventilation, removing $CO_2$.
- **Buffers:** the body's weak acids and bases, guarding against sudden pH shifts. Most important in blood: carbonic acid and bicarbonate.

## Normal values and disorders

- **Normal arterial values:** pH 7.35-7.45, pO2 80-100 mmHg, pCO2 35-45 mmHg, HCO3 22-28 mEq/L.
- **Disorders:** metabolic acidosis, renal tubular acidosis, metabolic alkalosis, respiratory acidosis, respiratory alkalosis.

| Disorder | Primary change | Compensation |
|---|---|---|
| Metabolic acidosis | HCO3 falls | pCO2 falls |
| Metabolic alkalosis | HCO3 rises | pCO2 rises |
| Respiratory acidosis | pCO2 rises | HCO3 rises |
| Respiratory alkalosis | pCO2 falls | HCO3 falls |

## Metabolic acidosis

- **Increased-anion-gap causes:** increased acid load (lactic acidosis, diabetic ketoacidosis); ethylene glycol, methanol and salicylate intoxication.
- **Alkali indication:** pH below 7.2 or HCO3 below 10.
- **Sodium bicarbonate dose:** 0.4 times body weight times bicarbonate deficit.

## Metabolic alkalosis

- **Definition:** HCO3 above 26 mEq/L.
- **Causes:** excessive vomiting, certain diuretics, severe dehydration, excessive alkali drugs, heavy antacid ingestion, endocrine disorders.
- **Treatment:** mainly volume and electrolyte replacement and treating the cause; in severe cases occasionally dilute acid as intravenous ammonium chloride.

# 16 BPH

**Other topics in this book**

## Lasers in BPH

▸ **Holmium laser**

- **Fibre:** end-firing or side-firing; flexible.
- **Disadvantage:** high cost.

▸ **Thulium laser**

- **Generators:** 70 W or 120 W.
- **Fibres:** rigid, 550 µm or 880 µm.
- Smoother incision than holmium.
- Less tissue scarring, because thermal energy scatters less.
- Continuous-wave thulium is still investigational for stone disease.

▸ **KTP/LBO laser**

- **Modes:** continuous and pulsed.
- **Fibres:** rigid, side-firing.
- Photoselective vaporisation has a simple (short) learning curve.

▸ **Nd:YAG laser**

- **Modes:** continuous or pulsed (Q-switched).
- **Historical prostate techniques:** laser ablation and interstitial laser coagulation.

## Green light laser and photoselective vaporisation of the prostate (PVP)

▸ **Green light generator**

- A laser diode pumps the Nd:YAG through a coupling lens. The beam passes an input monitor, the Nd:YAG rod, the KTP crystal and an output monitor, then emerges as green light.
- Photoselective vaporisation creates a cavity in the prostate.

▸ **PVP technique**

- Sweep the fibre from inside out in a painting motion.
- **Fibre too close:** contact vaporisation, which damages the fibre.
- **Fibre too far:** inefficient vaporisation and more coagulation.
- **No large bubbles:** more coagulation, a longer operation, and more postoperative dysuria and LUTS.

## Irrigation fluids in TURP

▸ **Uses and ideal properties**

- **Uses:** better visualisation; distension of the bladder and prostatic urethra; flushing out blood and tissue debris.
- **Ideal fluid (further properties):** non-metabolisable, easy to sterilise, rapidly excretable, non-combustible.

▸ **Individual fluids**

| Fluid | Points |
|---|---|
| Glycine | Standard 1.5%; also available as 1.2% and 2.2% |
| Distilled water | Haemolysis and haemoglobinuria |
| Ringer's lactate | Iso-osmolar, less risk of TUR syndrome; ionised, so cannot be used for monopolar resection |
| Mannitol 5% | Osmotic diuresis, high cost, acute intravascular volume expansion |
| Sorbitol | Lactic acidosis, hyperglycaemia, osmotic diuresis |
| Sorbitol plus mannitol (Cytal) | Main disadvantage is expense |

## Prostatic urethral lift (UroLift)

▸ **Indications and selection**

- An option for patients with multiple comorbidities not fit for regional or general anaesthesia. Other options for unfit men are a long-term catheter or prostatic artery embolisation.
- **Inclusion criteria:** IPSS 12 or more; Qmax 15 mL/s or less; post-void residual less than 350 mL.
- **Not suitable:** large median lobe, high bladder neck.
- Short learning curve.

▸ **Technique**

- Cystoscopy with a 2.9 mm 0 degree telescope through a 20 Fr sheath.
- The cystoscope bridge is replaced with the UroLift system headpiece, and the telescope is passed through the headpiece.
- A wrongly placed implant can be removed with a grasper or biopsy forceps.

# 17 Prostate Cancer

## Ch 153 · Epidemiology, Aetiology and Prevention of Prostate Cancer

### Epidemiology

▸ **Incidence and ranking**

- **Worldwide:** second commonest cancer (lung cancer is first); 6th leading cause of cancer deaths.
- **US men:** most common non-cutaneous malignancy.
- **PSA screening effect:** incidence rose sharply from 1989 to 1992, then fell steeply from 1992 to 1995 (the "cull effect").
- **Stage migration:** pathological stage at diagnosis is moving downward, partly because of PSA screening. About 80% of newly diagnosed men have localised disease.

▸ **Age**

- Only 2% are diagnosed in men younger than 50 years.
- Average age at death from prostate cancer: 77 years.

▸ **Race and geography**

- **Highest risk:** African Americans and Jamaican men of African descent.
- **Gradient:** White Americans highest, then men of Asian descent living in the US, then similar men living in Asia.

### Genetic risk factors

- About 15% of all prostate cancer is caused by germline mutation.
- **Susceptibility mutations:** chromosomes 2, 3, 4, 5, 6, 7, 8, 10, 11, 12, 13, 17, 19, 22 and X.
- **HOXB13:** on chromosome 17.
- **Affected second-degree relative:** relative risk 1.68.

### Aetiological and lifestyle factors

▸ **Inflammation and infection**

- **Proposed triggers of epithelial damage:** infection, oestrogens and cooked meat.
- **Sequence:** epithelial damage, then inflammatory infiltrates and proliferative inflammatory atrophy, then oxidant stress, then DNA damage, and finally cancer.
- **Propionibacterium acnes:** predominant bacterial species in both benign and malignant prostatic tissue; associated with prostatic inflammation.
- No infectious agent has been proven to cause prostate cancer. No single sexually transmitted agent is directly implicated; infections may raise risk through inflammation.

▸ **Metabolic syndrome and obesity**

- Of the single components, increased waist circumference and hypertension carry a significantly greater risk.
- Obesity is associated with higher rates of biochemical failure.

▸ **Diet**

- **Fried foods:** increase risk.

- **Phytoestrogens:** decreased risk.
- **Dietary fat:** no relationship with risk.

▸ **Vitamin D**

- Low levels are associated with a higher risk.

▸ **Drugs and hormones**

- **Statins:** no association with overall incidence, but a protective association with advanced prostate cancer. This suggests an effect in the late stages of carcinogenesis.
- **Testosterone in hypogonadal men:** no increased risk.
- **Oestrogens:** direct and indirect effects on prostatic growth and development; probably play a role in initiation and progression.

▸ **Smoking**

- **Proposed mechanisms:** cadmium exposure, raised circulating androgen levels and cellular oxidative stress.
- **Current smokers:** higher risk of biochemical failure, metastasis and prostate cancer-specific mortality.

▸ **Other factors**

- **Protective:** frequent ejaculation; circumcision.
- **Higher risk:** night-shift work.

## Chemoprevention

▸ **Definition and goal**

- **Definition:** use of natural or synthetic agents that reverse, suppress or prevent the carcinogenic process, so that clinically evident cancer does not develop.
- **Goal:** lower the incidence of a given cancer while also reducing treatment-related side effects and morbidity.

▸ **Levels of prevention**

- **Primary:** the general population of healthy individuals at risk.
- **Secondary:** individuals with a premalignant lesion such as HGPIN (high-grade prostatic intraepithelial neoplasia), to prevent progression to frank cancer.
- **Tertiary:** prevent a second primary cancer in individuals who already have one.

▸ **Rationale and challenges**

- **Clinical rationale:** the established risk factors (age, ethnicity and family history) cannot be modified.
- **Biologic rationale:** premalignant changes appear as long as 20–30 years before carcinoma, giving time for lifestyle change (diet, smoking cessation, exercise) or chemoprevention.
- **Challenges:** finding an effective intervention with acceptable toxicity, and identifying a population at high enough risk for chemoprevention to be appropriate and cost-effective.

▸ **Prevention framework**

- Intensity of intervention rises with cancer risk, set by age, family history, ethnicity, baseline PSA and genetic susceptibility.
- **Steps:** lifestyle and dietary change, then vitamins and micronutrients, then pharmacological agents, and finally prophylactic surgery.

## Chemoprevention trials

▸ **PCPT (Prostate Cancer Prevention Trial)**

- First large-scale chemoprevention trial for prostate cancer.
- **Entry:** men aged 55 years or older with a normal DRE and PSA of 3 ng/ml or less.

▸ **REDUCE**

- **Entry:** negative prior biopsy (6–12 cores) within the previous 6 months; prostate volume 80 ml or less.
- The excess of Grade Group 8–10 cancers with dutasteride appeared in years 3 and 4.

▸ **FDA reassessment of 5-alpha reductase inhibitors**

- For every 150–200 men treated, one additional man would be diagnosed with a high-grade cancer in order to avert 3–4 low-grade cancers.

▸ **Toremifene citrate**

- Oestrogen receptor modulator, FDA-approved for breast cancer.
- Its RCT failed to show a reduction in prostate cancer.

▸ **SELECT**

- Planned to run for 12 years.

▸ **Green tea catechins**

- Induce apoptosis and inhibit prostate cancer in vitro.
- In a small RCT of 60 men with HGPIN, associated with a lower risk of prostate cancer than placebo; confirmatory trials are needed.

## Screening trials (ERSPC and PLCO)

| Feature | ERSPC | PLCO |
|---|---|---|
| Size and contamination | About twice as large as PLCO, with less contamination | Smaller |
| Age | Core age group 55–69 years | |
| Screening | PSA cut-off 3 ng/ml; DRE added in the first and second rounds | Annual DRE and PSA for 6 years; PSA cut-off 4 ng/ml |

# Ch 154 · Prostate Cancer Biomarkers

## Biomarker development

- **Early Detection Research Network (EDRN):** set up to identify, develop and validate promising biomarkers and technologies for early cancer detection.

| Phase | Content |
|---|---|
| 1 | Biomarker discovery in biomarker development laboratories |
| 2 | Sensitivity and specificity; ability to tell cases from controls |
| 3 | Ability to detect preclinical disease; lead time before clinical presentation |
| 4 | Indications for applying the biomarker clearly defined |
| 5 | Cancer control studies: does it reduce the burden of disease? Reduced mortality is the successful outcome |

## Assessment of biomarker performance

### ▸ Sensitivity and specificity

- **Sensitivity:** proportion of people with the disease who test positive. TP divided by (TP plus FN), times 100.
- **Specificity:** proportion of people without the disease who test negative. TN divided by (TN plus FP), times 100.

### ▸ Accuracy and the ROC curve

- **Accuracy:** (TP plus TN) divided by the total population.
- **Area under the ROC curve (AUC):** measures accuracy. 1 is an ideal test; 0.5 is a poorly performing biomarker.
- **Axes:** sensitivity on the vertical axis, 1 minus specificity on the horizontal axis.

### ▸ Predictive values

- **PPV:** probability that a positive test indicates disease. TP divided by (TP plus FP).
- **NPV:** probability that a negative test indicates absence of disease. TN divided by (TN plus FN).
- NPV and PPV are more relevant to clinical practice than sensitivity and specificity.

### ▸ Likelihood ratio

- How much a new biomarker raises or lowers the probability of disease before biopsy.

## Regulatory status

- **FDA-approved:** PSA, free-to-total PSA ratio, PHI and PCA3.
- **CLIA certification (Clinical Laboratory Improvement Amendments):** allows a test to be offered commercially, but does not guarantee rigorous clinical evaluation.

## Prostate-specific antigen (PSA)

### ▸ Biology

- Glycoprotein secreted by prostatic epithelial cells in zymogen form.
- **Molecular weight:** 33 kD.
- **Function:** liquefies semen by acting on semenogelin and fibronectin.
- **Seminal fluid level:** 0.5–5.0 mg/mL.

### ▸ Precursors and molecular forms

- **Sequence:** pre-pro PSA, then inactive proPSA, then active PSA after hK2 cleaves the 7-amino-acid leader sequence in the lumen.
- **proPSA isoforms:** minus 7, minus 4 and minus 2 proPSA. Partial cleavage of minus 7 proPSA gives minus 4 and minus 2 proPSA.
- **Free PSA subforms:** proPSA, BPSA and other free PSA (such as intact PSA), circulating in roughly equal concentrations.
- **Epitopes:** free PSA exposes all epitopes (A to E). Alpha-1-antichymotrypsin (ACT) blocks only the E epitope, so PSA–ACT remains measurable.
- In prostate cancer, both proPSA and bound PSA rise in serum.

▸ **Factors affecting PSA**

- **Age:** becomes detectable at puberty, when LH and testosterone rise. Expression peaks twice, at 0–6 months and after about 10 years of age, in line with testosterone levels.
- **BMI:** the fall in PSA with raised BMI may be due to haemodilution.
- **Assay:** can cause pseudo-acceleration or pseudo-deceleration.
- **Genetics:** may influence levels.
- **DRE:** the change falls within the error of the assay.
- **Long-distance cycling:** about 10% rise after rides of more than 55 km.
- **Urinary retention:** variable degree of rise.

▸ **Percentage free PSA**

- Prognostic value is debatable.
- Use is unclear when total PSA is below 4 ng/mL.

▸ **PHI (Prostate Health Index)**

- Approved for men aged 50 years or older with PSA 4–10 ng/mL and a negative DRE.

▸ **BPSA**

- Generally limited to the transition zone; highly expressed in BPH; correlates with prostate volume; unlikely to distinguish cancer from BPH.

▸ **PSA density and volume-based derivatives**

- Volume-based derivatives also include complexed PSA density and PSA transition zone density.
- Usefulness of PSA density has not been confirmed in all studies.

▸ **PSA velocity**

- Some studies use a cut-off of 0.4 ng/mL/year and apply velocity to men with PSA below 4 ng/mL as well as 4–10 ng/mL.
- Above 2 ng/mL/year: increased mortality after radiotherapy or radical prostatectomy.

## Prostate-specific membrane antigen (PSMA)

- Type II transmembrane glycoprotein in the cell membrane of prostatic epithelial cells; the extracellular C-terminus binds glutamate and glutamate-like structures.
- **Gene:** chromosome 11p.
- **Other sites:** central nervous system and intestine.
- **Uses:** diagnosis, prognosis and imaging; greatest current use is targeted imaging and therapeutics.
- Raised levels probably indicate a worse prognosis.

## Human kallikrein 2 (hK2) and other kallikreins

- 80% amino acid homology with PSA; similar specificity for prostatic tissue; regulated by androgens.
- **Concentration:** less than 2% of total PSA in prostatic tissue, seminal fluid and serum.
- **Serum forms:** bound and free, mostly free.
- **Expression:** varies independently of PSA; less intense than PSA in benign tissue and more intense in cancer.

- Correlates with more aggressive pathological features.
- **Other kallikreins:** KLK4, KLK14 and KLK15, found mainly in tissue rather than serum.

## Endoglin

- CD105; transmembrane glycoprotein and cell-surface coreceptor for TGF-beta 1 and TGF-beta 3, on vascular endothelial cells; key role in angiogenesis.
- **Pre-operative plasma level:** associated with node-positive disease; predicts biochemical recurrence.
- **Urinary level:** correlates with tumour volume.
- Used alone or with other markers; clinical utility needs further study.

## Circulating tumour cells (CTCs)

- **Definition:** CD45-negative cells positive for EpCAM and/or cytokeratin.
- No single gold-standard identification method.
- **CellSearch:** the only FDA-approved method. Antibodies against EpCAM, CD45 (negative) and cytokeratins 8, 18 and 19 (positive).
- **Prognosis:** 5 or more CTCs per 7.5 mL of blood is poor prognosis in prostate, colorectal and breast cancer.
- **CTC-chip:** developed at Harvard Medical School; highest sensitivity; detects a single EpCAM-positive cell among 1 billion blood cells.

## Autoimmune responses

- The body's autoimmune response to cancer (autoimmune signatures) can be a target for biomarker development.

## Urine-based biomarkers

### ▸ PCA3

- **Gene:** chromosome 9q.
- Not expressed outside the prostate; levels far higher in malignant than benign tissue.
- **Commercial assay:** Progensa (Hologic).
- **Attentive DRE:** three firm strokes on each lobe towards the median sulcus; collect the first 20–30 mL of urine within 1 hour.
- **Cut-offs:** 10, 25 and 35 proposed, with different sensitivity and specificity.
- Reported to lead to better health outcomes.

### ▸ Mi-Prostate Score

- Gives the likelihood of detecting prostate cancer on biopsy and the probability of detecting high-grade cancer.

### ▸ Metabolomics

- Exploits differences in metabolic processes between cancer cells and benign cells.
- **Sarcosine:** the most prominent metabolite; a metabolite of glycine found in the urine of patients with prostate cancer.
- **Prostarix:** commercial panel run on a post-DRE sample; not FDA approved.

## Other serum and tissue biomarkers

▸ **Annexin A3**

- Calcium- and phospholipid-binding protein; level inversely related to presence of prostate cancer; used alone or with other markers.

▸ **MicroRNA**

- Single-stranded RNAs that regulate mRNA, with diagnostic and prognostic roles. miR-141 is a marker for prostate cancer.

▸ **AMACR (alpha-methylacyl coenzyme A racemase)**

- **Gene:** chromosome 5; upregulated in prostate cancer.
- **Function:** beta-oxidation of fatty acids.
- **In biopsy tissue:** 97% sensitivity and 100% specificity.

▸ **Epigenetic modifications**

- Include histone acetylation as well as DNA methylation.
- **Key hypermethylated genes:** retinoic acid receptor beta 2 (RAR-beta-2), GSTP1, APC and RASSF1A.

# Ch 155 · Prostate Biopsy: Techniques and Imaging

## Ultrasonographic anatomy

▸ **Zones and capsule**

- **Corpora amylacea:** age-related calcifications along the surgical capsule, highlighting the plane between peripheral and transition zones.

▸ **Urethra and internal sphincter**

- The sagittal plane shows the prostatic urethra along its entire course.
- **Internal sphincter:** extends from the bladder neck around the urethra to the verumontanum. Appears as a hypoechoic ring around the upper prostatic urethra, giving a funnelled appearance proximally as it arises from the bladder neck.

▸ **Seminal vesicles and vasa**

- **Seminal vesicles:** smooth saccular appearance; should be symmetrical.
- **Vasa deferentia:** in the transverse plane, course just above the seminal vesicles before diving caudally towards the prostate near the midline.

## TRUS equipment and settings

▸ **Probes**

- **Commonest frequency:** 7 MHz (range 6–10 MHz).
- **6–10 MHz:** high-resolution image, focal range 1 to 4 cm from the transducer; best for the peripheral zone, where most cancers arise.
- **Lower frequency:** lower resolution, but more accurate volume measurement and better anterior delineation of large glands.
- **End-fire versus side-fire:** no significant difference in positive biopsy rate. Side-fire may give better patient tolerance (check).

▸ **Coupling medium**

- Water-density medium, because soft tissue acoustic properties resemble water and useful ultrasound energy does not propagate through air.

▸ **Machine settings**

- **Magnification:** low for prostate measurements, high during biopsy.
- **Brightness:** optimal when normal peripheral zone is medium grey; this is the reference for echogenicity.
- **Echogenicity relative to normal peripheral zone:** hypoechoic is darker, isoechoic the same, hyperechoic lighter, anechoic completely black.

▸ **Other uses of TRUS**

- Treatment planning of radiofrequency ablation and microwave therapy.

## Probe manipulation

▸ **Transverse imaging**

- **Cephalad into the rectum:** prostate base, seminal vesicles and bladder neck.
- **Caudal towards the anal sphincter:** prostate apex and proximal urethra.

▸ **Sagittal imaging**

- **Rotation:** clockwise shows the left side; counterclockwise shows the right side.
- **Angling:** towards the floor shows the right side; towards the ceiling shows the left side.

## TRUS findings

▸ **Prostate cancer**

- 60% hypoechoic, 1% hyperechoic, remaining 39% isoechoic.
- **Other hypoechoic lesions:** granulomatous prostatitis, lymphoma and prostatic infarct.
- Include all hypoechoic lesions in the biopsy, although they are not pathognomonic of cancer.

▸ **Treated prostate**

- **After EBRT:** volume decreases by 6 months.
- **Brachytherapy seeds:** hyperechoic with posterior shadowing.
- **After radical prostatectomy:** normal if the bladder neck tapers smoothly to the urethra. A nodule anterior to the anastomosis is the ligated dorsal vein complex.

## Preparing the patient

- **Bladder:** a small amount of urine helps by demonstrating the prostate–bladder junction.
- **Enema:** decreases faeces in the rectum and gives a superior acoustic window.
- **Antibiotic prophylaxis:** oral TMP-SMX is one choice.
- **Endocarditis risk:** preoperative IV ampicillin plus gentamicin, followed by 2–3 days of oral fluoroquinolone.

## Analgesia and positioning

▸ **Local anaesthesia**

- **Periprostatic lidocaine site:** the hyperechoic fat pad marking the junction of seminal vesicles and prostate, near the bladder base.

- Direct intraprostatic injection can augment periprostatic injection.

▸ **Positioning**

- **Left lateral decubitus:** knees and hips flexed 90 degrees; pillow between the knees; buttocks flush with the end of the table.
- **Right lateral decubitus:** can also be used.
- **Lithotomy:** preferred for fiducial gold marker placement, brachytherapy planning and transperineal biopsy.

## Biopsy technique

- **Biopsy gun:** advances the needle 0.5 cm and samples the next 1.5 cm; the tip extends 0.5 cm beyond the area sampled.
- **Fixation:** 10% formalin.
- **Sextant biopsy:** one core from base, mid-gland and apex on each side, in the parasagittal plane.
- **Seminal vesicle biopsy:** recommended by some when PSA is above 30.

## Repeat biopsy

- **PIN at multiple sites:** 20–30% cancer risk; an indication for repeat biopsy.
- **Third or fourth biopsy:** justified by high suspicion of cancer, and also by poor prognostic factors on the first or second biopsy.
- Direct attention towards the apex.

## Transperineal and transurethral biopsy

▸ **Transperineal**

- Dorsal lithotomy with an end-fire transducer.
- Seminal vesicles and focal hypoechoic peripheral-zone lesions are hard to see.
- Minimum 6 cores, 3 from each side.

▸ **Transurethral**

- Not done, because TRUS biopsy samples the transition zone adequately.
- Solitary transition-zone cancers without peripheral-zone tumour are rare (under 5%).

## Complications of biopsy

- **Further risk factors for infection:** non-white race, more comorbidities, prostatomegaly and number of previous biopsies.
- Prostatitis and epididymitis are listed complications.
- Erectile dysfunction after biopsy is incompletely characterised.

## Advanced and investigational techniques

▸ **Colour and power Doppler**

- Aim to detect the neovascularity of prostate cancer.
- Unenhanced Doppler cannot show cancer microvessels, which are 10–15 micrometres in diameter.

▸ **Microbubble contrast agents**

- IV bubbles of air or a high-molecular-weight gas, encapsulated so that they last longer.
- **Size:** 1–10 micrometres.
- **Capsule:** albumin or polymer hard shell, or a lipid or surfactant coat.

▸ **Elastography**

- Measures tissue stiffness; reduced elasticity suggests malignancy. May prove better than colour Doppler at finding malignant areas.

▸ **Prostate HistoScanning**

- Evolving technology that differentiates, characterises and displays prostate tissue by analysing back-scattered ultrasound.

▸ **Computerised templates**

- Turn 2D ultrasound data into a 3D model for targeted imaging (TargetScan, Artemis).

## mpMRI-targeted biopsy

- **Commercial MRI/TRUS fusion platforms:** UroNav and Artemis (with ProFuse software).

# Ch 156 · Pathology of Prostatic Neoplasia

## Prostatic intraepithelial neoplasia

▸ **Evidence that HGPIN is a precursor**

- More HGPIN is linked to a greater number of multifocal carcinomas.
- HGPIN and cancer share the same biomarker and molecular changes.

▸ **Clinical significance**

- Risk of cancer on a further biopsy within a year of HGPIN: 26.4%.
- PIN on its own does not raise PSA.

▸ **Repeat biopsy**

- **Unifocal HGPIN:** after 3 years.
- **PINATYP:** within 6 months.

## Gleason system

- Grading is done at relatively low magnification, 10x.
- The original score ranges from 2 to 10.
- **Scores 2–4 are not given on needle biopsy:** experts grade most of these tumours higher on review; reproducibility is poor; they do not reliably mean favourable findings at RP.

| Pattern | Features |
|---|---|
| 1 | Well-circumscribed nodule of closely packed, medium-sized, round to oval glands |
| 2 | Like pattern 1 and fairly circumscribed, but the nodule edge may show minimal infiltration; glands looser and less uniform |
| 3 | Glands usually smaller than in patterns 1 and 2, vary markedly in shape and size, infiltrate in and among non-neoplastic acini |
| 4 | Includes hypernephromatoid glands |
| 5 | Includes comedocarcinoma: central necrosis surrounded by papillary, cribriform or solid masses |

## Gleason grading of variants

| Variant | Pattern |
|---|---|
| Signet-ring cell carcinoma | 5 |
| Sarcomatoid carcinoma | 5 |
| Xanthomatous carcinoma | 3 or 4 |
| Atrophic carcinoma | 3 (3/2) |
| Pseudohyperplastic carcinoma | 2 or 3 |

## Clinical significance of Gleason grade

- **Gleason score 6:** more homogeneous tumours; RP is predicted to cure them.
- **Grade group 4 or 5 in more than 5% of prostatic volume at RP:** high chance of biochemical recurrence.
- **Gleason score predicts:** side-specific extraprostatic extension and extension into the neurovascular bundle; prognosis after cryotherapy and after HIFU.
- Helps set intervention criteria during active surveillance.

## Spread of prostate cancer

- **Rectum:** rarely involved; can be hard to tell from a rectal primary.
- **Lung:** metastases commonly found at autopsy.
- **After nodes and bone:** bladder, liver and adrenal gland are the next commonest sites.

# Ch 157 · Diagnosis and Staging of Prostate Cancer

## Screening for prostate cancer

- **Advantage:** lowers the rate of advanced disease at diagnosis.
- **EAU 2018 (weak recommendation):** risk-adapted strategy based on initial PSA. Men initially at risk (PSA above 1 ng/ml at 40 years or above 2 ng/ml at 60 years) are followed 2-yearly; for men not at risk, follow-up is postponed to 8 years. The other EAU early-detection statements are strong recommendations.

## Diagnosis

▸ **DRE**

- Only about 18% of cancers are detected by DRE alone.
- Detects tumour only at a volume of 0.2 ml or more, so misses a substantial portion of early cancers.
- Can overestimate or underestimate tumour extent.

▸ **PSA**

- Better predictor of cancer than DRE alone or TRUS alone.
- **Cancer risk by PSA (PCPT):** 17% at 1.1–2.0, 24% at 2.1–3.0 and 27% at 3.1–4.0 ng/ml.

▸ **PSA derivatives**

- Complexed PSA, alongside free PSA, PSA isoforms, hK2, PHI and the 4K panel.

## Staging systems

| Whitmore–Jewett stage | Description |
|---|---|
| A | Found incidentally; not clinically detectable |
| B | Found on physical examination; not spread outside the capsule |
| C | Extended through the capsule |
| D | Spread to other organs |

# Ch 158 · Active Management Strategies for Localised Prostate Cancer

## Evaluation before treatment of localised cancer

- Men planning surgery should have a fuller work-up (bone scan, CT or MRI, fitness tests), as higher-risk men do.
- Monoclonal antibody scans and MR spectroscopy are not widely used but may prove useful in future.

# Ch 159 · Active Surveillance of Prostate Cancer

## Active surveillance

▸ **Rationale and outcomes**

- In low-grade disease, progression and mortality increase slowly up to 15 years but do not increase rapidly beyond that.
- Overdiagnosis and overtreatment place a financial burden on the patient and the healthcare system.
- Disease-specific and all-cause survival compare favourably with long-term outcomes after radical prostatectomy or radiotherapy in similar-risk patients.
- 25–50% develop objective evidence of tumour progression.

▸ **Problems**

- Risk of losing the window of opportunity for cure of early localised disease.
- Patient must accept a strict follow-up protocol.
- Biopsy-related infections and erectile dysfunction; nerve-sparing surgery may be difficult after repeated biopsies.
- Less established in patients with a long life expectancy.

▸ **Counselling and triggers**

- Candidates should understand they carry a risk of extraprostatic extension and lymph node metastasis.
- **Trigger for intervention:** tumour extension on DRE.

## Watchful waiting

- **Monitoring:** history, examination with DRE, PSA and creatinine every 6 months; bone scan yearly.
- **Palliative treatment (usually ADT):** started on symptomatic progression, upper tract obstruction or metastatic disease.

## Patient selection and evaluation

▸ **General selection**

- **Upper age limit:** generally 76 years.
- **Ideal candidate:** tumour that is biologically significant and completely resectable.
- **Advantage of RP:** treatment failure is easier to detect.
- **Staging tests not widely used but possibly useful in future:** monoclonal antibody scans, PET scans, MR spectroscopy and lymphotropic MRI.

▸ **Neoadjuvant therapy**

- **Hormones:** make nerve-sparing surgery more difficult.
- **Chemotherapy:** rarely produces a complete pathological response.

▸ **Nerve-sparing selection**

- **Nerve sparing is of doubtful value with:** poor preoperative erections; conditions causing ED (diabetes, hypertension, neurological disorders); no current or expected sexual relationship.
- **Counselling:** nerve sparing is not always possible; men may need PDE5 inhibitors, intraurethral or intracorporeal vasodilators, vacuum erection devices, venous flow constrictors or a penile prosthesis.

## Surgical approaches

▸ **Perineal RP**

- Shorter operating time than retropubic RP.
- Higher risk of rectal injury, with faecal incontinence.

▸ **Retropubic RP**

- **Advantages:** familiar anatomy; low risk of rectal injury; wide exposure with access to pelvic nodes; preservation of the neurovascular bundles; lower risk of cancer at the surgical margins.

▸ **Laparoscopic RP**

- Higher risk of severe complications.
- Difficult haemostasis, with injury to the neurovascular bundle and cavernous nerves.
- Bowel injury and urinary ascites.
- **Cancer control uncertain:** no long-term studies; early studies showed more positive margins.

## Open RP: preparation and operative steps

▸ **Preoperative preparation**

- **Bowel preparation:** clear liquid diet the day before, magnesium citrate in the evening, enema on the morning of surgery.
- **Antibiotics:** 1 hour before surgery, continued for a total of 24 hours.
- **Anaesthesia:** general, with systolic blood pressure kept below 100 mmHg.

- **Position:** supine, with or without table flexion at the anterior superior iliac spine; 16 Fr Foley catheter to drainage.

▸ **Operative steps**

- **Endopelvic fascia:** incised where it is transparent and shows the underlying levator ani.
- **Superficial branch of the dorsal vein:** divided with the puboprostatic ligaments.
- **Apical dissection:** the most complex part of the operation.
- **Neurovascular bundle:** usually found at the apex, but dissection can also start at the base.
- **Lateral pedicles:** thick, so divided lateral to the seminal vesicles in superficial, middle and deep layers.
- **Vas:** the ampullary portion is dissected with the seminal vesicles.

▸ **Positive margins**

- **Commonest sites, in order:** apex, posterior surface, posterolateral surface.

## Modifications of classical RP

▸ **Nerve-sparing RP**

- Before excising one NVB, first free the contralateral NVB from the prostate, starting at the apex, to avoid traction injury.
- **High anterior release technique:** incise the levator fascia over the anterior apex and carry it distally along the lateral edge of the apex. Expose and release the NVB before transecting the urethra, without entering the underlying prostatic fascia.
- **Drawback of high anterior release:** less soft tissue at the apex, so positive margins could increase.

▸ **Bladder neck sparing RP**

- **Aim:** hasten return of urinary control.
- **Continence:** clinically significant advantage at 6 months, which largely disappears by 1 year.
- **Margins:** little difference in positive margin rates.
- Bladder neck contracture rates may be high, so not done routinely.
- **Heidelberg technique:** preserves the circular fibres of the bladder neck, with a urethro-urethral anastomosis.

▸ **Seminal vesicle sparing RP**

- Most men with low-risk disease can undergo it.
- Non-oncological risks and benefits are uncertain and compromised cancer control is a concern; randomised trials are needed, so not done routinely.

## Postoperative care

- **Ambulation:** with assistance on the evening or afternoon of surgery.
- **Catheter removal:** 3–21 days, depending on anastomotic integrity. Removal before 7 days gives retention in 15–20%.
- **Pad:** used until complete urinary control.
- **First PSA:** 1 month after RP.

## Cancer control and recurrence

- **Independent prognostic factors:** preoperative PSA and treatment given, along with stage and Gleason score.
- **Adverse pathological factors:** lymphovascular invasion, non-organ-confined disease, extracapsular extension, positive margins, seminal vesicle invasion and nodal metastases.
- **After a PSA rise:** only 34% go on to metastasis or cancer death.
- **Neuroendocrine tumours:** exception to PSA follow-up, as PSA rises little; DRE is used to detect palpable recurrence.
- **Follow-up duration:** recurrence keeps rising for at least 15 years, and mortality risk may rise for 25 years or more, so follow-up must be long.

## Continence and erectile function

▸ **Incontinence**

- **Causes:** intrinsic sphincter deficiency (usually older men); injury during ligation and division of the DVC; damage to urethral smooth muscle; preoperative overactive bladder from bladder outlet obstruction.

▸ **Potency**

- **Definition:** erections rigid enough for penetration and intercourse, with or without a PDE5 inhibitor.
- **Determinants:** age, preoperative potency, extent of nerve sparing and era of surgery.
- **Rates:** satisfactory erections in about 95% of men in their 40s and 50% in their 70s.
- **Timing:** partial erections return first, at 3–6 months.
- Erections are usually not as good as before surgery.

## Complications of RP

- **Early complications:** under 10% in experienced hands.

▸ **Bladder neck contracture**

- **Causes:** inadequate mucosal approximation, haematoma and urinary extravasation.
- **Recalcitrant cases:** cold-knife incision, then triamcinolone 200 mg in 5 mL injected into the bladder neck.

▸ **Obturator nerve injury**

- **Options:** tension-free primary repair; nerve graft (cutaneous or genitofemoral nerve); or no repair, with physical therapy.

▸ **Lymphocele**

- Perioperative heparin thromboprophylaxis may predispose to lymphoceles.

# Ch 161 · Laparoscopic and Robotic-Assisted Laparoscopic Radical Prostatectomy and Pelvic Lymphadenectomy

## Laparoscopic and robotic prostatectomy

▸ **Preparation and positioning**

- **Bowel preparation:** magnesium citrate or bowel prep the day before, or only a Fleet enema on the morning of surgery.

- **Steep Trendelenburg:** legs spread slightly apart to allow access to the rectum.

▸ **Ports**

- **Robotic (RALP):** 12 mm camera port; 12 mm and 5 mm assistant ports; three 8 mm robotic ports.
- **Laparoscopic:** two 10–12 mm ports and two 5 mm ports.

▸ **Transperitoneal versus extraperitoneal approach**

- A urine leak is intraperitoneal with the transperitoneal approach and contained with the extraperitoneal approach.

▸ **Identifying the bladder neck**

- Further methods: retracting the bladder dome cephalad, and bimanual palpation (pinching) of the bladder neck with the two instruments.

▸ **Pelvic lymph node dissection**

- Generally recommended with intermediate- or high-risk features, such as PSA above 10, a large palpable nodule or Gleason score 7 or more.

▸ **Complications and limitations**

- **Position-related:** optic neuropathy and raised intraocular pressure.
- **Open conversion:** under 2%.
- **Robotic systems:** very bulky.

## Ch 162 · Radiation Therapy for Prostate Cancer

### Prognostic factors for localised disease

| Risk group | 5-year PSA-failure-free survival with RT |
|---|---|
| Low | About 85% |
| Intermediate | About 50% |
| High | About 33% |

- **Favourable versus unfavourable intermediate risk:** percentage of positive cores, number of unfavourable factors, primary Gleason pattern, and also pretreatment PSA velocity and mpMRI findings.

### External beam radiotherapy (EBRT): techniques

▸ **Conventional (pre-CT)**

- Portals set by skeletal anatomy (pubic symphysis and femoral heads); prostate position inferred from the Foley balloon, bladder contrast and a rectal marker.

▸ **3D conformal radiotherapy**

- CT-based 3D tumour visualisation began in the 1990s; increased precision and lowered dose to surrounding organs.

▸ **IMRT**

- With inverse planning, the prostate can receive about 1.8 Gy daily while most of the adjacent bladder receives less than half that dose.

▸ **Image guidance**

- Reduces interfraction and intrafraction motion; previously the beam was enlarged to accommodate prostate motion.
- **Interfraction motion:** daily ultrasound localisation just before treatment, or fiducial markers with software to locate the prostate each day.
- **Intrafraction motion:** intrarectal balloon pinning the prostate against the pubic bone, or real-time tracking with the Calypso system.

▸ **Dose**

- Current dose 76–80 Gy. Above 80 Gy is not beneficial; rectal toxicity limits the dose.

## EBRT side effects

- **Mechanism:** injury to the microvasculature of bladder, rectum, striated sphincter muscle and urethra.
- **Acute proctitis and cystitis:** about one-third of patients, usually when dose exceeds 50 Gy. Most settle after treatment; 5–10% have permanent symptoms.
- **Erectile dysfunction:** about 50%, from injury to the vasculature of the cavernous nerves and corpora cavernosa; usually begins about 1 year after treatment.

## Heavy particle beams and SBRT

▸ **Heavy particle beam therapy**

- Includes neutron and proton beams. Limited availability and expensive; perceived higher complication rates without much gain in tumour control.

▸ **Bragg peak**

- The peak on the Bragg curve, which plots energy loss of ionising radiation as it travels through matter. The proton beam is widened (spread out) to treat a larger tumour volume.

▸ **SBRT (CyberKnife)**

- Linear accelerator on a robotic arm, giving large doses in a small number of fractions.

▸ **Extreme hypofractionation**

- Often given on fewer than 5 days a week to allow normal tissue recovery; the whole course can be completed within 2 weeks.

## Brachytherapy (LDR and HDR)

▸ **Technique**

- **Early brachytherapy:** free-hand implantation, with lower success than today's image-guided implantation.
- **HDR:** iridium-192 source (1–10 curie) applied for seconds to minutes.
- **ADT to shrink large glands:** delays any PSA rise that would show tumour persistence or recurrence.

▸ **Toxicity**

- Depends on implant technique and quality.
- **Predisposing factors:** age, comorbidities (diabetes, peripheral vascular disease) and tobacco use.
- **TURP:** needed in about 10%; only a channel TURP should be done.

- **Impotence causes:** underlying small vessel disease, cavernous nerve damage and damage to vascular tissue around the penile bulb.

## Radiobiology: alpha-to-beta ratio

- **Definition:** the dose at which cell killing from the linear component equals that from the quadratic component.
- Early-responding tissues typically have a high ratio; late-responding tissues a low one.
- **Head and neck cancers:** 13–16 Gy.
- **Bladder:** 5 Gy.

## PSA follow-up and biochemical failure after RT

### ▸ Monitoring

- PSA every 6 months until nadir.

### ▸ PSA nadir

- **Source:** residual benign epithelium, residual local cancer cells or subclinical disseminated micrometastases.
- Level and timing both correlate with distant metastasis and cause-specific survival; some studies use 18 months as the time-to-nadir cut-off.
- A significant predictor of outcome, but no absolute threshold can or should define cure; the lower the nadir, the higher the chance of success.
- **After conventional-dose RT (median nadir):** 0.4–0.5 ng/ml with no evidence of disease; above 1 ng/ml with local failure; above 2 ng/ml with distant failure.
- After neoadjuvant hormonal therapy, PSA is often already undetectable at the start of RT, so true nadir and time to nadir cannot be determined.

### ▸ Post-nadir doubling time

- **Under 3–6 months:** suggests distant failure.
- **1 year or more:** suggests local failure.

### ▸ PSA bounce

- **Causes:** radiation prostatitis, delivery of a sublethal radiation dose, instrumentation and testosterone recovery after ADT.

### ▸ Rising PSA after brachytherapy

- Within 30 months: if PSA is 10 ng/ml or more, look for metastasis.
- If below 10 ng/ml but beyond 30 months, biopsy.
- Otherwise continue careful follow-up.

### ▸ Definitions of biochemical recurrence

- **Old ASTRO:** three consecutive PSA rises measured 6 months apart.
- **Phoenix:** a rise of 2 ng/ml often takes considerably longer, so failure is called later.
- **Locally advanced disease:** about 50% ultimately develop biochemical failure within 10 years of treatment.

## Post-radiation therapy biopsy

- **Problems:** timing after treatment, interpretation of indeterminate biopsies and sampling error.
- RT markedly alters gland architecture, the basis of Gleason scoring, so post-RT biopsies are often falsely reported as high grade.
- **Basal cells:** labelled by an anti-cytokeratin monoclonal antibody; present in benign tissue and absent in malignant tissue.
- **Proliferation markers:** Ki-67 and proliferating cell nuclear antigen (PCNA) can detect residual tumour.

## Molecular therapies and radiation

▸ **Rationale**

- Target vulnerable aspects of cancer growth or of the cancer's ability to repair injury, in a targeted way to minimise effects on normal tissue.

▸ **siRNA radiosensitisation**

- Also sensitises non-cancerous cells; delivery to specific cell types is the challenge.

▸ **MHC class I defects**

- Seen in 85% of primary prostate cancers and all metastatic tumours, so evasion of host immunity may be critical in prostate cancer development.

▸ **Radio-gene therapy**

- Based on enzyme/prodrug systems, oncolytic viruses and cytotoxins, causing cytolytic or apoptotic cell death; combining with RT may give superior cell killing.
- **Enzyme/prodrug therapy:** an enzyme vectored only to tumour cells converts an inactive prodrug, so the active drug stays away from normal cells.
- **HSV thymidine kinase plus ganciclovir:** suicide gene therapy, combined with a replication-competent adenovirus and radiation; recently tested in prostate cancer and appears safe.

▸ **Oncolytic viruses**

- Have a lytic phase in their normal life cycle that is lethal to the host cell.
- **Candidates:** adenovirus, vaccinia, Newcastle disease virus, HSV, influenza virus and mumps virus.
- **Adenovirus suits prostate cancer:** low pathogenicity for normal tissue, not tumourigenic, only mild to moderate symptoms when infection occurs.

# Ch 163 · Focal Therapy of Prostate Cancer

## Localisation of disease

▸ **Ultrasound-based modalities**

- **Options:** TRUS, contrast-enhanced TRUS (CE-TRUS), elastography and Prostate HistoScanning.
- **Limitation:** poor at localising disease, assessing tumour burden and detecting small tumour foci.

▸ **Saturation biopsy**

- Only a small additional benefit unless carried out with a template mapping technique.

▸ **MRI**

- **T2-weighted:** water is bright and fat is dark.
- **MR spectroscopy (MRS):** cancer shows an increased ratio of choline plus creatine to citrate.

## Cryotherapy

▸ **Mechanism of injury**

- **Freezing:** intracellular dehydration with pH changes contributes to cell death.
- **Warming:** osmotic cellular swelling and vascular hyperpermeability.
- **Later repair phase:** fibrosis, calcification and hyalinisation.

▸ **Devices and technique**

- **Recommended freeze:** rapid freezing down to minus 40 °C, held for at least 3 minutes, as part of a double freeze–thaw cycle.
- **Third-generation devices:** gas-based systems giving rapid freeze–thaw cycles.
- **Multiprobe systems:** smaller individual probes allow more precise freezing and reduce toxicity.
- **Cryoneedles:** inserted through the perineum, via a brachytherapy grid or freehand.
- **Thermocouples:** in the treatment area, Denonvilliers' fascia and other key areas such as the rhabdosphincter and neurovascular bundles.
- **Cystoscopy:** at the start, to verify needle position.

▸ **COLD (Cryo On-Line Database) registry**

- 3-year BCR-free survival 75.7%; new erectile dysfunction 41.9%; incontinence 1.6%.

## High-Intensity Focused Ultrasound (HIFU)

▸ **Physics**

- **Ultrasound:** mechanical vibration above the threshold of human hearing (16 kHz).
- **Generation:** an alternating voltage makes piezoelectric crystals (lead zirconate titanate) oscillate at the frequency of the current; the waves propagate as alternating cycles of compression and rarefaction.
- **Frequency:** diagnostic 1–20 MHz; therapeutic HIFU 0.8–3.5 MHz.
- **Low intensity (0.125–3 W/cm²):** can stimulate normal physiological responses to injury and accelerate drug transport across the skin.
- **High intensity:** damages tissue during the high-pressure (compression) phase.

▸ **Tissue effects**

- 56 °C for 1 second causes irreversible cell death by coagulative necrosis.
- **Histology:** coagulative necrosis, then inflammation, then granulation tissue (immature fibroblasts and new capillaries) at the edge of the necrotic area at about 1 week, then proliferative repair tissue at the edge by 2 weeks.

▸ **Procedure and selection**

- **Anaesthesia:** spinal or general, with a suprapubic or per-urethral catheter.

- **Guidance:** TRUS (commonest), TRUS–MR fusion or in-gantry MRI.
- Patient movement prevents successful treatment.
- **Suitable men:** clinically localised (T1–T2) disease, low PSA (less than 15 ng/mL) and low prostate volume.

## Photodynamic therapy

- A photosensitising drug is activated by light of a specific wavelength after a defined drug–light interval.
- Some photosensitisers, such as aminolevulinic acid (ALA), accumulate preferentially in tumour.

| | Vasculature-activated drugs | Tissue-activated drugs |
|---|---|---|
| Drug–light interval | Short (minutes) | Long (hours to days) |
| Sessions | Single session | Drug and light in separate sessions |
| Light precautions | None after a few hours | Slow clearance and skin accumulation; avoid sunlight for a few weeks to prevent sunburn |

### ▸ Light delivery

- **Transperineal:** hollow plastic needles placed under TRUS; optical fibres carry low-power laser light through them.
- **Other routes:** transurethral, or open insertion of fibres at laparotomy.

## Focal photothermal (laser) therapy

- Uses neither oxygen nor photosensitisers, unlike photodynamic therapy.
- **Planning and placement:** MRI for planning, TRUS for fibre placement; an open-ended catheter is passed transperineally into the lesion and the laser fibre is inserted through it.
- **Anaesthesia:** sedation or general.

## Focal irreversible electroporation (IRE)

- Needle electrodes go at the boundaries of the lesion, not within the treatment area.
- Once the needles are placed, treatment usually takes less than 5 minutes.

## Radiofrequency ablation (RFA)

### ▸ Pathology

- Coagulative necrosis with cell membrane disruption, protein denaturation and vascular thrombosis.

### ▸ Types

- **Monopolar or bipolar:** bipolar has less risk of accidental burns.
- **Wet or dry:** wet RFA uses constant saline infusion to reduce charring.

## Complications and cancer control

- **Rectal toxicity:** rectal bleeding and diarrhoea, besides perineal pain and rectourethral fistula.

- **Trifecta outcome:** achieved in 50–89%.

## Follow-up and failure after focal therapy

### ▸ PSA density and kinetics

- BPH contributes about 0.3 ng/mL per gram and cancer about 3.5 ng/mL per $cm^3$; this may be used in future.
- The role of PSA velocity and PSA doubling time needs future trials.

### ▸ HIFU failure

- **Stuttgart definition of biochemical failure:** PSA nadir plus 1.2 ng/mL.
- **Management:** no clear-cut guideline; options are salvage radiotherapy and salvage radical prostatectomy.

# Ch 164 · Treatment of Locally Advanced Prostate Cancer

## Incidence

- About 10% of men with newly diagnosed prostate cancer have locally advanced disease.

## Staging, imaging and novel markers

### ▸ TRUS

- Understages prostate cancer and has high interobserver variability for local staging.
- Colour and power Doppler TRUS for local staging are still under investigation.

### ▸ CT or MRI of the pelvis

- **Indications:** cT3 or cT4 disease, or a calculated probability of lymph node invasion above 10%.

### ▸ Novel markers

- Needed because traditional clinical and pathological parameters predict the local extent of tumour poorly.
- **TMPRSS2–ETS gene fusion:** associated with high pathological stage.
- **Sarcosine:** associated with metastatic prostate cancer.

## Radical prostatectomy (RP)

### ▸ Outcomes

- Cancer-specific survival about 60% at 15 years; overall survival about 75% at 10 years.

### ▸ Frozen section of pelvic nodes (EAU)

- Do not use intraoperative frozen section to decide whether to complete the RP; complete the RP regardless.

### ▸ Neoadjuvant (preoperative) ADT

- Does not alter the rate of lymph node metastases.
- Might benefit high-risk disease by lengthening the time to biochemical recurrence.
- Nerve sparing may be difficult afterwards.

## Adjuvant therapy after RP

▸ **Adjuvant radiotherapy**

- **Indication:** includes extensive extracapsular extension.
- **Field:** prostatic fossa.
- **Timing:** after urinary function recovers, at about 3 months after surgery.
- **Timing (also given as):** typically 4–6 months after RP, while PSA is still undetectable and once continence is regained.

▸ **Adjuvant chemotherapy**

- Only within clinical trials.

▸ **pN1 disease**

- **Observation after extended lymph node dissection (EAU):** 2 or fewer nodes, microscopic involvement only, no extranodal extension, and PSA below 0.1 ng/mL.
- **Most likely to benefit from adjuvant RT:** low-volume nodal disease (fewer than 3 nodes) with Gleason 7–10, pT3–pT4 or R1 margins; possibly also 3–4 positive nodes.
- **RT field:** no recommendation on extent, though whole-pelvis RT appears beneficial.

## Primary radiotherapy and other treatments

▸ **ADT with radiotherapy**

- **Neoadjuvant ADT:** starts 2–3 months before RT.
- **Adjuvant ADT:** starts at the onset of RT.

▸ **Brachytherapy**

- Less studied as primary treatment in high-risk and locally advanced cases.

▸ **Focal ablative therapy**

- May have a role when combined with ADT or other systemic therapy.

▸ **Intermittent androgen deprivation protocol**

- Total androgen deprivation until PSA falls below 4.0 ng/mL (usually within 24–32 weeks).
- Then withhold treatment and monitor PSA.
- Restart when PSA returns to the pretreatment level or rises above 10–20 ng/mL.

## Management of post-operative biochemical recurrence (BCR)

▸ **Timing of recurrence**

- 50% recur within 3 years and 99% within 10 years; recurrence is rare after 15 years.

▸ **Salvage RT**

- Start before PSA reaches 0.5 ng/mL.
- **Favourable response:** PSA recurrence long after surgery, slowly rising PSA, low-grade tumour, no seminal vesicle or lymph node involvement, and possibly positive margins (check).

▸ **Surveillance or delayed RT**

- **Suitable men:** pathological stage below pT3a, time to recurrence over 3 years, PSA doubling time over 12 months and Gleason 7 or less.

## LHRH agonists

▸ **Flare**

- Seen with all preparations; can cause severe, life-threatening exacerbation of symptoms.
- Some authorities start the antiandrogen 1 week before the agonist.

▸ **Leuprolide**

- **Brands:** Eligard (subcutaneous) and Lupron Depot (intramuscular).
- **Doses:** 3.75 mg or 7.5 mg monthly; 11.25 mg or 22.5 mg 3-monthly; 30 mg 4-monthly; 45 mg 6-monthly.

▸ **Monitoring**

- PSA in the first few weeks; LH, FSH and serum testosterone after 4 weeks.

▸ **Adverse events**

- **More than 10%:** hot flushes, cardiovascular changes or ischaemia, fatigue, pain and peripheral oedema.
- **1–10%:** asthenia, gynaecomastia, headache, testicular atrophy, anorexia, anaemia, nausea, vomiting, constipation and urinary frequency.
- **Less often:** impotence and haematuria.

## LHRH antagonists

- **Members:** degarelix, abarelix and cetrorelix.

▸ **Abarelix**

- First-generation; causes a severe allergic reaction not seen with third- and fourth-generation antagonists.
- Observe for 30 minutes after injection.

▸ **Degarelix (Firmagon)**

- **Supply:** 80 mg and 120 mg powder for injection.
- No systemic allergic reactions of the abarelix type.
- **Adverse effects over 10%:** hot flushes, injection-site erythema and pain, weight gain and raised liver enzymes.
- **Adverse effects 1–10%:** arthralgia, back pain, fatigue, hypertension and injection-site swelling or induration.

## Oestrogens and androgen synthesis inhibitors

▸ **Diethylstilbestrol (DES)**

- Oestrogen that suppresses testosterone; about a thousand-fold more potent than testosterone at suppressing LH and FSH.
- No bone loss, but severe thromboembolism even at low doses and cardiovascular toxicity, so not standard first-line treatment.

▸ **Androgen synthesis inhibitors**

- Aminoglutethimide, ketoconazole and abiraterone.

## Steroidal antiandrogens

- **Members:** cyproterone acetate, megestrol acetate and medroxyprogesterone acetate; all have poor overall efficacy.

### ▸ Cyproterone acetate

- **Action:** blocks the androgen receptor directly; its progestational action crosses the blood–brain barrier and inhibits LH centrally, rapidly lowering testosterone by 70–80%.
- **Side effects (hypogonadal):** loss of libido, erectile dysfunction, lassitude, gynaecomastia (under 20%) and rare fulminant hepatotoxicity.

## Non-steroidal antiandrogens

- No hypogonadism, so potency is preserved; long-term preservation in about 20%.

### ▸ Flutamide

- The first pure antiandrogen.
- **Half-life:** 6 hours; active metabolite 2-hydroxyflutamide, excreted by the kidneys.
- No fluid retention or thromboembolism.
- Overall survival significantly shorter than with DES.

### ▸ Bicalutamide

- Unaffected by age, renal insufficiency or moderate hepatic impairment.
- Inferior to DES.
- 50 mg/day monotherapy is inferior to orchidectomy in metastatic or locally advanced disease.

### ▸ Nilutamide

- **Half-life:** 56 hours; cleared by the liver.
- **Side effects:** alcohol intolerance, nausea, and pneumonitis that can progress to interstitial fibrosis.

## Combined androgen blockade (CAB)

- Serum testosterone does not fall to zero after surgical or medical castration, showing additional (adrenal) androgen sources.
- **Largest trial (1286 patients):** no difference between surgical castration with or without flutamide.

## Intermittent androgen deprivation therapy (IADT)

- **Rationale:** in animal models, intermittent deprivation lengthened the time to refractory growth; and the benefit of continuous ADT was increasingly questioned because of its profound side effects.
- **Possible benefits:** off-treatment periods relieve side effects, improving quality of life and compliance; delayed progression to CRPC; efficacy similar to continuous ADT, especially with lower disease burden.
- **Candidates:** PSA below 4 during a 5–7 month induction, poor tolerance of continuous ADT, non-metastatic disease or nodal-only metastases, locally advanced disease, older age, Gleason 7 or more, and longer PSA doubling time.

## Bilateral orchidectomy

- Subcapsular orchidectomy does not improve cancer control over simple orchidectomy.

## Complications of androgen deprivation

### ▸ Bone loss

- Risk of osteopenia, osteoporosis and fracture rises with time on ADT; 4 years puts an average man in the osteopenia range.
- **Risk factors:** older age, family history, Asian or Hispanic ethnicity, corticosteroid use, pre-existing bone loss and longer ADT.
- **Measurement:** bone mineral density at the hip by DEXA.
- Transdermal estradiol increases bone mineral density.

### ▸ Skeletal complications and bone evaluation

- **Complications:** fracture, uncontrolled pain, spinal cord compression, nerve entrapment and hypercalcaemia.
- **Blood and urine markers:** bone-specific ALP; urine N-telopeptide (NTX) for bone turnover; TRAP (tartrate-resistant acid phosphatase) for bone resorption.
- **Bone scintigraphy:** best sensitivity but low specificity for bone metastases.
- **Plain X-ray:** sensitivity only 40–50%; equivocal cases need X-ray, CT or MRI, and PET or PSMA PET.
- **Bone biopsy:** accurate but usually unnecessary.

### ▸ Hot flashes

- **Definition:** subjective warmth in the upper torso and head followed by objective perspiration.
- **Mechanism:** raised hypothalamic adrenergic concentrations, with altered beta-endorphins and calcitonin gene-related peptide, acting on the hypothalamic thermoregulatory centre.
- **Course:** intensity and frequency fall over time but may persist; treat only men who find them bothersome.

| Treatment | Dose | Effect and drawbacks |
|---|---|---|
| Megestrol acetate | 20 mg twice daily; can reduce to 5 mg twice daily | Appetite stimulation |
| Oestrogens (low-dose DES, transdermal estradiol) | Not given | Up to 90% partial or complete relief; painful gynaecomastia and thromboembolism |
| Venlafaxine (antidepressant) | 12.5 mg twice daily | Reduces hot flashes in more than 50% |

### ▸ Sexual dysfunction

- 10–17% keep erections adequate for intercourse; up to 20% keep some sexual activity.
- **Libido:** more affected than erection, worse with longer ADT, and extremely difficult to treat.
- **Erectile dysfunction:** oral PDE5 inhibitors or intracavernosal alprostadil, though perhaps not long term.

### ▸ Cognitive function

- Reduced verbal fluency, which improves with testosterone; other cognitive symptoms do not.
- Also memory loss, anxiety, psychological distress and decline in complex information processing.

### ▸ Body habitus, diabetes and cardiovascular effects

- Weight rises by 1.8–3.8% after 1 year.
- Increased appetite; low testosterone is linked to raised insulin and increased abdominal girth.
- Subcutaneous fat accumulation and a rise in HDL.
- Diabetes or metabolic syndrome in more than 50% on long-term ADT.
- Cardiovascular morbidity and mortality arise from effects on lipids, glucose and body habitus, and are more pronounced with lower-risk cancer.

### ▸ Gynaecomastia

- Prophylactic breast radiotherapy gives no benefit once it has begun.
- **Established gynaecomastia:** liposuction and subcutaneous mastectomy.

### ▸ Anaemia

- Very common; thought to be due to loss of testosterone stimulation of erythropoietin, reflecting ineffective erythropoiesis.

## Biochemical recurrence after radical prostatectomy

### ▸ Natural history

- Progression from BCR to metastasis and death depends on PSA doubling time, pathological Gleason score and time from surgery to BCR, and also on age and advanced tumour stage.

### ▸ Ultrasensitive PSA (uPSA)

- Measures PSA below 0.1 ng/mL.
- May cause anxiety over clinically meaningless rises and lead to overtreatment.

### ▸ Imaging

- **Bone scan:** positive in only about 4% when PSA is below 10 ng/mL.
- **TRUS and TRUS-guided fossa biopsy:** reasonable sensitivity for local recurrence at low PSA.
- **Indium-111 capromab pendetide (ProstaScint):** IgG monoclonal antibody to the intracellular domain of PSMA, exposed only in apoptotic or necrotic cells, which limited its use.
- **J591 and newer anti-PSMA antibodies:** bind the external (extracellular) domain.
- Sensitivity and specificity of these antibody scans only about 60–70%.
- **18F-FDG PET:** low sensitivity, because benign tissue and post-operative scar show similar uptake.
- **Better PET tracers:** acetate, choline and fluorocholine; choline PET is US FDA approved for detecting recurrence.

### ▸ Salvage RT

- At PSA 0.5 ng/mL or less, progression-free survival is 48%.
- Improves all-cause mortality.
- **Causes of failure:** persistent local disease, local recurrence, persistent metastasis and new metastatic disease.
- **May be delayed:** stage pT3a or lower, PSA doubling time over 12 months and Gleason 7 or less.

### ▸ ADT

- Almost 60% still receive ADT, though it is not curative.
- One theoretical benefit of adding ADT to salvage RT is sterilising micrometastatic disease.

## Recurrence after radiotherapy

### ▸ General approach

- Men usually fail after 65 years of age and have comorbidities, so options range from watchful waiting and delayed systemic ADT to local salvage therapy.
- Over 90% are managed with watchful waiting and delayed systemic ADT.
- About 93.5% receive ADT as second-line therapy after radiation failure.

### ▸ Imaging

- Difficult because of fibrosis and prostate shrinkage.
- **TRUS:** limited ability to identify suspicious lesions.
- **T2-weighted MRI:** sensitivity only 27%, but high specificity.

### ▸ Salvage radical prostatectomy

- The most definitive salvage treatment; 10-year cancer-specific survival 70–80%.
- About 50% had major complications in older series.
- Morbidity reduced by better surgical planning, meticulous attention to anatomy and improved radiation delivery.

### ▸ Salvage cryotherapy

- **Incontinence:** as high as 72% in earlier series, now 4–6.5%.
- **Other complications:** rectal pain, scrotal pain or oedema, and haematuria.

### ▸ Salvage HDR brachytherapy

- Can implant the seminal vesicles and treat extracapsular extension, while conformal delivery limits dose to nearby organs.

### ▸ Salvage HIFU

- **Complications include:** urinary retention, urinary tract infection and urethrocutaneous fistula.

## Recurrence after primary cryotherapy

- **Definition:** no single accepted one; used definitions are PSA above 0.5 ng/mL, PSA above 1 ng/mL, the ASTRO definition, and nadir plus 2 ng/mL.
- **Management:** not well established; repeat cryotherapy, salvage radiotherapy (the feasible choice) or radical prostatectomy.

## Trials in hormone-sensitive disease

### ▸ STAMPEDE

- **Name:** Systemic Therapy in Advancing or Metastatic Prostate Cancer: Evaluation of Drug Efficacy.
- **Design:** multi-arm, multi-stage (MAMS) randomised controlled trial in a tertiary-care setting.
- **Participants:** men starting long-term hormone therapy.
- **Hypothesis:** research interventions would improve survival over standard of care alone.

- **Primary outcomes:** overall survival (definitive) and failure-free survival (intermediate).
- **Secondary outcomes:** toxicity, quality of life, cost-effectiveness and symptomatic skeletal events.

| Arm | Treatment | Result |
|---|---|---|
| A (control) | Standard of care: ADT, with or without RT, with or without docetaxel | Not applicable |
| B | Plus zoledronic acid | No increase in overall survival |
| C | Plus docetaxel | Overall survival up by 10 months; more side effects |
| D | Plus celecoxib | No change in overall survival |
| E | Plus zoledronic acid and docetaxel | Increased overall survival |
| F | Plus zoledronic acid and celecoxib | No change in overall survival |
| G | Plus abiraterone acetate and prednisolone | Increased overall survival, progression-free survival and time to radiographic progression |
| J | Plus abiraterone and enzalutamide | In follow-up |
| K | Plus metformin | Recruiting |
| L | Transdermal oestradiol, with or without RT, with or without docetaxel | Recruiting |

▸ **CHAARTED**

- **Design:** phase III randomised; 790 patients enrolled July 2006 to November 2012.
- **Arms:** 1:1 to ADT alone or ADT plus docetaxel; docetaxel started within 4 months of starting ADT.
- **Primary end point:** overall survival.

▸ **LATITUDE**

- **Design:** phase III, double-blind, randomised.
- **Arms:** 1:1 to ADT plus abiraterone 1000 mg once daily plus prednisolone 5 mg daily (n = 597), or ADT plus placebo (n = 602).
- **Co-primary end points:** overall survival and radiographic progression-free survival.
- **Secondary end points:** time to pain progression, PSA progression, next symptomatic skeletal event, chemotherapy and subsequent prostate cancer therapy.
- **Results:** progression-free survival and all secondary end points significantly improved; safety consistent with earlier studies in metastatic castration-sensitive disease.

## Ch 166 · Management of Castration-Resistant Prostate Cancer

### Definition and states of CRPC

- Symptomatic progression alone is not enough to diagnose CRPC (EAU).
- **Non-metastatic CRPC:** about one third develop metastases within 2 years.
- **Bone metastasis in mCRPC:** circulating tumour cells arrest in cortical and medullary bone spaces, adhere to bone surfaces and expand, causing pain and fracture.

## Mechanisms of castration resistance

- **Classic models:** hypersensitive AR pathway, AR promiscuity, outlaw AR pathway, bypass AR pathway and lurker cell model.
- **Hypersensitive:** AR is activated by even lower androgen levels; about one third of tumours show AR gene amplification.
- **Outlaw:** growth factors such as EGF and IGF-1 increase AR transcriptional activity without androgen.
- **Lurker cell:** a small pre-existing population of possible epithelial stem cells grows out under androgen deprivation (inherent, not acquired).

## Antiandrogen withdrawal phenomenon

- **PSA fall:** within 4 weeks of stopping flutamide; within 6 weeks for bicalutamide or nilutamide.
- Objective response can also improve, but is less common than a PSA fall.
- Cause unclear; commoner in men who had rapid PSA responses to androgen ablation.

## Androgen synthesis inhibitors

### ▸ Aminoglutethimide

- **Action:** blocks side-chain cleavage enzyme, so cholesterol is not converted to pregnenolone.
- Causes medical adrenalectomy (blocks aldosterone and cortisol too), so fludrocortisone and cortisone replacement are needed.
- **Dose:** 1000 mg/day, the same as abiraterone.
- **Side effects:** anorexia, skin rash, lethargy, vertigo, nystagmus and hypothyroidism.

### ▸ Ketoconazole

- **Action:** oral broad-spectrum azole antifungal that blocks two cytochrome P450 pathways: lanosterol to cholesterol (inhibiting 14-methylation) and C21 to C19 steroids (blocking 17,20-desmolase).
- Stops adrenal and Leydig-cell steroid synthesis, so hydrocortisone 20 mg twice daily is given with it.
- **Dose:** 400 mg three times daily, continuously, as effects reverse immediately.
- In hormonally intact men, testosterone rises again and can reach low-normal within 5 months.
- **Current use:** second-line hormonal agent in non-metastatic CRPC.
- **Side effects:** gynaecomastia, weakness and lethargy, hepatic dysfunction, nausea and visual disturbance.

### ▸ Abiraterone (Zytiga)

- **Action:** steroidal anti-androgen and irreversible inhibitor of CYP17, the key regulator of extragonadal androgen synthesis; also a partial AR antagonist.
- **Other enzymes inhibited:** 3-beta-hydroxysteroid dehydrogenase (3βHSD), CYP11B1 (steroid 11-beta-hydroxylase) and other CYP450 enzymes.
- **Metabolism:** abiraterone acetate is the prodrug; the more potent metabolite delta-4-abiraterone (D4A) can lower testosterone below 1 ng/dL.

- **Hormone changes, rise:** deoxycorticosterone about 10-fold, corticosterone about 40-fold, 17-OH progesterone about 3-fold, 11-deoxycortisol about 4-fold.
- **Hormone changes, fall:** cortisol about 2-fold, aldosterone about 1.5-fold, DHEA about 3-fold; androstenedione below 2 ng/dL and testosterone below 1 ng/dL.
- Mineralocorticoid excess suppresses renin, lowering aldosterone.
- **First-line mCRPC (with prednisolone):** ECOG 0 or 1 and asymptomatic or mildly symptomatic disease.
- **Very common (over 10%):** hypokalaemia, oedema, diarrhoea, hypertension and urinary tract infection.
- **Common (1–10%):** cardiac failure, angina, arrhythmias including atrial fibrillation and tachycardia, hypertriglyceridaemia, raised transaminases, dyspepsia, fractures, haematuria and sepsis.
- **Uncommon (0.1–1%):** adrenal insufficiency, myopathy, rhabdomyolysis and allergic alveolitis.

## Androgen receptor antagonists

### ▸ Enzalutamide (Xtandi)

- **Action:** strong AR antagonist in the castration-resistant state even with AR overexpression; also impairs AR binding to the transcriptional complex.
- **Dose:** 160 mg as four capsules once daily, swallowed whole, not chewed.
- **Side effects:** asthenia, back pain, arthralgia, hypertension, diarrhoea, hot flushes and neutropenia.
- **CNS side effects:** visual hallucinations, cognitive disorder, memory impairment, vertigo, insomnia and seizures, as it crosses the blood–brain barrier and inhibits the GABA-A receptor.

### ▸ Apalutamide (Erleada, ARN-509)

- Approved February 2018 on the SPARTAN trial; first drug specifically approved for non-metastatic CRPC.
- **Action:** non-steroidal pure AR antagonist; also inhibits AR nuclear translocation and DNA binding.
- **Dose:** 240 mg as four 60 mg tablets once daily.
- **Side effects:** fatigue, nausea, abdominal pain, diarrhoea, breast tenderness and seizures.
- **Contraindications:** pregnancy and history of seizures.

### ▸ SPARTAN trial

- **Design:** multicentre, placebo-controlled, double-blind, phase 3 randomised trial of apalutamide safety and efficacy; duration about 43 months.
- **Arms:** 1027 patients, 2:1 to apalutamide plus ADT or placebo plus ADT; apalutamide 240 mg once daily by mouth as 4 tablets.
- **Secondary outcomes:** time to metastasis, progression-free survival, time to symptomatic progression, overall survival and time to chemotherapy.
- Also improved progression-free survival, time to symptomatic progression and time to PSA progression.

## Chemotherapy

### ▸ Docetaxel

- Also blocks anti-apoptotic signalling by suppressing BCL-XL.
- **mCRPC dose:** 75 mg/m2 3-weekly with prednisolone 5 mg twice daily, up to 10 doses.
- **Other uses:** breast, lung, stomach, ovarian and head and neck cancers.
- **Additional toxicities:** anaemia and thrombocytopenia (myelosuppression), nausea, vomiting, diarrhoea and stomatitis.

### ▸ Cabazitaxel

- Causes fatigue and peripheral neuropathy, the neuropathy less than with docetaxel.

### ▸ Mitoxantrone

- **Action:** semi-synthetic anthracycline and topoisomerase II inhibitor; intercalates between DNA bases, disrupting DNA synthesis and repair.
- **Dose:** 12 mg/m2 3-weekly as a palliative third-line agent.
- Useful for docetaxel- or cabazitaxel-refractory disease or intolerance.
- **Side effects:** nausea and vomiting, hair loss, immunosuppression and irreversible cardiomyopathy.

### ▸ Abiraterone or docetaxel in mCRPC

- Choice depends on patient preference, cost, availability and side effects.

| | Docetaxel | Abiraterone |
|---|---|---|
| Favoured by | Hypertension, diabetes, hypokalaemia | Neuropathy, neutropenia |
| Course and cost | Finite course of injections, lower cost | Prolonged oral therapy, higher cost |

## Immunotherapy

### ▸ Sipuleucel-T (Provenge)

- Personalised vaccine from the patient's own CD54-positive dendritic cells (the major antigen-presenting cells), collected at leukapheresis.
- Not for men needing narcotic analgesia for cancer pain.
- **Side effects:** mild infusion fevers and rigors, plus bladder pain, swelling of face, arms or feet, bloody or cloudy urine, body aches, chest pain, confusion, cough, diarrhoea and sleeplessness.
- Trials are testing it with other AR-directed therapies.

### ▸ PROSTVAC-VF

- **TRICOM costimulatory molecules:** LFA-3, ICAM-1 and B7.1.
- **Schedule:** subcutaneous, 1 priming and 6 boosting doses; generates PSA-specific T cells that kill PSA-expressing cells.
- Relatively inexpensive.
- **Phase III trial:** PROSPECT, in asymptomatic or minimally symptomatic mCRPC.

### ▸ Ipilimumab

- Anti-CTLA-4 monoclonal antibody; promising in mCRPC but not FDA approved for prostate cancer.
- **Schedule:** IV every 3 weeks for 12 weeks, then every 12 weeks.

- **Best benefit:** no visceral metastases, normal haemoglobin and normal ALP.
- **Side effects:** fatigue, nausea, constipation and pruritus.
- **Immune-related:** adrenal insufficiency, other endocrinopathies, dermatitis and vitiligo.

## Targeted treatments

### ▸ PI3K/Akt/mTOR pathway

- Rapamycin, everolimus and temsirolimus failed in mCRPC, alone and with bicalutamide.
- **Under investigation:** BEZ235, MK2206, BKM120 and PX866.

### ▸ Anti-angiogenic agents

- **Bevacizumab:** no overall survival gain; serious toxicity: hypertension, GI perforation, haemorrhage, mucositis and pneumonitis.
- **Aflibercept:** VEGF trap, a decoy receptor binding circulating VEGF before it reaches cellular receptors; no significant OS or PFS difference.
- **Sunitinib:** increased PFS but not OS; not FDA approved for this use.
- **Tasquinimod:** inhibits angiogenesis driven by HIF-1-alpha upregulation and aberrant VEGF expression, and inhibits S100A9 (involved in differentiation and cell-cycle progression); reached phase III.
- **Tasquinimod side effects:** fatigue, nausea, constipation, anorexia, raised lipase and amylase, anaemia and DVT.

### ▸ MET pathway: cabozantinib

- **c-MET:** cell-surface receptor for hepatocyte growth factor; high expression in about 50% of primary cancers at diagnosis and universal in bone metastases.
- **Cabozantinib (XL184):** oral tyrosine kinase inhibitor of c-MET, VEGFR2 and RET.
- **Side effects:** diarrhoea, fatigue, anorexia, rash, hand-foot syndrome, mucositis and raised liver enzymes.

### ▸ Apoptosis pathway: custirsen

- IV antisense oligonucleotide inhibiting clusterin, an anti-apoptotic protein; may reverse taxane resistance.
- **Trials:** SYNERGY and AFFINITY.

## Bone-targeted therapy

### ▸ Bisphosphonates

- **Action:** anti-resorptive; inhibit osteoclast activity and proliferation, maintaining or increasing bone density and strength.
- **Other indications:** bone metastases from breast and lung cancer, postmenopausal osteoporosis, Paget disease of bone and multiple myeloma.
- **Other uses:** fibrous dysplasia, Gaucher disease, paediatric osteogenesis imperfecta, giant cell tumour of the appendicular skeleton and elderly men with non-traumatic fracture.
- **Other agents:** pamidronate, etidronate and clodronate; risedronate and alendronate (oral daily or weekly); ibandronate (oral monthly).

### ▸ Zoledronate

- First approved for hypercalcaemia and for decreased bone density in postmenopausal women.

- **Dose:** 4 mg IV in 100 mL normal saline every 4 weeks for several months (Indian brand Zobone, Sun Pharma).
- **Side effects:** fatigue and anaemia; osteonecrosis of the jaw is unusual and its cause is poorly understood.

▸ **Denosumab (Xgeva)**

- Fully human monoclonal antibody against RANKL.
- 120 mg, marketed as Xgeva; FDA approved November 2010.
- **Side effects:** muscle and joint pain, skin rash, nausea, diarrhoea, constipation, breathlessness, flu-like symptoms, back pain, fatigue, eye changes and rare allergy.
- **Other indications:** bone metastases from breast, kidney and lung cancer; osteoporosis unresponsive to other drugs; unresectable giant cell tumour of bone.

▸ **Radiopharmaceuticals**

- Unsealed-source or molecular radiotherapy.
- Alpha particles are about 7000 times heavier than beta particles: 1–2 hits kill a cell, against 100–1000 hits for beta.
- **Older beta emitters:** strontium-89, samarium-153, phosphorus-32 and rhenium-186, used for multiple painful bone metastases; problem is marrow suppression.
- **Radium-223:** US FDA approved 2013; has replaced strontium-89 and samarium-153; used in men ineligible for chemotherapy or with docetaxel-refractory disease.

## Palliation

▸ **Bone metastases**

- **Pain:** localised, often continuous and unrelenting at any site.
- **Diagnosis:** examination, plain X-ray and bone scan; CT or MRI if suspected but X-ray and bone scan are negative.
- **Prophylactic fixation (orthopaedic referral):** intramedullary lytic lesion of 50% or more of the cross-sectional diameter, or a lytic lesion involving a cortical length equal to or greater than the cross-sectional diameter, or more than 2.5 cm axially.
- Pathological fracture is relatively infrequent with blastic prostate metastases.

▸ **Epidural spinal cord compression**

- Fairly common and potentially devastating; needs early diagnosis and treatment.
- Causes paraplegia and autonomic dysfunction; pain is the main symptom (95%) and precedes diagnosis by about 4 months.
- Function rarely returns once paraplegia occurs.
- **Surgery before radiotherapy:** also indicated for compression by bone and when tissue diagnosis is unknown.
- Chemotherapy is rarely used.

**Other topics in this book**

## Biases in PSA screening

▸ **Lead-time bias**

- **Compensation:** include the lead time when comparing screened and unscreened cohorts, and measure disease-specific mortality rather than survival time.

▸ **Length-time bias**

- Aggressive tumours have a short potential screening period (between possible screen detection and symptoms); slow-growing tumours have a longer one.
- Aggressive tumours therefore tend to present clinically between rounds, so screen-detected tumours appear to survive better; this reflects biology but is wrongly credited to screening, and is stronger with regular-interval screening.
- Overdiagnosis is its extreme form.

▸ **Cost-effectiveness**

- No level 1 evidence that PSA screening is cost-effective in reducing prostate cancer mortality.

## SPARTAN trial: adverse effects

- Fracture, hypothyroidism and rash were more common with apalutamide.

## Multiparametric MRI of the prostate

▸ **Components**

- **Anatomical:** T2-weighted imaging.
- **Functional:** diffusion-weighted and dynamic contrast-enhanced imaging, with MR spectroscopy as the fourth component.
- **Diffusion-weighted:** qualitative and quantitative assessment of aggressiveness.
- **Dynamic contrast-enhanced:** cancer enhances early and strongly; used for detection and monitoring response.
- **MR spectroscopy:** cancer shows low citrate and high choline; usable for detection and response monitoring, but routine use not suggested.

▸ **Clinical applications**

- **Clinically significant cancer:** 0.5 mL or more.
- Poor sensitivity for low-volume Gleason 3+3 cancer, arguably an advantage, as these indolent tumours drive overdiagnosis and overtreatment.
- **Biopsy:** in-bore MRI-guided biopsy is time consuming; MRI/TRUS fusion can be done at the bedside.
- **Active surveillance:** can substitute for, or guide, repeat biopsy.
- **After biopsy:** delay MRI by 6 weeks because haemorrhage and inflammation cause artefacts.

## PET and PET-CT

▸ **Basic principles**

- **Advantage:** gives details of the nervous system.
- **Disadvantages:** ionising radiation, radioactive substances, and poor resolution when used alone.

- **18F-FDG:** a cyclotron produces 18F, which is attached to FDG; hexokinase phosphorylates it to FDG-6-phosphate, trapping it in the cell.

▸ **PET-CT compared with PET**

- CT provides the data for attenuation correction.
- Distinguishes normally high metabolic activity from abnormally increased activity.
- Accurately localises small areas of uptake; excellent spatial resolution, whereas PET alone is poor.

▸ **Preparation**

- **Contraindications:** pregnancy; caffeine, tobacco or alcohol in the past 24 hours; sedatives; drugs that induce or inhibit metabolism.
- Nil by mouth for 4–6 hours, avoiding caffeine and alcohol; avoid strenuous exercise.
- **Blood glucose:** below 150 mg/dL desirable for FDG PET-CT.
- **Bowel opacification:** oral water-soluble iodinated contrast, except in head and neck studies.
- **Dose:** 10 millicurie IV; limit activity for 20 minutes after injection; scan starts at 60 minutes.

▸ **Tracers in prostate cancer**

- **18F-FDG:** higher sensitivity than bone scan, but does not detect soft tissue metastases.
- **11C-choline:** low urinary excretion, so high-quality pelvic imaging.
- **11C-acetate:** detects local recurrence, nodal and distant metastasis; lower detection rate at low PSA.
- **PSMA:** short scanning time and low background signal; main drawback is lack of availability.

## PSMA

▸ **Biology**

- Type II transmembrane protein with intracellular, transmembrane and extracellular portions; gene on the short arm of chromosome 11 (11p).
- Overexpressed 100–1000 fold in prostate cancer, and expressed at all stages.
- Expression rises with tumour aggressiveness, androgen-refractory disease, metastasis and recurrence.
- **Location:** apical side of normal epithelium; in neoplasia or dysplasia it moves from the apical to the luminal surface, with higher expression on less differentiated cells.

▸ **PSMA PET imaging**

- **Agent groups:** new-generation antibodies (64Cu- or 89Zr-labelled), aptamers (64Cu-labelled) and low-molecular-weight PSMA inhibitors (11C-, 18F-, 68Ga-, 64Cu- or 86Y-labelled).
- **Promising agents:** 68Ga-HBED-CC conjugate (HBED-CC is the chelator), 18F-DCFBC and BAY 1075553.
- **Potential indications:** marker of response to ADT; local tumour detection after a previous negative biopsy.

▸ **PSMA-based targeted therapy**

- **Targeted RNA therapy:** siRNA (small interfering RNA) or shRNA (short hairpin RNA) aptamers bind PSMA and are internalised; the RNA interferes with protein synthesis and inhibits DNA repair, increasing radiation-induced cell death (radiosensitisation).

- **Antisense siRNA:** blocks mRNA translation, inhibiting protein synthesis and causing cell death.
- **Radionuclide therapy:** lutetium-177 bound to the J591 monoclonal antibody targets PSMA-expressing cells in bone and soft tissue.

| | Radium-223 (alpha) | Lutetium-177 (beta) |
|---|---|---|
| Destructive range | About 8 cells | 0.25 mm, about 125 cells |
| Cell-killing power | Higher | Lower |
| Imaging | Cannot be imaged | Gamma camera |
| Side effects | Oedema and GI effects | Not given |